L2 AND L3 ACQUISITION OF THE PORTUGUESE STRESSED VOWEL INVENTORY BY NATIVE SPEAKERS OF ENGLISH

by

Miriam Díaz Granado

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Miriam Díaz Granado

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DEDICATION

A mis abuelos

TABLE OF CONTENTS

LIST OF FIGURES 10

LIST OF TABLES 15

ABSTRACT 22

CHAPTER 1 INTRODUCTION 24
1.1 Presentation 24
1.2 Learning to pronounce a new language 25
1.3 Research on L2 phonetic acquisition 26
1.3.1 Models of cross-linguistic sound perception and production . . 28
1.3.2 Phonetics and the acquisition of an L3 34
1.4 This dissertation 40

CHAPTER 2 NATIVE VOWEL INVENTORIES 45
2.1 Method 46
2.1.1 Materials 46
2.1.2 Participants 47
2.1.3 Procedure 49
2.1.4 Data analysis 53
2.2 Results 54
2.2.1 Specific vowel comparisons and statistical analysis of the data 55
2.2.2 The diphthongization of mid vowels 74
2.3 Summary of Results and Discussion 96
2.4 Concluding Remarks 100

CHAPTER 3 VOWELS OF L2 AND L3 LEARNERS OF PORTUGUESE . 102
3.1 Predictions for L2 learning 104
3.1.1 Predictions for the learning of the Portuguese high front vowel 104
3.1.2 Predictions for the learning of the Portuguese high back vowel 105
3.1.3 Predictions for the learning of the Portuguese low vowel . . . 107
3.1.4 Predictions for the learning of the Portuguese mid-back vowels 110
3.1.5 Predictions for the learning of the Portuguese mid-front vowels 112
3.2 Method 114
3.2.1 Materials 114
3.2.2 Participants 115

TABLE OF CONTENTS – *Continued*

3.2.3 Procedure 118
3.2.4 Data analysis 119
3.3 Results 120
3.3.1 Learners vs. Native speakers' Portuguese vowels 120
3.3.2 Learners vs. Native speakers' Portuguese vowels: an alternative analysis 167
3.4 Summary of results and discussion of the predictions for L2 learning . 178

CHAPTER 4 PERCEPTION OF PORTUGUESE /e/-/ɛ/ BY L2 AND L3 LEARNERS 184
4.1 Experiment 4.1 185
4.1.1 Method 185
4.1.2 Results 187
4.2 Experiment 4.2 190
4.2.1 Method 190
4.2.2 Results 200
4.3 Summary of overall results and Discussion 210

CHAPTER 5 DISCUSSION AND CONCLUSIONS 213
5.1 Introduction 213
5.2 Summary of Results 213
5.2.1 Comparison between native vowel inventories 213
5.2.2 L2 and L3 speakers' production of Portuguese vowels 214
5.2.3 L2 and L3 speakers' perception of Portuguese /e/-/ɛ/ 215
5.3 The importance of input in L2 and L3 speech learning 216
5.4 Speech perception, production and signs of learning in the acquisition of a new sound inventory 220
5.5 Searching for a model of L2 and L3 phonetic acquisition 222
5.6 Vowel dispersion and L3 speech learning 223
5.7 Acquiring multiple languages in different orders 224
5.8 Pedagogical applications of current research 227
5.9 Future research 228
5.10 Final words 230

APPENDIX A LANGUAGE BACKGROUND QUESTIONNAIRE 232

APPENDIX B RESULTS FROM CROSS-LINGUISTIC COMPARISONS OF MID VOWELS OF ENGLISH SPANISH AND PORTUGUESE AT DIFFERENT POINTS OF MEASUREMENT 236

TABLE OF CONTENTS – *Continued*

REFERENCES 240

LIST OF FIGURES

2.1 Vowels of English, Spanish and Portuguese as produced by the seven native speaker participants in each group. 56
2.2 High front vowels of English, Spanish and Portuguese, as produced by the three native speaker groups. 59
2.3 High back vowels of English, Spanish and Portuguese as produced by the three native speaker groups. 62
2.4 Low vowels of English, Spanish and Portuguese as produced by the three native speaker groups. 64
2.5 Mid-back vowels of English, Spanish and Portuguese as produced by the three native speaker groups. 68
2.6 Mid-front vowels of English, Spanish and Portuguese as produced by the three native speaker groups. 72
2.7 Movement of the mid-back vowels of English, Portuguese and Spanish in height through time (Diff1). English /o/ is referred to as "E.o," Spanish /o/ as "S.o" and Portuguese /ɔ/ and /o/ as "P.O" and "P.o" respectively. Negative Diff1 values represent raising of the vowels in the vowel space while positive values represent lowering. The graph shows that (a) English /o/ is the most diphthongizing mid-back vowel of all moving upwards through the length of the vowel, (b) Portuguese /ɔ/ moves down, (c) Portuguese /o/ moves down to a greater degree than Portuguese /ɔ/, and (d) Spanish /o/ does not show any movement at all. 77
2.8 Movement of the mid-back vowels of English, Portuguese and Spanish in backness through time (Diff2). English /o/ is referred to as "E.o," Spanish /o/ as "S.o" and Portuguese /ɔ/ and /o/ as "P.O" and "P.o" respectively. Negative Diff2 values represent movement of the vowels backwards in the vowel space through time, while positive values represent fronting. The graph shows that (a) English /o/ moves backwards, (b) the two Portuguese vowels move backwards through the length of the vowel, and (c) Spanish /o/ remains stable. . 78
2.9 Movement of mid-back vowels of English, Spanish and Portuguese through time (top=25%, middle=midpoint, bottom=75% of vowel). English /e/ and /ɛ/ are graphed using a solid line, Spanish /e/ using a dashed line, and Portuguese /e/ and /ɛ/ use a dotted line. 82

LIST OF FIGURES – *Continued*

2.10 Movement of the mid-front vowels of English, Portuguese and Spanish in height through time (Diff1). English /e/ and /ɛ/ are referred to as "E.e" and "E.E" respectively, Portuguese /e/ and /ɛ/ as "P.e" and "P.E," and Spanish /e/ as "S.e" in that order. Negative Diff1 values represent raising of the vowels in the vowel space while positive values represent lowering. The graph shows that (a) English /e/ undergoes the most amount of movement in height, rising through time; (b) English /ɛ/ moves upwards in the vowel space very slightly; (c) the two Portuguese mid-front vowels move lower in the vowel space through time, and (d) Spanish /e/ is the most stable of all five. . . . 86

2.11 Movement of the mid-back vowels of English, Portuguese and Spanish in backness through time (Diff2). English /e/ and /ɛ/ are referred to as "E.e" and "E.E" respectively, Portuguese /e/ and /ɛ/ as "P.e" and "P.E," and Spanish /e/ as "S.e" in that order. Negative Diff2 values represent movement of the vowels backwards in the vowel space through time, while positive values represent fronting. The graph shows that (a) English /e/ moves clearly backwards; (b) English /ɛ/ shows some variability in its degree of backwards movement in the vowel space with its mean nearing zero; (c) the two Portuguese vowels move backwards through the length of the vowel; and (d) Spanish /e/ remains stable. 87

2.12 Movement of mid-front vowels of English, Spanish and Portuguese through time (top=25%, middle=midpoint, bottom=75% of vowel). English /e/ and /ɛ/ are graphed using a solid line, Spanish /e/ using a dashed line, and Portuguese /e/ and /ɛ/ use a dotted line. 93

3.1 Portuguese vowels produced by Portuguese native speakers, L2 group and L3 group. 121

3.2 Portuguese /u/ and English /u/ vowels of both L2 and L3 groups at their midpoint (top = logF1, bottom = logF2). 'E' and 'P' represent the 'language' factor, L2 and L3 represent the 'group' factor. 128

3.3 English /ɑ/ (E.a), English /æ/ (E.ae) and Portuguese /a/ (P.a) vowels of both L2 and L3 groups at their midpoint (top = logF1, bottom = logF2). 134

3.4 English vowels of all three Native, L2 and L3 groups. Vowels produced by the Native group are graphed using a solid line, those produced by the L2 group use a dashed line and those produced by the L3 group use a dotted line. 135

LIST OF FIGURES – *Continued*

3.5 Density plots of logF2 values for Portuguese /a/ tokens produced by each of the three participant groups: Native group (top), L2 group (bottom left) and L3 group (bottom right). 138
3.6 Density plots of logF2 values for Portuguese /a/ tokens produced by each of the participants in the L3 group: from left to right, participants T03, T05, T09 (top), T13, T17 and T19 (middle) and T20 (bottom). 140
3.7 Density plots of logF2 values for Portuguese /a/ tokens produced by each of the participants in the L2 group: from left to right, participants B01, B02, and B03 (top), B04, B05 and B06 (middle), and B07 (bottom). 141
3.8 Density plots of logF2 values for Portuguese /a/ tokens produced by each of the participants in the Native Portuguese speaking group: from left to right, participants N02, N07, N10 (top), N11, N12, N13 (middle) and N14 (bottom). 142
3.9 Mean logF1 values of the Portuguese /o/ (P.o) and Portuguese /ɔ/ (P.O) as produced by Native speakers, L2 group and L3 group. Graphed values are those measured at the midpoint of the vowel. . . 144
3.10 Mean logF2 values of the Portuguese /o/ (P.o) and Portuguese /ɔ/ (P.O) as produced by Native speakers, L2 group and L3 group. Graphed values are those measured at the midpoint of the vowel. . . 145
3.11 Degree of diphthongization in height of English /o/ (E.o), Portuguese /o/ (P.o) and Portuguese /ɔ/ (P.O) as produced by the L2 and L3 groups. Values graphed correspond to mean Diff1 values. 149
3.12 Degree of diphthongization in backness of English /o/ (E.o), Portuguese /o/ (P.o) and Portuguese /ɔ/ (P.O) as produced by the L2 and L3 groups. Values graphed correspond to mean Diff2 values. . . . 150
3.13 Diphthongization in height of the Portuguese /o/ (P.o) and Portuguese /ɔ/ (P.O) produced by the Native, L2 and L3 groups. Values graphed correspond to mean Diff1 values. 151
3.14 Diphthongization in backness of the Portuguese /o/ (P.o) and Portuguese /ɔ/ (P.O) produced by the Native, L2 and L3 groups. Values graphed correspond to mean Diff2 values. 152
3.15 Mean logF1 values of the Portuguese /e/ (P.e) and Portuguese /ɛ/ (P.E) as produced by the Native, L2 and L3 groups. Graphed values are those measured at the midpoint of the vowel. 155
3.16 Mean logF2 values of the Portuguese /e/ (P.e) and Portuguese /ɛ/ (P.E) as produced by the Native, L2 and L3 groups. Graphed values are those measured at the midpoint of the vowel. 156

LIST OF FIGURES – *Continued*

3.17 Mean logF1 values of English /e/ (E.e), English /ε/ (E.E), Portuguese /e/ (P.e) and /ε/ (P.E) produced by the L2 and L3 groups. Graphed values are those measured at the midpoint of the vowel. 157
3.18 Mean logF2 values of English /e/ (E.e), English /ε/ (E.E), Portuguese /e/ (P.e) and /ε/ (P.E) produced by the L2 and L3 groups. Graphed values are those measured at the midpoint of the vowel. 158
3.19 Vowel space of participants in L2 group with vowels of English and Portuguese in it. 161
3.20 Vowel space of participants in L3 group with vowels of English, Spanish and Portuguese in it. 162
3.21 Degree of diphthongization in height of English /e/ (E.e), English /ε/ (E.E), Portuguese /e/ (P.e) and Portuguese /ε/ (P.E) as produced by the L2 and L3 groups. Values graphed correspond to mean Diff1 values. 164
3.22 Degree of diphthongization in backness of English /e/ (E.e), English /ε/ (E.E), Portuguese /e/ (P.e) and Portuguese /ε/ (P.E) as produced by the L2 and L3 groups. Values graphed correspond to mean Diff2 values. 165
3.23 Degree of diphthongization in height of Portuguese /e/ (P.e) and Portuguese /ε/ (P.E) produced by the Native, L2 and L3 groups. Graphed values are mean Diff1 values. 168
3.24 Degree of diphthongization in backness of Portuguese /e/ (P.e) and Portuguese /ε/ (P.E) produced by the Native, L2 and L3 groups. Graphed values are mean Diff2 values. 168
3.25 Means of Euclidean Distances from the centroid for each of the Portuguese vowels produced by the participants in the study and separated by group, where B refers to the L2 group, N refers to the NP group and T refers to the L3 group. The graph shows a significant group by vowel interaction. 173

4.1 Proportion of Portuguese /e/ (labeled 'e' and represented by dark blue bars) and /ε/ tokens (labeled 'EE' and represented by green bars) identified by the native English (NE), L2 group and L3 group as the English /ε/ in the word 'bet.' A score of 1.0 indicates perfect fit between the Portuguese vowel and English /ε/ while a score of 0.5 indicates insensitivity to similarities between the Portuguese vowel heard in each case and English /ε/. 189

LIST OF FIGURES – *Continued*

4.2 Proportion of mean accurate responses in change trials for each vowel contrast-pair (see list in Table 4.3) separated by language group (L2, L3, NE and NP). Contrast-pairs, which follow, left to right, the same order as the vowel pairs in Table 4.3 above, are labeled according to the vowel categories and languages they consist of. A score of 1.0 indicates perfect discrimination, and a score of 0.33 indicates insensibility to a contrast. 202
4.3 Proportion of mean accurate responses in change trials for each language group (L2, L3, NE and NP). Numbers inside bars represent Mean accurate response scores for each group. A score of 1.0 indicates perfect discrimination while a score of 0.33 indicates insensitivity to a contrast. No statistically significant differences were found among groups. 203
4.4 Proportion of accurate responses in change trials by the L2 (in blue) and L3 group (in green), separated by trials with (right side) and without (left side) the Spanish mid-front vowel involved. Graph shows that both groups obtained significantly higher accuracy scores when discriminating between vowel pairs which did not contain the Spanish vowel than when discriminating between pairs that involved Spanish. 206
4.5 Proportion of accurate responses in no-change trials consisting of each of the five mid-front vowels of English, Portuguese and Spanish as provided by the NP, NE, L2 and L3 groups. A score of 1.0 indicates perfect identification of the stimulus-strings as representing a single vowel category while 0.33 represents chance level. 209

LIST OF TABLES

2.1 English target words with vowels /ɪ/, /i/, /ɛ/ and /e/ used in elicitation materials. 46
2.2 English target words with vowels /a/, /ɑ/, /o/ and /u/ used in elicitation materials. 47
2.3 Spanish target words used in elicitation materials. 48
2.4 Portuguese target words with /i/, /e/ and /ɛ/ used in elicitation materials. 49
2.5 Portuguese target words with /a/, /ɔ/, /o/ and /u/ used in elicitation materials. 50
2.6 Participants' biographic and linguistic background where Age is the mean age at time of testing, AOD represents the age of departure of participants from their country of origin when applicable, and M/F represents their gender; L1UseGen represents amount of L1 use in general in their daily lives and L1UseFam amount of L1 use in their families; ranges are presented in parentheses. 51
2.7 English vowels: first and second formant data (in Hz). 54
2.8 Spanish vowels: first and second formant data (in Hz). 54
2.9 Portuguese vowels: first and second formant data (in Hz). 55
2.10 Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the high front vowels of English, Spanish and Portuguese, with Portuguese /i/ as the intercept. 58
2.11 Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the high front vowels of English, Spanish and Portuguese, with Spanish /i/ as the intercept. 60
2.12 Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the high front vowels of English, Spanish and Portuguese, with English /i/ as the intercept. 60
2.13 Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the high back vowels of English, Spanish and Portuguese, with Portuguese /u/ as the intercept. 61
2.14 Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the high back vowels of English, Spanish and Portuguese, with Spanish /u/ as the intercept. 63
2.15 Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the low vowels of English, Spanish and Portuguese, with Portuguese /a/ as the intercept. 65

LIST OF TABLES – *Continued*

2.16 Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the low vowels of English, Spanish and Portuguese, with English /ɑ/ as the intercept. 65
2.17 Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the low vowels of English, Spanish and Portuguese, with Spanish /a/ as the intercept. 66
2.18 Summary of statistically significant differences among low vowels of English, Spanish and Portuguese. 67
2.19 Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the mid-back vowels of English, Spanish and Portuguese, with Portuguese /ɔ/ as the intercept. 68
2.20 Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the mid-back vowels of English, Spanish and Portuguese, with Spanish /o/ as the intercept. 69
2.21 Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the mid-back vowels of English, Spanish and Portuguese, with Portuguese /o/ as the intercept. 70
2.22 Summary of statistically significant differences among mid-back vowels of English, Spanish and Portuguese. 70
2.23 Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the mid-front vowels of English, Spanish and Portuguese, with Portuguese /ɛ/ as the intercept. 71
2.24 Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the mid-front vowels of English, Spanish and Portuguese, with Spanish /e/ as the intercept. 72
2.25 Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the mid-front vowels of English, Spanish and Portuguese, with English /e/ as the intercept. 73
2.26 Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the mid-front vowels of English, Spanish and Portuguese, with Portuguese /e/ as the intercept. 74
2.27 Summary of statistically significant differences among mid-front vowels of English, Spanish and Portuguese. 75
2.28 Results from a linear mixed-effects model on mid-back vowel diphthongization data in the three languages, with Portuguese /ɔ/ as the intercept. 79
2.29 Results from a linear mixed-effects model on mid-back vowel diphthongization data in the three languages, with Spanish /o/ as the intercept. 80

LIST OF TABLES – *Continued*

2.30 Results from a linear mixed-effects model on mid-back vowel diphthongization data in the three languages, with English /o/ as the intercept. 80
2.31 Summary of statistically significant differences among mid-back vowels of English, Spanish and Portuguese after a set of statistical analyses on the logF1 and logF2 values at each of the three individual points of measurement through the duration of the vowels. 83
2.32 Results from a linear mixed-effects model on mid-front vowel diphthongization data in the three languages, with Spanish /e/ as the intercept. 88
2.33 Results from a linear mixed-effects model on mid-front vowel diphthongization data in the three languages, with English /ɛ/ as the intercept. 89
2.34 Results from a linear mixed-effects model on mid-front vowel diphthongization data in the three languages, with English /e/ as the intercept. 90
2.35 Results from a linear mixed-effects model on mid-front vowel diphthongization data in the three languages, with Portuguese /ɛ/ as the intercept. 90
2.36 Summary of results from multiple linear mixed-effects models on English, Spanish and Portuguese mid-front vowel diphthongization data. 91
2.37 Summary of statistically significant differences among mid-front vowels of English, Spanish and Portuguese after analysis of differences in time. 94

3.1 Summary of participants' biographic and age of acquisition data where Age is the mean age of the participants at time of testing; M/F represents their gender; AOASpan and AOAPort represent the Age Of Acquisition of Spanish and/or Portuguese respectively; ranges are presented in parentheses. 115
3.2 Participants' language use and study abroad experience where PortUse5m and PortUse5w represent the percentage of self-reported Portuguese use during the 5 months or 5 weeks previous to the time of testing and StAbSpan and StAbPort represent the amount of time (in months) spent in a foreign country where each of the target languages were spoken. 117
3.3 Native Portuguese speakers' Portuguese vowels: F1 and F2 data in Hz. 122
3.4 L2 group's Portuguese vowels: F1 and F2 data in Hz. 122
3.5 L3 group's Portuguese vowels: F1 and F2 data in Hz. 123

LIST OF TABLES – *Continued*

3.6 Results from a linear mixed-effects model with native speakers' Portuguese /i/ as the intercept. 123
3.7 Results from a set of linear mixed-effects models on the Portuguese /i/ and English /i/ produced by the L2 and L3 groups, with the Portuguese /i/ produced by the L2 group as the intercept. 125
3.8 Results from a linear mixed-effects model on the Portuguese /u/ and English /u/ produced by the Native Portuguese, L2 and L3 groups with the native speakers' Portuguese /u/ as the intercept. 126
3.9 Results from a linear mixed-effects model on the Portuguese /u/ and English /u/ produced by the L2 and L3 groups with English /u/ produced by the L2 group as the intercept. 127
3.10 Results from multiple linear mixed-effects models on the effect of 'language' on the Portuguese /u/ and English /u/ produced by the L2 and L3 groups separately. English /u/ was taken as the intercept in all cases. 130
3.11 Results from a linear mixed-effects model on the Portuguese /a/ produced by the Native Portuguese, L2 and L3 groups. 131
3.12 Results from a linear mixed-effects model on the Portuguese /a/ and English /æ/ and /ɑ/ produced by the L2 and L3 groups with L2 English /æ/ as the intercept. 132
3.13 Results from a linear mixed-effects model on the Portuguese /a/ and English /æ/ and /ɑ/ produced by the L2 and L3 groups with L2 Portuguese /a/ as the intercept. 132
3.14 Results from four linear mixed-effects models on SD (in Hz) values at the midpoint of Portuguese /a/ tokens as produced by all three Native Portuguese, L2 and L3 groups, with different intercepts. . . . 136
3.15 Results from two linear mixed-effects models on logF1 and logF2 values of Portuguese /o/ and /ɔ/ as produced by the native, L2 and L3 groups with L2 Port. /ɔ/ as the intercept. 143
3.16 Results from two 2x3 linear mixed-effects models on Diff1 and Diff2 values of English /o/, Portuguese /o/ and Portuguese /ɔ/ as produced by the L2 and L3 groups with L2 English /o/ as the intercept. 148
3.17 Results from two linear mixed-effects models on Diff1 and Diff2 values of Portuguese mid-back vowels as produced by the native, L2 and L3 groups with the L2 Portuguese /ɔ/ as the intercept. 149
3.18 Results from two linear mixed-effects models on logF1 and logF2 values of Portuguese /e/ and /ɛ/ as produced by the native, L2 and L3 groups with the L2 group's Portuguese /e/ as the intercept. 154

LIST OF TABLES – *Continued*

3.19 Results from two linear mixed-effects models on logF1 and logF2 values of English /e/, English /ɛ/, Portuguese /e/ and /ɛ/ as produced by the L2 and L3 groups with L2 English /e/ as the intercept. 157
3.20 Results from two linear mixed-effects models on logF1 and logF2 values of English /e/, English /ɛ/, Portuguese /e/ and /ɛ/ as produced by the L2 and L3 groups with L2 Portuguese /ɛ/ as the intercept. . . 159
3.21 Results from two linear mixed-effects models on Diff1 and Diff2 values of English /e/, English /ɛ/, Portuguese /e/ and /ɛ/ as produced by the L2 and L3 groups with L2 English /e/ as the intercept. 163
3.22 Results from two linear mixed-effects models on Diff1 and Diff2 values of English /e/, English /ɛ/, Portuguese /e/ and /ɛ/ as produced by the L2 and L3 groups with L2 Portuguese /ɛ/ as the intercept. 166
3.23 Results from two linear mixed-effects models on Diff1 and Diff2 values of Portuguese /e/ (="15") and /ɛ/ (="16") as produced by the native (="N"), L2 (="B") and L3 ("T") groups with L2 Portuguese /e/ as the intercept. 167
3.24 Results from a two-factor linear mixed-effects regression model on the Portuguese vowels produced by the NP, L2 and L3 groups, with 'distance to centroid' (ED) as the response, 'group' (NP, L2, L3) and 'vowel' (Portuguese /i/, /e/, /ɛ/, /a/, /ɔ/, /o/, /u/) as fixed factors, and 'speaker' and 'word' as random factors. The L2 group's Portuguese /i/ was the intercept. 172
3.25 Results from a linear mixed-effects regression model testing for differences in distance of Portuguese /i/ from the centroid among the three groups of participants (NP, L2, L3). 174
3.26 Results from a linear mixed-effects regression model testing for differences in distance of Portuguese /e/ from the centroid among the three groups of participants (NP, L2, L3). 175
3.27 Results from a linear mixed-effects regression model testing for differences in distance of Portuguese /ɛ/ from the centroid among the three groups of participants (NP, L2, L3). 175
3.28 Results from a linear mixed-effects regression model testing for differences in distance of Portuguese /a/ from the centroid among the three groups of participants (NP, L2, L3). 176
3.29 Results from a linear mixed-effects regression model testing for differences in distance of Portuguese /ɔ/ from the centroid among the three groups of participants (NP, L2, L3). 176

LIST OF TABLES – *Continued*

3.30 Results from a linear mixed-effects regression model testing for differences in distance of Portuguese /o/ from the centroid among the three groups of participants (NP, L2, L3). 177
3.31 Results from a linear mixed-effects regression model testing for differences in distance of Portuguese /u/ from the centroid among the three groups of participants (NP, L2, L3). 177

4.1 Participants' biographic and linguistic background where M/F represents their gender; Age is the mean age at time of testing; SpanAOA represents age of acquisition of Spanish and AOAPort represents age of acquisition of Portuguese. 186
4.2 Participants' biographic and linguistic background where M/F represents their gender; Age is the mean age at time of testing; SpanAOA represents age of acquisition of Spanish and AOAPort represents age of acquisition of Portuguese. 191
4.3 Contrast pairs tested in the category discrimination test in Experiment 4.2. 194
4.4 Architecture for the stimulus-strings in Experiment 4.2. where capital letters refer to the first ("A") or second ("B") member of the vowel pair, numbers ("1", "2" or "3") refer to the Speaker producing the specific vowel token, and lowercase letters ("a" or "b") refer to the specific token produced by the Speaker in question. 12 change trials were built for each contrast pair, 6 with base A (1-6) and 6 with base B (7-12). 6 more no-change trials were built for each contrast pair, 3 for each member of the pair. 195
4.5 First step into building an architecture for the stimulus-strings representing the change trials for the English /e/ - English /ɛ/ vowel pair in Experiment 4.2. 196
4.6 Second step into building an architecture for the stimulus-strings representing the change trials for the English /e/ - English /ɛ/ vowel pair in Experiment 4.2. 196
4.7 Architecture for the stimulus-strings representing the change trials in the category discrimination test in Experiment 4.2 for the within English vowel pair, where English /ɛ/ was the odd-item out. 197
4.8 Architecture for the stimulus-strings representing the change trials in the category discrimination test in Experiment 4.2 for the within English vowel pair, where English /e/ was the odd-item out. 198
4.9 Architecture for the stimulus-strings representing the no-change trials in the category discrimination test in Experiment 4.2. 199

LIST OF TABLES – *Continued*

B.1 Results from a set of linear mixed-effects models on each of the logF1 and logF2 measures of the mid back vowels of English, Spanish and Portuguese, at the three points of measurement, with Portuguese /ɔ/ as the intercept. 236
B.2 Results from a set of linear mixed-effects models on each of the logF1 and logF2 measures of the mid back vowels of English, Spanish and Portuguese, at the three points of measurement, with Spanish /o/ as the intercept. 237
B.3 Results from a set of linear mixed-effects models on each of the logF1 and logF2 measures of the mid back vowels of English, Spanish and Portuguese, at the three points of measurement, with Portuguese /o/ as the intercept. 237
B.4 Results from a set of linear mixed-effects models on each of the logF1 and logF2 measures of the mid front vowels of English, Spanish and Portuguese, at the three points of measurement, with Portuguese /ɛ/ as the intercept. 238
B.5 Results from a set of linear mixed-effects models on each of the logF1 and logF2 measures of the mid front vowels of English, Spanish and Portuguese, at the three points of measurement, with Spanish /e/ as the intercept. 238
B.6 Results from a set of linear mixed-effects models on each of the logF1 and logF2 measures of the mid front vowels of English, Spanish and Portuguese, at the three points of measurement, with English /e/ as the intercept. 239
B.7 Results from a set of linear mixed-effects models on each of the logF1 and logF2 measures of the mid front vowels of English, Spanish and Portuguese, at the three points of measurement, with Portuguese /e/ as the intercept. 239

ABSTRACT

This dissertation explores the potential differences in the acquisition of the sound system of a second language (L2) versus a third language (L3), building on recent research on adult L2 speech learning and testing the hypothesis that new category acquisition is available across the lifespan. On the one hand, recent influential theories of L2 speech learning predict that new sound categories will be difficult to acquire due to complex interactions among the phonetic categories residing in one same perceptual space. On the other, there exists the common assumption that the more sound categories ones native language contains, the less difficult it will be to acquire new ones in a native-like fashion. My work provides additional evidence to this discussion, while providing speech data from the following five different speaker groups: (1) native speakers of American English; (2) native speakers of Mexican Spanish; (3) native speakers of Brazilian Portuguese; (4) native speakers of English learning Portuguese as their L2, and (5) native speakers of English learning Portuguese as their L3, who speak Spanish as their L2.

The dissertation consists of three content chapters. The first of the content chapters describes the vowel systems of Mexican Spanish, American English and Brazilian Portuguese as informed by production data of the three native speaker groups above (1, 2 and 3). The second and third chapters describe and contrast the vowel system(s) of the two learner groups (4 and 5) as informed by their production and perception of Portuguese vowels respectively. The results from the analysis of the data from the various production and perception experiments performed as part of this dissertation provide evidence for (a) phonetic category assimilation and dissimilation processes in post-L1 speech learning and phonetic category interactions in general; (b) the relevance of quality and quantity of input in language learning; (c) the discussion on the relation between perception and production in post-L1

speech learning; and (d) the need for an extension of current models of L2 speech learning and cross-linguistic speech perception in order for these to address post-L2 speech learning processes, among other topics.

CHAPTER 1

INTRODUCTION

1.1 Presentation

The ability to speak multiple languages proficiently is a highly desirable skill in today's increasingly global world. Still, the phenomenon that we call "foreign accent" is an aspect of language that affects both speakers and listeners in both perception and production, and, consequently, in social interaction.

The effect that previously learned languages may have in the process of learning a new language is known as cross-linguistic influence. Much of the research on cross-linguistic phonetic influence in the past few decades has pinpointed the fact that the speakers' knowledge of their L1 seems to warp one way or another their accurate perception and production of foreign sounds, and a large part of the studies have focused on how language users perceive and produce foreign speech and what factors affect the degree of foreign accent in Second Language Acquisition (SLA).

With the increasing demand of a multilingual population, however, the field of SLA has recently experienced a need to expand. The results from studies on second language (L2) phonetic acquisition do not seem to provide the answers to questions arising from cross-linguistic influence when more than two languages are involved in the process. Thus, it seems only reasonable to believe that the study of cross-linguistic influence in third language (L3) speech learning is potentially more complex than the study of cross-linguistic influence in second language (L2) speech learning and that the mechanisms involved in L3 speech learning might include those implicated in L2 speech learning as well as additional processes rooted in the multifaceted relationships between the multiple languages in question.

The present research study addresses the topic of the acquisition of a non-native phonetic system by bringing to the table the potential difference between the ac-

quisition of a new sound system as an L2 versus as an L3. Thus, this dissertation reports on a series of phonetic experiments that specifically investigate the acquisition of the stressed vowel system of Portuguese by two groups of native speakers of English. The first group learned Portuguese as an L2, while the second group learned Portuguese as an L3, having had previous and more extensive experience with Spanish as an L2.

This way, this dissertation seeks to provide evidence that will, first, help advance our knowledge on the topic of L2 phonetic acquisition, and second, expand on currently available theoretical models with consequences for second language speech learning by examining the effect of having an L2 vowel system in one's inventory when acquiring an L3.

1.2 Learning to pronounce a new language

It is widely known that individuals who learned an L2 often speak it with a foreign accent. Many studies in the past decade or two have found evidence that "earlier is better" as far as learning an L2 is concerned, with individuals who started learning an L2 in childhood resembling native speakers of the target language more than those whose started learning the L2 in adolescence or early adulthood do. This, of course, is not a categorical statement. There are a number of individuals who acquired an L2 in adulthood who have been found to show native-like proficiency in the target language. This, however, is not the norm. From all possible linguistic aspects (syntax, morphology, vocabulary) in which a user of a second/foreign language could differ from the native norm, pronunciation has been known as the most revealing in giving the speaker's non-nativeness away (Saville-Troike (2006)).

Many research studies have investigated the effects of different factors in the relative success in SLA. Especially in the area of the acquisition of L2 speech patterns, Flege and his colleagues have done extensive research to establish the relevant factors affecting degree of foreign accent in an L2, many of which are often confounded with "age" (Flege (2007); Flege et al. (1995b)). More specifically, Flege and his

colleagues identify the amount and quality of L2 input (Flege & Liu (2002)), as well as amount of continued L1 use (Flege et al. (1999b); Guion et al. (2000b))[1], as the most relevant factors in determining likelihood of success in the acquisition of a native-like L2 pronunciation. When faced with the study of L3 acquisition, the factors identified by Flege and his colleagues remain relevant, nevertheless, the interactions between different kinds and amounts of input in each language and age of acquisition of each language, among other potentially significant factors, turn the study of L3 phonetic acquisition into a different kind of creature.

The most recent theoretical models with consequences for the acquisition of the sound system of a new language begin from the idea that it is the learners' perception of the similarities and differences between the sounds in the L2 and the most similar sounds in their L1 that determines how successful they will be in mastering the L2 phonetic system. Among these theoretical frameworks, the Speech Learning Model (SLM hereafter), proposed by Flege (1995), and the Perceptual Assimilation Model (PAM hereafter), proposed by Best (1995), are crucial to understanding the theoretical bases for the present study. Both models, which address different aspects of cross-linguistic sound perception or production, will be discussed in this introductory chapter.

1.3 Research on L2 phonetic acquisition

Although not especially designed to address the process of acquisition of an L2, the most current model addressing L1 phonetic acquisition, the Native Language Magnet Model (NLM), also makes predictions for SLA. The NLM (Kuhl (1993, 2000)) assumes a "blank state" situation at time of birth, and proposes that language experience produces a change in infants' discriminative abilities and listening preferences. Thus, according to the NLM, infants' perception of speech sounds is

[1]These two factors were, before Flege's latest work on the topic, represented by the factor of Age of Arrival in the foreign country (AOA) for the immigrant populations that were studied. In his latest work, he finds evidence that AOA is representative of the combination of 'amount and quantity of L2 input' and 'amount of continued L1 use.'

attuned to their native language during the first 6-12 months of life, creating a filter through which language sounds are perceived.

The NLM argues for the notion of prototypical sounds, which are here interpreted to be equivalent to the best exemplar of a phonetic category. These prototypical sounds in the L1 would attract other category-like sounds, building up the perceptual space of phonetic categories. This magnet effect of prototypical sounds happens, according to the NLM, in infants' L1 as they are exposed to their native language. Thus, the perceptual space of phonetic categories is established, producing a complex network through which language is perceived, and constraining the perception of L2 speech sounds in the future. In adults, the distortion of perception caused by language experience is clearly observable in L2 learners past the critical period. However, the NLM proposes that constraints on the perception of L2 speech sounds arise from prior experience, not from loss of neural plasticity, as findings supporting the Critical Period Hypothesis have implied (Johnson & Newport (1989); Scovel (1988); Patkowski (1980, 1990)).

The NLM seems to be supported by the findings from four different studies testing differences between native Spanish learners of Catalan and native speakers of Catalan in Barcelona. These studies assessed the perception of Catalan /e/ and /ɛ/ vowels by young native Spanish adults who had begun learning Catalan in Barcelona by the age of 6 years. These two Catalan vowels occur in a portion of vowel space that is occupied by a single Spanish vowel, Spanish /e/[2]. In all of these studies, the Spanish early learners of Catalan were found to perform worse than native Catalan speakers in perceiving the Catalan vowel contrast (Bosch & Costa (2000); Pallier et al. (1997, 2001); Sebastián-Gallés & Soto-Faraco (1999)). The findings from these studies agree with the predictions of the NLM for L2 acquisition in that they suggest that phonemic categories not only are acquired early in life but their acquisition can be seriously compromised despite early and extensive exposure to L2.

[2]Bosch & Costa (2000) describes Spanish /e/ as having both [e] and [ɛ] allophones and as occurring near the perceptual boundary of Catalan /e ɛ/

Contrary to the fatalistic prognosis for L2 perceptual learning proposed by the NLM and supported by the four Catalan /e ɛ/ studies in Barcelona, a vast amount of research has focused on differences between the L2 speech of L2 learners of different languages, with differing amounts of L1 daily use, and different ages of acquisition. The majority of these studies conclude, concurring with common knowledge, that 'earlier is better' for L2 acquisition (Flege et al. (1995b, 1999b); Flege (1992); Flege et al. (1999a); Piske et al. (2001); Flege (1991); MacKay et al. (2001b,a)), with some of them reporting native-like attainment of specific phonetic cues by speakers of the language as an L2, as in the case of early Spanish speaking learners of English who produced word-initial English /t/ with mean VOT values similar to those of native English speakers (Flege (1991)).

Be it as it may, the difficulties in the process of acquisition of an L2 are explained by NLM through a constraining attunement of our perceptual attention to language specific cues during our first months of life. This explanation is closely related to the first of the two common assumptions behind PAM and SLM (see below), namely that the learnability of L2 phonetic segments is based on the perceived phonetic distance between them and the most similar segments in the L1 phonetic inventory.

1.3.1 Models of cross-linguistic sound perception and production

The two most relevant models in the area of L2 phonetic acquisition, namely SLM and PAM, agree on the assumptions that (1) all phonetic categories reside in a common space in the speaker's mind (Flege (1995), Flege et al. (1999a), Flege et al. (2002), Best (1995), Best & Tyler (2007), Flege (2007)), and (2) the learnability of L2 phonetic segments is perceptual in nature and depends on the perceived phonetic distance between them and the most similar segments in the L1 phonetic inventory (Flege et al. (2003), Flege (1995), Flege (2007)). Despite the common assumptions, these two models address different aspects of cross-linguistic phonetic contact. SLM was designed to address production (and to a lesser degree perception) of L2 speech by L2 learners with considerable experience in the L2. PAM is a model that was designed specifically to explain nonnative speech perception by naïve listeners of the

sounds of an unfamiliar language. Thus, while PAM makes its predictions based on perception data, SLM mostly, although not exclusively, bases its hypotheses for L2 speech learning on production data. Although these two models address different questions, I will use the predictions and hypotheses drawn from my understanding of the tenets of them both to interpret the findings from the phonetic experiments on Portuguese stressed vowel acquisition by the two learner groups mentioned above.

PAM

PAM (Best (1995)), a model of cross-linguistic speech perception, starts from the observation that certain pairs of sounds from an unfamiliar foreign language are easier to discriminate than other pairs. According to PAM, foreign language sounds are perceived according to their similarities to, or discrepancies from, the L1 sounds that are closest to them in the speakers' phonological space. PAM proposes that the L2 learners will judge the foreign language sounds to be similar or different from the L1 sounds based on articulatory properties of the sounds, and that the perceived distance between them will affect their discriminability[3] Thus, PAM proposes that foreign language sounds are assimilated to native ones in varying ways due to the perceived articulatory similarity of the foreign sounds to the native ones. If the learners perceive an L2 sound to be very similar to the closest sound in the L1, then the L2 sound will assimilate to the L1 sound category and a new phonetic category for the L2 sound will not be established.

According to PAM, a non-native phone may be perceptually assimilated to the

[3]PAM is founded on the ecological direct-realist premise that the actual object of speech perception is the information about the distal articulatory events that produced the speech signal, and consequently, that the information in the acoustic signal specifies the gestures that form it. This assumption is related to the Motor Theory of speech perception but differs from it in that it does not imply a specialized decoder of gestural information since, according to the direct-realist view, this information is in the acoustic signal itself. PAM's ecological direct-realist view of speech perception is compatible with the principles of Articulatory Phonology, which postulates "that the dynamic parameters of articulatory gestures in speech production serve as the primitives for (language-specific) phonology" (Best & Tyler (2007), p.22).

native system in one of three ways: (a) as a Categorized (C) exemplar of some native phoneme, for which its goodness of fit may range from excellent to poor; (b) as an Uncategorized (U) speech sound that falls somewhere in between native phonemes; and (c) as a Non-assimilable non-speech sound that does not resemble any native phoneme. Adult speakers' discrimination of each non-L1 sound pair is predicted to depend, under PAM, on how each of the members of the pair individually assimilate to the L1 inventory. Thus, each pattern of contrast assimilation entails expectations about discrimination performance:

- Very good to excellent discrimination is expected in Two Category (TC) assimilation pairs, where the two nonnative phones are perceived as acceptable exemplars of two different native phonemes.
- Poor discrimination is predicted for Single Category (SC) assimilation pairs, where the two nonnative phones are judged as equally good or bad tokens of the same native phoneme.
- Discrimination is intermediate for a Category Goodness (CG) difference, in which both the nonnative phones are judged as tokens of a single native phoneme but they differ in goodness of fit to that phoneme.

Alternatively, if at least one of the members of the nonnative contrast pair is assimilated to the native system as an Uncategorized speech sound, a different set of assimilation types and discriminability degrees is proposed by PAM:

- Very good discrimination is expected between the members of an Uncategorized-Categorized (UC) assimilation pair, where one of the members of the pair is heard as an Uncategorized speech sound.
- Poor to moderate discrimination is expected from Uncategorized-Uncategorized (UU) assimilation, where both members of the pairs are perceived as Uncategorized.

- Good to excellent discrimination is predicted between the two members of a Non-Assimilable pair, which contains two sounds so deviant from the articulatory properties of the listener's native phonemes that they are not perceived as speech sounds.

PAM considers both phonological and phonetic levels in explaining how the L1 system constrains the perception of nonnative sounds that are completely unfamiliar to the listener. According to PAM, both the native phonology and the goodness of fit of the nonnative phones will determine the kind of perceptual assimilation the pair will undergo, and thus, how well they will be discriminated.

Several research studies support PAM's predictions. Best et al. (2001) examined twenty-two native speakers of American English on the discrimination and identification of a TC, CG and SC contrasts in Zulu. The contrasts were voiceless and voiced lateral fricatives, voiceless and ejective velar stops and plosive and implosive bilabial stops respectively. The TC contrast was well discriminated, followed by the CG contrast and the SC contrast, complying with the gradient discriminability scale predicted by PAM. The subjects identification of the contrasts showed the same assimilation pattern as the discrimination test, confirming the predictions made by PAM. Harnsberger (2001) tested native speakers of Malayalam, Marathi, Punjabi, Tamil, Oriya, Bengali and American English on the identification and discrimination of nasal consonants varying in place of articulation in Malayam, Marathi and Oriya. Based on the results of the identification experiment, each contrast was classified as a TC, UC, UU, CG or SC contrast. The results showed that the mean percent correct discrimination scores corresponded to the predictions made by the PAM.

Despite the fact that PAM was designed to address only cross-linguistic perception of nonnative sounds by naïve listeners, an extension of the model that I will be referring to as PAM-L2 (Best & Tyler (2007) discusses the possible applications and predictions of PAM to the perception of L2 speech, and presents testable hypotheses as a guide for future research. These hypotheses will be presented, discussed, used and tested in Chapters 2 and 3 in this dissertation.

SLM

SLM (Flege (1995); Flege et al. (1999a, 2002); Flege (2003a)) focuses on changes across the life span in L2 speech learning and the development through time of this phonetic acquisition process. SLM has two basic hypotheses: (1) the speech learning mechanisms, such as those allowing for new speech category formation, remain intact across the lifespan and remain accessible to L2 learners of all ages; and (2) L2 learners will be more likely to establish new phonetic categories for those L2 sounds that they perceive to be different from their closest L1 sound than for those that they do not perceive to be different from their closest L1 sounds.

According to the SLM, "L1 and L2 phonetic subsystems of bilinguals mutually influence one another, and [...] the relative strength of L1 to L2 and L2 to L1 interference effects may depend on bilingual balance" (Flege (2007), p.366). L1 to L2 effects will be stronger in L1 dominant speakers (which would most probably be late learners of L2), while L2 to L1 effects will be stronger in L2 dominant speakers (most probably early learners of L2). In addition, the SLM's *interaction hypothesis* (Flege (1992), Flege et al. (1999b), Walley & Flege (1999)) holds that the interaction of L1 and L2 also depends on the stage of development of the speakers' L1, predicting that they are less likely to interact in younger than in older learners because early learners' L1 phonetic categories are malleable while late learners' L1 categories are fully developed.

SLM proposes two possible mechanisms through which the phonetic categories in the two languages interact:

- *Phonetic category assimilation.* Phonetic information perceived as related to both L1 and a close L2 sound merges together in one single wider phonetic category (MacKay et al. (2001a))

- *Phonetic category dissimilation.* Phonetic categories in the L1 reorganize to accommodate the new L2 categories, thus maximizing the distinguishability among them. In this case, both L1 and L2 categories pull away from one another in the phonetic space and the vowel categories of these bilingual speakers

tend to be slightly different from the vowels produced by monolingual speakers (Flege & Eefting (1987); Flege et al. (2003)).

Which of the two mechanisms will operate depends, according to SLM, on whether category formation has or has not taken place for L2 sounds, or, in SLM terms, whether the differences perceived by the learner between the L2 and the closest L1 sounds are big enough.

Flege & Eefting (1987) found evidence in support of category dissimilation in their study of the production of word-initial /p t k/ (as indicated by measuring their VOT values) by native speakers of Spanish who had learned English as children. They found that these early bilinguals did produce significantly shorter VOTs for Spanish /p t k/ than monolingual Spanish speakers and interpreted this finding as the bilingual speakers' resource to keep their English and Spanish /p t k/ separate in their category space. As early bilinguals they had likely established separate categories for English /p t k/ and needed to maximize phonetic contrast. MacKay et al. (2001a) found evidence for phonetic category assimilation in their study of native Italian speakers in Canada. They found evidence of category assimilation in bilinguals' productions of English /b d g/ as indicated by their VOT values, which differed both from English monolinguals productions as well as from Italian monolinguals' productions, and having an intermediate VOT value both when they produced Italian /b d g/ as well as when they produced English /b d g/. More recently, Flege et al. (2003) found acoustic evidence supporting phonetic category dissimilation in a study examining Italian speakers' production of English /ei/. They found that bilinguals in the early-low groups produced this sound with exaggerated tongue movement in order to differentiate it from Italian /e/ and so were more likely than late bilinguals to have established a new phonetic category for English /ei/. They therefore had to exaggerate it (dissimilate it, move it apart) to ensure phonetic contrast.

1.3.2 Phonetics and the acquisition of an L3

Fewer studies have been published on the acquisition of the phonetic system of an L3. While the studies on third language acquisition (TLA) have proliferated during the past couple of decades, these studies cover a wide range of aspects related to the phenomenon of TLA. Some researchers have investigated the effect of bilingualism on general proficiency in the L3 as compared with monolinguals learning the same target language as the L2 (Lasagabaster (2000), Sanz (2000), Sagasta Errasti (2000)) with results that evidence the bilinguals' advantage in this respect. Other studies have focused on specific areas of language acquisition such as the acquisition of specific aspects of syntax (Klein (1995)), pragmatics (Safont Jordà (2003)) and semantics (Ahukanna et al. (1981)) or lexical transfer (Cenoz et al. (2001), Ringbom (2001), Tremblay (2006)), with similar conclusions.

From these studies, several factors have been identified as salient in processes related to cross-linguistic interference and TLA. Among them is the perceived linguistic distance between languages (also referred to as 'psychotypology'). It seems to be the case that, when the L2 and the L3 are perceived as similar regardless of whether there exists an actual close linguistic relationship between them, the L2 has greater interference in the L3 than the L1 does (Cenoz (2005), Ecke (2001), Ringbom (2005)). 'Recency of use of the L2' is also identified as an important factor of cross-linguistic interference in TLA, with interference from the L2 being found to be likelier in the L3 when the L2 has been used recently by the learner than when the L2 has been dormant for a while (Williams & Hammarberg (1998), citetflynn2004cumulative, and Grosjean (2001), Zampini & Green (2001) for SLA). 'Level of proficiency in the target language' (L3) has been found to have an effect too, with learners of the L3 at early stages of acquisition being more likely to suffer more cross-linguistic interference than more advanced speakers of the L3 (Williams & Hammarberg (1998)). In addition, 'L2 status,' also known as 'the foreign language effect,' has been reported to influence the kind of cross-linguistic interference in L3 acquisition, as previously learned non-native languages seem to affect L3 to

a greater degree than the learners' L1 (Hammarberg (2001)). Last but not least, the learners' perception of correctness of an L3-target word has been identified as one of the factors that may affect cross-linguistic interference in TLA (De Angelis & Selinker (2001), Ecke (2001), Hall & Ecke (2003)).

As mentioned above, research studies on the acquisition of the phonetic system of an L3 are scarce. Nevertheless, some of the same variables identified as relevant in the process of L3 acquisition in general have also been reported to influence the acquisition of some aspect of L3 speech learning. These variables include typological distance (Llama et al. (2010)), L2 status (Wrembel (2010), Llama et al. (2010)), and the order in which the languages are learned (Smith & Baker (2007)).

In one of the oldest studies on third language acquisition of vowels in the literature, Llisterri & Poch (1987) examined the phonetic interference in Catalan(L1)-Spanish(L2) bilinguals' learning of an L3 (French), and found that bilingual speakers tend to follow the distributional pattern of their L1 in the vowel space and, in the authors' own words, "the position of this same space for the L2 vowels does not seem to interfere with their L3 productions" (p.137). However, Llisterri & Poch (1987)'s conclusions seem to be drawn from graphed vowel clouds only, and no statistical analysis of the numerical data drawn from F1 and F2 values is presented.

Iverson & Evans (2007) studied the perception of English vowels in a variety of situations (in quiet, in noise, etc.) by individuals with a wide range of L1 vowel systems (Spanish, French, German and Norwegian) to test whether the different groups showed use of different phonetic cues (e.g. formant movement and duration) when perceiving English vowels. They found that individuals with larger and more complex L1 vowel systems (German and Norwegian) were more accurate at recognizing English vowels than individuals with smaller L1 vowel systems (Spanish and French). The results also showed that, despite this finding, all groups showed a similar learning pattern and cue weighting when responding to the English vowel stimuli. Despite this being a study of L2 perception and not L3, these results support the claim that order of acquisition is important in the acquisition of multiple languages, with individuals with richer L1 phonetic systems having an advantage

over individuals with simpler L1 phonetic inventories.

Smith & Baker (2007) examined the perception of L3 vowels by inexperienced English(L1)-French(L2) bilinguals with German as the L3, and by inexperienced English(L1)-German(L2) bilinguals with French as the L3. None of the participants had experience in their respective L3 before the experiment took place. After examining the cross-language similarity between English and French vowels, and between English and German vowels, the authors found that both groups of participants perceived French vowels as more distinct from English vowels than German vowels were with reference to English ones. From this finding they predicted that it should be easier for English-French bilinguals to identify and discriminate French vowels than it should be for English-German bilinguals to identify and discriminate German ones. They also predicted that, since English-French bilinguals had been exposed to a language with a vowel system that is less similar to English, they should be more accurate in identifying and discriminating German vowels than English-German bilinguals did French vowels. Both predictions were found to be true and the results showed that the English-French bilinguals did better at perceiving vowels in an L3 than English-German bilinguals did. They even found that English-French bilinguals were as good as the English-German bilinguals at discriminating German vowels. This study, then, supports, on the one hand, the idea that order of acquisition of previous languages influences the perception of L3 sounds, and on the other hand, the SLM and PAM's assumptions that the learnability of new sounds depends on the 'distance' perceived between the new sounds and those already known to the learner.

Hammarberg & Hammarberg (2009) presents the result of a case study of an individual with British English as his L1, a very advanced level of German as his L2, and Swedish as the L3, who was living in Sweden during the course of the research study. The research project was initially planned with lexical and grammatical investigations in mind, but recording sessions were carried out and production data was collected for phonetic investigations as well. In this study, Hammarberg & Hammarberg (2009) did not find significant effects of the relative typological sim-

ilarity of L1 and L2 to L3 in the subject's pronunciation of the L3, which they attribute to the fact that the sound systems of these languages are not generally perceived as typologically distant from each other. Nevertheless, after examining the subject's development over time, these authors report that the effect of the L2 (German) on the pronunciation of the L3 (Swedish) seemed to be very strong especially at earlier stages of L3 learning. The same conclusion was reached by Wrembel (2010), who conducted a study consisting of foreign accent judgements performed by a group of expert judges who were presented recordings of L3 English produced by individuals with Polish as their L1 and an advanced command of German as their L2. The participants were classified as either beginner/elementary or pre-intermediate/intermediate based on previously completed language-learning biographies (language background questionnaires). The results from the foreign accent judgement data supported the idea that the influence of the L2 on L3 phonology is stronger at earlier stages of L3 acquisition, and it decreases with higher proficiency in the L3.

Also recently, Llama et al. (2010) studied the influence of typology and L2 status on the acquisition of Spanish voiceless stops by two groups of French-English bilinguals in Canada, who varied in order of L1 and L2 acquisition. In their study, typological distance is determined by the distribution of short and long lag VOT across the voiceless stops of the three languages, with French and Spanish voiceless stops been characterized by having a short lag VOT (unaspirated) and English ones showing long lag VOT (aspirated). The L2 status was once again defined in relation to the idea of having a "non-mother tongue" space in the speakers' mind where all post-L1 languages would be stored and one which the speaker would access when speaking any non-L1 language. Thus, if typology were to override L2 status, French-like VOT values would be observed in the Spanish voiceless stops regardless of whether French was the L1 or the L2 of these participants. If L2 status were to override typology, then both groups would transfer mainly from their L2. In this case, the French(L1) group would show more English-like VOT values in their Span-

ish voiceless stops, while the English(L1) group would show more French-like[4] VOT values. A third possibility was that the L1 would be the main potential for transfer into the L3, regardless of order of acquisition of L1 and L2. They found that both groups of participants produced L2 voiceless stops with VOT values intermediate to those of monolingual English and French speakers regardless of which their L1 and L2 were, and that they seemed to transfer this compromise to Spanish, everyone's L3. Typology was not found to be a determinant factor in these participants' Spanish productions and L2 status was found to be the most influential factor in the participants' L3 productions. These findings partially support Llisterri & Poch (1987)'s conclusions in that they place L1 as the main source of interference both for L2 and L3.

Finally, Gut (2010) investigated the possible sources and directions of cross-linguistic influence on vowel reduction and speech rhythm in data produced by four trilingual speakers. The speakers were of different ages (Speaker 1 was 30 years old at the time of testing, Speaker 2 was 50, Speaker 3 was 46, and Speaker 4 was 31 years old), they had different L1s (Polish, Russian, Hungarian and Spanish respectively) and they had either English or German as L2 or L3 (Speaker 1: English(L2)-German(L3); Speaker 2: English(L2)-German(L3); Speaker 3: German(L2)-English(L3), and Speaker 4: German(L2)-English(L3)). Participants had also learned their respective L2 and L3 at different ages (12, 12, 10 and 12 years old (L2) and 13, 18, 46 and 29 years old (L3) respectively) and in different contexts (at school, living in the country where the language is spoken, at work, a combination of different contexts, etc.). The participants' productions of both their L2 and L3 were analyzed for fluency, vowel reduction and speech rhythm and compared to English and German native speakers' data. Distinct differences were found between the participants' and the native speakers' data in vowel reduction and speech rhythm. However, there was no indication in the data that the participants' L1 had any influence on their L3 in terms of vowel reduction or speech

[4]that is, target-like, since French and Spanish are typologically similar as far as VOT is concerned

rhythm. When considering the possibility of the L2 influencing the L3, the results were mixed, the author suggesting that no conclusion could be drawn from the available data in this respect. Although the potential effect of proficiency in both their L2 and L3 as well as language similarity are discussed, Gut (2010) did not report any evidence of the effect of these factors in her data.

The results from this and other studies reviewed above support the idea that the study of cross-linguistic influence in TLA is intrinsically more complex than the study of cross-linguistic influence in SLA. While the factors affecting second or foreign language acquisition are still relevant in TLA, the interactions between the languages are exponentially more complex when more than two linguistic systems are in play. While in SLA the cross-linguistic interaction has been studied in two directions (L1 to L2, and L2 to L1), the possibilities in the case of TLA are manifold (L1 to L3, L2 to L3, L2 to L1, L3 to L2, L3 to L1, etc. and/or any combination of them). In addition, the linguistic background of trilingual individuals is inherently less uniform than that of bilinguals, with cases of simultaneous bilinguals learning an L3 (Llisterri & Poch (1987)) at potentially different ages (Llama et al. (2010)), or consecutive bilinguals differing in age of acquisition of the L2 learning an L3, which they might have learned at different ages and in different contexts (Smith & Baker (2007)), or even cases of trilinguals who may differ in the order of acquisition of their two non-native languages but who had, to begin with, different L1s (Gut (2010)), making the recruiting of homogeneous population samples for scientific investigation of cross-linguistic influences in L3 a very challenging task. Order of acquisition of the different languages, the age at which each of them are acquired and the frequency of use of each language seem to be, among others, relevant factors that should be kept in mind when studying the speech of L3 speakers. While some of the phonetic studies on L3 speech reviewed above take these factors in mind when selecting their population samples doing their best to keep them as homogeneous as possible (Smith & Baker (2007), Llama et al. (2010)), others do not (Gut (2010)), or they conduct case studies instead (Hammarberg & Hammarberg (2009)).

While the research studies reviewed here constitute part of the pioneer work in

the study of L3 speech and TLA in general, the results and evaluation from these studies (as well as the findings from studies in L2 speech learning) suggest that this area of study is still at its beginnings. The review of the studies described here suggest, on the one hand, that more strict participant recruitment parameters need to be used in order to reach more generalizable and interpretable results in L3 speech learning research, and on the other hand, that there is a lot to be done in this area of study before we be able to create testable theories for speech learning that incorporate the learning of multiple languages.

1.4 This dissertation

This dissertation seeks to shed some light into the potential differences between the process of acquisition of an L2 versus an L3 phonetic system, while building on current knowledge of the L2 phonetic acquisition process. With this purpose in mind, the Portuguese vowels of a group of L1 English-L2 Spanish-L3 Portuguese speakers was compared to those of L1 English-L2 Portuguese speakers, in addition to being compared, when considered necessary, to monolingual (or L1-dominant) native speaker productions of each of the three languages involved. The study of such groups of participants allows for the investigation of L2 influence in L3 phonetic acquisition, in addition to potentially influential factors such as language typology, L2 status, cross-linguistic phonetic similarity and category interaction in trilinguals' phonetic inventories.

In order to control for the influence of factors already identified as relevant in the literature, this dissertation studies the Portuguese oral vowel productions of learners with comparable ages of acquisition for each language, similar amount of use of each language and the same order of acquisition. The typological distance between the three languages involved is assumed to be closer between Portuguese and Spanish than between any of these two and English. The influence of 'recency of learning the prior language' and the 'L2 factor' were not directly explored in this dissertation, as the L1 (English) would be the most recently learned language for the L1 English-L2

Portuguese group and the L2 (Spanish) would be the most recently learned language for the L1 English-L2 Spanish-L3 Portuguese group, thus invalidating the possibility to study the effect of the two variables in isolation. The possible effect of 'language mode' was controlled for by following a very strict methodological procedure in which a single researcher, proficient in the three languages involved in the study, had contact with each participant, using each language at pre-determined times during each stage of the data collection process.

This study is different from previous research in that not only it involves languages from different language families and with different phonetic inventory sizes, but it focuses on the process of learning a smaller post-L1 sound inventory. Thus, the speakers in the current study all have American English as their L1, with its 12 to 15 monophthongal vowel categories, and are in the process of learning Brazilian Portuguese, with 7 distinct oral vowels. Additionally, the group learning Portuguese as an L3 has Spanish as an L2, which only has 5 vowel categories in its inventory. Thus, the foreign language speech learning process of the latter group involves the need to reduce their native English vowel inventory to only 5 vowel categories and subsequently expanding it again to the 7 Portuguese categories. In addition, there are vowel category contrasts that exist in Portuguese that do not exist in Spanish but which have very similar distributions in English (see the description of each vowel inventory in Chapter 2 of this dissertation). The potential difficulties arising from learning an L2 with a much smaller vowel inventory, followed by an L3 with a slightly bigger but still smaller inventory than the L1, are, to my knowledge, still unexplored.

This dissertation project is also unique in that it not only includes data from the L2 and L3 languages of the bilingual and trilingual participants under study but it also incorporates data from their L1 speech, which has been documented to play an important part in L2 speech learning, be it because it is the source of cross-linguistic influence or affected by it, thus contributing to our knowledge of phonetic category interaction processes in L2/foreign language speech learning. Furthermore, the majority of the L3 speech research studies reviewed above focus

either on production data by L3 speakers alone (Llisterri & Poch (1987), Gut (2010), Llama et al. (2010), Wrembel (2010)), while one of them, namely Smith & Baker (2007), makes use of a perception task to establish degree of perceived similarity between languages before proceeding to the main production experiments. This dissertation, on the other hand, comprises both the study of production data by L2 and L3 speakers of Portuguese in the different languages they speak, but also an investigation of their perception of specific L3 sounds and the assimilation patterns to their native English (L1) vowel categories. By exploring both aspects of the process of speech learning, the relation between perception and production of speech sounds is also examined.

Therefore, the current study seeks to expand our knowledge of the L2 speech learning process by (a) adding learning L3 to the equation, (b) implementing a strict selection of participants that compose an, if not perfect, close-to-homogeneous representative sample, while (c) incorporating data from all their L1, L2 and L3, as well as (d) including both production and perception data into the analysis.

Chapter 2 in this dissertation includes both a description and cross-linguistic comparison of the stressed vowel systems of either monolingual or L1-dominant (1) native speakers of American English, (2) native speakers of Mexican Spanish, and (3) native speakers of Brazilian Portuguese. Thus, data from three production experiments will be presented, one for each native speaker group, which will represent the native vowel inventories in each of the three languages under study. This description of the three vowel inventories will be followed by a cross-linguistic comparison of each of the vowel groups according to their traditional phonological description: high front vowels, high back vowels, low vowels, mid front vowels and mid back vowels. Vowel categories in each of these groups will be compared with reference to different phonetic measures, such as first and second formant values at different points through the duration of the vowel, and degree and direction of diphthongization when relevant. These comparisons will establish which vowel categories within each vowel group are cross-linguistically phonetically similar or different from each other, thus forming the base for the creation of hypotheses for the learning of

Portuguese as an L2 or as an L3 that will be tested in the following chapter.

Chapter 3 consists of the analysis of the Portuguese productions of the two experimental groups in this dissertation: the learners of Portuguese as an L2 (L2 group) and the learners of Portuguese as an L3 who had previous and extensive experience with Spanish as an L2 prior to the onset of learning of Portuguese (L3 group). Both experimental groups are composed of native English speakers. The chapter starts with the development of hypotheses, based on my interpretation of SLM and PAM, for the acquisition of the Portuguese vowel categories by native speakers of English, including predictions for potential differences expected between the two experimental groups. After the predictions have been established, the results from three additional production experiments are reported, one for each of the languages under study. The L2 group participated in the English and Portuguese production experiments, while the L3 group participated in all three the English, the Spanish and the Portuguese production experiments. The three production experiments reported in this chapter are identical to those reported for the three native speaker groups in Chapter 2, since it was crucial that the comparisons of the acoustic data from the five participant groups in this dissertation be reliable and the conclusions drawn from them be valid. While the main focus of comparisons in Chapter 3 is to compare the Portuguese native speaker production data reported in Chapter 2 with the Portuguese speech data produced by the two experimental groups in this chapter, the English vowel data produced by the two experimental groups will be used, when necessary, to explore the possibility of phonetic category interaction processes in the experimental groups' vowel inventories.

After investigating the two experimental groups' production of Portuguese as an L2 or as an L3, Chapter 4 presents the results from two perception experiments. Both experiments look into the perception of the Portuguese /e/-/ɛ/ vowel pair by four of the five participant groups whose production data was analyzed in Chapters 2 and 3, as this vowel pair proves to be one of the most difficult distinctions to produce by the experimental groups in Chapter 3. The native Spanish speaking group did not participate in the perception experiments reported in this chapter.

The first of the two experiments explores the perceptual assimilation patterns of the Portuguese mid-front vowels to the English mid-front vowel categories through a goodness of fit task. The second experiment tests the discrimination of all the mid-front vowels involved in this dissertation (five in total: two for English, one for Spanish, and two for Portuguese), via a 4-item oddity task. While the goodness of fit experiment seeks to expand on the issue of perceived phonetic similarity between Portuguese and English, the cross-linguistic vowel discrimination task investigates the learners' ability to ignore phonetically irrelevant auditory information as well as the ability to recognize phonetically relevant cues that are necessary for the perceptual discrimination of the mid-front vowels in these three languages.

Chapter 5 consists of a discussion of some of the different topics for which the results of this study are relevant. The hypotheses made in Chapter 2 for the acquisition of Portuguese vowels by the two experimental groups are reviewed and discussed in relation to the results from the different experiments reported in previous chapters, including an evaluation of the applicability of the SLM and PAM models for L3 speech research and a discussion of the relation between speech perception and production in L2/L3 speech learning. The relevance of quality and quantity of input is addressed and the signs of learning in the acquisition of a new sound inventory in general are considered. More interestingly, the differences between learning Portuguese as an L2 versus as an L3 reported in the previous chapters are discussed, especially with reference to additional literature on the correlation between vowel inventory size and vowel dispersion. A discussion of the effect of factors such as perceived cross-linguistic similarity (and/or typology) and order of acquisition of the different languages in this study follows, as well as possible pedagogical applications of the findings and suggestions for further L3 speech learning research.

CHAPTER 2

NATIVE VOWEL INVENTORIES

The aim of this chapter is to describe the native stressed oral vowel inventory in all three languages: American English, Mexican Spanish and Brazilian Portuguese. The primary objective of this chapter is, thus, to determine the extent to which the vowels of these three languages overlap in the phonetic space. In this chapter, thus, I will compare and contrast native speaker productions of the stressed vowels of these three languages in an attempt to establish the acoustic similarities and differences among them. The findings from this chapter will guide the creation of the hypotheses for the acquisition of Portuguese vowel categories by native speakers of English who may or may not have had prior and more extensive experience with Spanish speech patterns.

Thus, in this chapter, I present the results from three independent production experiments: an English production experiment, a Spanish production experiment, and a Portuguese production experiment. The participants in these experiments were native speakers of each of these languages and are assumed to represent the monolingual speech patterns in each language. The acoustic data gathered from these three production experiments are presented, statistically analyzed and described by vowel group. Once the significant cross-linguistic acoustic distinctions are established, these findings will inform the creation of hypotheses and predictions for the acquisition of the phonetic categories of Portuguese by native speakers of English which will be presented and tested in subsequent chapters of this dissertation.

2.1 Method

2.1.1 Materials

The English, Spanish and Portuguese words in Tables 2.1 and 2.2, 2.3, and 2.4 and 2.5 were selected to provide examples of the vowels of each of the languages in stressed position. For Spanish, all of / i e a o u / were represented in the list (in Table 2.3). For Portuguese, all of / ɪ e ɛ a ɔ o u / were exemplified (in Tables 2.4 and 2.5). For English, / ɪ e ɛ a ɑ ɔ o u / were represented (in Tables 2.1 and 2.2). 8 examples of each stressed vowel were included in each list, with a total of 40 target words for Spanish, 56 for Portuguese and 64 for English.

Table 2.1: English target words with vowels /ɪ/, /i/, /ɛ/ and /e/ used in elicitation materials.

ɪ	i	ɛ	e
discipline	teaser	television	dater
tilted	leader	telegram	calculated
participate	liter	delegate	belated
cinema	seated	molested	tasted
litter	seeded	sesame	laces
sitter	cheater	tested	elated
sinner	cheated	setter	translatable
listed	proceeded	attested	places

The target vowels were followed by a syllable containing "schwa" in the English words, and by a syllable headed by the vowel /a/ both in the Spanish and the Portuguese words. Note that all target vowels were surrounded by coronal consonants to control for the effect of place of articulation in the onset and offset of the stressed vowel in each case.

The target words in Tables 2.1, 2.2, 2.3, and 2.4 and 2.5 were inserted into carrier phrases. The carrier phrases were the following:

For English:

- What did X say? X said — twice.

Table 2.2: English target words with vowels /a/, /ɑ/, /o/ and /u/ used in elicitation materials.

a	ɑ	o	u
satellite	dollop	soda	suitable
classified	philosophy	stolen	reduces
salad	philosopher	notable	tulip
plaza	clotted	notify	loses
anatomy	plotted	toasted	tooted
banana	solidly	doses	looter
unanimous	anonymous	floated	looted
tatted	technology	noses	suited

For Spanish:

- ¿Qué dijo X? X dijo — para ti. (*What did X say? X said — for you.*)

For Portuguese:

- O que o X diz? O X diz — para ti. (*What does X say? X says — for you.*)

2.1.2 Participants

Seven male native English (NE) speakers, seven male native Spanish (NS) speakers and seven male native Portuguese (NP) speakers participated in the study. All participants were required to have been born in the US, Mexico or Brazil respectively, and to have lived in the country until at least 12 years of age (see Table 2.6).

The participants provided the answers to complete the Language Background Questionnaire (LBQ) in Appendix A. The participants' biographic and linguistic background information is summarized in Table 2.6.

Flege and his colleagues have done extensive work over the last three decades on phonetic category interaction in second language acquisition. They concluded that amount of L1 use and age of arrival (AOA) in the foreign country are the most relevant factors affecting (L1 and) L2 speech (Flege et al. (1995b), Flege et al.

Table 2.3: Spanish target words used in elicitation materials.

i	e	a	o	u
sal**i**da	mad**e**ra	pel**a**da	pel**o**ta	pel**u**da
exist	*wood*	*peeled*	*ball*	*hairy*
sol**i**ta	gal**e**sa	mir**a**da	cel**o**sa	verd**u**ra
alone	*Welshwoman*	*look*	*jealous*	*vegetable*
med**i**da	bol**e**ra	bar**a**ta	dud**o**sa	med**u**sa
measurement	*bowling alley*	*cheap*	*doubtful*	*jellyfish*
gor**i**la	pal**e**ta	mul**a**ta	far**o**la	cult**u**ra
gorilla	*lollipop*	*of mixed race*	*streetlamp*	*culture*
ret**i**ra	lad**e**ra	vel**a**da	des**o**la	pel**u**sa
withdraw	*slope*	*soirée*	*devastate*	*fuzz*
vis**i**ta	ter**e**sa	mor**a**da	cas**o**ta	sal**u**da
visit	*Theresa*	*dwelling*	*big house*	*greets*
del**i**ra	car**e**ta	pos**a**da	gol**o**sa	bas**u**ra
is delirious	*mask*	*inn*	*sweet toothed*	*trash*
mel**i**sa	tut**e**la	des**a**ta	val**o**ra	mes**u**ra
lemon balm	*guardianship*	*unties*	*values*	*moderation*

(1999a), Flege et al. (1999b), Guion et al. (2000b), Piske et al. (2001), Flege (2002), Flege & Liu (2002), Flege & MacKay (2004) etc).

Due to the inherent difficulty in finding true monolingual speakers of the three languages under study in the available community in Tucson, AZ, native speakers of each language who were highly dominant in their first language were selected as participants in this study. The dominance in their L1 was determined by the researchers' interpretation of their responses to the LBQ (Linguistic Background Questionnaire) administered prior to the experiments. The most important factor determining their dominance in their L1 was having been born and raised in an L1 speaking environment/country until they were at least 12 years old, thus delaying as much as possible the effect of learning an L2 in their L1 phonetic inventory. The participants were also chosen according to amount of L1 currently used here represented by both amount of general L1 use in their lives and amount of L1 use in a family environment. As shown in Table 2.6, the participants in the NS and

Table 2.4: Portuguese target words with /i/, /e/ and /ɛ/ used in elicitation materials.

i	e	ɛ
vis**i**ta	s**e**da	t**e**la
visit	*silk*	*fabric*
hes**i**ta	d**e**sça	n**e**ta
doubts	*descend*	*granddaughter*
coz**i**da	d**e**va	ac**e**lera
cooked	*owe*	*accelerates*
desc**i**da	c**e**ra	tol**e**ra
slope	*wax*	*tolerate*
infin**i**ta	il**e**sa	gal**e**ra
infinite	*unharmed*	*galley*
un**i**da	cass**e**ta	s**e**ta
united	*??*	*arrow*
osc**i**la	plan**e**ta	n**e**sta
swings	*planet*	*in this one*
vac**i**la	camis**e**ta	pr**e**ssa
hesitates	*T-shirt*	*hurry*

NP groups all use their L1 frequently in their daily life, although this might vary from participant to participant as indicated by the range of responses. However, the participants also used their L1 frequently in a family context, which is here interpreted as the participants being active members of their L1 linguistic communities and still dominant in their L1. Although monolingual speakers of each of the three languages would be ideal for this kind of study, despite the variability in linguistic experience among the native speakers selected for participation in this study, it is concluded here that the participants selected constitute a representative sample of L1 dominant speakers of the three languages under study.

2.1.3 Procedure

One male native speaker of American English was recorded reading the carrier phrases that included the words listed in Tables 2.1 and 2.2, one male native speaker

Table 2.5: Portuguese target words with /a/, /ɔ/, /o/ and /u/ used in elicitation materials.

a	ɔ	o	u
l**a**ta	cal**o**sa	t**o**da	l**u**sa
tin	*with caluses*	*whole*	*Portuguese*
d**a**ta	n**o**ta	t**o**la	literat**u**ra
date	*note*	*dumb*	*literature*
n**a**ta	an**o**ta	t**o**sca	gross**u**ra
cream	*notes down*	*thickness*	*crude*
s**a**la	s**o**la	gar**o**ta	cabel**u**da
room	*sole*	*girl*	*hairy*
ass**a**da	n**o**ssa	profess**o**ra	cond**u**ta
roasted	*ours*	*teacher*	*conduct*
sal**a**da	gost**o**sa	lavad**o**ra	entit**u**la
salad	*tasty*	*washer*	*entitles*
cal**a**da	pel**o**ta	edit**o**ra	est**u**da
quiet	*ball*	*editor*	*studies*
mul**a**ta	vaid**o**sa	calculad**o**ra	mass**u**da
of mixed race	*vain*	*calculator*	*corpulent*

of Mexican Spanish was recorded reading the carrier phrases with the words listed in Table 2.3, and one female native speaker of Brazilian Portuguese was recorded reading the phrases with the words listed in Tables 2.4 and 2.5. The American English speaker was a native speaker of English born and raised in Dearborn, MI, US. The Mexican Spanish speaker was a native speaker of Spanish born and raised in the town of Tanganhuato, in the state of Guerrero, Mexico. The Brazilian Portuguese speaker was a native speaker of Portuguese born and raised in Bauru, in the state of Sao Paulo, Brazil. Note that the speakers read each of the phrases preceded by a question. The speakers read the list twice. The second set of productions was used to create the stimulus materials for the experiments reported in this chapter.

The productions of the carrier phrases containing the target words were recorded in a Macintosh desktop computer using a MOTU UltraLite-mk3 audio interface and an AKG C520 condenser head-mounted microphone inside a sound proof booth.

Table 2.6: Participants' biographic and linguistic background where Age is the mean age at time of testing, AOD represents the age of departure of participants from their country of origin when applicable, and M/F represents their gender; L1UseGen represents amount of L1 use in general in their daily lives and L1UseFam amount of L1 use in their families; ranges are presented in parentheses.

Group	Age	AOD	M/F	L1UseGen	L1UseFam
NE	24 (19-28)	N.A.	7M	84% (50-100)	90% (60-100)
NS	30 (20-38)	21 (15-28)	7M	71% (55-80)	94% (80-100)
NP	25 (23-32)	22 (14-30)	7M	25% (5-60)	87% (20-100)

Recordings were sampled at 16-bit rate and 44.1KHz.

A native speaker of Spanish with advanced proficiency in English and intermediate to advanced proficiency in Portuguese interacted with the participants. The researcher gave all instructions and asked all questions using the participants' native language as a means of communication in each case. Participants were first asked a series of biographical and linguistic background questions orally while the researcher completed the Language Background Questionnaire (LBQ) with the information provided by the participants.

The participants were then asked to listen and repeat the sentences containing the target words in the carrier phrases mentioned above, following a delayed repetition technique developed by Flege et al. (1995b). For each language, the question preceding the carrier phrase in each case was duplicated so that the participants heard a sequence of question-answer-question for each target sentence they were instructed to repeat. An example of the question-answer-question sequence for the English experiment is included here for illustrative purposes:

- –What did Mary say?
- –Mary said "banana" twice.
- –What did Mary say?

The interval between the end of the carrier phrase (the answer in the question-

answer-question sequence) and the repetition of the question was .5 seconds. For each sequence of question-answer-question the participants heard, they were asked to repeat the complete answer, which contained the target word in the controlled phonetic environment ('Mary said "banana" twice' in the example above, where "banana" is the target word).

The presentation of the auditory material after the stimulus (repetition of the question) was designed as an attempt to minimize the possibility of mimicry. According to Guion (2003), the demands placed on the processing system by the intervening perception and production tasks should make it likely that the target word be lexically coded before repetition, thus activating a speaker's long-term memory representation of the word. While the potential for the short-term representation of the word to influence their elicitation of the target sentence does not cease to exist, the inclusion of the intervening question between the auditory presentation of the target sentence and their repetition is believed to allow for the activation of the long-term memory representation of the target word as well, potentially minimizing the direct effect of mimicry. This elicitation method is commonly used in this field (see Flege et al. (1995b), Flege et al. (2006), Flege et al. (2003), Guion (2003), among others).

All participants in each language group heard the same tokens presented over high-quality headphones. Presentation of the stimuli was regulated via PRAAT 5.1.09 by Paul Boersma and David Weenink (2009) on a personal computer. The participants practiced with 3 randomly chosen items before the experiment began. During the practice, the volume for the headphones and the recording was adjusted for each participant. The tokens were presented in two separate randomized blocks. The two sets of stimuli were presented as different experiments and the participants were allowed to take a break between the two. The first part of the experiment allowed the participants not only to listen to the question-answer-question sequences but also to see the target word in the auditory stimulus on a computer screen. The second part of the experiment only included the auditory stimuli. The participants' repetitions were digitally recorded onto a desktop computer. Only the recordings

from the second part of the experiment were used for phonetic analysis.

2.1.4 Data analysis

The audio files from each participant's production of the carrier phrases were analyzed using PRAAT 5.1.09. Acoustic analyses of the target vowels were conducted by scrolling through the audio files to locate the target words and identify the target vowels. The onset and offset of each vowel was located by searching for the second and second-to-last pitch periods showing first and second formant (F1 and F2) stability in the spectrograms of the target vowels. The beginning and end of the vowel were set between the second and third periodic pulses and between the third-to-last and the second-to-last periodic pulses on the waveform.

For all vowel tokens, F1 and F2 measures were taken at three different points through the vowels, namely 25%, 50% and 75% into each vowel, as defined by the boundaries described above. To minimize as much as possible the errors found by the formant tracking algorithm used by PRAAT, a simplified version of the formant data extraction method proposed by Escudero et al. (2009) was used. This method allows for formant ceiling optimization for each vowel of each speaker. Note that each vowel was produced 8 times by each speaker. For each of these individual tokens, F2 was determined 20 times, namely for all ceilings between 4500 and 6500 in increments of 100 Hz, rendering 160 total F2 values by vowel by speaker. From the 20 ceilings calculated, the optimal ceiling for each speaker for each vowel was the one that yielded the lowest standard deviation of F2 over the 20 measured standard deviation values. The F1 and F2 measures at the optimal ceilings were extracted and converted into logarithmic values (LOG).

Log-transformation of the raw F1 and F2 values was implemented to approximate perception. Although a great body of studies use Bark-transformation with the same objective, the decision to use log-transformation here was based on previous literature, including research on the acoustics of Portuguese (Escudero et al. (2009)), Spanish (Morrison & Escudero (2007)) and English (Miller (1984), Miller (1989)) vowels, which are the three languages investigated in this dissertation. Since using

similar vowel-representation methods allows for a more direct comparison of previous work, log-transformation was considered the most appropriate option.

First and second formant information of the English, Spanish and Portuguese vowels produced by each of the the native speaker participant groups in these experiments can be found in Tables 2.7, 2.8 and 2.9 respectively.

Table 2.7: English vowels: first and second formant data (in Hz).

		ɪ	e	ɛ	æ	ɑ	ɔ	o	u
	N	53	55	57	54	56	57	54	56
F1 (in Hz)	Mean	295	419	417	567	655	639	455	320
	Median	291	426	410	573	667	646	449	320
	SD	30	70	65	63	92	88	52	39
F2 (in Hz)	Mean	2236	1607	1968	1483	1564	1136	1233	1908
	Median	2213	1666	1967	1484	1588	1134	1240	1870
	SD	206	187	257	167	229	139	217	357

Table 2.8: Spanish vowels: first and second formant data (in Hz).

		i	e	a	o	u
	N	54	56	56	56	56
F1 (in Hz)	Mean	281	452	661	473	324
	Median	277	450	656	470	324
	SD	34	36	62	43	42
F2 (in Hz)	Mean	2209	1857	1426	1022	978
	Median	2229	1834	1421	1029	977
	SD	177	154	144	152	134

2.2 Results

The mean, median and standard deviation (SD) of the formant values at the midpoint of the vowels produced by the native English, native Spanish and native Portuguese speakers are shown in Tables 2.7, 2.8 and 2.9 respectively, while the mean

Table 2.9: Portuguese vowels: first and second formant data (in Hz).

		i	e	ɛ	a	ɔ	o	u
	N	56	56	57	53	57	56	55
F1 (in Hz)	Mean	292	393	544	683	556	398	304
	Median	295	398	542	686	553	402	306
	SD	27	34	47	61	61	37	24
F2 (in Hz)	Mean	2060	1849	1699	1352	1039	990	931
	Median	2082	1841	1688	1350	1028	993	925
	SD	164	106	117	108	75	75	72

values are plotted in Figure 2.1. Some of the tokens recorded were discarded either because the participant produced an erroneous target word or because of inadequate recording quality. From a total of 448 possible productions of the English vowels, 442 were kept for acoustic analysis (98.7%), from a total of 280 possible Spanish productions 278 (99.3%) were kept, and from a total of 392 Portuguese productions 390 (99.5%) were kept for analysis.

2.2.1 Specific vowel comparisons and statistical analysis of the data

Specific vowels were selected for cross-language comparison based on their traditional phonological description. A series of linear mixed-effects regression models were carried out to establish the statistically significant differences among the 20 vowels analyzed. While both ANOVAs and linear mixed-effects analyses allow for the extraction of the contribution of a large number of effects and their interactions, the main advantage of using the linear mixed-effects model is that it can handle more than one random factor ('word' and 'subject'), while ANOVAs can only handle one[1]. Thus, in the linear mixed-effects regression models performed in this dissertation, the dependent variables (otherwise called 'response') were both first and second formant measurements (logF1 and logF2 values in Hz). 'Vowel' was the fixed-factor

[1]For further information on the advantages of using linear mixed-effects regression models when analyzing data with crossed random effects for items and subjects see Baayen et al. (2008)

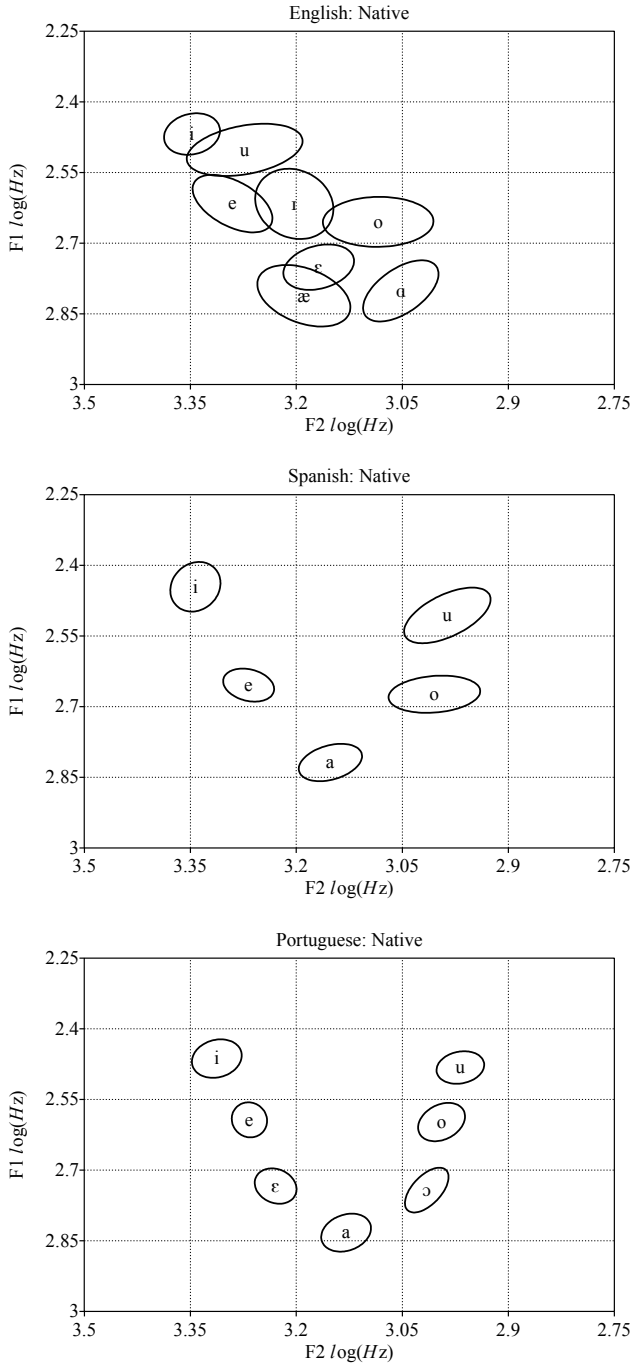

Figure 2.1: Vowels of English, Spanish and Portuguese as produced by the seven native speaker participants in each group.

in all cases, while 'speaker' and 'word' were treated as random intercepts.

Please note that, for the purposes of this dissertation, and as suggested by Baayen (2008) (p.248), a t-statistic >2 or <-2 will be assumed to represent a statistically significant difference between the intercept and the object of comparison in these types of statistical models. In the different tables presenting results from linear mixed-effects regression models, an asterisk will be placed next to the t-statistics that represent a statistically significant difference. Also, all response data used in the statistical analyses described from here on correspond to measurements taken at the midpoint of the vowel unless stated otherwise.

The statistical comparisons presented in the following subsections seek to establish whether the vowels in each group are indeed phonetically different entities by describing the acoustic similarities and differences between the different vowels compared[2].

High front vowels

In the case of the high front vowels of the three languages under consideration here, the logF1 and logF2 measurements of the Portuguese /i/ vowel were used as the intercept in the first of the three linear mixed-effects models presented below. The results, as shown in Table 2.10, show that Portuguese /i/ is statistically significantly different in logF1 from English /ɪ/. The intercept was also found to statistically differ in logF2 from English /ɪ/. These results suggest that English /ɪ/ is statistically different from Portuguese /i/. For the purposes of this dissertation, a difference that is rendered statistically significant will be interpreted as indicating that the two objects of comparison are two distinct phonetic entities. Under this assumption, Portuguese /i/ and English /ɪ/ have been found to be two different phonetic categories.

In addition, Spanish /i/ and English /i/ were not found to be significantly different from the intercept, Portuguese /i/ in any of the two measures. Since no sta-

[2]This information will be used later to make the predictions about second and third language learning that will be tested in following chapters of this dissertation.

Table 2.10: Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the high front vowels of English, Spanish and Portuguese, with Portuguese /i/ as the intercept.

		β	t
logF1	Intercept (= Portuguese /i/)	2.463	
	Spanish /i/	-0.017	-0.73
	English /i/	0.004	0.17
	English /ɪ/	0.153	6.35*
logF2	Intercept (= Portuguese /i/)	3.312	
	Spanish /i/	0.032	1.61
	English /i/	0.035	1.71
	English /ɪ/	-0.108	-5.27*

tistically significant difference was found between the Portuguese high front vowel and either the Spanish or the English /i/, it is concluded that Portuguese /i/ is not acoustically a different vowel category from Spanish /i/ or English /i/. These differences can also be appreciated in Figure 2.2.

Although the differences, or lack thereof, between Portuguese /i/ and the other three high front vowels have been established in the linear mixed-effects analysis model in Table 2.10, the potential differences between additional vowel pairs are yet to be tested. The remaining pairs are (a) Spanish /i/ and English /i/; (b) Spanish /i/ and English /ɪ/; and (c) English /i/ and /ɪ/. For such purpose, two additional linear mixed-effects models were run. The first of these two models took Spanish /i/ as the intercept.

As shown in Table 2.11, Spanish /i/ is only statistically different from English /ɪ/ both in logF1 and logF2. Spanish /i/ was not rendered statistically different either from English /i/ nor from Portuguese /i/. The finding that Spanish /i/ is not significantly different from Portuguese /i/ is consistent with results from the previous model in Table 2.10 where the same vowel pair contrast was first tested.

One last linear mixed-effects model was run with English /i/ as the intercept to test the last of the vowel pairs, namely, that of English /i/ and English /ɪ/. The

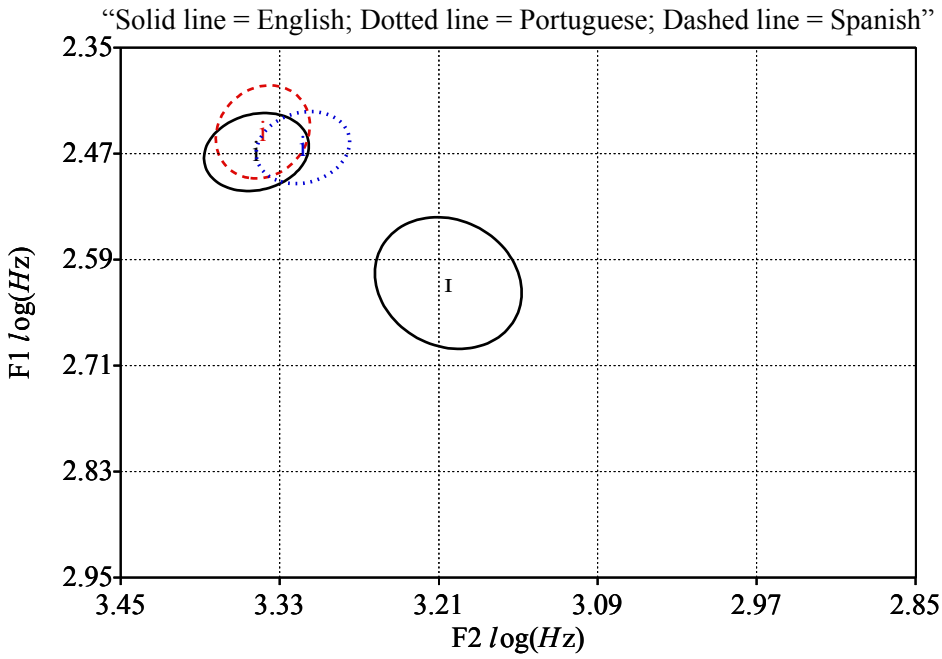

Figure 2.2: High front vowels of English, Spanish and Portuguese, as produced by the three native speaker groups.

results from this model are shown in Table 2.12, where the differences between the two English high front vowels are rendered statistically significant in both the logF1 and the logF2 dimensions.

To sum up, the three linear mixed-effects models presented above show that there is no statistically significant difference between Spanish /i/, Portuguese /i/ and English /i/ native vowels, that is, they are not different phonetic entities, while the three of them are different from English /ɪ/. This relationship among the high front vowels of English, Portuguese and Spanish can also be appreciated in Figure 2.2, where the overlapping in the vowel space of English /i/, Spanish /i/ and Portuguese /i/ and their distance to English /ɪ/ supports the results from the statistical analyses performed in this section.

Table 2.11: Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the high front vowels of English, Spanish and Portuguese, with Spanish /i/ as the intercept.

		β	t
logF1	Intercept (= Spanish /i/)	2.445	
	English /i/	0.021	0.89
	English /ɪ/	0.170	7.07*
	Portuguese /i/	0.017	0.73
logF2	Intercept (= Spanish /i/)	3.343	
	English /i/	0.003	0.13
	English /ɪ/	-0.140	-6.85*
	Portuguese /i/	-0.032	-1.61

Table 2.12: Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the high front vowels of English, Spanish and Portuguese, with English /i/ as the intercept.

		β	t
logF1	Intercept (= English /i/)	2.467	
	English /ɪ/	0.149	10.02*
	Portuguese /i/	-0.004	-0.17
	Spanish /i/	-0.021	-0.89
logF2	Intercept (= English /i/)	3.346	
	English /ɪ/	-0.143	-11.54*
	Portuguese /i/	-0.035	-1.71
	Spanish /i/	-0.003	-0.13

High back vowels

The logF1 and logF2 measurements of the three high back vowels studied in this dissertation were sent to a linear mixed-effects regression model for a statistical analysis of the potential differences among them. The results, shown here in Table 2.13, indicate that, when taking Portuguese /u/ as the intercept, Spanish /u/ does not show any statistically significant difference in any of the two measurements, while Portuguese /u/ and English /u/ are statistically different in logF2 (β= 0.305, t = 10.52). This statistical difference in logF2 but not logF1 is perceptible visually in Figure 2.3, where English /u/ is produced in a position further front (hence the difference in logF2) than both Spanish /u/ and Portuguese /u/.

Table 2.13: Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the high back vowels of English, Spanish and Portuguese, with Portuguese /u/ as the intercept.

		β	t
logF1	Intercept (= Portuguese /u/)	2.482	
	Spanish /u/	0.024	1.01
	English /u /	0.019	0.83
logF2	Intercept (= Portuguese /u/)	2.968	
	Spanish /u/	0.018	0.64
	English /u /	0.305	10.52*

One more vowel pair is to be tested for phonetic contrast among the high back vowels in Figure 2.3 so that all possible vowel pairs are tested for significant differences. The last pair to be tested is that of Spanish /u/ and English /u/. For such purpose, another linear mixed-effects model was run with Spanish /u/ as the intercept. The results from this model, in Table 2.14, show that Spanish /u/ and English /u/ are not significantly different in their height (logF1 dimension), but they are different in their backness (logF2 dimension). This result confirms the visually perceptible difference between these two vowels in Figure 2.3. Table 2.14 also replicates the result, shown before in Table 2.13, that there exists no statistically

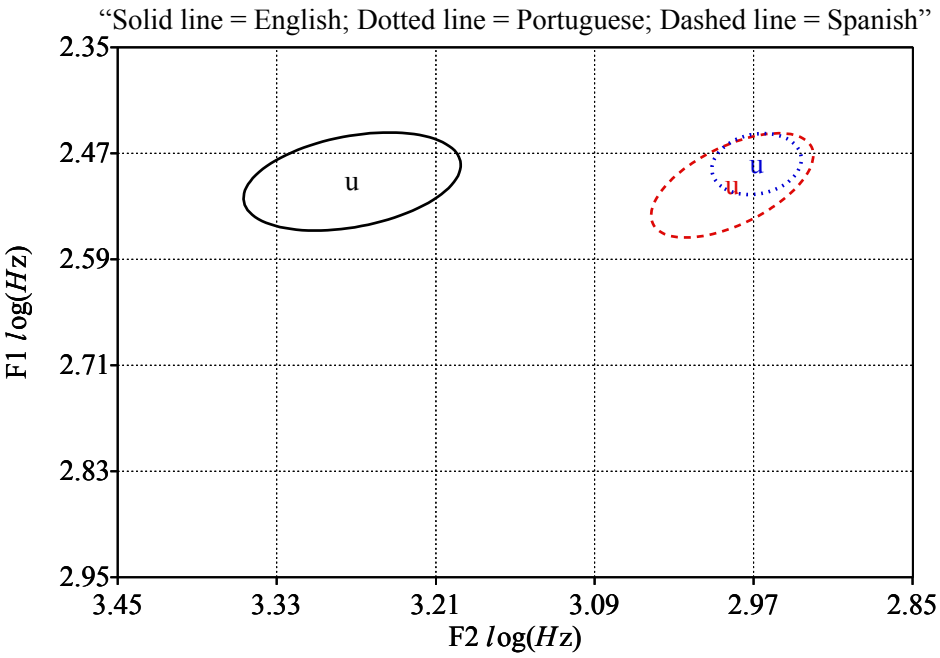

Figure 2.3: High back vowels of English, Spanish and Portuguese as produced by the three native speaker groups.

Table 2.14: Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the high back vowels of English, Spanish and Portuguese, with Spanish /u/ as the intercept.

		β	t
logF1	Intercept (= Spanish /u/)	2.506	
	English /u /	-0.004	-0.18
	Portuguese /u/	-0.024	-1.01
logF2	Intercept (= Spanish /u/)	2.986	
	English /u /	0.287	9.89*
	Portuguese /u/	-0.018	-0.64

significant difference between Spanish /u/ and Portuguese /u/.

Most surprisingly, it is interesting to observe that English /u/ was found to differ from both Spanish /u/ and Portuguese /u/ in its extremely fronted position in the vowel space. While /u/ phonemes are traditionally described as high back, the data from the native English speaker group in this dissertations suggests that, in American English, English /u/ is extremely fronted. In the topmost vowel chart of Figure 2.1 above, it is observable that English /u/ is almost as fronted as English /i/ in this variety of English[3].

As a summary, the results of the analyses performed on the natives' high back vowel data suggest that Spanish /u/ and Portuguese /u/ are not two separate phonetic categories, but that they both differ from English /u/, which is located in an extremely fronted position in the vowel space. These findings are interpreted as there being two different phonetic entities among the high back vowels of these three languages studied here, one of them comprises both Spanish and Portuguese /u/, which did not prove to be different categories, and the second is English /u/, which is produced in a more fronted position compared to Spanish and Portuguese /u/ (see Figure 2.3).

[3]Please refer to the Summary of Results and Discussion section in this chapter for a discussion of the relevance of these findings with reference to previous literature on this phenomenon.

Low vowels

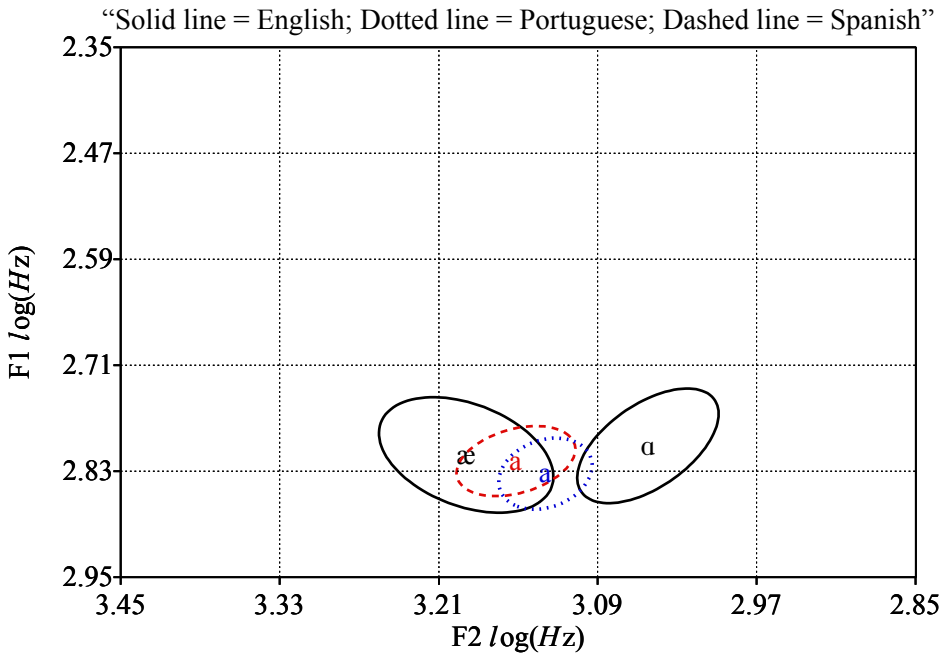

Figure 2.4: Low vowels of English, Spanish and Portuguese as produced by the three native speaker groups.

To establish the similarities and differences between the four low vowels in the vowel inventories of English, Spanish and Portuguese studied here, three separate linear mixed-effects models were run. The first model had Portuguese /a/ as the intercept. The results from this statistical analysis are shown in Table 2.15.

The results suggest that there are only two statistically significant differences between the Portuguese /a/ and the other three vowels under consideration. The first difference is in the logF2 measure between Portuguese /a/ and the same measure of the English low-back vowel /ɑ/. The second significant difference is also on the logF2 measure between Portuguese /a/ and English /æ/. None of the differences in logF1 tested by this model are rendered statistically significant.

A second linear mixed-effects model was run with the English low-back /ɑ/ as the intercept to statistically test (a) the assumption that the two low vowels of

Table 2.15: Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the low vowels of English, Spanish and Portuguese, with Portuguese /a/ as the intercept.

		β	t
logF1	Intercept (= Portuguese /a/)	2.832	
	Spanish /a/	-0.015	-0.70
	English /æ/	-0.021	-1.04
	English /ɑ/	-0.032	-1.56
logF2	Intercept (= Portuguese /a/)	3.129	
	Spanish /a/	0.022	0.90
	English /æ/	0.060	2.43*
	English /ɑ/	-0.078	-3.19*

English are indeed two different vowel categories, and (b) the contrast between the English low-back /ɑ/ and Spanish /a/, which was not tested in Table 2.15. Results from the second model are presented in Table 2.16.

Table 2.16: Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the low vowels of English, Spanish and Portuguese, with English /ɑ/ as the intercept.

		β	t
logF1	Intercept (= English /ɑ/)	2.800	
	Spanish /a/	0.018	0.85
	English /æ/	0.011	0.78
	Portuguese /a/	0.032	1.56
logF2	Intercept (= English /ɑ/)	3.051	
	Spanish /a/	0.101	4.04*
	English /æ/	0.138	8.38*
	Portuguese /a/	0.078	3.19*

As expected from the lack of significance in the results of the statistical model shown in Table 2.16, none of the logF1 differences between the lower back English vowel and each of the other three vowels tested here were rendered statistically

significant. When testing for logF2 differences, however, the difference between the English low-back vowel and the low-front vowel was found to be statistically significant, as well as that between English /ɑ/ and Spanish /a/, and between English /ɑ/ and Portuguese /a/. This last result is consistent with the results obtained from the first of the linear mixed-effects models run on the low vowel data shown in Table 2.15.

For the sake of completion, a third mixed-effects linear model tested the potential difference between Spanish /a/ and the rest of the low vowels in order to establish whether the difference between Spanish /a/ and English /æ/ was statistically significant, the only low vowel pair that was not tested by the previous two models. The results from this third model are shown in Table 2.17.

Table 2.17: Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the low vowels of English, Spanish and Portuguese, with Spanish /a/ as the intercept.

		β	t
logF1	Intercept (= Spanish /a/)	2.818	
	English /æ/	-0.007	-0.33
	Portuguese /a/	0.015	0.70
	English /ɑ/	-0.018	-0.85
logF2	Intercept (= Spanish /a/)	3.152	
	English /æ/	0.038	1.51
	Portuguese /a/	-0.022	-0.90
	English /ɑ/	-0.101	-4.04*

Table 2.17 shows that, as expected, none of the four low vowels show any statistical significance when their logF1 is tested. On logF2, however, Spanish /a/ is not statistically significantly different from neither English /æ/ nor Portuguese /a/, but it is statistically different from English /ɑ/. These results are consistent with the results of the previous two models and establish the lack of significant phonetic difference between Spanish /a/ and English /æ/, which suggest that Spanish /a/ is far enough front not to statistically differ from English /æ/.

Table 2.18: Summary of statistically significant differences among low vowels of English, Spanish and Portuguese.

Vowel pair compared	Different phonetic categories?	logF1	logF2
Portuguese /a/ vs. Spanish /a/	No	n.s.	n.s.
Portuguese /a/ vs. English /æ/	Yes	n.s.	*
Portuguese /a/ vs. English /ɑ/	Yes	n.s.	*
English /ɑ/ vs. Spanish /a/	Yes	n.s.	*
English /ɑ/ vs. English /æ/	Yes	n.s.	*
Spanish /a/ vs. English /æ/	No	n.s.	n.s.

Table 2.18 sums up the results from all vowel pair comparisons performed in the three linear mixed-effects models described above. The summary table shows that, although none of the four vowels differ from each other in height, when tested on the backness measure, all three, Portuguese /a/, English /æ/ and English back /a/ are found to be different phonetic categories. Spanish /a/, on the other hand, shows that it is a different category from English /ɑ/, but not different from neither Portuguese /a/ nor English /æ/.

Mid-back vowels

The similarities and differences between the four mid-back vowels in the three stressed vowel inventories studied here are tested by running three separate linear mixed-effects models. The first model had Portuguese /ɔ/ as the intercept. The results from this statistical analysis, shown in Table 2.19, suggest that Portuguese /ɔ/ is significantly different from English /o/ in both height (logF1 axis) and backness (logF2 axis), while Spanish /o/ and Portuguese /o/ only differ from the intercept, Portuguese /ɔ/, in height. Since Portuguese /ɔ/ in this statistical model was found to statistically differ from all three mid-back vowels to which is being compared at least in one of the two measures, the results suggest that Portuguese /ɔ/ is indeed a vowel category different from each of them. The results from this test are shown in Table 2.19. The differences can be visually appreciated in Figure 2.5.

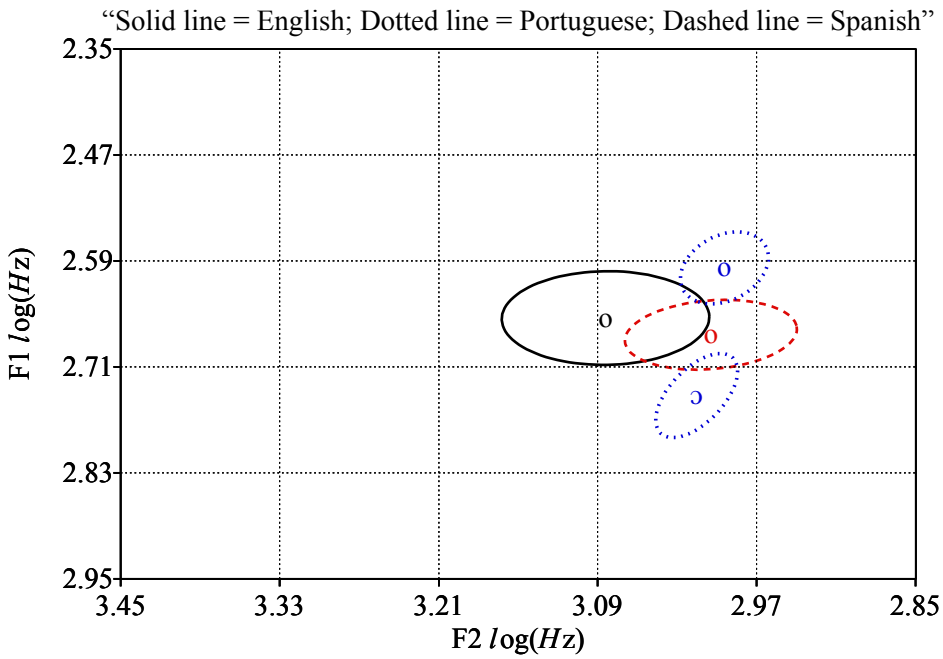

Figure 2.5: Mid-back vowels of English, Spanish and Portuguese as produced by the three native speaker groups.

Table 2.19: Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the mid-back vowels of English, Spanish and Portuguese, with Portuguese /ɔ/ as the intercept.

		β	t
logF1	Intercept (= Portuguese /ɔ/)	2.741	
	Portuguese /o/	-0.144	-17.68*
	Spanish /o/	-0.068	-3.43*
	English /o/	-0.086	-4.33*
logF2	Intercept (= Portuguese /ɔ/)	3.015	
	Portuguese /o/	-0.021	-1.37
	Spanish /o/	-0.010	-0.40
	English /o/	0.069	2.63*

Table 2.20: Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the mid-back vowels of English, Spanish and Portuguese, with Spanish /o/ as the intercept.

		β	t
logF1	Intercept (= Spanish /o/)	2.673	
	English /o/	-0.018	-0.90
	Portuguese /ɔ/	0.068	3.43*
	Portuguese /o/	-0.076	-3.82*
logF2	Intercept (= Spanish /o/)	3.004	
	English /o/	0.0079	3.02*
	Portuguese /ɔ/	0.010	0.40
	Portuguese /o/	-0.010	-0.39

A second linear mixed-effects model analyzes the differences between Spanish /o/, the intercept, and each of the other three native mid-back vowels. The results from this analysis are shown in Table 2.20. The results show that Spanish /o/ differs from the other three mid-back vowels in different ways, but it does indeed differ from each and all of them. Spanish /o/ differs from English /o/ in backness but not in height, while it differs from Portuguese /o/, as well as from Portuguese /ɔ/, in height but not in backness. The results from this model regarding the differences between Spanish /o/ and Portuguese /ɔ/ replicate the results from the previous model shown in Table 2.19.

In order to complete all possible comparisons between the mid-back vowels here, a third linear mixed-effects model tests the differences between Portuguese /o/ and English /o/. The results, as shown in Table 2.21, show that these two vowels differ both in the logF1 and logF2 dimensions. The other two vowel pairs replicated in this third statistical model show results consistent with previous results testing the same differences.

To sum up the results from the statistical analyses performed on the mid-back vowels of English, Spanish and Portuguese in this section, Table 2.22 is included here. Table 2.22 shows that each of the mid-back vowels in the native speakers'

Table 2.21: Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the mid-back vowels of English, Spanish and Portuguese, with Portuguese /o/ as the intercept.

		β	t
logF1	Intercept (= Portuguese /o/)	2.598	
	Spanish /o/	0.076	3.82*
	English /o/	0.058	2.92*
	Portuguese /ɔ/	0.144	17.68*
logF2	Intercept (= Portuguese /o/)	2.994	
	Spanish /o/	0.010	0.39
	English /o/	0.089	3.41*
	Portuguese /ɔ/	0.021	1.37

Table 2.22: Summary of statistically significant differences among mid-back vowels of English, Spanish and Portuguese.

Vowel pair compared	Different phonetic categories?	logF1	logF2
Portuguese /ɔ/ vs. Portuguese /o/	Yes	*	n.s.
Portuguese /ɔ/ vs. Spanish /o/	Yes	*	n.s.
Portuguese /ɔ/ vs. English /o/	Yes	*	*
Spanish /o/ vs. English /o/	Yes	n.s.	*
Spanish /o/ vs. Portuguese /o/	Yes	*	n.s.
Portuguese /o/ vs. English /o/	Yes	*	*

speech analyzed in this chapter is a distinct phonetic entity that significantly differs from each of the other three at least on one dimension, be it height, backness or both.

Mid-front vowels

Three separate linear mixed-effects models were performed to test the differences between each possible pair of vowels among the five mid-front vowels studied here, namely English /e/ and /ɛ/, Spanish /e/, and Portuguese /e/ and /ɛ/.

The first of the three statistical analyses was run with Portuguese /ɛ/ as the in-

Table 2.23: Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the mid-front vowels of English, Spanish and Portuguese, with Portuguese /ɛ/ as the intercept.

		β	t
logF1	Intercept (= Portuguese /ɛ/)	2.734	
	Portuguese /e/	-0.141	-13.59*
	Spanish /e/	-0.079	-4.28*
	English /ɛ/	0.017	0.93
	English /e/	-0.118	-6.34*
logF2	Intercept (= Portuguese /ɛ/)	3.229	
	Portuguese /e/	0.038	3.13*
	Spanish /e/	0.038	1.98
	English /ɛ/	-0.061	-3.16*
	English /e/	0.060	3.10*

tercept. Results from this analyses are shown in Table 2.23. These results suggest that Portuguese /e/ and English /e/ both differ from Portuguese /ɛ/ in both measures (height and backness), while Spanish /e/ differs from the intercept in height but not backness (although it does show a t-statistic nearing significance: t = 1.98), and English /ɛ/ differs from the intercept in backness but not in height. For an image of the five mid-front vowels compared in this section see Figure 2.6.

A second linear mixed-effects model took Spanish /e/ as the intercept (see Table 2.24 for results). The results show that each of the four mid-front vowels compared to Spanish /e/ here statistically differ from it in height, with English /ɛ/ differing from it in backness as well. English /e/ and Portuguese /e/ do not show a statistically significant difference from the intercept in logF2, while, as already shown with the previous statistical analysis, Spanish /e/ and Portuguese /ɛ/ show a near significant difference in the logF2 dimension.

A third statistical analysis of similar structure was run on the mid-front vowel data with English /e/ as the intercept in order to test for significant differences between two vowel pairs not tested by the previous analyses, namely (a) English

Table 2.24: Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the mid-front vowels of English, Spanish and Portuguese, with Spanish /e/ as the intercept.

		β	t
logF1	Intercept (= Spanish /e/)	2.654	
	English /ɛ/	0.097	5.21*
	English /e/	-0..038	-2.06*
	Portuguese /ɛ/	0.080	4.28*
	Portuguese /e/	-0.061	-3.31*
logF2	Intercept (= Spanish /e/)	3.267	
	English /ɛ/	-0.099	-13.59*
	English /e/	0.021	1.11
	Portuguese /ɛ/	-0.038	-1.98
	Portuguese /e/	-0.000	-0.03

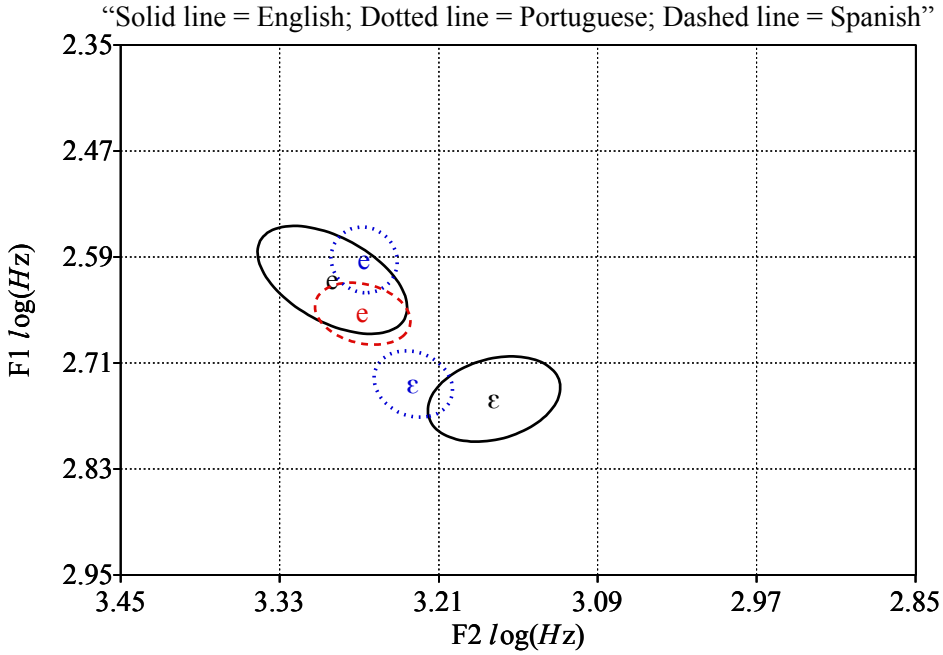

Figure 2.6: Mid-front vowels of English, Spanish and Portuguese as produced by the three native speaker groups.

Table 2.25: Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the mid-front vowels of English, Spanish and Portuguese, with English /e/ as the intercept.

		β	t
logF1	Intercept (= English /e/)	2.615	
	Spanish /e/	0.038	2.06*
	English /ɛ/	0.135	12.83*
	Portuguese /ɛ/	0.118	6.34*
	Portuguese /e/	-0.023	-1.24
logF2	Intercept (= English /e/)	3.289	
	Spanish /e/	-0.021	-1.11
	English /ɛ/	-0.121	-9.78*
	Portuguese /ɛ/	-0.060	-3.10*
	Portuguese /e/	-0.022	-1.15

/e/ and Portuguese /ɛ/ and (b) English /e/ and Portuguese /e/. The results of this statistical analysis, presented in Table 2.25, show that English /e/ is different from all three Spanish /e/, English /ɛ/ and Portuguese /ɛ/ in the logF2 dimension (backness), but is only different from English /ɛ/ and Portuguese /ɛ/ in the logF1 dimension (height). As English /e/ and Portuguese /ɛ/ were found to differ in both measures, they are assumed to be two distinct phonetic entities. Nevertheless, the differences between English /e/ and Portuguese /e/ are not found to be statistically significant in any of the two measures. Thus, English /e/ and Portuguese /e/ were not found to be two distinct phonetic entities.

Only one mid-front vowel pair remains to be tested for differences in height or backness, namely English /ɛ/ and Portuguese /e/. The results of one last linear mixed-effects model, which are presented in Table 2.26, show that, as traditional phonological description of these two vowel sounds suggest, the two mid-front vowels of Portuguese are indeed two distinct phonetic entities, statistically differing in both height and backness. The additional three vowel pairs tested in this analysis are consistent with results from previous analyses of the same pairs above.

Table 2.26: Results from a linear mixed-effects model on the logF1 and logF2 values (in Hz) of the mid-front vowels of English, Spanish and Portuguese, with Portuguese /e/ as the intercept.

		β	t
logF1	Intercept (= Portuguese /e/)	2.593	
	Spanish /e/	0.061	3.31*
	English /ɛ/	0.158	8.53*
	English /e/	0.023	1.24
	Portuguese /ɛ/	0.141	13.59*
logF2	Intercept (= Portuguese /e/)	3.267	
	Spanish /e/	0.000	0.03
	English /ɛ/	-0.099	-5.16*
	English /e/	0.022	1.15
	Portuguese /ɛ/	-0.038	-3.13*

The results from the four statistical analyses performed on the mid-front vowels of English, Spanish and Portuguese in this section are summarized in Table 2.27. Table 2.27 shows that all mid-front vowels of English, Spanish and Portuguese are different phonetic categories from each other except English /e/ and Portuguese /e/, which did not show any statistically significant differences neither in height nor in backness. This lack of phonetic difference is illustrated in Figure 2.6, where it is observable that Portuguese /e/ occupies part of the perceptual space taken by English /e/.

2.2.2 The diphthongization of mid vowels

While the findings in the previous section are revealing, it is important to notice that conclusions drawn from such a limited amount of data would only be interpretable under the assumption that all relevant differences in vowel quality among the vowels under analysis must be present at that single point of measurement (the midpoint of the vowel in this case). Traditionally, however, the mid-back vowel of American English is known to be described as diphthongizing, while the Spanish

Table 2.27: Summary of statistically significant differences among mid-front vowels of English, Spanish and Portuguese.

Vowel pair compared	Different phonetic categories?	logF1	logF2
Portuguese /ɛ/ vs. Portuguese /e/	Yes	*	*
Portuguese /ɛ/ vs. Spanish /e/	Yes	*	n.s.
Portuguese /ɛ/ vs. English /ɛ/	Yes	n.s.	*
Portuguese /ɛ/ vs. English /e/	Yes	*	*
Spanish /e/ vs. English /ɛ/	Yes	*	*
Spanish /e/ vs. English /e/	Yes	*	n.s.
Spanish /e/ vs. Portuguese /e/	Yes	*	n.s.
English /e/ vs. English /ɛ/	Yes	*	*
English /e/ vs. Portuguese /e/	No	n.s.	n.s.
Portuguese /e/ vs. English /ɛ/	Yes	*	*

and Portuguese mid-back vowels are not. This fact brings a new perspective into the question of whether these vowels are different phonetic entities or not. If it is true that certain mid vowels of English move in the vowel space through time, it might be the case that the significant differences identified in the previous section at midpoint are not enough to establish the phonetic contrasts among the mid vowels in these three languages. It is thus worth describing the movement of formants through time in the mid vowels studied in this dissertation, as well as the differences among the diphthongization patterns among them.

The following section intends to shed some light on this issue by first quantifying and comparing the amount of diphthongization in both the mid-back vowels and the mid-front vowels in the three languages studied here, and performing additional analyses of the logF1 and logF2 values of these vowels at three different points of measurement (25%=beginning, 50%=middle, and 75%= end) throughout the length of the vowels.

Diphthongization in mid-back vowels

In order to quantify the amount of diphthongization in the four mid-back vowels studied here, Diff1 and Diff2 values were calculated for each vowel by each partici-

pant, where Diff1 equals the subtraction of logF1 at the 25% of the vowel from the same value at the 75% point through the length of the vowel. Similarly, Diff2 equals the subtraction of logF2 at the 25% point of the vowel from logF2 at the 75% of the vowel. Diff1 is taken here as a measure of movement of the vowels in the F1 axis, that is, the raising or lowering of the vowels in the vowel space through time, while Diff2 is a measure of their movement in the F2 axis, that is, the fronting or retracting of the vowels through time. The mean Diff1 and Diff2 values for each of the four vowels studied here are shown here in Figures 2.7 and 2.8.

Note that logF1 values are bigger for low vowels than for high vowels in the vowel space and logF2 values are bigger for front vowels than for back vowels. Thus, the average values graphed in Figures 2.7 and 2.8 represent the amount of movement of the first and second formants respectively from the 25% of the vowel to the 75% of the vowel. In Figure 2.7, which represents the movement of the vowel in height through its duration, a box that is located below 0 represents a movement of the vowel upwards in the vowel space, while a positive value represents a lowering movement of the vowel through time. In Figure 2.8, a box located on the negative side represents a retraction of the vowel, that is, a movement of the vowel backwards through time. In these kinds of box and whiskers plots, 50% of the data is represented inside the box, the whiskers extend up to a total of 75% of the data and the dark line inside the box represents the median. Figures 2.7 and 2.8 show that English /o/ moves backwards and up, while the two Portuguese mid vowels move slightly lower and get more fronted through the length of the vowel. The Spanish vowel is the least diphthongizing and most stable of the four.

In order to statistically test these potential differences in diphthongization, a linear mixed-effects model was run with formant change in Portuguese /ɔ/ as the intercept. The results from the statistical model are shown in Table 2.28.

The results from the linear mixed-effects model in Table 2.28 show that, indeed, the difference in formant movement between Portuguese /ɔ/ and English /o/ is statistically significant for both F1 and F2. The difference between Portuguese /ɔ/ and Spanish /o/ is also statistically significant as for change in vowel backness only.

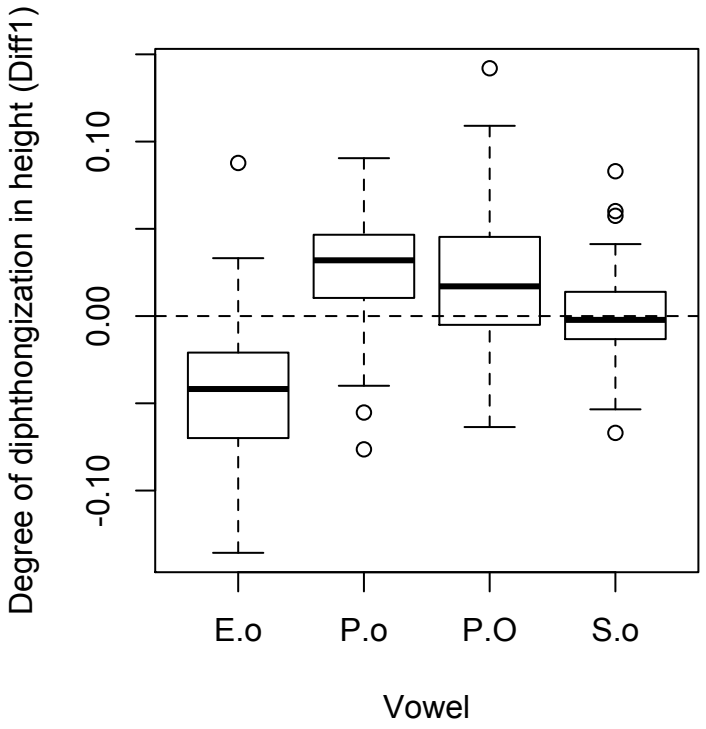

Figure 2.7: Movement of the mid-back vowels of English, Portuguese and Spanish in height through time (Diff1). English /o/ is referred to as "E.o," Spanish /o/ as "S.o" and Portuguese /ɔ/ and /o/ as "P.O" and "P.o" respectively. Negative Diff1 values represent raising of the vowels in the vowel space while positive values represent lowering. The graph shows that (a) English /o/ is the most diphthongizing mid-back vowel of all moving upwards through the length of the vowel, (b) Portuguese /ɔ/ moves down, (c) Portuguese /o/ moves down to a greater degree than Portuguese /ɔ/, and (d) Spanish /o/ does not show any movement at all.

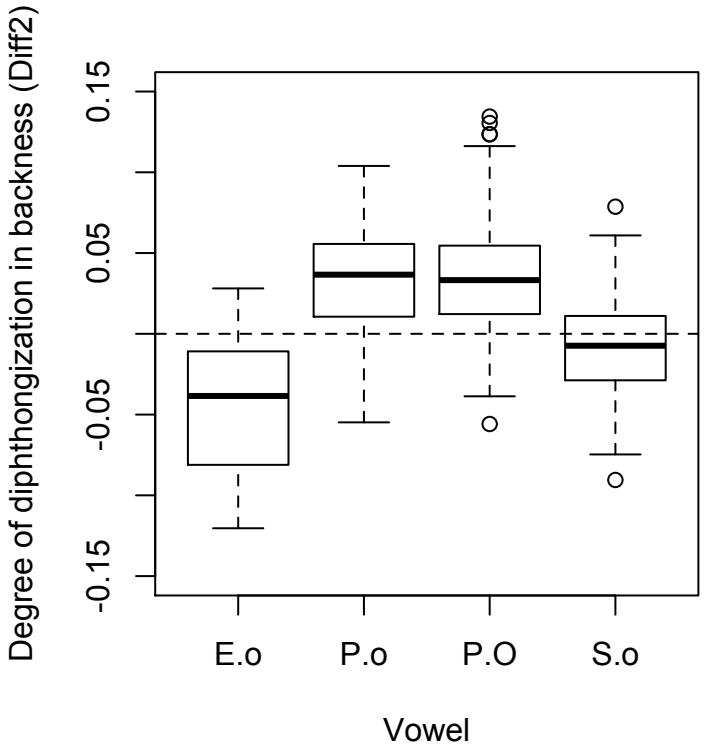

Figure 2.8: Movement of the mid-back vowels of English, Portuguese and Spanish in backness through time (Diff2). English /o/ is referred to as "E.o," Spanish /o/ as "S.o" and Portuguese /ɔ/ and /o/ as "P.O" and "P.o" respectively. Negative Diff2 values represent movement of the vowels backwards in the vowel space through time, while positive values represent fronting. The graph shows that (a) English /o/ moves backwards, (b) the two Portuguese vowels move backwards through the length of the vowel, and (c) Spanish /o/ remains stable.

Table 2.28: Results from a linear mixed-effects model on mid-back vowel diphthongization data in the three languages, with Portuguese /ɔ/ as the intercept.

		β	t
Diff1	Intercept (= Portuguese /ɔ/)	0.017	
	Portuguese /o/	0.009	0.92
	Spanish /o/	-0.016	-1.36
	English /o/	-0.060	-5.00*
Diff2	Intercept (= Portuguese /ɔ/)	0.036	
	Portuguese /o/	-0.003	-0.20
	Spanish /o/	-0.044	-2.39*
	English /o/	-0.086	-4.68*

However, the difference in formant movement between the two Portuguese mid-back vowels did not reach significance in any of the two measures of diphthongization.

A second model was run to establish whether differences in diphthongization between Spanish /o/ and the rest of the mid-back vowels were significant. The results from this model are shown in Table 2.29.

The results from the second statistical model show that the differences in formant movement between Spanish /o/and English /o/, and between Spanish /o/ and Portuguese /o/ are statistically significant in both the logF1 and logF2 dimensions. The differences between Spanish /o/ and Portuguese /ɔ/ are also significant in Diff2, but not in Diff1, as was rendered by the first of the two statistical models on the diphthongization measures.

One last linear mixed-effects model was run to test the difference in diphthongization between English /o/ and Portuguese /o/, the only vowel pair not tested by the previous two models. The results from this last model, which confirm that these two vowels are significantly different in diphthongization, are shown in Table 2.30.

Despite the revealing results from the analysis of the diphthongization measure through Diff1 and Diff2, the significant differences found through this kind of comparison only add limited information as for the cross-linguistic differences between

Table 2.29: Results from a linear mixed-effects model on mid-back vowel diphthongization data in the three languages, with Spanish /o/ as the intercept.

		β	t
Diff1	Intercept (= Spanish /o/)	0.000	
	English /o/	-0.044	-3.64*
	Portuguese /ɔ/	0.016	1.36
	Portuguese /o/	0.026	2.15*
Diff2	Intercept (= Spanish /o/)	0.036	
	English /o/	-0.042	-2.29*
	Portuguese /ɔ/	0.044	2.39*
	Portuguese /o/	0.018	2.19*

Table 2.30: Results from a linear mixed-effects model on mid-back vowel diphthongization data in the three languages, with English /o/ as the intercept.

		β	t
Diff1	Intercept (= English /o/)	-0.043	
	Spanish /o/	0.044	3.64*
	Portuguese /ɔ/	0.060	5.00*
	Portuguese /o/	0.070	5.79*
Diff2	Intercept (= English /o/)	-0.050	
	Spanish /o/	0.042	2.29*
	Portuguese /ɔ/	0.086	4.68*
	Portuguese /o/	0.083	4.49*

vowels at specific points in time. While speakers and listeners might pay attention to relevant differences in formant information at the midpoint of the vowel as well as to differences in amount of diphthongization to identify different vowel categories, it is also important to compare these vowels at different points in time.

For example, vowel pairs such as English /o/ and Portuguese /o/ have been found to differ in logF1 and logF2 measures at their midpoint and show statistically significant differences in both amount of diphthongization both in height and in backness. However, despite Portuguese /o/ being located higher and in a more retracted position in the vowel space than English /o/, Portuguese /o/ shows a tendency to move towards the front and lower while English /o/ moves backwards. This movement of the vowels in opposite directions in the vowel space opens up the possibility of these two vowels merging into a single phonetic category at, in this specific case, the last portion of the vowel in time. A comparison of the spectral differences between the mid vowels of these three languages at different points through time might reveal significant differences between members of a vowel pair that might have been invisible under the previous comparative method.

To illustrate the movement of these vowels in the perceptual space throughout their production, the data from the three distinct points of measurement have been plotted in Figure 2.9. The sequence of vowel plots in Figure 2.9 visually describes the movement through time of these vowels, which correlates with the interpretation of Figures 2.7 and 2.8 above. The observation of the three figures as a chronological sequence suggests that English /o/ moves backwards and up through time, Spanish /o/ remains stable, and the two Portuguese vowels move slightly lower and get more fronted through the length of the duration of the vowel.

The raw logF1 and logF2 values of each individual mid-back vowel produced by the participants in each of the three languages were tested for statistical differences with nine separate linear-mixed effects models. The first three models took Portuguese /ɔ/ as the intercept and tested the differences between this and the other three mid-back vowels at each of the three different points of measurement separately. The second set of models took Spanish /o/ as the intercept and the third

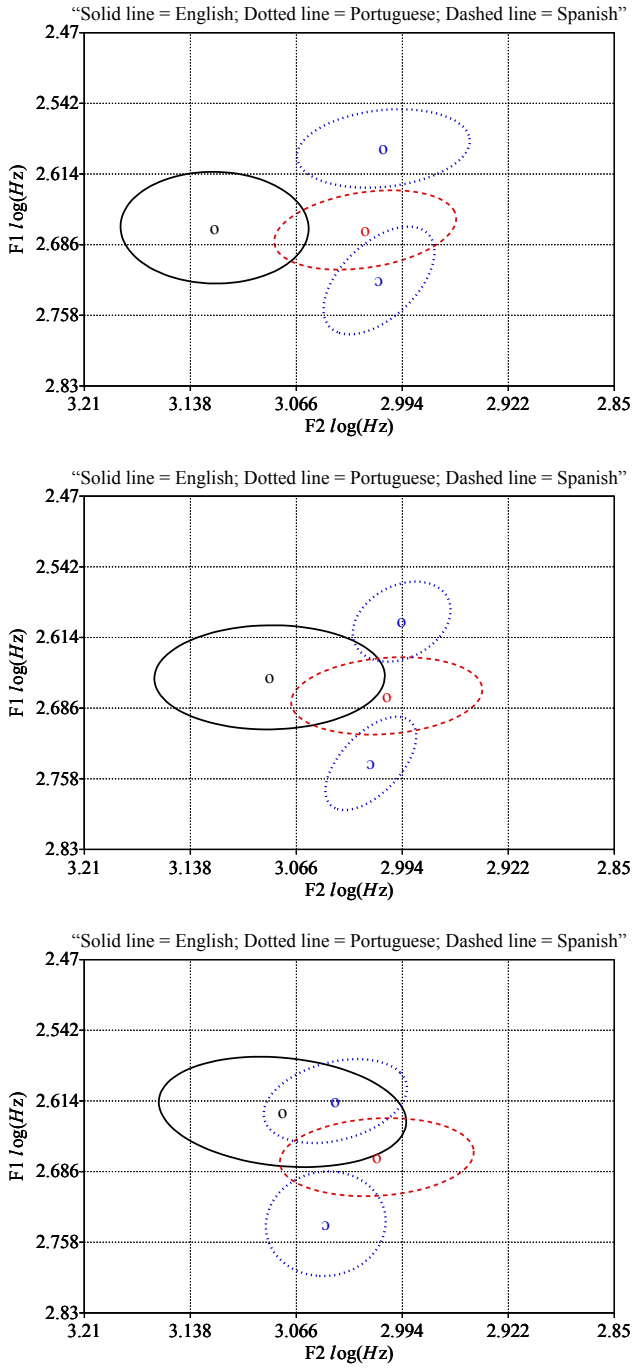

Figure 2.9: Movement of mid-back vowels of English, Spanish and Portuguese through time (top=25%, middle=midpoint, bottom=75% of vowel). English /e/ and /ɛ/ are graphed using a solid line, Spanish /e/ using a dashed line, and Portuguese /e/ and /ɛ/ use a dotted line.

Table 2.31: Summary of statistically significant differences among mid-back vowels of English, Spanish and Portuguese after a set of statistical analyses on the logF1 and logF2 values at each of the three individual points of measurement through the duration of the vowels.

Vowel pair compared	Contrast	25%		50%		75%	
	before/after	logF1	logF2	logF1	logF2	logF1	logF2
Port. /ɔ/ vs. Port. /o/	Yes/Yes	*	n.s.	*	n.s.	*	n.s.
Port. /ɔ/ vs. Span. /o/	Yes/Yes	*	n.s.	*	n.s.	*	n.s.
Port. /ɔ/ vs. Engl. /o/	Yes/Yes	*	*	*	*	*	n.s.
Span. /o/ vs. Engl. /o/	Yes/Yes	n.s.	*	n.s.	*	*	*
Span. /o/ vs. Port. /o/	Yes/Yes	*	n.s.	*	n.s.	*	n.s.
Port. /o/ vs. Engl. /o/	Yes/Yes	*	*	*	*	n.s.	n.s.

set of models took Portuguese /o/ as the base of the comparison. Due to the fact that the presentation of these results would be too lengthy if done exhaustively, only the summary of the results will be presented here. However, the results from each individual test are included in Appendix B for the readers' reference if needed.

In order to summarize the findings from the multiple statistical analyses performed on the mid-back vowels of the three languages studied in this dissertation, Table 2.31 was created. Table 2.31 includes the results from the comparisons of all mid-back vowel pairs on both measures at all points of measurement.

In Table 2.31, the column headed 'Contrast' shows whether the two vowels in each pair are found to be two different phonetic categories or not according to the multiple statistical analyses performed. The 'Contrast' column includes a before and after to refer to the difference in interpretation of results before comparing the logF1 and logF2 measures at all three points of measurement, that is, when only the midpoint was analyzed versus after taking the effects of potential diphthongization into account. Since all mid-back vowel pairs were found to be phonetically contrastive even before comparing them at all three points of measurements, this column does not add any new information at this point. However, note that the inclusion of the findings from the statistical analyses performed on logF1 and logF2

through time does indeed add information as to where and when the vowel pairs contrast with each other and how, and not only in what amount, they differ in their diphthongization patterns.

The findings from this section on the diphthongization of the mid-back vowels of English, Spanish and Portuguese, thus, show that each of the four mid-back vowels studied in this section are separate phonetic categories. The differences among them are summarized as follows:

- Portuguese /ɔ/ and Portuguese /o/ differ in height at all points of measurement but not in backness, and do not differ in diphthongization amount or direction.

- Portuguese /ɔ/ and Spanish /o/ differ in height at all points of measurement but not in backness, but they do differ in diphthongization amount along the logF1 axis, since Spanish /o/ is the most stable of all four vowels compared.

- Portuguese /ɔ/ and English /o/ differ at all points of measurement on both height and backness, and they also differ in diphthongization degree and direction.

- Spanish /o/ and English /o/ differ mainly in backness but not in height, except at the end of the vowel when English /o/ has raised, and thus they differ in diphthongization along the logF2 axis.

- Spanish /o/ and Portuguese /o/ differ in height at all times but not in backness, and they diphthongize at different degrees.

- English /o/ and Portuguese /o/ differ in both height and backness at the beginning and through the midpoint of the vowels in time, but they seem to merge into a similar phonetic category after the midpoint. These two vowels diphthongize in different degree and show opposite direction in diphthongization in both the logF1 and logF2 axis.

Diphthongization in mid-front vowels

The case of the mid-front vowels is similar to the mid-back vowels described in the previous section in that the mid-high front vowel of English /e/ has traditionally been described as diphthongizing. In order to observe this potential diphthongization pattern in this and the rest of the mid-front vowels in English, Spanish and Portuguese, Diff1 and Diff2 values were calculated for each mid-front vowel by each participant, where Diff1 and Diff2 again represent the difference in formant values from the 25% of the vowel to the 75%. Diff1 represents the movement of the vowels in the logF1 axis, while Diff2 measures their movement in the logF2 axis. The mean Diff1 and Diff2 values for each of the five vowels are shown in Figures 2.10 and 2.11.

Figure 2.10 represents the movement of the vowels in the logF1 axis. In Figure 2.10 a box that is located on the negative side of the chart represents the movement of a vowel upwards in the vowel space, while the positive values represent a lowering of the vowel through time. In Figure 2.11 the movement of vowels on the logF2 axis is represented. Thus, a box located on the positive side of the chart represents fronting, while the negative side represents retraction, that is, movement of the vowels backwards in the vowel space. The further away from 0 the boxes are, the more the vowels diphthongize.

Figures 2.10 and 2.11 confirm the traditional description of these front mid vowels showing Spanish /e/ as the least diphthongizing of the five and English /e/ as the most diphthongizing one. It is however interesting to observe that English /ɛ/ also shows a bit of formant movement, as it seems to move upwards in the logF1 axis throughout the length of the vowel. This movement, however, is small enough that it might not differ from the movement undergone by Spanish /e/, which is the most stable of all five vowels tested here. The results from such comparison will be presented in Table 2.32 below. It is also interesting to observe that the Portuguese vowels also show formant movement as they move, in both cases, lower and backwards in the vowel space.

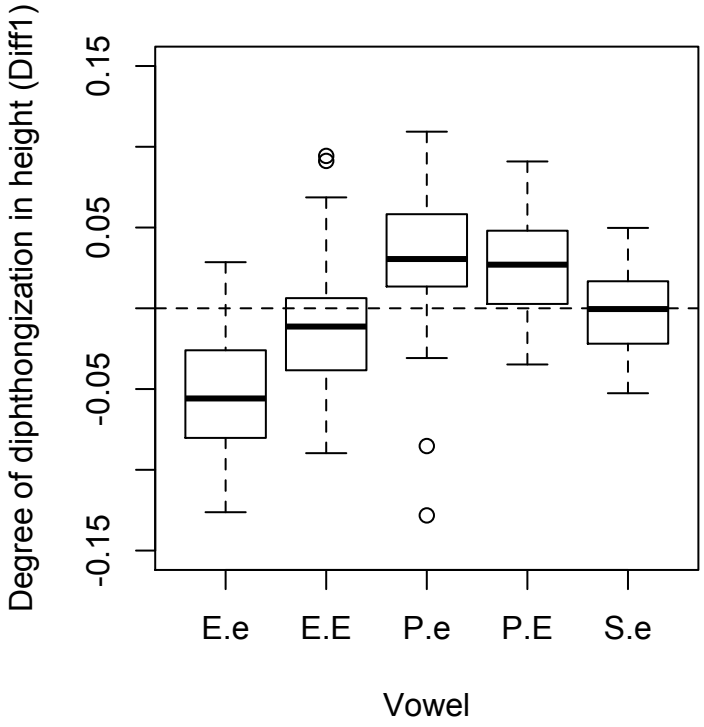

Figure 2.10: Movement of the mid-front vowels of English, Portuguese and Spanish in height through time (Diff1). English /e/ and /ɛ/ are referred to as "E.e" and "E.E" respectively, Portuguese /e/ and /ɛ/ as "P.e" and "P.E," and Spanish /e/ as "S.e" in that order. Negative Diff1 values represent raising of the vowels in the vowel space while positive values represent lowering. The graph shows that (a) English /e/ undergoes the most amount of movement in height, rising through time; (b) English /ɛ/ moves upwards in the vowel space very slightly; (c) the two Portuguese mid-front vowels move lower in the vowel space through time, and (d) Spanish /e/ is the most stable of all five.

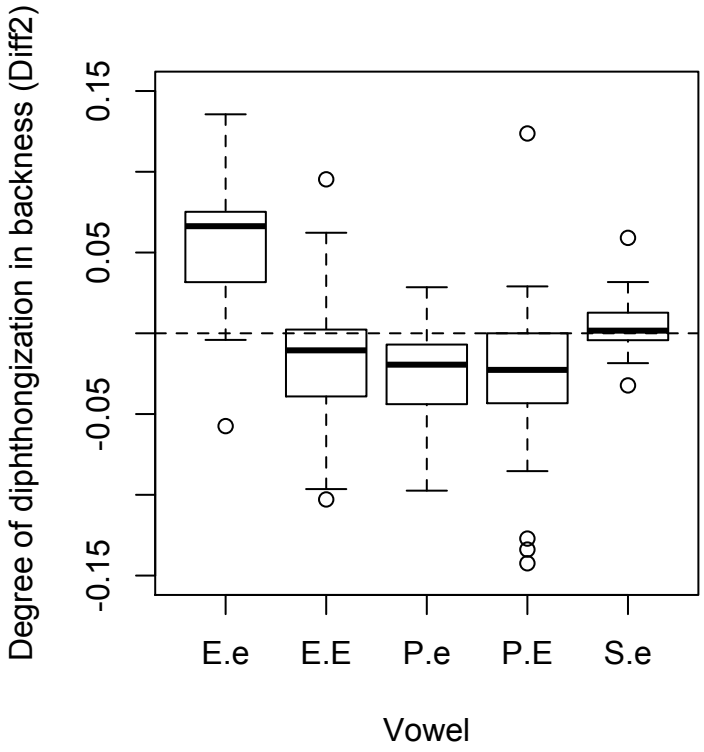

Figure 2.11: Movement of the mid-back vowels of English, Portuguese and Spanish in backness through time (Diff2). English /e/ and /ɛ/ are referred to as "E.e" and "E.E" respectively, Portuguese /e/ and /ɛ/ as "P.e" and "P.E," and Spanish /e/ as "S.e" in that order. Negative Diff2 values represent movement of the vowels backwards in the vowel space through time, while positive values represent fronting. The graph shows that (a) English /e/ moves clearly backwards; (b) English /ɛ/ shows some variability in its degree of backwards movement in the vowel space with its mean nearing zero; (c) the two Portuguese vowels move backwards through the length of the vowel; and (d) Spanish /e/ remains stable.

Table 2.32: Results from a linear mixed-effects model on mid-front vowel diphthongization data in the three languages, with Spanish /e/ as the intercept.

		β	t
Diff1	Intercept (= Spanish /e/)	-0.039	
	English /ɛ/	-0.010	-0.67
	English /e/	-0.048	-3.30*
	Portuguese /ɛ/	0.032	2.18*
	Portuguese /e/	0.036	2.49*
Diff2	Intercept (= Spanish /e/)	0.000	
	English /ɛ/	-0.013	-0.95
	English /e/	0.059	4.22*
	Portuguese /ɛ/	-0.027	-1.92
	Portuguese /e/	-0.026	-1.88

In order to compare these diphthongization patterns, a series of linear mixed-effects models were run on the Diff1 and Diff2 data. The first of these models took Spanish /e/ as the intercept and compared its diphthongization pattern to each of the other four mid-front vowels. The results from this first statistical model are shown in Table 2.32. These results show that movement of Spanish /e/ on both dimensions (height and backness) is different from the movement of English /e/. Portuguese /ɛ/ and Portuguese /e/ also diphthongize differently from Spanish /e/ in the logF1 axis, but not in the logF2 axis, although they do show t values approaching significance. The difference in diphthongization between Spanish /e/ and English /ɛ/ is not rendered significant on any of the two measures, as hypothesized in the previous paragraph.

A second linear mixed-effects model tested the difference between the diphthongization of English /ɛ/ and the other four vowels. The results from this model show that the diphthongization of English /ɛ/ significantly differs from that of English /e/ in both dimensions. However, English /ɛ/ only differs from Portuguese /ɛ/ and Portuguese /e/ in its movement along the logF1 axis, but not along the logF2 axis. The results from this second model are presented in Table 2.33. Diphthongization

Table 2.33: Results from a linear mixed-effects model on mid-front vowel diphthongization data in the three languages, with English /ɛ/ as the intercept.

		β	t
Diff1	Intercept (= English /ɛ/)	-0.014	
	English /e/	-0.038	-3.19*
	Portuguese /ɛ/	0.042	2.85*
	Portuguese /e/	0.046	3.16*
	Spanish /e/	0.010	0.67
Diff2	Intercept (= English /ɛ/)	-0.013	
	English /e/	0.072	6.03*
	Portuguese /ɛ/	-0.013	-0.97
	Portuguese /e/	-0.013	-0.92
	Spanish /e/	0.013	0.95

of Spanish /e/ is not rendered significantly different from that of English /ɛ/, as it was already established in Table 2.32.

Three more vowel pairs are yet to be compared for diphthongization. The first two, namely English /e/ and Portuguese /e/, and English /e/ and Portuguese /ɛ/ are tested with a third linear mixed-effects model that took English /e/ as the intercept. The results from this third model are presented in Table 2.34. The last vowel pair, namely that of Portuguese /e/ and /ɛ/, is tested for difference in diphtongization with a fourth and last model. The results from this model, which took Portuguese /ɛ/ as the intercept, are presented in Table 2.35.

Results in Table 2.34 show that both English /e/ and Portuguese /e/, and English /e/ and Portuguese /ɛ/ significantly differ in diphthongization, both in logF1 as well as logF2 movement. The results for the third pair, namely that composed of both Portuguese mid-front vowels, are shown in Table 2.35. These results show that these two vowels do not differ in their diphthongization in neither of the two measures.

To summarize the results from the statistical tests on the diphthongization of the five mid-front vowels studied in this section, Table 2.36 is included here. Table

Table 2.34: Results from a linear mixed-effects model on mid-front vowel diphthongization data in the three languages, with English /e/ as the intercept.

		β	t
Diff1	Intercept (= English /e/)	-0.052	
	Portuguese /ɛ/	0.080	5.48*
	Portuguese /e/	0.085	5.79*
	Spanish /e/	0.048	3.30*
	English /ɛ/	0.038	3.19*
Diff2	Intercept (= English /e/)	0.059	
	Portuguese /ɛ/	-0.085	-6.14*
	Portuguese /e/	-0.085	-6.10*
	Spanish /e/	-0.059	-4.22*
	English /ɛ/	-0.072	-6.03*

Table 2.35: Results from a linear mixed-effects model on mid-front vowel diphthongization data in the three languages, with Portuguese /ɛ/ as the intercept.

		β	t
Diff1	Intercept (= Portuguese /ɛ/)	0.028	
	Portuguese /e/	0.004	0.37
	Spanish /e/	-0.032	-2.18*
	English /ɛ/	-0.042	-2.85*
	English /e/	-0.080	-5.48*
Diff2	Intercept (= Portuguese /ɛ/)	-0.026	
	Portuguese /e/	0.000	0.05
	Spanish /e/	0.027	1.92
	English /ɛ/	0.013	0.97
	English /e/	0.085	6.14*

Table 2.36: Summary of results from multiple linear mixed-effects models on English, Spanish and Portuguese mid-front vowel diphthongization data.

	Diff1	Diff2
Spanish /e/ vs. Portuguese /ɛ/	*	n.s.
Spanish /e/ vs. Portuguese /e/	*	n.s.
Spanish /e/ vs. English /ɛ/	n.s.	n.s.
Spanish /e/ vs. English /e/	*	*
English /ɛ/ vs. English /e/	*	*
English /ɛ/ vs. Portuguese /ɛ/	*	n.s.
English /ɛ/ vs. Portuguese /e/	*	n.s.
English /e/ vs. Portuguese /ɛ/	*	*
English /e/ vs. Portuguese /e/	*	*
Portuguese /e/ vs. Portuguese /ɛ/	n.s.	n.s.

2.36 shows that the only two vowel pairs that do not show any significant difference in the amount of diphthongization through time are Spanish /e/ and English /ɛ/ on the one hand, and Portuguese /e/ and Portuguese /ɛ/ on the other.

These findings lead to a new question. It has been shown that specific vowels differ significantly in degree of diphthongization (see Table 2.36) as well as in specific formant measurements at the midpoint. However, the diphthongization data suggests that there might be differences among the mid-front vowels in these three languages at specific points of measurement that might affect our previous interpretation as to whether these vowels are different phonetic entities and how they differ when they do. Particularly, English /e/ and Portuguese /e/ were found to differ in degree and direction of diphthongization but did not show any significant spectral differences at their midpoint, neither in height nor in backness, meaning that these two vowels are one single phonetic category at their midpoint. Moreover, this was the only mid-front vowel pair that was found not to differ in any of the two measures at their midpoint. On the other hand, these two vowels were found to differ in degree and direction of diphthongization, which opens the question of whether they happen to overlap at their midpoint alone or they are actually one single phonetic category all through their duration in time.

In order to investigate the relationship between diphthongization and spectral differences through time that might have consequences for the phonetic categorization of all these vowels, new linear mixed-effects models were conducted comparing, now taking into account all three points of measurement, all possible pairs of mid-front vowels. These statistical models take not only the midpoint of the vowel as the dependent variable that defines the nature of the vowel, but also the 25% and 75% of the vowel. The differences in value at each point of measurement both in logF1 and logF2 provide additional information to complete the phonetic description of these vowels.

To visualize the movement of the five mid-front vowels in the perceptual space through time, the three points of measurement have been plotted in Figure 2.12. The sequence of vowel plots in Figure 2.12 visually confirm the description of the formant movement of these five vowels. While Spanish /e/ is the most steady of all five, the two Portuguese vowels move down and back in the vowel space, English /ɛ/ moves back and slightly up, and English /e/ moves front and up in the space.

In order to test the phonetic differences at each of these three points of measurement among the five vowels, four additional sets of linear mixed-effects models were performed. Again, each set of models took a different vowel as the intercept so that the differences between all possible vowel pairs would be tested for statistical significance. The independent variables in all cases were the logF1 and logF2 values at each of the three points of measurement (25%, 50% and 75% into the vowel) and the models took 'speaker' and 'word' as random intercepts. The detailed results from these four sets of linear mixed-effects models are included in Appendix B, for the readers' reference.

Nevertheless, as a review of the results from the multiple statistical analysis on all potential differences at all points of measurement between all possible pairs of mid-front vowels, Table 2.37 is included here. These results, together with the findings in differences in Diff1 and Diff2 among them, are summarized in the following list below:

- Portuguese /ɛ/ and /e/ differ at all points of measurement both in height and

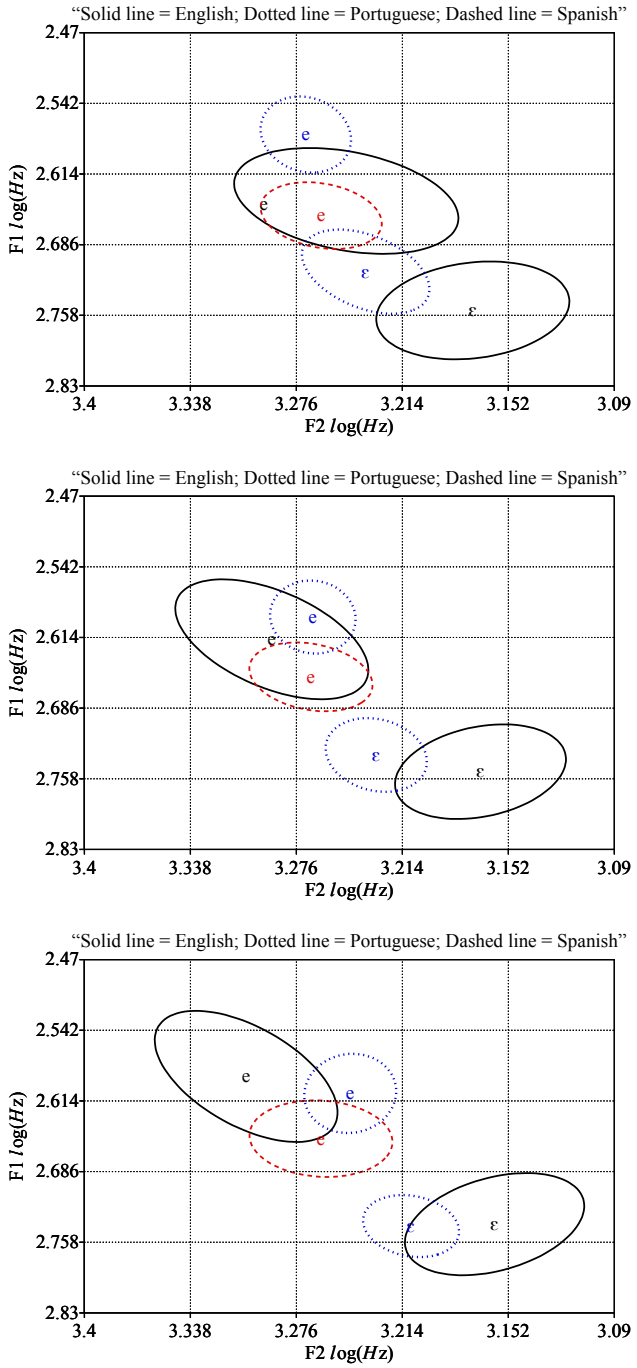

Figure 2.12: Movement of mid-front vowels of English, Spanish and Portuguese through time (top=25%, middle=midpoint, bottom=75% of vowel). English /e/ and /ɛ/ are graphed using a solid line, Spanish /e/ using a dashed line, and Portuguese /e/ and /ɛ/ use a dotted line.

Table 2.37: Summary of statistically significant differences among mid-front vowels of English, Spanish and Portuguese after analysis of differences in time.

Vowel pair compared	Contrast	25%		50%		75%	
	before/after	logF1	logF2	logF1	logF2	logF1	logF2
Port. /ɛ/ vs. Port. /e/	Yes/Yes	*	*	*	*	*	*
Port. /ɛ/ vs. Span. /e/	Yes/Yes	*	n.s.	*	n.s.	*	*
Port. /ɛ/ vs. Engl. /ɛ/	Yes/Yes	*	*	n.s.	*	n.s.	*
Port. /ɛ/ vs. Engl. /e/	Yes/Yes	*	n.s.	*	*	*	*
Span. /e/ vs. Engl. /ɛ/	Yes/Yes	*	*	*	*	*	*
Span. /e/ vs. Engl. /e/	Yes/Yes	n.s.	n.s.	*	n.s.	*	*
Span. /e/ vs. Port. /e/	Yes/Yes	*	n.s.	*	n.s.	*	n.s.
Engl. /e/ vs. Engl. /ɛ/	Yes/Yes	*	*	*	*	*	*
Engl. /e/ vs. Port. /e/	No/Yes	*	n.s.	n.s.	n.s.	n.s.	*
Port. /e/ vs. Engl. /ɛ/	Yes/Yes	*	*	*	*	*	*

backness, even though they diphthongize to a similar degree and in the same direction.

- Portuguese /ɛ/ and Spanish /e/ differ in height at all times, as they do in degree of movement across the logF2 dimension through time. The differences between them are mainly in height, but they become different in backness as the vowels progress due to the difference in amount of diphthongization.

- Portuguese /ɛ/ and English /ɛ/ differ at all points of measurement even though they become similar in height by the midpoint. The differences between them are mainly in backness and in degree of movement through time on the logF2 axis, English /ɛ/ sliding backwards to a greater degree than Portuguese /ɛ/.

- Portuguese /ɛ/ and English /e/ differ at all points of measurement in mostly both height and backness as well as in degree and direction of diphthongization, since they move in opposite directions through time.

- Spanish /e/ and English /ɛ/ differ at all points of measurement on both height and backness, although they were not found to be different in the way and

degree they diphthongize.

- Spanish /e/ and English /e/ start off as similar phonetic categories and they differ as the vowels move through time. Since the former was found to be the least diphthongizing of all vowels compared here, and the latter the most diphthongizing one, the spectral differences arise as English /e/ becomes more fronted and raises in the vowel space through time.

- Spanish /e/ and Portuguese /e/ differ in height, but not in backness, at all points of measurement. They also differ in diphthongization pattern along the logF1 dimension.

- English /e/ and /ɛ/ differ in all measures and points of measurement, as well as in diphthongization degree and direction.

- English /e/ and Portuguese /e/ show similar spectral characteristics at their midpoint but opposite direction of diphthongization, which causes them to differ both at the beginning (in height) and end (in backness) of their movement in time.

- Portuguese /e/ and English /ɛ/ differ at all points of measurement in both height and backness, as well as in diphthongization direction along the logF1 dimension.

In addition, it is interesting to observe that English /e/ was found to be the most diphthongizing of all mid-front vowels studied in this subsection while Spanish /e/ was, once again, found to be the most stable of all.

In conclusion, the Table shows that all mid-front vowel pairs studied here are interpreted to be phonetically contrastive due to a significant difference found in one or both of the measures (height and/or backness) at at least one of the points of measurement (25%, 50%, and 75% into the vowel).

Note, however, that the only vowel pair that did not contrast acoustically in previous comparisons, namely that composed by English /e/ and Portuguese /e/,

is now found to be contrastive after finding differences at the beginning (25%) and end (75%) throughout the vowels. This finding is most revealing as it suggests that traditional cross linguistic vowel comparisons based on spectral measurements at the vowel midpoint might have missed information that is relevant for their phonetic representation and categorization.

2.3 Summary of Results and Discussion

The multiple statistical analyses of logF1 and logF2 values of the stressed vowels of English, Spanish and Portuguese performed in this chapter can be summarized as follows:

Statistical comparisons of the logF1 and logF2 values of Spanish /i/, Portuguese /i/, English /i/ and English /ɪ/ at their midpoint resulted in the finding that Spanish /i/, Portuguese /i/ and English /i/ are all acoustically the same vowel category, which differ in both height and backness from English /ɪ/.

Similar comparisons at the midpoint of the vowel between the three high back vowels compared rendered the result that Portuguese /u/ and Spanish /u/ are acoustically the same sound category but they differ from English /u/ in backness.

As for the four low vowels compared, no differences were found in height among them. In backness, however, Portuguese /a/ was found to differ from both English /æ/ and /ɑ/ and the two English low vowels were found to differ from each other. As for Spanish /a/, this vowel was found to be acoustically similar to both Portuguese /a/ and English /æ/ while differing from English /ɑ/.

The comparisons of the four mid-back vowels of these three languages showed that these four vowels not only differ in spectral differences at their midpoint but also at different points of measurement through time and in degree and direction of diphthongization. Portuguese /ɔ/ and Portuguese /o/ acoustically differ in height but neither in backness nor in diphthongization patterns. Portuguese /ɔ/ and Spanish /o/ differ in height but not in backness. In addition, due to the stability of Spanish /o/ through time and the slight movement of Portuguese /ɔ/ along the logF2 axis,

these two vowels also differ in diphthongization. Portuguese /ɔ/ and English /o/ differ both in height and backness all through the duration of the vowel, as well as in degree and direction of diphthongization, as Portuguese /ɔ/ lowers and gets more centralized and English /o/ raises and gets more fronted in time. Spanish /o/, being the most stable of all four, and English /o/, being the most diphthongizing, differ in backness but not in height except at the end of the vowel after English /o/ has finished its movement upwards, where the difference in height is also revealed. Spanish /o/ and Portuguese /o/ differ in height at all times but not in backness, and they diphthongize at different degrees. Finally, English /o/ and Portuguese /o/ differ in both height and backness at the beginning and through the midpoint of the vowels in time, but they seem to merge into one single phonetic category after the midpoint. These two vowels diphthongize in different degree and show opposite direction in diphthongization in both the logF1 and logF2 axis. After all the statistical analyses, each of the four mid-back vowels studied in this chapter are interpreted as being acoustically different from the rest and thus their own distinct phonetic category.

The spectral comparison at all three points of measurement between the mid-front vowels rendered interesting results confirming traditional phonological distinction between vowels within a language, and revealing new information about the acoustic comparisons cross-linguistically. Among the within-language distinctions confirmed, Portuguese /e/ and /ɛ/ were found to differ acoustically in both height and backness but not in diphthongization, although they were both found to move backwards and lower in the vowel space through time. English /ɛ/ and /e/ were also found to be acoustically distinct both in height and fronting at all points of measurement. Moreover, the greater degree and direction of diphthongization of English /e/ was confirmed, namely its raising and fronting through time, which was found to contrast with the backwards movement of English /ɛ/.

Among the cross-linguistic acoustic differences found, Spanish /e/ was found to be the least diphthongizing of all five vowels, differing from all four vowels in diphthongization pattern. Spanish /e/ was also found to differ in spectral quality

from Portuguese /e/ in height, but not in backness; from Portuguese /ɛ/ mainly in height, although the latter shows a tendency to move apart from Spanish /e/ in the logF2 axis by the end of the vowel; from English /ɛ/ at all points of measurement and in both height and backness; and starts to differ from English /e/ in both height and backness as they reach the midpoint of the vowel in time.

In addition, Portuguese /ɛ/ and English /ɛ/ were found to be distinct acoustic entities differing in height and in amount of movement along the logF2 axis; and Portuguese /ɛ/ and English /e/ are also distinct phonetic categories in all features as they are located close but at different locations in the vowel space and move in opposite directions through time; and Portuguese /e/ and English /ɛ/ were also found to differ in most spectral and diphthongization measures.

Finally, and most relevantly, English /e/ and Portuguese /e/ were found to show significant spectral differences in both height and backness only after comparing them at different points of measurement, as they seem to start out at different locations in the vowel space and cross each other as they diphthongize, thus overlapping in spectral quality at their midpoint. If the diphthongization pattern and spectral quality of these two vowels at the beginning and end of their production had not been analyzed in this chapter, these significant spectral differences would not have been found.

With regards to how the vowels by these three native speaker groups compare to those reported by previous studies, it is important to note that no direct comparative analysis was made because different studies either described different varieties of these languages, different elicitation materials were used (varying, for example, in the phonetic context of the target vowels) and/or different methods of data analysis were implemented. Nevertheless, after cursory inspection of the results from previous studies, no apparent mismatches were found between the new data presented in this chapter and the data presented by previous studies.

For Portuguese, Escudero et al. (2009) acoustically describe the seven oral vowels produced by two groups of native speakers. The first group contained 10 male speakers of Brazilian Portuguese from the region of Sao Paulo. As mentioned above,

although the values for F1 and F2 (in Hz) reported for the native Brazilian Portuguese speakers in the current study do not perfectly match those reported by Escudero et al. (2009) for Sao Paulo speakers, no substantial mismatches were found between the two groups.

For Spanish, Morrison & Escudero (2007) carry out a cross-dialect comparison of peninsular and Peruvian Spanish vowels. Although the raw data in Hz for the analyzed vowels is not available, the graphed results of the vowel space of the peninsular and Peruvian Spanish speakers show a similar general pattern of distribution and the same characteristic stability of the Spanish vowels produced by the Mexican Spanish speakers who participated in the current study. Additionally, various studies present data from Castilian Spanish male speakers' vowels (Martínez Celdrán & Fernández Planas (2007), Bradlow (1995)). This data is also consistent with the data from Mexican Spanish vowels presented in this chapter.

As far as American English vowels is concerned, Hillenbrand et al. (1995) provides first and second formant data for 45 male American English speakers' vowels. It is interesting to observe that, while most of the F1 and F2 values reported by Hillenbrand et al. (1995) do not seem inconsistent with the findings in this chapter, these authors' reported mean F2 values for their male participants' English /u/ (997Hz) differs considerably from our current results (1908Hz). In other words, the English /u/ reported by Hillenbrand et al. (1995) is located considerably further back in the speakers' vowel space than the English /u/ reported in the current investigation (see Figure 2.3 above).

The fronting of English /u/ reported in this chapter is, however, not a feature specific to the participants in this study. The fronting of English /u/ has been reported in the past as a characteristic of various North American dialects, including specific varieties of New World English (see Habick (1991) for central Illinois; Thomas (1989) for central Ohio; Labov (1994) for Pennsylvania; Labov et al. (1972) for Texas; Clarke et al. (1995) for particular parts of Canada, or Thomas (2001) and Feagin (2003) for parts of the Southern USA). Hinton et al. (1987) reports, however, that back vowel fronting was rarely present in the West before the 1980s, when it

started to be documented. Since then, back vowel fronting has also been documented in areas of the Western USA including Portland, Oregon (Ward (2003)), Utah (Di Paolo & Faber (1990)), and more recently Arizona (Hall-Lew (2004)). It has often being interpreted as a relatively recent sound change representative of specific social or geographic strata and which affects younger generations of speakers in particular. While Hillenbrand et al. (1995) does not report the ages of his male adult speakers, the participants in the current study were males between 19 and 28 years of age, that is, males who were born after the first documentations of the back vowel fronting in American English. This being the case, the difference between Hillenbrand et al. (1995)'s reported F2 values and the current results for English /u/ might be interpreted as supporting evidence for the Western Vowel Shift.

On an additional note, according to Hall-Lew (2004), /u/ fronting in American English is usually (although not always) accompanied by fronting of /o/. In our case, the native English speakers in this study did produce an English /u/ that is much more fronted than expected from its traditional phonological description, while their English /o/ also showed signs of centralization, both of which are consistent with previous accounts of the Western Vowel Shift phenomenon in the literature. However, Hall-Lew (2004) quotes Stevens & House (1963) when clarifying that a preceding coronal consonantal context considerably favors the fronting of /u/, while no such accounts have been reported for the fronting of /o/. It is worth noting, thus, that all the vowels analyzed in this dissertation were located in between coronal consonants, and that this context might have triggered English /u/ productions that are characteristic of this and not all phonetic contexts. Altogether, however, the back vowels produced by the English native speaker participants in this study are consistent with the previous literature on back vowel fronting in American English.

2.4 Concluding Remarks

The results from the multiple cross-linguistic vowel comparisons described in the previous sections show that the relationships among the stressed vowels of English,

Spanish and Portuguese are very complex. While some vowel categories cannot be distinguished from one another with respect to the parameters studied in this dissertation, others are found to differ in varying measures and degrees. The cases of the English, Spanish and Portuguese /i/, the Spanish and Portuguese /u/, or Spanish and Portuguese /a/ are three of the cases where these vowels did not prove to be acoustically distinct. The comparisons among the mid-back vowels in these three languages on the one hand, and the mid-front vowels on the other, revealed a diversity of ways in which these vowels differ. More specifically, amount and directionality of formant movement, as well as differences and similarities in formant structure at different points through the vowel, were found to be relevant factors interacting to create significant phonetic category differentiation.

Subsequent chapters in this dissertation seek to investigate the effects of these acoustic similarities and differences in the acquisition of new phonetic categories by adults learning a foreign language. Thus, Chapter 3 will present a description and analysis of the Portuguese vowels produced by two groups of learners of Portuguese whose L1 is English. Chapter 4 will present the results from a perception study on the assimilation and discriminability of the Portuguese mid-front vowels described in this chapter by the learners of Portuguese whose vowel productions will be described and analyzed in Chapter 3. Ultimately, the findings in the current chapter will serve to create the predictions, based on this author's interpretation of the tenets of the two models of either second language speech learning (the SLM) or cross-linguistic speech perception (PAM) discussed in Chapter 1, for the learners' acquisition of the oral vowels of Portuguese.

CHAPTER 3

VOWELS OF L2 AND L3 LEARNERS OF PORTUGUESE

The results from the multiple cross-linguistic vowel comparisons described in the previous chapter show that the cross-linguistic relationships among the stressed vowels of English, Spanish and Portuguese are very complex. While some vowel categories cannot be distinguished from one another with respect to the parameters studied in this dissertation, others are found to differ in varying measures and degrees. The cases of the English, Spanish and Portuguese /i/, the Spanish and Portuguese /u/, or Spanish and Portuguese /a/ are three of the cases where these vowels did not prove to be cross-linguistically acoustically distinct. The comparisons among the mid-back vowels in these three languages on the one hand, and the mid-front vowels on the other, revealed a diversity of ways in which these vowels differ. More specifically, amount and directionality of formant movement, as well as differences and similarities in formant structure at different points through the vowel, were found to be relevant factors interacting to create significant phonetic category differentiation.

This chapter describes the Portuguese stressed vowel inventory as produced by two learner groups with different foreign language experience: (a) native American English speakers at an intermediate level of Portuguese as their L2, (the L2 group hereafter); and (b) native American English speakers at the same level of proficiency in Portuguese as their L3, who learned Spanish as their L2 before starting to learn Portuguese (the L3 group hereafter). The primary aim of this chapter is to compare and contrast the stressed vowels of Portuguese of these two groups as they relate to the native Portuguese stressed vowel inventory described in the previous chapter, as well as between each other with the objective of investigating phonetic category interactions in the perceptual space of the two Portuguese learners under study. A secondary objective is to investigate the effect that learning a second language, Spanish in this case, has in learning a third language, Portuguese.

As it was described in Chapter 1 of this dissertation, there are two models of phonetic acquisition that address either cross-linguistic speech perception or L2 learning, namely PAM-L2 and SLM. These two models agree on the assumptions that (1) all phonetic categories reside in a common space in the speaker's mind (Flege (1995), Flege et al. (1999a), Flege et al. (2002), Best (1995), Best & Tyler (2007), Flege (2007)), and that (2) the learnability of L2 phonetic segments is perceptual in nature and depends on the perceived phonetic distance between them and the most similar segments in the L1 phonetic inventory (Flege et al. (2003), Flege (1995), Flege (2007)).

Despite the commonalities in these two models, it is important to note that PAM (and PAM-L2) is a model that was designed specifically to explain nonnative speech perception by naïve listeners, while SLM was designed to address production/perception of L2 speech by L2 learners with considerable experience with the L2. While PAM makes its predictions based on previously available perception data, SLM mostly, although not exclusively, bases its hypotheses for L2 speech learning development on acoustic data. In no manner is it assumed here that both models address the same exact questions, neither it is assumed that there is a direct correlation between perceptually relevant differences and acoustically relevant ones. Nevertheless, several research projects have suggested that perceptual skill level is positively correlated with accuracy in producing L2 vowels (Flege et al. (1997), Flege et al. (1995a), Ingram & Park (1998)). Based on this precedent, this dissertation will address the topic of L2 and L3 phonetic learning by making use of the results of the analysis of acoustic data provided by native speakers of English, Spanish and Portuguese in the previous chapter, in order to make predictions for L2 and L3 production of Portuguese vowels by native speakers of English.

Having researched the acoustic similarities and differences among the vowel categories of the three languages in the previous chapter, predictions will be made in the following section for the acquisition of Brazilian Portuguese vowels by native speakers of American English, who may or may not have previous experience with Spanish speech patterns. These predictions will be created by using the principles of

both PAM/PAM-L2 and SLM. The predictions will be tested in this and subsequent chapters, where the production and discriminability of Portuguese stressed vowels by L2 and L3 learners of the latter will be investigated.

3.1 Predictions for L2 learning

In order to determine the predictions for the learning of the Portuguese stressed vowels by native speakers of English, each of the vowel groups analyzed and described in the previous chapter will be addressed separately. The effect of having been exposed to Spanish speech patterns for a longer period and prior to the onset of exposure to Portuguese will also be discussed. The predictions for learning specific vowel groups will be based on the principles of the two theoretical models reviewed above and they will ultimately be tested by analyzing production data gathered from the two Portuguese learner groups under study.

3.1.1 Predictions for the learning of the Portuguese high front vowel

The data analysis in the previous chapter demonstrated that Portuguese /i/ is not acoustically statistically different from English /i/, but it is significantly different from English /ɪ/. The fact that English /i/ was not found to be different from Portuguese /i/ suggests that native English speaking learners of Portuguese will do one of two things when producing the target Portuguese /i/: (1) they will use their native English /i/ to produce the target Portuguese /i/, or (2) they will have learned to produce a native-like Portuguese /i/. Thus, it would be impossible to tell from these speakers' Portuguese production data whether native-like learning of a new phone has actually occurred or whether the segment produced is the result of transfer from an existing sound in their L1 inventory.

Be it as it may, no significant spectral differences are expected to be found between Portuguese /i/ of either the learner groups or the native target /i/. In addition, no spectral difference between the L2 and L3 groups is expected in their productions of Portuguese /i/, since Spanish /i/ was not found to differ acoustically

from neither the native or target, that is, Portuguese, high front vowel either.

3.1.2 Predictions for the learning of the Portuguese high back vowel

Portuguese /u/ was found to be acoustically distinct from English /u/. According to the SLM, the more salient the perceived differences between the L1 and the target vowel, the more likely it is that a new separate phonetic category will be established for the foreign language sound. Portuguese /u/ being acoustically so different from English /u/, the establishment of a new phonetic category is expected for both groups of Portuguese learners.

In addition, PAM-L2 (Best & Tyler (2007)) creates its predictions for L2 speech perception by taking into account not only phonetic differences but also phonological ones. As explained in Chapter 1, PAM (Best (1995)) was specifically developed to explain non-native speech perception by naïve listeners. As such, PAM assumes that naïve perceivers are unaware of which phonetic distinctions constitute phonological differences in the unfamiliar target language and, consequently, they could not possibly differentiate phonetic and phonological levels in nonnative sounds.

Under this perspective, despite the phonetic differences between them, Portuguese /u/ and English /u/ should be perceived as similar at the phonological (functional, lexical) level by the learners studied in this chapter. In this kind of case, PAM-L2 (Best & Tyler (2007), p.23) predicts that these learners will perceive these Portuguese /u/ and English /u/ phones as phonologically equivalent but phonetically different, perceiving Portuguese /u/ as deviant with respect to the closest corresponding L1 phonetic category (English /u/ in this case) and for which these learners should create a distinct phonetic category under the common phonological category (here hypothesized to be /u/). Given that the present production experiments and data analyses were only designed to capture acoustically relevant differences, the hypothesis created for the acquisition of Portuguese /u/ from the author's understanding of PAM and PAM-L2 is coherent with that created on the base of the author's interpretation of the SLM, namely that the establishment of a new phonetic category is expected for both groups of English speaking learners of

Portuguese.

In order to make predictions about the potential differences between the L2 and L3 group in the production of Portuguese /u/, it is important to remember that Portuguese /u/ and Spanish /u/ were not found to acoustically differ from each other. This finding suggests that participants in the L3 group, who were familiar with Spanish speech patterns for years before being exposed to Portuguese ones, have been exposed to a vowel category of non-distinct acoustic characteristics for a longer period (first through Spanish and then through Portuguese, versus only through Portuguese for the L2 group). This earlier and longer exposure to a vowel of similar acoustic characteristics as the target Portuguese vowel might have consequences for these learners' production of their target /u/.

Previous research by Cobb (2009) found evidence for a scale in unstressed Spanish vowel centralization by (a) native speakers of the language and (b) two English speaking groups of learners of Spanish with differing levels of proficiency. Cobb (2009) concludes that native speakers of Spanish centralize more than advanced L2 learners, who centralize more than the intermediate learners. Additional evidence found by Simonet et al. (2011) shows that all three groups centralize their unstressed Spanish /u/ to different degrees, with the advanced learners reaching native-like (in F1) or close to native-like (in F2) targets while intermediate learners only fronted their targets (in F2). For unstressed Spanish /o/, on the other hand, both learner groups centralized less than the native speakers.

On a related note, Flege (1987) studied the French /y/ and /u/ productions of monolingual native French speakers and native English speaking learners of French with differing degrees of experience in their L2. He found that more experienced learners of French did not produce French /y/ differently from monolingual speakers, while all learners, regardless of experience level, differed significantly (in F2) from monolingual speakers in their productions of French /u/. Flege (1987) concluded that it is possible for advanced experienced learners to achieve native-like pronunciation in their L2, but that the likelihood of this happening depends on "equivalence classification." In this case, Flege hypothesizes that "equivalence classification pre-

vents experienced L2 learners from producing similar L2 phones, but not new L2 phones, authentically" (Flege (1987), p. 47). In the case of French vowels studied by Flege, French /u/ would constitute a "similar" phone to one existing in the learners L1 (English /u/), while French /y/ would constitute a "new" phone because it is presumed to be too different from any phone in the learners' L1.

The results from the above mentioned studies are relevant for the creation of hypotheses for the two learner groups in the present study. On the one hand, since Spanish and Portuguese /u/ were not found to be different acoustic entities for native speakers of those two languages, and since both learner groups started learning Portuguese at similar ages (mean AOA of L2 group = 19, mean AOA of L3 group = 20) with the L3 group having been exposed to Spanish speech patterns for a few years longer (mean AOA = 13), it is assumed here that participants in the L3 group have more experience with the Portuguese /u/ (=Spanish /u/) phone than the L2 group. Thus, L3 learners would behave more alike more experienced learners of Portuguese /u/, while the L2 group would constitute a less experienced learner group. On the other hand, given the acoustic differences found between English /u/ and Portuguese /u/ described in the previous chapter, and assuming it is the case that acoustic differences relate to perceptual differences of the kinds described by PAM, it is very likely that both groups have created a new category for the target Portuguese /u/. Nevertheless, due to the difference in amount of input received by each group, a difference in spectral quality of the Portuguese /u/ produced by the two learner groups studied here is expected. It is hypothesized here that L3 learners, having received more input that the L2 group regarding Portuguese /u/, will have created a new category that is closer or non-distinct from the native speakers' Portuguese /u/ than the one produced by the L2 group.

3.1.3 Predictions for the learning of the Portuguese low vowel

Chapter 2 showed that Portuguese /a/ is located between English /ɑ/ and /æ/ in F2, and that it is different form both. The SLM predicts that, if the learners do not perceive these differences, it is likely that they will assimilate the Portuguese vowel

to one of the two native vowel categories, and if they do, they are likely to create a new category for the target vowel sound.

It is however relevant to observe that in the case of the high vowels of English or Portuguese already discussed above, there was a target vowel that either differed substantially from the English vowel (high back vowel /u/) or overlapped with the English vowel in the learners' perceptual space (high front vowel /i/). In the case of the low vowel of Portuguese, on the other hand, the target vowel falls right in between the two possible English vowels with which it could possibly be "equated," namely English /æ/ and /ɑ/. This kind of relationship between L1 and target vowels is described by Escudero & Boersma (2002) as a Multiple Category Assimilation (MCA), and it adds to the non-native speech perception patterns described by PAM. MCA involves perceiving a binary contrast in an L2 as two or more categories in the L1. It is proposed here that the case of the Portuguese and English low vowels falls into MCA, as some of the target Portuguese /a/ tokens might be perceived by the English speakers as tokens of the front low vowel /æ/ while others might be perceived as tokens of the low back /ɑ/.

From the MCA account offered by Escudero & Boersma (2002), it is deduced that the learners of Portuguese might perceive Portuguese /a/ as one English low vowel sometimes, and as the other English low vowel some other times. Moreover, learners of Portuguese might use tokens of both their L1 low vowel categories to produce the target Portuguese /a/ at the early stages of learning, while reducing the phonetic variability in their productions of this sound as their level of experience in the language, and thus the amount of input received, increases. This way, both learner groups are expected to create a new phonetic category for Portuguese /a/ with mean F1 and F2 values that fall in between the two English low vowels. Due to MCA, however, this new category is likely to show a wider phonetic distribution than the Portuguese native /a/.

The deduction following Escudero & Boersma (2002) that the acoustic distribution of newly created categories in MCA cases will decrease as L2 experience increases has consequences for the creation of hypotheses for the learning and pro-

duction of Portuguese /a/ by the L2 and L3 groups. In the previous chapter, Portuguese /a/ was found not to differ significantly from Spanish /a/. The latter, however, was found to be different from English /ɑ/ but not sufficiently different from /æ/. This new four-way comparison might have both facilitatory and non-facilitatory effects for the learning of Portuguese /a/ for the L3 group.

On the one hand, since Spanish and Portuguese /a/ have not been found to differ from each other, additional experience with this Spanish sound might be of advantage to the L3 group in that they will have received more input of this new category, which falls, just as Portuguese /a/ does, right in between their two native ones. This way, more input from Spanish /a/ is likely to facilitate the creation of a new phonetic category, different from the English two vowels, for the L3 group. On the other hand, the L3 learners were exposed for a longer period to a version of the target Portuguese /a/ (namely Spanish /a/) that was closer to English /æ/ than to English /ɑ/. The closer relationship of Spanish /a/ with English /æ/ than with English /ɑ/ might affect the learners' perception of the Portuguese by causing them to assimilate the target /a/ with English /æ/. It is hypothesized here that this additional input is likely to affect the Portuguese /a/ productions of the L3 group by making their newly acquired category for Portuguese /a/ more fronted in comparison with the one produced by the L2 group, whose only input was from a Portuguese /a/ located right in between the English low vowels and differing from both.

Thus, it is expected that both groups of learners of Portuguese will have created a new category for Portuguese /a/ whose mean logF2 values fall somewhere in between the two native English low vowels. This new category is predicted to have a wider phonetic distribution than the target native Portuguese /a/. In addition, a difference in the way the L2 and L3 group produce Portuguese /a/ is expected. While the L2 group is predicted to produce a Portuguese /a/ that is different from both native English low vowels and falls more or less in between these two on the logF2 axis, the L3 group is predicted to produce a Portuguese /a/ with mean logF1 and logF2 values that are slightly more fronted in the vowel space than the one

produced by the L2 group.

3.1.4 Predictions for the learning of the Portuguese mid-back vowels

In the case of the Portuguese mid-back vowels, the two target vowels were found to differ considerably from the one mid-back vowel of English, both at each point of measurement as in amount of diphthongization. According to this author's interpretation of the SLM, and assuming that statistically significant acoustic differences are likely to be perceivable, if they are perceived, then two things may happen: (1) a new category will be created for each of the Portuguese mid-back vowels provided that the differences between them are perceived by the learners; or (2) a single new category, which will show an approximation in acoustic features towards the targets, will be created.

PAM-L2 (Best & Tyler (2007)) also makes predictions about the discriminability and learning of the new phonetic contrast of Portuguese. PAM determines the discriminability of a non-native sound contrast depending on articulatory contrast and goodness of fit with the native categories. This way, predictions can be made, based on PAM, about the discriminability of the new two mid-back vowels of Portuguese by speakers of English as an L1. Since the two Portuguese mid-back sounds have been found to deviate equally from the native English /o/ both in amount of diphthongization as in punctual differences in logF1 and logF2 in time, under the assumption that acoustically relevant differences are also perceivable differences, this target pair would be classified as a case of a Single Category (SC) assimilation under PAM. For this kind of assimilation, PAM-L2 hypothesizes that:

> the learner will initially have trouble discriminating these two phones, which would be assimilated both phonetically and phonologically to the single L1 category, and minimally contrasting L2 words would be perceived as homophones. [...] Whether or not L2 listeners can learn to perceive a difference between single-category assimilated L2 phones may depend on whether they are perceived as good or as poorer exemplars

> of the L1 phoneme. However, we would not expect most learners to be able to do so very well. To the extent that a learner *could* perceptually attune to this type of L2 contrast, we hypothesize that they would first have to perceptually learn a new phonetic category for at least one of the L2 phones before they could establish a new phonological category or categories. (Best & Tyler (2007), p. 29-30)

Assuming that PAM-L2 is right in its predictions and the two Portuguese mid-back vowels are difficult to discriminate for learners, then it is predicted that it will not be the case that two new independent categories will be created for the two mid-back vowels of Portuguese. While at least one new category is expected to be created by the learners due to perceived differences with the native English /o/, the question remains of whether (a) this new category will assimilate to one of the two target Portuguese mid-back categories, or (b) the new category will assimilate to both target Portuguese categories with its mean located somewhere in between the two target categories.

Previous experience with the Spanish sound system in the learning of the mid-back vowels of Portuguese by these speakers might, once again, have both a facilitatory and a non-facilitatory effect. On the one hand, having had experience with a mid-back category that differs from the English /o/ category in the same direction (in logF2) as the target Portuguese categories might have a facilitatory effect in the establishment of a new category, one that is different from English /o/. On the other hand, the acoustic differences found between the Spanish /o/ and Portuguese /o/ and /ɔ/ suggest that experience with Spanish /o/ may not have a facilitatory effect on the learning of the /o/-/ɔ/ Portuguese contrast.

Thus, it is expected that both learner groups will create one new phonetic category for the Portuguese /o/-/ɔ/ pair, with a potential difference in spectral structure between the L2 and L3 group due to additional input received from a related phonetic category in Spanish by the L3 group.

3.1.5 Predictions for the learning of the Portuguese mid-front vowels

The acoustic similarities and differences between the mid-front vowels of Portuguese and English were found to be very complex. The four vowels differ in all of amount of diphthongization, direction of diphthongization and formant pattern at specific points throughout the vowels, but they also are similar to each other in relative position in the vowel space, phonological status and tendency to undergo formant movement.

Given the multiple acoustic similarities between the mid-front vowel categories of Portuguese and the two mid-front categories of English, it is believed by the author that, under PAM, the two Portuguese categories would be categorized as articulatorily similar to those in English. In the absence of available perception data of the Portuguese /e/-/ɛ/ contrast by English native listeners at this point, conclusions drawn from acoustic data discussed in the previous chapter comparing the two vowel systems will be used to interpret the principles of PAM and make predictions about the assimilation and discriminability of these two sounds of Portuguese by native English speakers.

Under this author's interpretation of PAM, it is hypothesized here that due to the fact that the two English mid-front vowels are separated by native phonological boundaries, native phonology should aid in the discrimination of the Portuguese /e/-/ɛ/ contrast. Assuming that the two Portuguese mid-front vowel categories are perceived as acceptable exemplars of the two different native English mid-front vowels, then, under PAM, the Portuguese mid-front vowel contrast pair would be considered a Two-Category assimilation pair (TC), where each of the members of the nonnative language assimilates to a different phonological category in the L1. PAM predicts that TC pairs will be discriminated well by naïve listeners, so the Portuguese /e/-/ɛ/ pair is predicted to be discriminated well by native speakers of English learning Portuguese. Based on this interpretation of PAM and its application to the available acoustic data, it is predicted here that the learners of Portuguese under study in this dissertation will assimilate their target mid-front vowels of Portuguese

to their native English mid-front vowel categories and will produce two separate phonetic categories that will approximate native Portuguese /e/ and /ɛ/.

Alternatively, based on the same principles of PAM and PAM-L2, there also exists the possibility of the Portuguese /e/-/ɛ/ pair being perceived as more similar to one, and not to two distinct, English mid-front vowels. If the two Portuguese vowels are perceived as being close to each other in phonological space, according to PAM-L2, then the listener should find them difficult to discriminate. In this case, PAM-L2 predicts that "a single, new phonological category would be learned, encompassing the two L2 phones" (Best & Tyler (2007), p.30).

On a related note, the distribution of acoustic cues relevant to the distinction of each of the native and target mid-front vowels is not obvious. It has been shown that each of the English native mid-front vowels diphthongizes in a different direction and degree, while the two Portuguese mid-front vowels move in the same degree and direction, in a pattern that approximates the diphthongization pattern of one of the English vowels, namely /ɛ/. Moreover, spectral differences among the four vowels change considerably through time, as the vowels change spectral structure through time. Although Portuguese /e/ seems to be relatable to English /e/ at all points during the length of the vowel (possibly because of a lack of additional surrounding vowels), Portuguese /ɛ/ seems to be located in a position in between English /e/ and /ɛ/ at the beginning, with the potential risk of being "equated" with English /e/ instead of English /ɛ/, only to move closer to and in the same direction as the latter through the duration of the vowel.

Thus, in SLM terms, the learners are likely to create new categories for the Portuguese sounds due to the multiple acoustic differences between the target and the native sounds, which they are likely to perceive. However, the complex patterns of similarities and differences between the target and native vowels make it a hard task for the learners to be able to perceive them accurately, which makes it unlikely for the learners, at least at an intermediate to advanced level of acquisition, to be able to produce them in a native-like fashion.

As for the potential difference between the L2 and the L3 group in producing

the two Portuguese mid-front vowels, it is relevant to observe that Spanish /e/ is the most stable of the five categories so it occupies a position in the perceptual space that is shared (a) with English /e/ most of the time, (b) with portuguese /ɛ/ during the first part of the duration of the vowel, and (c) with Portuguese /e/ at the middle and end of the vowel, although it ultimately differs from all three. It is not clear whether having been exposed to Spanish /e/ prior and for longer than to Portuguese /e/-/ɛ/ might have a facilitatory or a non-facilitatory effect on the acquisition of the new Portuguese categories. On the one hand, the potential prior existence of a category for Spanish /e/ in the L3 participants' perceptual space might act as a facilitator in the creation of new categories for these learners, who have been exposed to different inputs for longer and have had to undergo the necessary process of "equivalence classification" (in SLM terms) before. On the other hand, the multiple acoustic differences between the Spanish /e/ and both English /e/ and /ɛ/ on the one hand, and Portuguese /e/ and /ɛ/ on the other, might cause further interference in the accurate perception of the two Portuguese categories, causing the L3 group to produce Portuguese /e/ and /ɛ/ categories differently from those produced by the L2 group.

To sum up, the learners of Portuguese under study are expected to have created two new phonetic categories for Portuguese /e/-/ɛ/. These two categories are expected to show similar spectral and diphthongization patterns as English /e/ and /ɛ/ respectively. The Portuguese /e/ and /ɛ/ produced by the two groups of learners are also expected to differ acoustically from each other.

3.2 Method

3.2.1 Materials

The materials used for this production experiment are the same as were used for collecting the production data of the three native speaker groups in Chapter 2. However, since the participants in this experiment were speakers of more than one language, the participants in the L2 group took part in both the American English

Table 3.1: Summary of participants' biographic and age of acquisition data where Age is the mean age of the participants at time of testing; M/F represents their gender; AOASpan and AOAPort represent the Age Of Acquisition of Spanish and/or Portuguese respectively; ranges are presented in parentheses.

Group	Age	M/F	AOASpan	AOAPort
L2	24 (21-29)	7M	N.A.	19 (18-20)
L3	22 (19-26)	7M	13 (8-15)	20 (18-26)

and the Brazilian Portuguese production experiments, and the participants in the L3 group participated in all three: the English, the Spanish and the Portuguese production experiments.

3.2.2 Participants

7 male native speakers of American English who were L2 learners of Portuguese (L2 group) and 7 male native speakers of American English who had learned Spanish as an L2 and were learners of Portuguese as their L3 (L3 group) participated in these experiments. All participants were required to have extensive foreign language learning experience, as indicated by their living/study abroad experience and years of study of the foreign language(s), and were required not to be able to speak / have learnt any other language other than the three studied in this research project.

The participants in both groups also provided the answers to complete the Language Background Questionnaire (LBQ) that the native speaker participants in Chapter 2 completed. These participants' biographic and linguistic background information is summarized in Table 3.1.

Due to the nature of the topic under study, namely phonetic accuracy in L2 and L3 learning, it was difficult to control for all possible variables that might affect the participants performance in the target language[1]. These variables, such as age

[1]As reviewed in Chapter 1 of this dissertation, Flege and his colleagues have thoroughly researched the factors that potentially affect a speakers' degree of foreign accent in numerous articles (Flege et al. (1995b), Flege et al. (1999a), Flege et al. (1999b), Guion et al. (2000b), Piske et al.

of acquisition (AOA) of each of the foreign languages (when appropriate), amount of time spent in the foreign country, foreign language learning experience, order of acquisition of the different languages, and amount of L1, L2 and/or L3 used, were controlled for when at all possible. The selection of the 7 participants in each of these groups responds to an attempt to create two learner groups that show as much homogeneity as possible in linguistic background and experience from an original pool of 9 participants in the L2 group and 20 participants in the L3 group. Since initial participants differed considerably in their linguistic experience and language use, priority was given to such factors as AOA of each of the languages and amount of language learning experience. Amount of language learning experience was determined by this author's evaluation of the data provided by the participants in the LBQ regarding percentage of use of each of the languages in daily life, age of acquisition, age at time of testing, and study abroad experience. The relevant biographical and linguistic experience information of the participants selected as part of the L2 and L3 groups is included in Tables 3.1 and 3.2.

In addition, as suggested by Simonet et al. (2011) and Flege (1987), level of proficiency in the foreign language in general might also have consequences for degree of foreign accent, and thus for the likeliness of a native-like production of specific phones in the target language. This way, it is important to establish the similarity in Portuguese proficiency levels across the learner groups described in this dissertation for comparability puposes.

As shown in Tables 3.1 and 3.2, participants in both groups started learning Portuguese at a similar age (19 years old) while the L3 group started learning Spanish a few years earlier (13 years old). This difference in exposure to foreign language is relevant because it shows that the L3 group was exposed to a foreign language at an earlier age on the one hand, and for a longer period on the other, which might be interpreted as an indication of the L3 group potentially showing a

(2001), Flege (2002), Flege & Liu (2002), Flege & MacKay (2004)). They concluded that age of arrival in the foreign country, as related to age of acquisition and amount of input received, is one of the most relevant factor affecting second language speakers' degree of foreign accent

Table 3.2: Participants' language use and study abroad experience where PortUse5m and PortUse5w represent the percentage of self-reported Portuguese use during the 5 months or 5 weeks previous to the time of testing and StAbSpan and StAbPort represent the amount of time (in months) spent in a foreign country where each of the target languages were spoken.

	Participant	PortUse5m	PortUse5w	StAbSpan	StAbPort
	L201	5%	5%	N.A.	24
	L202	4%	10%	N.A.	8
	L204	20%	0%	N.A.	30
	L205	1%	1%	N.A.	24
	L206	5%	5%	N.A.	24
	L207	2%	2%	N.A.	24
	L208	2%	2%	N.A.	24
Averages	L2	5.5%	3.5%	N.A.	22.5
	L303	30%	30%	0	1.8
	L305	35%	5%	1.5	1.5
	L309	10%	15%	7	0
	L313	5%	30%	4	0
	L317	3%	3%	36	0
	L319	10%	10%	2	0
	L320	0%	0%	18	0
Averages	L3	13%	13%	9.8	0.5

higher proficiency in Portuguese. However, the L2 group has had more Portuguese language experience abroad than the L3 group, who has none or very little experience living in a Portuguese speaking country, from which it can be deducted that the L2 group has had more meaningful input in Portuguese overall, supporting the potential for a lesser degree of foreign accent in the production of Portuguese in general. The L3 group, on the other hand, uses Portuguese more often than the L2 group. Amount of language use and amount and quality of input received being two of the most relevant factors identified by Flege and his colleagues in their extensive research on this topic, there are reasons to believe that these two groups might be at similar proficiency levels in Portuguese.

Homogeneity in the proficiency levels in Portuguese across groups was confirmed informally. Eight native speakers of Brazilian Portuguese did an informal oral questionnaire after they participated in a foreign accent rating task where they rated the L2 and L3 groups' Portuguese productions for foreign accentedness. Although the results from the foreign accent rating task will not be reported in this dissertation, all eight native speakers reported having perceived little variation in the degree of foreign accent encountered in the data they were exposed to during the task.

3.2.3 Procedure

As with the experiments in Chapter 2, the author, who is a native speaker of Spanish with advanced proficiency in English and in Portuguese, interacted with the participants. The participants answered a series of biographical and linguistic background questions orally in English, while the researcher filled out the LBQ with the information provided by them. The experiment was done by only one participant at a time, and each participant was required to do multiple experiments, one for each language spoken, in one single session. The session lasted between 1.5 and 2 hours. The location, materials and technology used to carry out the experiments were the same as those described in Chapter 2 for the native speakers' production experiments.

As for the languages used during the experiments described in this chapter, the

researcher greeted the participants and went through the LBQ (in Appendix B) and the English section of the production experiment with them in English. There was a short break between the two sections of each experiment and a longer pause in between experiments where the experiments tested the participants' production in different languages. Thus, the researcher used English through the LBQ and the two sections of the English experiment, and switched into Spanish/Portuguese (whichever appropriate) as soon as the participants finished the second part of the English production experiment. The break between languages, usually between 10-15 minutes, was spent conversing about the participants' language learning experience (among other informal topics) in the language in which the following experiment was going to be on. This change in the language used by the researcher and the participant was assumed to ease a switch in speech/language mode (Grosjean (2001)) in the participants, by providing an informal and less abrupt way for them to switch from one language to the next and be able to perform more appropriately in the next experiment minimizing the influence of the language previously tested. This way, the participants in the L2 group interacted in English with the researcher first, and they switched to Portuguese as soon as they finished the English production task. The participants in the L3 group interacted in English with the researcher through the English section of the experiment, then switched into Spanish as soon as they finished the English production task, and switched into Portuguese as soon as they finished the Spanish section. The L3 group, then, interacted in 3 different languages with the researcher (English, Spanish and Portuguese), while the L2 group interacted in only two of them (English, then Portuguese). The order in which the experiments were done by the participants (English [>>Spanish] >>Portuguese) paralleled the order of acquisition of the different languages by the participants.

3.2.4 Data analysis

The same data analysis procedures used in Chapter 2 for the native speakers' productions were used here to analyze the L2 and L3 groups production data. As in Chapter 2, both logF1 and logF2 values at the midpoint of the vowel were mea-

sured and graphed when appropriate. Amount of diphthongization and direction of formant movement were also measured and compared in the case of mid vowels.

3.3 Results

This section focuses on the comparison between the Portuguese vowels produced by the two groups of learners and those produced by the native Portuguese speakers in Chapter 2. Once the results from the multiple comparisons are presented, the hypotheses for L2 and L3 learning created above will be revisited.

Note that the same kind of statistical models run on Chapter 2 production data will be run in this chapter. Thus, multiple linear mixed-effects regression models will be run to establish the relationships among the Portuguese vowels produced by the three groups. In this chapter, a t-statistic of >2 or <-2 value in the linear mixed-effects models will still be the assumed level of statistical significance (Baayen (2008)). As it was done in Chapter 2, most of these models will take the logarithmic values of F1 and F2 in Hz as independent variables (logF1 and logF2 respectively), while 'speaker' and 'word' will be treated as random intercepts. In this occasion, the logF1 and logF2 values tested for significant differences will be those of the values at their midpoint. In the case of the mid vowels of Portuguese, differences in degree and direction of diphthongization will also be tested for statistical significance.

3.3.1 Learners vs. Native speakers' Portuguese vowels

In this section, the Portuguese vowels produced by both learner groups (L2 and L3 groups) and the native speaker group from Chapter 2 will be compared to establish whether the learners have acquired the Portuguese vowel sounds in a native-like fashion, and how the findings from such comparisons relate to the predictions for the acquisition of Portuguese vowels by native speakers of English made at the beginning of the current chapter.

To illustrate the relationship between the Portuguese vowels of each of the three groups compared in this section, Figure 3.1 was included. The following subsections

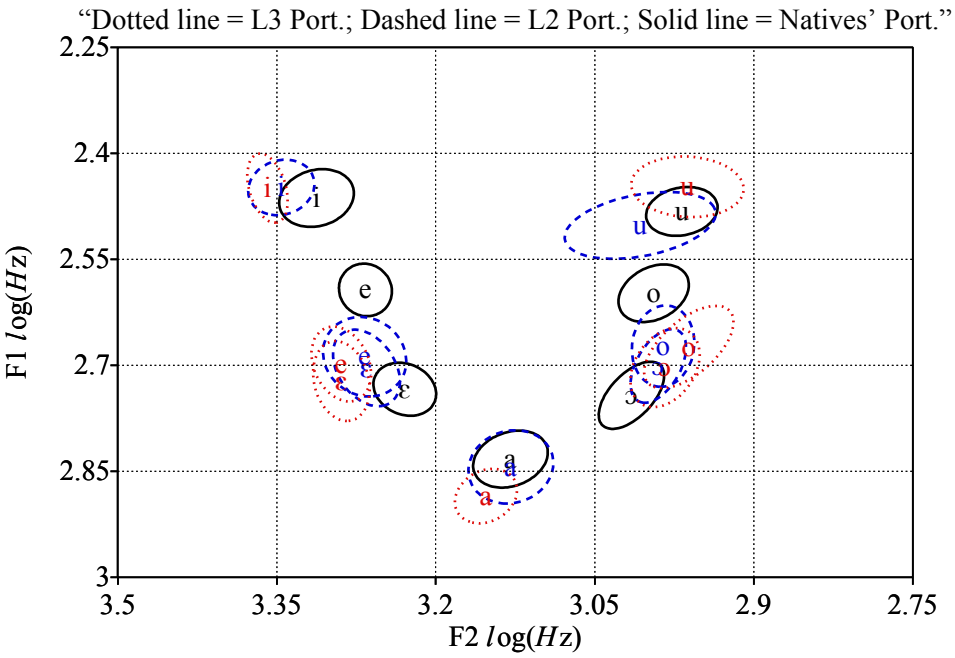

Figure 3.1: Portuguese vowels produced by Portuguese native speakers, L2 group and L3 group.

Table 3.3: Native Portuguese speakers' Portuguese vowels: F1 and F2 data in Hz.

		i	e	ɛ	a	ɔ	o	u
	N	56	56	57	63	57	56	55
F1 (in Hz)	Mean	292	393	544	683	556	398	304
	Median	295	398	542	686	553	402	306
	SD	27	34	47	61	61	37	24
F2 (in Hz)	Mean	2060	1849	1699	1352	1039	990	931
	Median	2082	1841	1688	1350	1028	993	925
	SD	164	106	117	108	75	75	72

Table 3.4: L2 group's Portuguese vowels: F1 and F2 data in Hz.

		i	e	ɛ	a	ɔ	o	u
	N							
F1 (in Hz)	Mean	282	491	509	702	506	474	319
	Median	281	490	499	692	494	468	312
	SD	36	58	62	84	63	63	26
F2 (in Hz)	Mean	2222	1857	1847	1352	979	969	1031
	Median	2235	1861	1829	1345	977	973	1018
	SD	155	170	136	120	58	66	187

will investigate the relationship among the native and non-native vowel categories in Figure 3.1 individually by target vowel. The mean, median and standard deviation (SD) of the F1 and F2 values at midpoint of the vowels plotted in Figure 3.1 are shown in Tables 3.3, 3.4 and 3.5.

Portuguese high front vowel

The high front Portuguese vowels produced by both the L2 and the L3 groups were compared to the ones produced by the native speakers of this language by means of two separate linear mixed-effects regression models. The first statistical model took the natives' /i/ as the intercept, while the second model took the L3 group's /i/ as the intercept. The results rendered by these two models are presented in Table 3.6.

Table 3.5: L3 group's Portuguese vowels: F1 and F2 data in Hz.

		i	e	ɛ	a	ɔ	o	u
	N							
F1 (in Hz)	Mean	283	502	532	770	509	476	281
	Median	277	488	525	775	505	467	275
	SD	34	63	69	66	66	64	29
F2 (in Hz)	Mean	2286	1950	1949	1424	966	919	924
	Median	2292	1949	1946	1439	976	908	904
	SD	100	127	120	94	72	90	111

Table 3.6: Results from a linear mixed-effects model with native speakers' Portuguese /i/ as the intercept.

		β	t
logF1	Intercept (= Native /i/)	2.463	
	L2 /i/	-0.016	-1.19
	L3 /i/	-0.015	-1.10
logF2	Intercept (= Native /i/)	3.312	
	L2 /i/	0.032	2.4*
	L3 /i/	0.047	3.5*
logF1	Intercept (= L3 /i/)	2.448	
	L2 /i/	-0.001	-0.08
logF2	Intercept (= L3 /i/)	3.360	
	L2 /i/	-0.015	-1.1

The results from these two mixed-effects models show that, although the L2 and the L3 groups do not seem to produce a statistically significant acoustically different vowel from one another, when compared to the Natives' /i/, both the L3 and the L2 groups were found to produce a high front vowel that is significantly more fronted than the target native Portuguese /i/. Since the Portuguese /i/ produced by the L2 group was not statistically different from that produced by the L3 group, it can be concluded that the two learner groups seem to be producing a similar, more fronted, Portuguese /i/. These differences are observable in Figure 3.1 above.

It had been predicted that no spectral difference was to be expected between any of the Portuguese high front vowels produced by the Natives, the L2 and the L3 group. Unexpectedly, the results from the statistical analyses of the data show that the L2 and L3 groups produce an /i/ that is further front in the vowel space than the target Portuguese /i/. Consequently, the question remains of whether the Portuguese /i/ the two groups of learners are producing is different from their own English /i/, which would suggest that phonetic category dissimilation has occurred in these learners' high front vowels.

In order to answer the question of whether phonetic category dissimilation has occurred between the learners' Portuguese /i/ and English /i/, an additional 2x2 linear mixed-effects model was run with 'language' and 'group' as the fixed factors and 'speaker' and 'word' as random intercepts. Each fixed factor had two levels: 'language' (English, Portuguese) and 'group' (L2 group, L3 group). The results from this linear mixed effects model is presented in Table 3.7.

As it is shown in Table 3.7, the 2x2 linear mixed-effects models did not render any 'group' by 'language' interaction. The effect of neither the 'language' nor 'group' fixed factors were rendered significant. From these results it is interpreted here that, on the one hand, these two learner groups are behaving similarly in their production of the high front vowels of English and Portuguese, and on the other, that their Portuguese /i/ and their English /i/ are acoustically non-distinct. Thus, the results from these analyses suggest that there has not been any phonetic category

Table 3.7: Results from a set of linear mixed-effects models on the Portuguese /i/ and English /i/ produced by the L2 and L3 groups, with the Portuguese /i/ produced by the L2 group as the intercept.

		β	t
logF1	Intercept (= L2 Port. /i/)	2.422	
	Group	0.031	1.91
	Language	0.024	1.45
	Group x Language	-0.029	-1.28
logF2	Intercept (= L2 Port. /i/)	3.347	
	Group	0.002	0.3
	Language	-0.002	-0.2
	Group x Language	0.127	0.9

dissimilation between the high front vowels of these two groups of learners.[2]

Portuguese high back vowel

The high back vowels of Portuguese produced by the Native, L2 and L3 groups were compared by means of two separate mixed-effects models in order to investigate whether statistically significant differences in either height (logF1) and/or backness (logF2) were found among the three. The results, in Table 3.8, showed that the native portuguese /u/ differs in different ways from the L2 group's and the L3 groups'. The Natives' /u/ differs from the L2 group's in backness, the L2 group's /u/ being more central (= fronted) than the high back vowel produced by the Natives. The Natives' /u/ differs from the L3 group's in height, the L3 groups' being produced higher than the Natives'. The L2 and L3 groups' Portuguese high back vowels differ from each other in both dimensions. These differences are also observable in Figure 3.1 above.

Going back to the predictions for learning at the beginning of the chapter, it was predicted, based on the author's interpretation of both the SLM and PAM-L2, that

[2]These results show that the English /i/ produced by the L3 group is also more fronted than the English /i/ produced by the Native English speaking group described in Chapter 3 of this dissertation.

Table 3.8: Results from a linear mixed-effects model on the Portuguese /u/ and English /u/ produced by the Native Portuguese, L2 and L3 groups with the native speakers' Portuguese /u/ as the intercept.

		β	t
logF1	Intercept (= Native /u/)	2.482	
	L2 /u/	0.019	1.16
	L3 /u/	-0.035	-2.08*
logF2	Intercept (= Native /u/)	3.312	
	L2 /u/	0.039	2.48*
	L3 /u/	-0.006	-0.37
logF1	Intercept (= L3 /u/)	2.501	
	L2 /u/	-0.054	-3.24*
logF2	Intercept (= L3 /u/)	3.007	
	L2 /u/	-0.004	-2.85*

due to significant acoustic differences between the English /u/, which is extremely fronted, and the target Portuguese /u/, both learner groups would establish a new phonetic category for the target Portuguese /u/. In addition, it was also predicted that the new category created by the L3 group would approximate the target native Portuguese /u/ more than the L2 groups', as the L3 group was assumed to have received more input of the new target phonetic category than the L2 group (due to more extensive exposure to Spanish /u/, which was found to be acoustically indistinguishable from Portuguese /u/).

The results from the statistical analyses performed here show that, although both groups have succeeded in creating a separate new category for the target vowel in question, none of them seems to have acquired a native-like Portuguese /u/. The L2 group seems to have created a category that coincides with the target in height but not in backness, while the L3 seems to have created a category that matches the target in backness but not in height. Although these results do not seem to perfectly match the predictions, it could be hypothesized that the L3 group has undergone phonetic category dissimilation in their high back vowel categories, since they seem to have overshot their target Portuguese /u/ in height.

Table 3.9: Results from a linear mixed-effects model on the Portuguese /u/ and English /u/ produced by the L2 and L3 groups with English /u/ produced by the L2 group as the intercept.

		β	t
logF1	Intercept (= L2 Engl. /u/)	2.476	
	Group	0.013	0.78
	Language	0.025	1.33
	Group x Language	-0.067	-2.79*
logF2	Intercept (= L2 Engl. /u/)	3.191	
	Group	0.057	2.45*
	Language	-0.184	-7.13*
	Group x Language	-0.101	-3.09*

To test for dissimilation effects in the high back vowels of these two groups, data from the L2 and L3 groups' English /u/ and Portuguese /u/ productions were sent to a 2x2 linear mixed-effects model with 'language' and 'group' as the fixed factors and 'speaker' and 'word' as random intercepts. Each fixed factor had two levels: 'language' (English, Portuguese) and 'group' (L2 group, L3 group). The results from this linear mixed effects model are presented in Table 3.9.

Table 3.9 shows that there is an interaction between 'group' and 'language' on both the logF1 and the logF2 measure. The data from all the high back vowels of English and Portuguese produced by the two groups was plotted in Figure 3.2. Two separate sets of linear mixed-effects models were run to test the differences between the two levels of the 'language' factor in each group separately.

The first set of statistical models tested the effect of 'language' in the logF1 measure of the high back vowels produced by each of the learner groups separately. The model that tested the effect of language in the L2 group showed that there was no statistically significant difference in height between their Portuguese /u/ and their English /u/. The model that tested for the same language effects in the L3 group, however, did render a statistically significant difference between their English /u/ and their Portuguese /u/. The interaction between 'group' and 'language' on

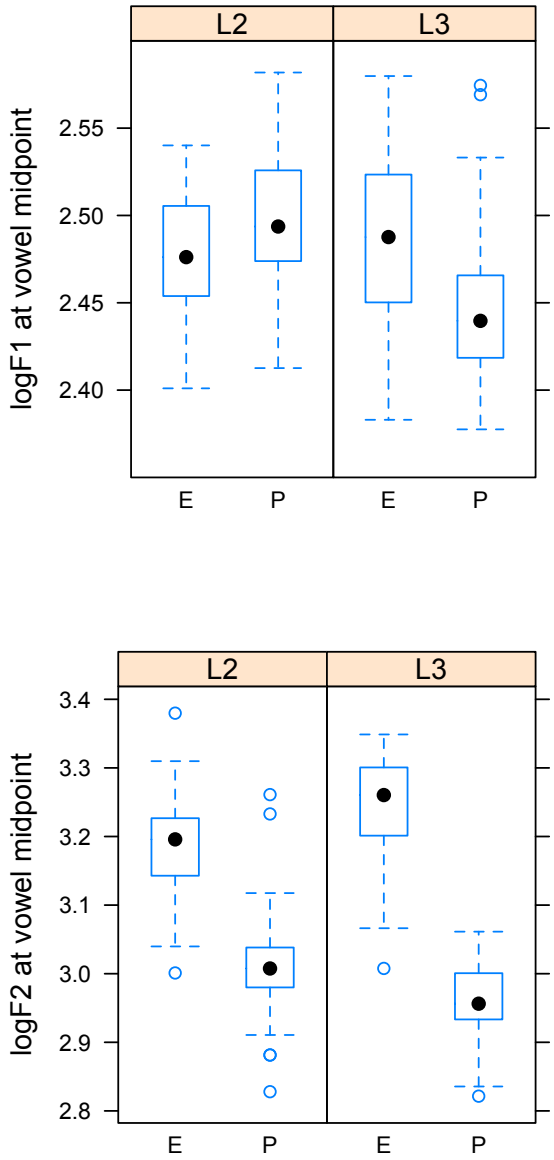

Figure 3.2: Portuguese /u/ and English /u/ vowels of both L2 and L3 groups at their midpoint (top = logF1, bottom = logF2). 'E' and 'P' represent the 'language' factor, L2 and L3 represent the 'group' factor.

the logF1 measure is thus explained: while the L2 group produces a Portuguese /u/ vowel at the same height as their own English /u/ vowel, the L3 group produces a Portuguese /u/ that is higher than their own English /u/. While the L2 group does not maintain a difference in height between the two, the L3 group does. This interaction is also visible in the top graph of Figure 3.2.

The second set of statistical models tested the effect of 'language' in the logF2 measure of the same high back vowels. These models found an effect of 'language' in the L2 group (β= -0.184, t = -6.95) as well as in the L3 group (β= -3.247, t = -10.85). Moreover, the difference in logF2 between the English /u/ and the Portuguese /u/ was bigger in the L3 group than in the L2 group, as indicated by the greater value of β for the former than for the latter [3]. The interaction found between 'group' and 'language' in the logF2 measure in Table 3.9 is thus explained: although both groups have created a new phonetic category for Portuguese that is more back into the vowel space than their native English /u/, the L3 group maintains a bigger difference between their English and their Portuguese high back vowel than the L2 group does. This interaction can be observed in the bottom graph of Figure 3.2.

The results from the two sets of linear mixed-effects models described above are presented in Table 3.10.

In summary, the results obtained from multiple statistical analyses on the logF1 and logF2 measures of the Portuguese and English high back vowels of the two learner groups under study suggest that there is a difference between the way the L3 group produces the high back vowels of these two languages and the way in which the L2 group does it. It has been shown that, on the logF1 dimension, the L3 group maintains a difference between the two high back vowels while the L2 group does not. It has also been shown that the L3 group maintains a bigger distance on

[3]the value of the estimate (= β) represents the difference between the intercept (English /u/) and the element of comparison (Portuguese /u/) within the group. Thus, a greater β value for the Portuguese /u/ produced by the L3 group shows that the Portuguese /u/ of the L3 group is at a greater distance from their own English /u/ than the Portuguese /u/ of the L2 group is from their own English /u/.

Table 3.10: Results from multiple linear mixed-effects models on the effect of 'language' on the Portuguese /u/ and English /u/ produced by the L2 and L3 groups separately. English /u/ was taken as the intercept in all cases.

	logF1		logF2	
	β	t	β	t
Intercept (= L2 Engl. /u/)	2.476		3.191	
L2 Port. /u/	0.025	1.44	-0.184	-6.95*
Intercept (= L3 Engl. /u/)	2.490		3.191	
L3 Port. /u/	-0.042	-2.12*	-3.247	-10.85*

the logF2 axis between their Portuguese /u/ and English /u/. These two findings, together with the initial finding that the L3 group produces a Portuguese /u/ that is produced higher than the one produced by native speakers of the language, seem to suggest that the L3 group has undergone phonetic category dissimilation, while the L2 group does not.

Portuguese low vowel

The differences between the Portuguese /a/ produced by all three the Native, L2 and L3 groups were tested by means of two linear mixed-effects regression models whose results are presented in Table 3.11. The results show that the native /a/ does not differ on any of the two dimensions from L2 groups' /a/, but it does differ from the L3 groups' /a/ in height, with L3 /a/ being produced lower than the target /a/. These differences are observable in Figure 3.1 above.

Regarding the predictions made for the Portuguese low vowel produced by the two groups of learners, the statistical analyses show that, as predicted, both learner groups have created a new phonetic category for their Portuguese /a/. This category is slightly different for each of the learner groups with the L3 group's falling lower in the vowel space than the one produced by both the L2 group and the natives.

The prediction, however, also included that the mean values of these new categories would fall somewhere between the speakers' two low English vowels. In order

Table 3.11: Results from a linear mixed-effects model on the Portuguese /a/ produced by the Native Portuguese, L2 and L3 groups.

		β	t
logF1	Intercept (= Native /a/)	2.833	
	L2 /a/	0.010	0.68
	L3 /a/	0.053	3.40*
logF2	Intercept (= Native /a/)	3.129	
	L2 /a/	-0.000	-0.04
	L3 /a/	0.023	1.66
logF1	Intercept (= L2 /a/)	2.843	
	L3 /a/	0.042	2.71*
logF2	Intercept (= L2 /a/)	3.129	
	L3 /a/	0.024	1.69

to specifically test for spectral differences between the learners' English /æ/ and /ɑ/ and their own Portuguese /a/, additional statistical models needed to be run.

Two 2x3 linear mixed-effects models were run with 'group' (L2 and L3 groups) and 'vowel' (Port. /a/, Engl. /æ/, Engl. /ɑ/) as the fixed factors, and 'speaker' and 'word' as the random intercepts. In the first model, the English /æ/ produced by the L2 group was taken as the intercept. In the second model, the intercept was the same group's Portuguese /a/. The results from these two models are shown in Tables 3.12 and 3.13.

As shown in Tables 3.12 and 3.13, no interactions were found to be significant. For logF1, the L2 group's English /æ/ was found to be statistically different from their own Portuguese /a/. As for the English /ɑ/ of the L2 group, this vowel was found to statistically differ from their own Portuguese /a/ too. As no interactions were rendered significant by the model, a similar relation is assumed between the L3 group's English /æ/ and their own Portuguese /a/, and between the L3 group's Portuguese /a/ and their own English /ɑ/.

With respect to logF2, no interactions were found to be significant either, so the same relation can be expected among the vowels produced within each group. The L2 group's English /æ/ was found to be more fronted than their own Portuguese

Table 3.12: Results from a linear mixed-effects model on the Portuguese /a/ and English /æ/ and /ɑ/ produced by the L2 and L3 groups with L2 English /æ/ as the intercept.

		β	t
logF1	Intercept (= L2 Engl. /æ/)	2.780	
	Portuguese /a/	0.063	2.36*
	L3 x Portuguese /a/	-0.011	-0.40
logF2	Intercept (= L2 Engl. /æ/)	3.182	
	Portuguese /a/	-0.053	-2.33*
	L3 x Portuguese /a/	-0.004	-0.16

Table 3.13: Results from a linear mixed-effects model on the Portuguese /a/ and English /æ/ and /ɑ/ produced by the L2 and L3 groups with L2 Portuguese /a/ as the intercept.

		β	t
logF1	Intercept (= L2 Port. /a/)	2.843	
	English /ɑ/	-0.067	-2.54*
	L3 x English /ɑ/	0.008	0.28
logF2	Intercept (= L2 Port. /a/)	3.129	
	English /ɑ/	-0.081	-3.55*
	L3 x English /ɑ/	-0.005	-0.20

/a/, and their Portuguese /a/ was found to be more fronted than their English /ɑ/. Moreover, the β value of the Portuguese /a/ when compared in logF2 to their English /ɑ/ is greater than the β value of their English /æ/ when compared in logF2 to their Portuguese /a/, which suggests that the Portuguese /a/ produced by both groups of learners is closer in the logF2 axis to their own English /æ/ than to their English /ɑ/. Both logF1 and logF2 comparisons are graphed in Figure 3.3.

To summarize these findings, the results from these statistical tests showed that both learner groups maintain a difference between their Portuguese /a/ and their two English low vowels both in height and backness, and that the two groups seem to be keeping similar difference patterns among their three low vowels. This finding contradicts the prediction that the L3 group and the L2 group would produce their Portuguese /a/ with different spectral quality as an effect of the extensive input from Spanish /a/ that the L2 group lacks.

Moreover, it is interesting to note that the first of the statistical models comparing the learners' Portuguese /a/ to the native speakers' (results in Table 3.12 above) found a significant difference between the Portuguese /a/ produced by the L2 and L3 group. This finding, together with the finding that the two learner groups maintain a similar pattern of difference between their three low vowels (English and Portuguese), leads to the conclusion that the L3 group's English low vowels must be located lower in the vowel space than the L2 group's English low vowels. This conclusion seems to be supported by the vowel plot containing all English vowels of the native speakers of English from Chapter 2, L2 and L3 groups in Figure 3.4.

There is however, one part of the prediction for the low vowel of Portuguese that has not yet been tested, namely the width of the distribution of the Portuguese /a/ vowel as produced by the two learner groups. It was predicted that both learner groups would show a wider phonetic distribution in their productions of Portuguese /a/ than the native speakers due to an effect of MCA. Moreover, due to prior and more extensive experience with Spanish /a/, the L3 group's Portuguese /a/ was expected to show a distribution that would approximate more that of the Native speakers' than the L2 group's Portuguese /a/ would. Thus, the Portuguese /a/

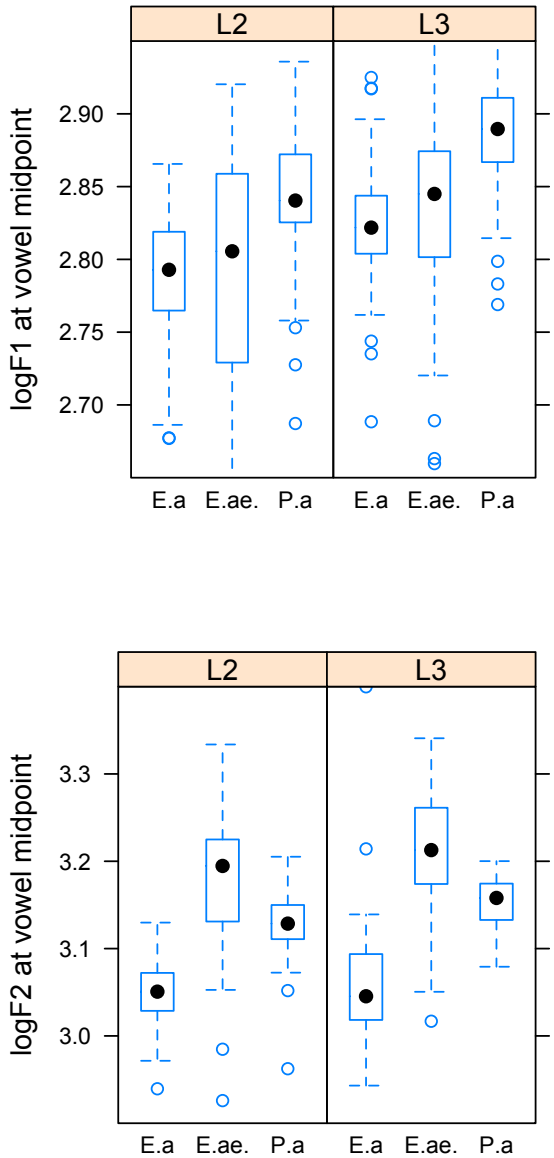

Figure 3.3: English /ɑ/ (E.a), English /æ/ (E.ae) and Portuguese /a/ (P.a) vowels of both L2 and L3 groups at their midpoint (top = logF1, bottom = logF2).

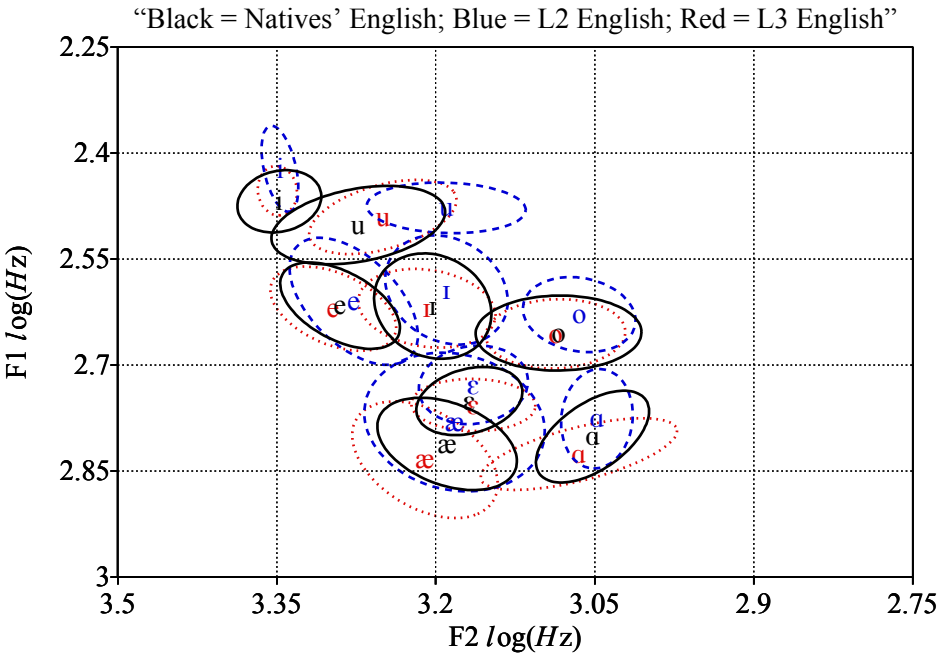

Figure 3.4: English vowels of all three Native, L2 and L3 groups. Vowels produced by the Native group are graphed using a solid line, those produced by the L2 group use a dashed line and those produced by the L3 group use a dotted line.

Table 3.14: Results from four linear mixed-effects models on SD (in Hz) values at the midpoint of Portuguese /a/ tokens as produced by all three Native Portuguese, L2 and L3 groups, with different intercepts.

	SDlogF1		SDlogF2	
	β	t	β	t
Intercept (= L2 Port. /a/)	0.040		0.027	
Native group	-0.011	-3.80*	-0.005	-2.43*
L3 group	-0.013	-4.38*	-0.006	-3.02*
Intercept (= Native Port. /a/)	0.029		0.022	
L3 group	-0.002	-0.58	-0.001	-0.58

produced by Natives was predicted to show a smaller width of distribution than that of the L3 group's, which in turn was predicted to show a narrower distribution in their Portuguese /a/ than the L2 group.

In order to explore this last question, the logF1 and logF2 standard deviation values of the Portuguese /a/ vowels produced by the two learner groups, together with the native speakers', were sent to two sets of 1x3 linear mixed-effects regression models. These models all had 'group' (Natives, L2, L3) as the fixed factor, 'speaker' as the random intercept, and 'SDlogF1' or 'SDlogF2' as the response. 'SDlogF1' represents the standard deviation (SD) of the logF1 values of the Portuguese /a/ vowel tokens produced by each speaker at their midpoint. 'SDlogF2' represents the SD of the logF2 values of the same vowel tokens at the same point of measurement. The first two tests took the Portuguese /a/ produced by the L2 group as the intercept, the second set of tests took the Native speakers' Portuguese /a/ as the intercept. The results from these tests are presented in Table 3.14[4].

The results presented in Table 3.14 show that a significant difference was found

[4]Although comparisons of distribution widths are not frequent in the literature, a similar analysis of SDs was carried out by Recasens & Espinosa (2009) in a comparison of the vowel categories of four different Catalan dialects. Although the statistical analysis performed in this chapter is different from the ANOVAs performed by Recasens & Espinosa (2009), I believe that the underlying procedure for comparison is preserved here.

among the SD values of the Portuguese /a/ produced by the L2 group and both the Native and L3 groups as it refers to both variability in logF1 and logF2. The L3 group, however, did not differ from the Native group in any of the two measures. These results seem to indicate that the L2 group produces a Portuguese /a/ with a wider distribution than that produced by the L3 and the Native groups.

Consequent with these findings, the conclusion could be made that the last part of the predictions made for the production of Portuguese /a/ by the two groups of learners was only partially confirmed, since the L3 group's Portuguese /a/ does not differ in distribution size from the Native group's, but both the L3 and Native groups' Portuguese /a/ show a narrower distribution than the L2 group's. However, there is a second way to look at the Portuguese /a/ production data by the three participant groups, namely looking at their Portuguese /a/ density plots to see whether their Portuguese /a/ productions actually are located around a single logF2 mean, indicating a native-like distribution, or whether the learners' Portuguese /a/ tokens are actually spread around two different means (a case of Multiple Category Assimilation, MCA, in Escudero & Boersma (2002)'s terms), which would suggest interference from their native English double low vowel system. These density plots, separated by group, are shown under Figure 3.5, where the density plots for the Native, L2 and L3 groups' Portuguese /a/ productions are presented.

Consistent with the results from the statistical analyses described above, the density plots in Figure 3.5 show that the L2 group's Portuguese /a/ has a wider distribution than that produced by either the Native or the L3 group. However, Figure 3.5 also shows that the L2 group's Portuguese /a/ distribution is almost perfectly unimodal (most of their Portuguese /a/ productions are located around a single mean logF2 value) approaching the Native's (top plot), while the L3 group's borders bimodality (their Portuguese /a/ productions might be located around two different means, as indicated by the secondary mode in the plot). Since both modality and width of distribution seem to be contributing to the reported difference across groups, the individual density plots for each participant were included here for the reader's reference. The density plots for each of the participants in the L3 group

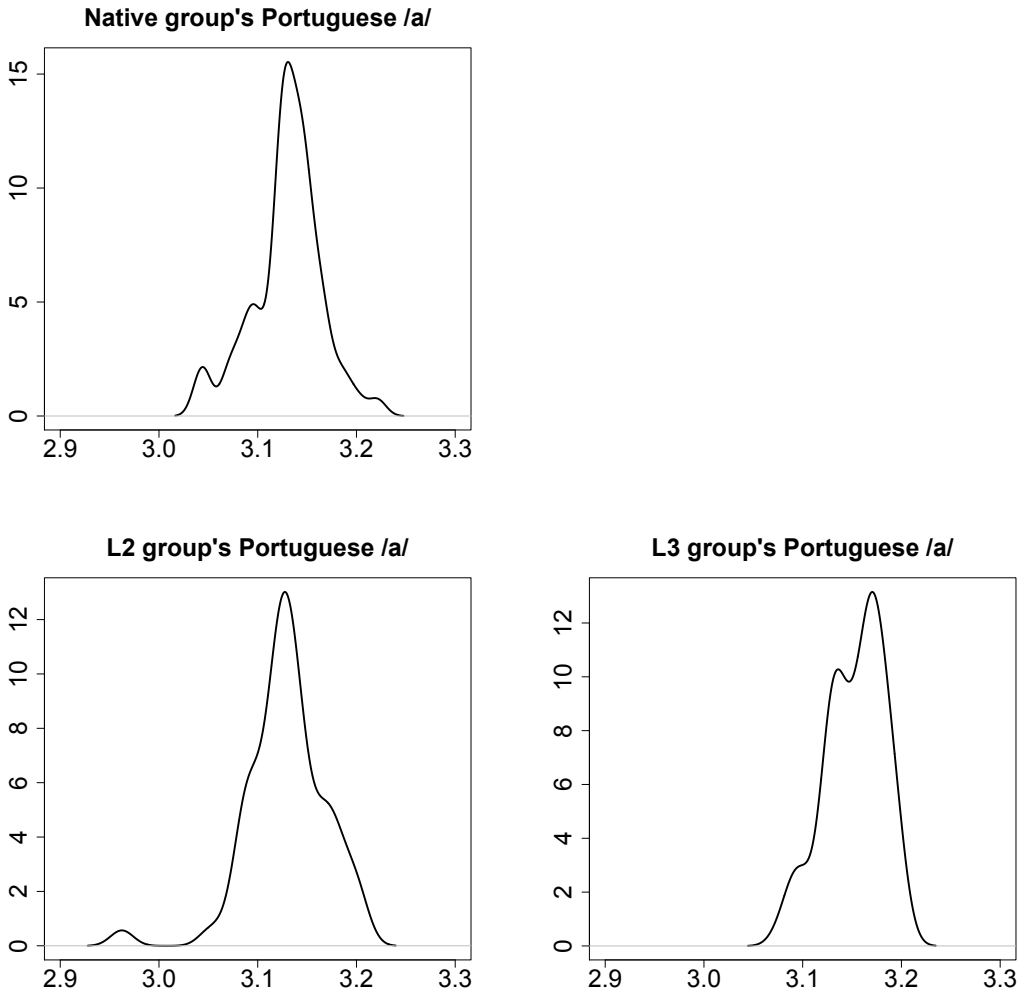

Figure 3.5: Density plots of logF2 values for Portuguese /a/ tokens produced by each of the three participant groups: Native group (top), L2 group (bottom left) and L3 group (bottom right).

are included in Figure 3.6, followed by the individual plots for each participant in the L2 group in Figure 3.7 and the Native group's in Figure 3.8 for the sake of completeness.

It is apparent from the plots in Figure 3.6 that individual speakers in the L3 group produce their Portuguese /a/ mostly around a single mean, but that this mean varies as far as how far back in the vowel space it is located, as it can be seen, for example, from the comparison between the plots for participants T13 and T09. The density plots in Figure 3.7 also offer valuable information in that they show how the speakers produce Portuguese /a/ vowels with a greater within category variation, which might be contributing to the reported difference in the production of Portuguese /a/ between groups.

To summarize the findings in this section, the results from a first series of tests showed that both the L2 and L3 groups maintain a clear difference between their Portuguese /a/ and their two English low vowels as for their respective location in the vowel space. This first set of tests also revealed that the two learner groups seem to be keeping similar difference patterns among their three low vowels. This finding was concluded to contradict the prediction that the two learner groups would produce their Portuguese /a/ with different spectral quality as an effect of the extensive input from Spanish /a/ that the L2 group lacks. In addition, the L2 group was found to produce a Portuguese /a/ that differed in distribution width from both the L3 and the Native groups', which did not differ from each other in this respect. A closer look at the density plots of the Portuguese /a/ tokens produced by the three participant groups, both by group and by speaker, illustrated this significant difference. All in all, these findings suggest that, as mentioned above, the L3 group produces a more target-like Portuguese /a/ than the L2 group, which is here claimed to be fostered by a more extensive experience with an acoustically similar phonetic category, that is, Spanish /a/.

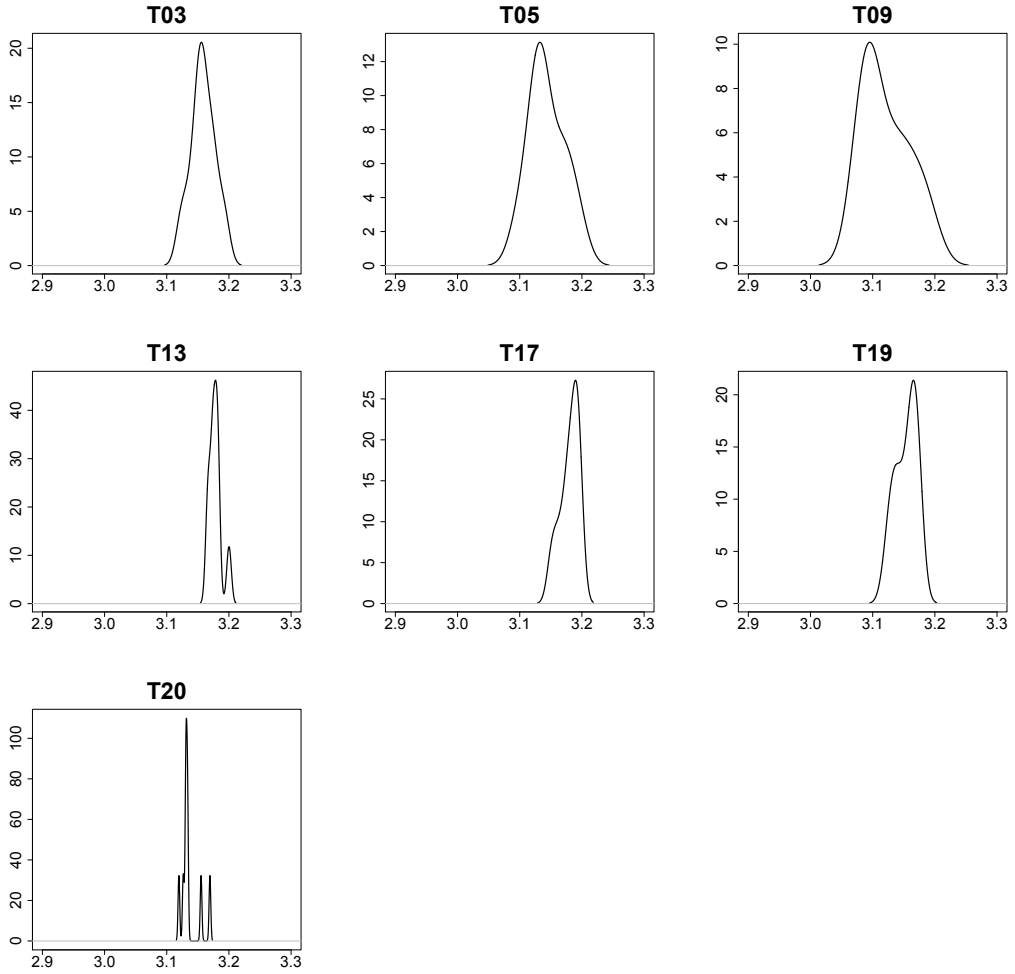

Figure 3.6: Density plots of logF2 values for Portuguese /a/ tokens produced by each of the participants in the L3 group: from left to right, participants T03, T05, T09 (top), T13, T17 and T19 (middle) and T20 (bottom).

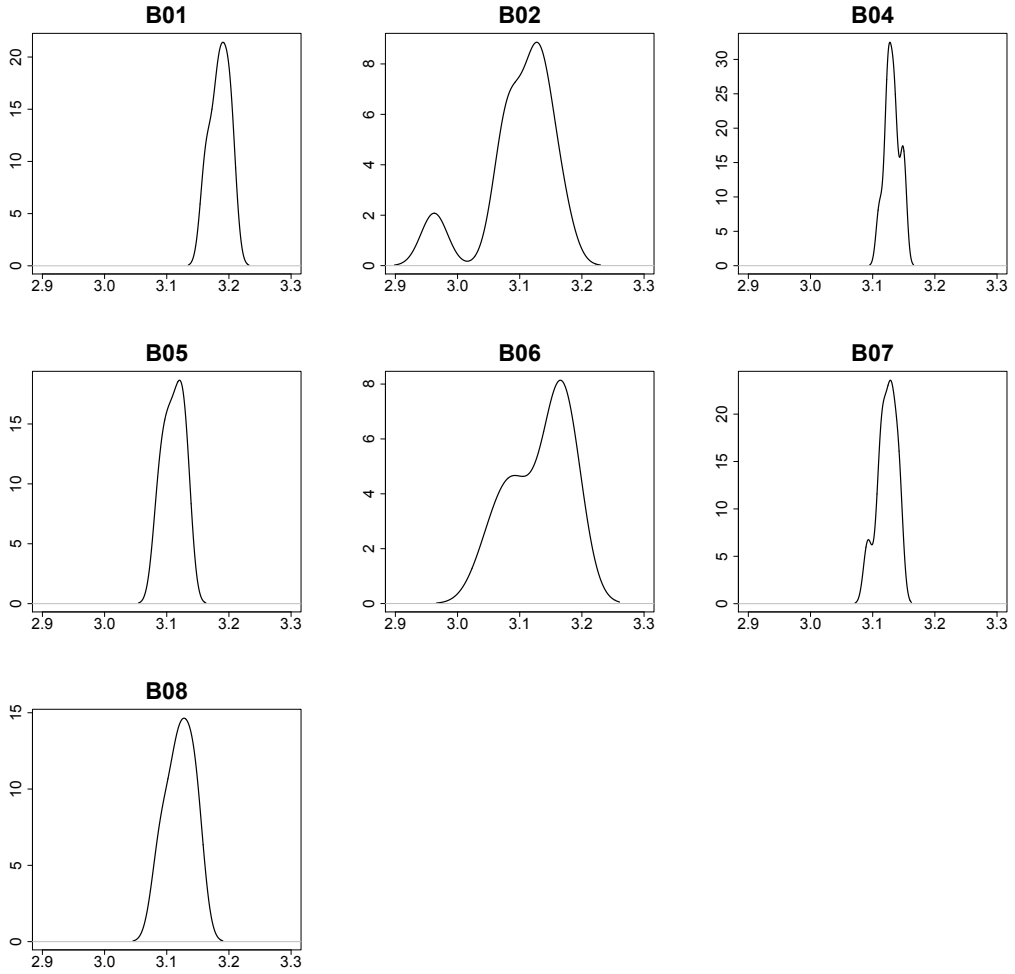

Figure 3.7: Density plots of logF2 values for Portuguese /a/ tokens produced by each of the participants in the L2 group: from left to right, participants B01, B02, and B03 (top), B04, B05 and B06 (middle), and B07 (bottom).

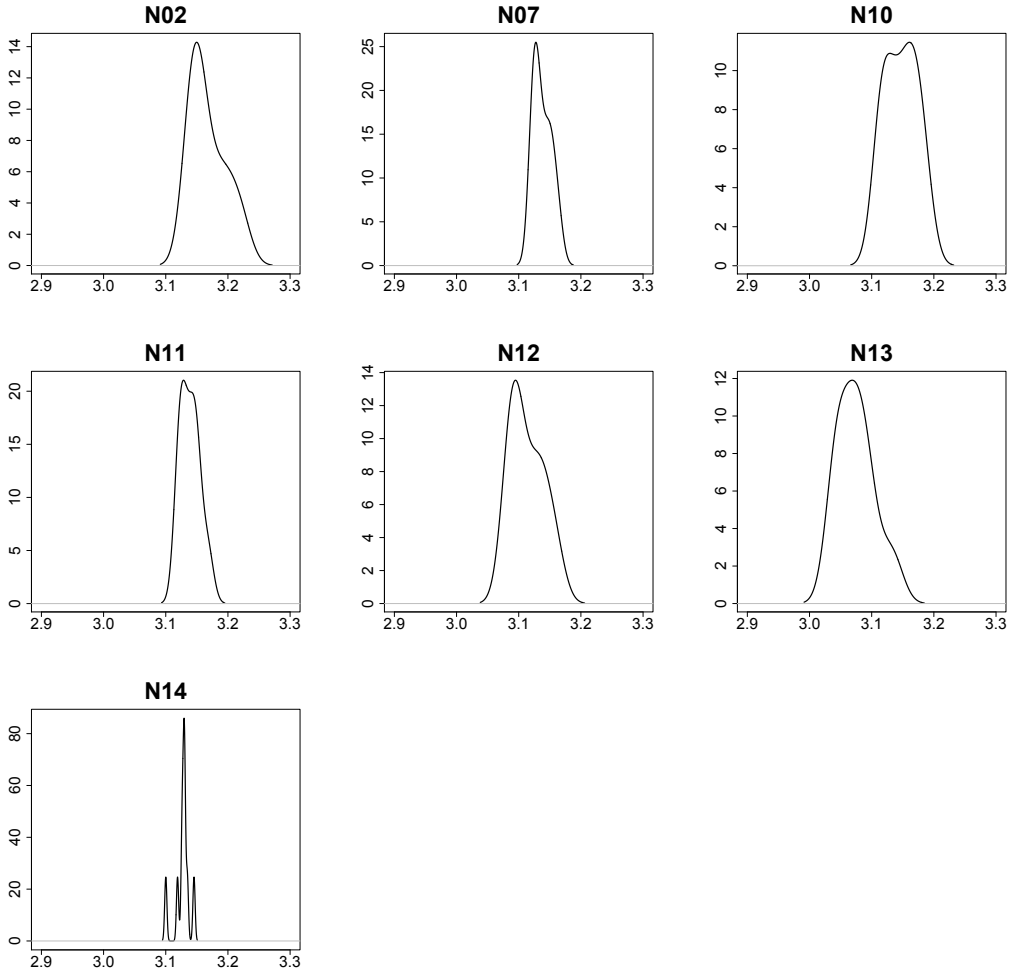

Figure 3.8: Density plots of logF2 values for Portuguese /a/ tokens produced by each of the participants in the Native Portuguese speaking group: from left to right, participants N02, N07, N10 (top), N11, N12, N13 (middle) and N14 (bottom).

Portuguese mid-back vowels

Data from native speakers of both English and Portuguese discussed in Chapter 2 confirmed that the mid vowels of these two languages tend to diphthongize in different degrees and direction. While English /e/ moves higher and becomes more fronted, English /ɛ/ moves back and higher, and Portuguese /e/ and /ɛ/ move back and lower into the vowel space through the duration of the vowel. This being the case, the mid vowels of Portuguese produced by all three groups of Portuguese speakers studied in this chapter will be investigated not only for significant spectral differences at their midpoint but also for potential differences in amount and direction of diphthongization.

The mid-back vowels of Portuguese produced by the L2, L3 and native speaker groups were analyzed by means of multiple linear mixed-effects regression models. The first set of linear mixed-effects models sought to investigate potential differences within each group. This first set of statistical tests consisted of a 2x3 linear mixed-effects model with "group" (Native, L2 and L3 groups) and "vowel" (Portuguese /o/, Portuguese /ɔ/) as the fixed factors, 'speaker' and 'word' as random intercepts, and logF1 and logF2 values at the midpoint of the vowel as the response. The intercept in this first set of tests was the L2 group's Portuguese /ɔ/.

Table 3.15: Results from two linear mixed-effects models on logF1 and logF2 values of Portuguese /o/ and /ɔ/ as produced by the native, L2 and L3 groups with L2 Port. /ɔ/ as the intercept.

	logF1		logF2	
	β	t	β	t
Intercept (= L2 Port /ɔ/)	2.070		2.990	
Port. /o/ x Native group	-0.115	-12.34*	-0.016	-2.3*
Port. /o/ x L3 group	-0.000	-0.02	-0.017	-2.5*

The results from this set of tests, presented in Table 3.15, show an interaction between 'group' and 'vowel,' both in logF1 as in logF2. The interaction in logF1 (t = -12.34) seems to be due to a different difference pattern between the two

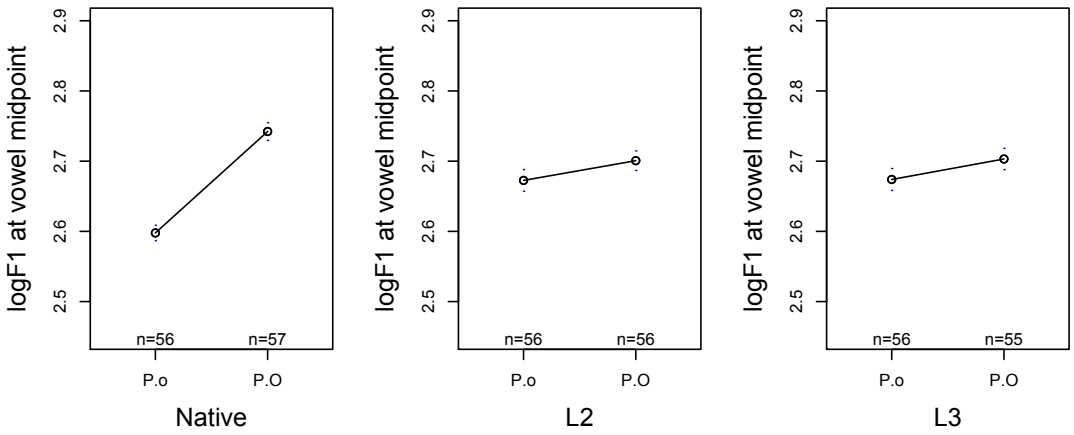

Figure 3.9: Mean logF1 values of the Portuguese /o/ (P.o) and Portuguese /ɔ/ (P.O) as produced by Native speakers, L2 group and L3 group. Graphed values are those measured at the midpoint of the vowel.

Portuguese vowels produced by the L2 group (= the intercept) as compared to the ones produced by the Natives. This different pattern seems to be apparent in Figure 3.9, where the logF1 values of the six vowels as produced by each of the groups have been graphed. Figure 3.9 suggests that, while the native speakers do maintain a considerable difference between their two vowels in height, the L2 group either does not, or maintains a difference that is much smaller than between the two vowels produced by the Native group. A follow-up set of linear mixed-effects models tested the differences in height between the Portuguese /o/-/ɔ/ produced by these three groups separately. The results from these three separate tests showed that all three groups maintain a difference in height between their Portuguese /o/ and /ɔ/ categories, while the Native speakers maintain a difference that is bigger (β = -0.144, t = -15.71) than both the L2 group (β = –0.028, t = -2.55) and the L3 group (β = -0.029, t = -2.13), as indicated by the differences in β values[5]. These differences are also observable in Figure 3.9.

[5]the results from the follow-up linear mixed-effects models by group are not reported explicitly in any Table, but they are reported in the text when appropriate

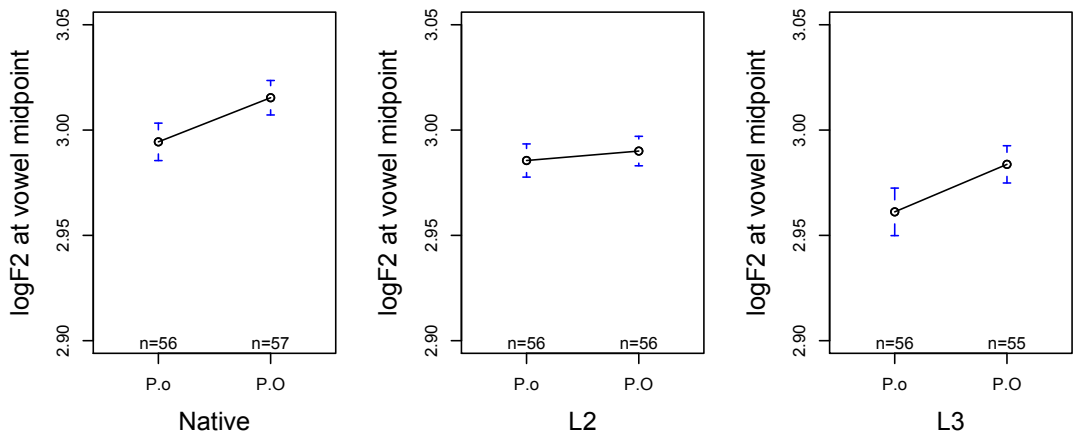

Figure 3.10: Mean logF2 values of the Portuguese /o/ (P.o) and Portuguese /ɔ/ (P.O) as produced by Native speakers, L2 group and L3 group. Graphed values are those measured at the midpoint of the vowel.

As for the interactions in logF2 shown in Table 3.15, both the Natives (t = -2.3) and the L3 group (t = -2.5) seem to show a different pattern in how they produce their Portuguese vowels as compared to the L2 group. The logF2 values of the six vowels as produced by the three groups are graphed in Figure 3.10. To investigate the source of the interactions further, three follow-up linear mixed-effects models were run with the data from each of the groups separately. The results show that the Natives keep a difference in logF2 between their Portuguese /o/ and /ɔ/ vowels (β = -0.021, t = -2.7), the L2 group does not (β = -0.004, t = -0.9), but the L3 group does (β = -0.022, t = -2.79)[6]. These patterns are also observable in Figure 3.10.

In conclusion, the results from this first set of statistical tests comparing the Portuguese /o/-/ɔ/ productions of the Native, L2 and L3 groups, showed that the two learner groups differ from the natives in the way they produce these vowels.

[6]Since the differences between the Portuguese native /o/ and /ɔ/ (see previous chapter for details) rely mostly on height, more attention will be given to differences in logF1 than to differences in logF2 between these two vowel categories as produced by the different groups in this chapter.

While the native speakers maintain an acoustic difference between their two mid-back vowels both in logF1 and logF2, the L2 group only maintain a difference between theirs in logF1, but not in logF2 and this difference is smaller than that maintained by the native speakers between theirs. As for the L3 group, they were found to maintain a difference between their Portuguese mid-back vowels both on logF1 and logF2, but the difference they maintain in logF1 is, once again, much smaller than the one maintained by the Natives.

It rests to statistically investigate the location of the Portuguese vowels produced by the two learner groups with respect to the ones produced by the native speakers. With this objective in mind, two additional sets of linear mixed-effects models were run with the two vowels of Portuguese separately. The first of these tests explored differences in logF1 across the Portuguese /ɔ/ vowels produced by the three groups. The intercept for this test was the L2 group. The results showed that the L2 group produces a Portuguese /ɔ/ that is not statistically different from that produced by the natives, although it approaches significance ($\beta = 0.040$, $t = 1.95$), while they produce a Portuguese /ɔ/ that does not differ from the one produced by the L3 in height. The second test explored the differences, also in logF1, between the Portuguese /o/ produced by each of the groups. Again, the L2 group's was found to be statistically different from the Natives' in height ($\beta = -0.074$, $t = -3.37$), but not from the one produced by the L3 group. These results also give information about the directionality of the differences: the learners' /ɔ/ is higher than the Natives', while their /o/ is lower than the Natives'. The finding that both Native Portuguese mid-back vowels differ from the learners', together with the directionality of the differences, suggest, as supported by Figure 3.1 above, that the L2 and L3 groups produce their Portuguese /o/ and /ɔ/ in a location between the two target native portuguese vowels.

The first set of statistical analyses presented in this subsection investigated differences in spectral quality between the Portuguese mid-back vowels produced by the L2 and L3 groups and the ones produced by the native speakers of the language. These spectral differences were explored by comparing the logF1 and logF2 values

of these vowels at the midpoint of the vowel. The cross-linguistic comparisons of these vowel categories as produced by native speakers of English and Portuguese in the previous chapter indicated, however, that the mid-back vowels of Portuguese also differ from that of English in their degree and direction of diphthongization. While English /o/ was found to move up and back in the vowel space through time, Portuguese /o/ and /ɔ/ move down and become more central. Just as the statistical analyses performed in Chapter 2 found differences in spectral quality between the English and the Portuguese mid-back vowels, it is also possible that the English native speakers of Portuguese under study in this chapter transfer the diphthongization patterns of their English /o/ to their Portuguese /o/ and /ɔ/.

In order to investigate the potential transfer of diphthongization patterns between the English /o/ and the Portuguese /o/ and /ɔ/ in the L2 and L3 groups' speech, a 2x3 linear mixed-effects model was run with 'group' (L2, L3) and 'vowel' (English /o/, Portuguese /o/, Portuguese /ɔ/) as the fixed factors, 'speaker' and 'word' as the random intercepts, and Diff1 and Diff2 values as the response.

Diff1 and Diff2 values were calculated based on logF1 and logF2 data. Diff1 was calculated by subtracting the value of logF1 at the first point of measurement (25% into the vowel) from the value of logF1 at the third point of measurement (75%) of the same vowel. Diff2 was calculated by means of the same subtraction of the logF2 at the first point of measurement from logF2 at the last point of measurement. This procedure is the same procedure used to study amount and degree of diphthongization of native vowels of English, Spanish and Portuguese in the previous chapter.

The results from the 2x3 linear mixed-effects factor are presented in Table 3.16. These results show that there is no interaction between the two fixed factors neither in Diff1 or Diff2. Thus, it can be assumed that the relationship among the three mid-back vowels tested is the same in each each of the two groups. These results also show that, while English /o/ moves backwards and up in the vowel space in both groups, Portuguese /o/ and /ɔ/ differ from English /o/ in direction of diphthongization, which means that it is not the case that they transfer the diphthongization patterns

Table 3.16: Results from two 2x3 linear mixed-effects models on Diff1 and Diff2 values of English /o/, Portuguese /o/ and Portuguese /ɔ/ as produced by the L2 and L3 groups with L2 English /o/ as the intercept.

	Diff1		Diff2	
	β	t	β	t
Intercept (= L2 English /o/)	-0.043		-0.045	
L3	-0.010	-0.73	0.005	0.29
Port. /ɔ/	0.050	3.18*	0.051	2.51*
Port. /o/	0.081	5.11*	0.059	2.86*
L3 x Port. /ɔ/	0.006	0.29	-0.005	-0.19
L3 x Port. /o/	0.002	0.08	-0.003	-0.11

from their L1 mid-back vowel to their Portuguese /o/ and /ɔ/.

In addition, the difference in the β values between the Portuguese /ɔ/ (β= 0.050 for Diff1, β= 0.051 for Diff2) and the Portuguese /o/ (β= 0.081 for Diff1, β= 0.059 for Diff2) suggest that the learners maintain a difference in degree of diphthongization between their two Portuguese mid-back vowels. These potential differences in diphthongization are observable, for Diff1, in Figure 3.11.

The potential difference in degree of diphthongization between the Portuguese /o/-/ɔ/ produced by the L2 and L3 group is unexpected. Since they could not have transferred this difference from their L1, it could be the case that they would have picked on this difference from the native speakers of this language. In order to investigate whether this is indeed feasible, one last set of 2x3 linear mixed-effects models was run including only the Portuguese /o/-/ɔ/ vowels of all three the Native, L2 and L3 groups. The intercept, both for the analysis of Diff1 and Diff2, was the /ɔ/ produced by the L2 group. The results of these tests are shown in Table 3.17.

The results in Table 3.17 show that there is an interaction between 'group' and 'vowel' in Diff1, suggesting that the two Portuguese vowels do not maintain the same relationship in diphthongization between each other across every group. In order to investigate where the differences in Diff1 between groups lied, three additional sets of linear mixed-effects models were run comparing the Diff1 values

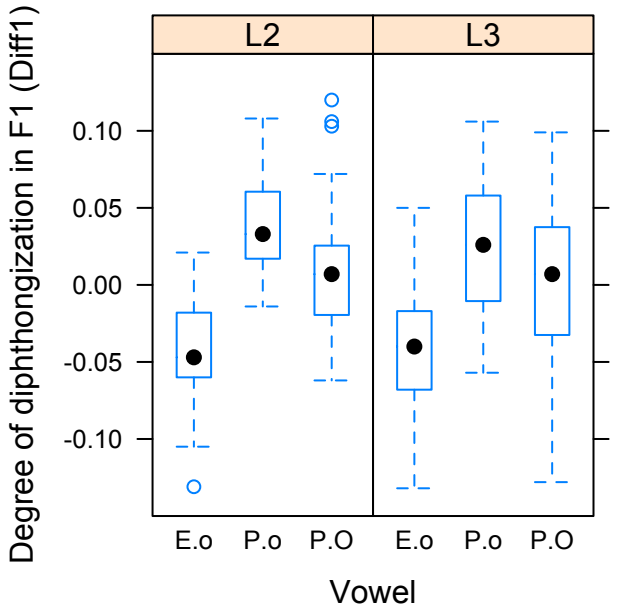

Figure 3.11: Degree of diphthongization in height of English /o/ (E.o), Portuguese /o/ (P.o) and Portuguese /ɔ/ (P.O) as produced by the L2 and L3 groups. Values graphed correspond to mean Diff1 values.

Table 3.17: Results from two linear mixed-effects models on Diff1 and Diff2 values of Portuguese mid-back vowels as produced by the native, L2 and L3 groups with the L2 Portuguese /ɔ/ as the intercept.

		Diff1		Diff2
	β	t	β	t
Intercept (= L2 Port. /ɔ/)	0.008		0.000	
Natives	0.009	0.68	0.000	2.43*
L3	-0.005	-0.36	0.000	0.01
Port. /o/	0.031	2.88*	0.000	0.53
Natives x Port. /o/	-0.021	-2.29*	-0.000	-0.89
L3 x Port. /o/	-0.004	-0.45	0.000	0.18

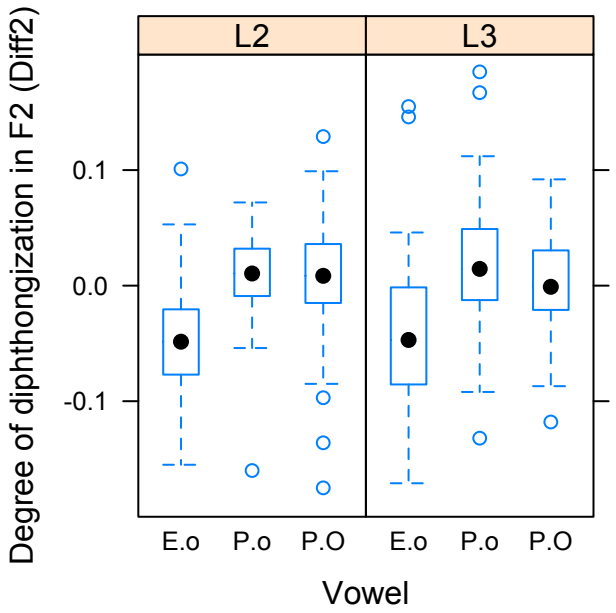

Figure 3.12: Degree of diphthongization in backness of English /o/ (E.o), Portuguese /o/ (P.o) and Portuguese /ɔ/ (P.O) as produced by the L2 and L3 groups. Values graphed correspond to mean Diff2 values.

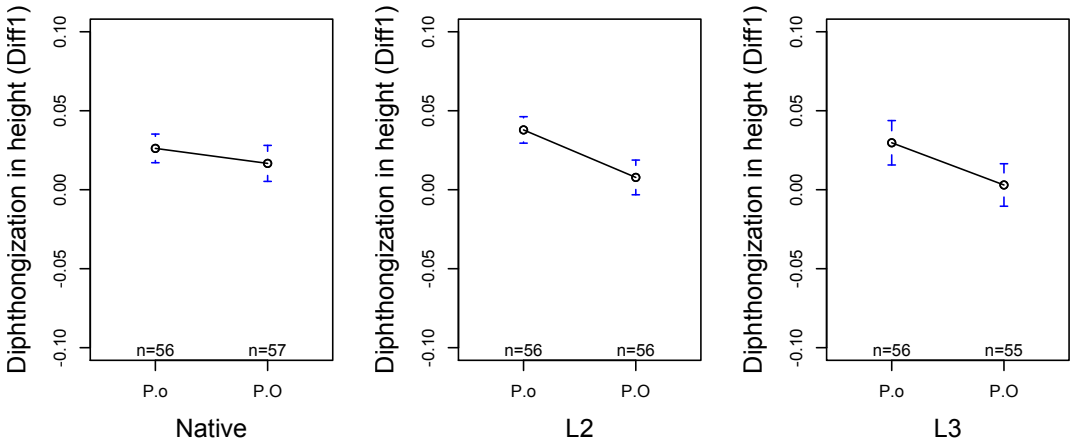

Figure 3.13: Diphthongization in height of the Portuguese /o/ (P.o) and Portuguese /ɔ/ (P.O) produced by the Native, L2 and L3 groups. Values graphed correspond to mean Diff1 values.

of the two Portuguese vowels of each group separately. These three models reported that while the natives' Portuguese /o/ and /ɔ/ do not differ in Diff1 (β= 0.010, t = 0.96), the L2 groups' do (β= 0.030, t = 3.78), and the L3 groups' show a tendency to do so (β= 0.027, t = 1.95). It is interpreted from these results that, while the native speakers produce both their /o/ and /ɔ/ with the same degree of spectral movement in height through time, the learner groups produce a Portuguese /o/ that raises more in height through time than their own /ɔ/. These differences can be observed in Figure 3.13. It is concluded here that the learners', more so in the case of the L2 group than in the L3 group, do not behave like the Natives as far as diphthongization in the logF1 axis is concerned.

With regards to Diff2, the results in Table 3.17 showed no interaction between 'group' and 'vowel' on this measure. This lack of interaction is assumed here to mean that all three groups maintain a similar difference, or lack thereof, in movement across the logF2 dimension through time between their two Portuguese vowels. Moreover, the L2 group's Portuguese /o/ and /ɔ/ do not differ in Diff2 from the L3 group's (β= 0.000, t= 0.01), but they do differ from the Natives' (β= 0.000,

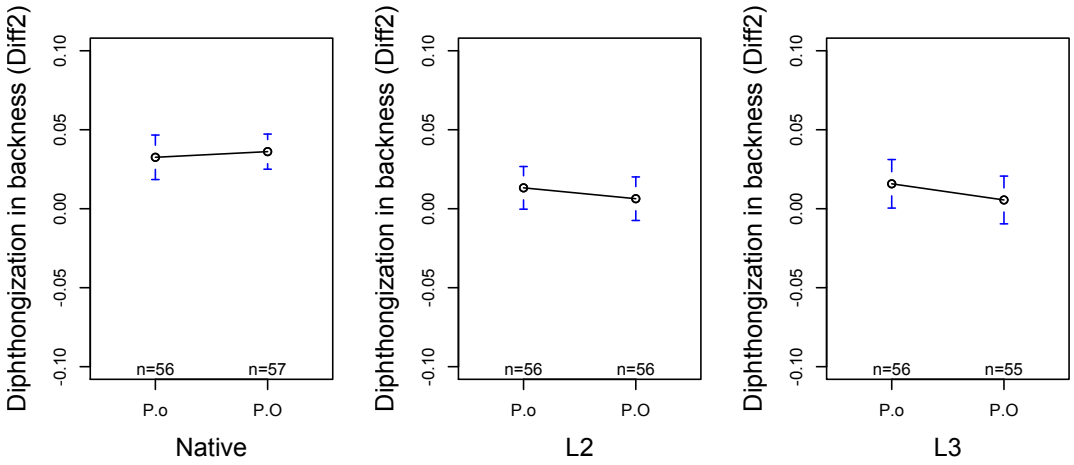

Figure 3.14: Diphthongization in backness of the Portuguese /o/ (P.o) and Portuguese /ɔ/ (P.O) produced by the Native, L2 and L3 groups. Values graphed correspond to mean Diff2 values.

t = 2.43). This is also visible in Figure 3.14. Thus, it is not possible that the learner groups have learned to diphthongize their two Portuguese mid-back vowels from the input received from native speakers of the language, and it is neither possible that they have transferred the diphthongization patterns they produce in their Portuguese /o/ and /ɔ/ from their L1.

To summarize the findings about the mid-back vowels of Portuguese discussed in this section, the two learner groups have both created two different categories for the Portuguese /o/-/ɔ/ pair that show indication of learning of the difference with their native English /o/ in the horizontal axis but not quite in height. Although the L3 group seems to maintain a difference in height between their Portuguese /o/-/ɔ/, the difference they maintain is much smaller than the one kept by the native speakers of the language. Additional evidence also suggests that the Portuguese /o/-/ɔ/ produced by the learners are located right in between the two target Portuguese mid-back vowels.

Statistical analyses of diphthongization data also suggest that the learners have noticed that Portuguese mid-back vowels diphthongize in different direction from

that of their L1. The analyses of the data also show that the learners have noticed that there is an acoustic difference between the two Portuguese vowels. However, they produce the distinction between their two Portuguese vowels by producing a difference in degree of diphthongization between them, while the native speakers' /o/ and /ɔ/ diphthongize in the same degree but differ in height. This being the case, it is interpreted here that the learners must have noticed an acoustic difference between the two categories of Portuguese but the feature they implement this acoustic distinction on is incongruent with the acoustic feature the native speakers use to distinguish these two mid-back vowels in Portuguese.

Portuguese mid-front vowels

As in the case of the mid-back vowels just discussed, results from the analysis of the production data from native speakers of both English and Portuguese presented in Chapter 2 confirmed that the mid-front vowels of these two languages tend to diphthongize in different degrees and direction. The analyses performed on native speaker data in the previous chapter provided acoustic evidence that the two mid-front vowels of English are two distinct phonetic categories, differing in spectral quality and amount and direction of diphthongization. While English /e/ diphthongizes by raising and becoming more fronted, English /ɛ/ diphthongizes, to a lesser degree, by raising and moving backwards in the vowel space. Portuguese /e/ and /ɛ/ also were also found to be two distinct phonetic categories, differing in spectral quality but diphthongizing backwards and downwards to the same degree. The Portuguese and English mid-front vowels, however, were found to differ in numerous aspects.

This being the case, the mid-front vowels of Portuguese produced by all three groups of Portuguese speakers (natives, L2 and L3 groups)[7] were investigated for potential differences in spectral quality at their midpoint as well as in degree and direction of diphthongization.

Once again, multiple linear mixed-effects regression models were performed to

[7]as well as their English mid-front vowels when necessary and appropriate

Table 3.18: Results from two linear mixed-effects models on logF1 and logF2 values of Portuguese /e/ and /ɛ/ as produced by the native, L2 and L3 groups with the L2 group's Portuguese /e/ as the intercept.

	logF1		logF2	
	β	t	β	t
Intercept (= L2 Port /e/)	2.687		3.268	
Native group x Port. /ɛ/	0.123	12.70*	-0.031	-5.9*
L3 group x Port. /ɛ/	0.004	0.46	0.004	0.8

analyze the mid-front vowels of Portuguese produced by the L2, L3 and native speaker groups. As in the analysis of mid-back vowels above, the first set of statistical models investigated potential spectral differences maintained by each of the groups between their two Portuguese vowels. This analysis sought to answer the question of whether the learners maintain the same kind of spectral difference between their two Portuguese vowels as the native speakers of the language do. Thus, a 2x3 linear mixed-effects model was run with 'group' (Native, L2, L3) and 'vowel' (Portuguese /e/, Portuguese /ɛ/) as the fixed factors, 'speaker' and 'word' as the random intercepts and logF1 and logF2 as the response. In these models, the Portuguese /e/ produced by the L2 group was used as the intercept. The results rendered by these models, presented in Table 3.18, show that there is an interaction, on both the logF1 and logF2 measures, between 'group' and 'vowel,' meaning that not all groups maintain the same spectral differences between their two Portuguese vowels. The data in Table 3.18, together with the graphed data in Figures 3.15 and 3.16, suggest that there might be a difference between the L2 group and the Natives in the way they maintain a spectral difference between their two Portuguese /e/ and /ɛ/ vowels.

In order to explore the nature of these interactions, a follow-up set of linear mixed-effects models were run on the Portuguese /e/ and /ɛ/ logF1 and logF2 data for each group separately.The results from these tests showed that the Natives speakers maintain a difference both in height (β= 0.140, t= 26.6) and backness (β=

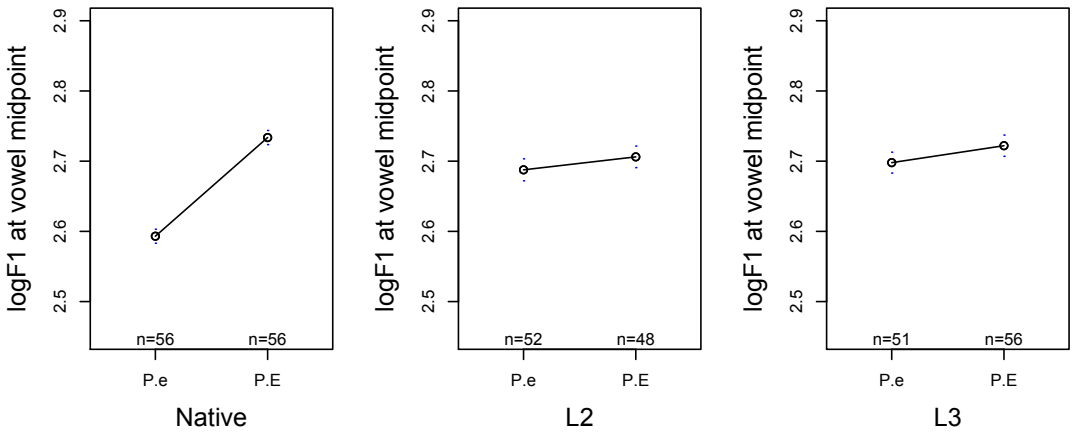

Figure 3.15: Mean logF1 values of the Portuguese /e/ (P.e) and Portuguese /ɛ/ (P.E) as produced by the Native, L2 and L3 groups. Graphed values are those measured at the midpoint of the vowel.

-0.037, t= -7.2) between their two mid-front vowels; the L2 group does not keep a difference in spectral quality between theirs on any of the two measures, and the L3 group maintains a difference between them in height (β= 0.022, t= 2.39) but not in backness. These interactions are observable in Figures 3.15 and 3.16.

From these findings the question is raised of whether the learners might be perceiving the Portuguese vowels as similar to any of their native English mid-front vowels, potentially causing them to not reach target-like pronunciation in Portuguese. To investigate this issue, a series of 4x2 linear mixed-effects models were performed on logF1 and logF2 data of the English /e/, English /ɛ/, Portuguese /e/ and Portuguese /ɛ/ produced by each of the learner groups. Once again, 'vowel' and 'group' were selected as the fixed factors and 'speaker' and 'word' were used as random intercepts. The data is graphed in Figures 3.17 and 3.18.

The first two tests used the L2 groups' English /e/ as the intercept. The results from these tests rendered no interaction between 'group' and 'vowel' on the logF1 measure, which leads to the assumption, supported by Figure 3.17, that the pattern of similarities and differences in height among the four mid-front vowels tested are

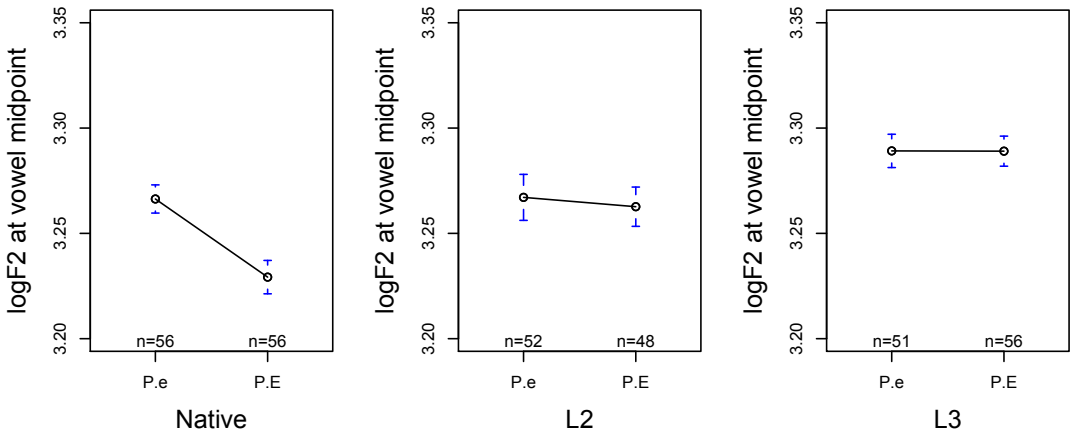

Figure 3.16: Mean logF2 values of the Portuguese /e/ (P.e) and Portuguese /ɛ/ (P.E) as produced by the Native, L2 and L3 groups. Graphed values are those measured at the midpoint of the vowel.

the same in both groups. The results show that the English /e/ produced by these learners is statistically different in height from their own English /ɛ/ (β= 0.118, t= 7.55), and both from their Portuguese /e/ (β= 0.077, t= 3.15) and /ɛ/ (β= 0.095, t= 3.83). These results, together with the results from similar tests performed on logF2 data, are shown in Table 3.19.

The results from the tests on logF2 data, on the other hand, rendered an interaction between 'group' and 'vowel' (β= -0.017, t= -2.18). From these results, and from the visual information on Figure 3.18, it is hypothesized that this interaction might come from the L2 and L3 groups maintaining a different distance in backness between their own two English mid-front vowels. This hypothesis is confirmed by the results from three additional linear mixed-effects models run on each of the three groups of speakers separately. These additional tests confirm the hypothesis that the L2 group (β= -0.113, t= -5.99) maintains a distance in backness between their English /e/ and /ɛ/ that is smaller than the distance maintained by the L3 group (β= -0.131, t= -5.48), as indicated in the size of their β values. The interaction found in the results of the linear mixed-effects model on logF2 data in Table 3.19

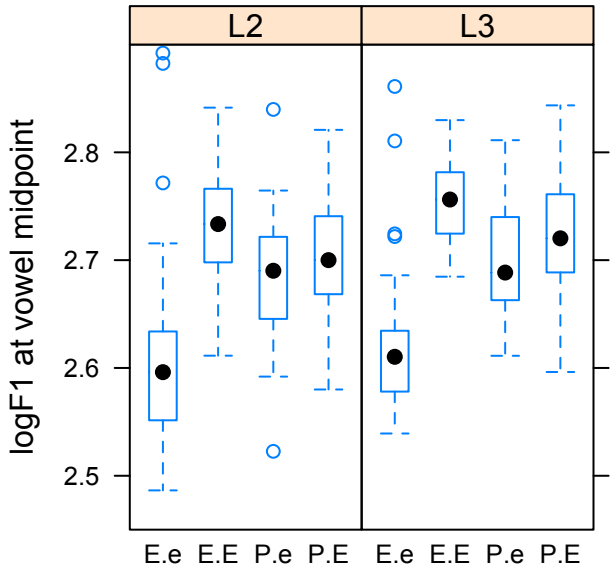

Figure 3.17: Mean logF1 values of English /e/ (E.e), English /ɛ/ (E.E), Portuguese /e/ (P.e) and /ɛ/ (P.E) produced by the L2 and L3 groups. Graphed values are those measured at the midpoint of the vowel.

Table 3.19: Results from two linear mixed-effects models on logF1 and logF2 values of English /e/, English /ɛ/, Portuguese /e/ and /ɛ/ as produced by the L2 and L3 groups with L2 English /e/ as the intercept.

	logF1		logF2	
	β	t	β	t
Intercept (= L2 Engl. /e/)	2.610		3.277	
L3 group	0.012	0.58	0.018	1.22
Engl. /ɛ/	0.012	7.55*	-0.014	-7.21*
Port. /e/	0.078	3.15*	-0.009	-0.43
Port /ɛ/	0.095	3.83*	-0.015	-0.71
L3 group x Engl. /ɛ/	0.016	1.45	-0.017	-2.18*
L3 group x Port. /e/	0.000	0.01	0.004	0.21
L3 group x Port. /ɛ/	0.005	0.19	0.008	0.41

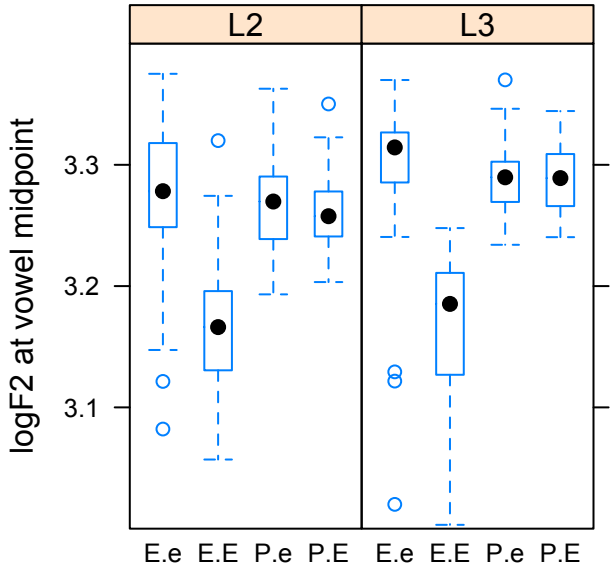

Figure 3.18: Mean logF2 values of English /e/ (E.e), English /ɛ/ (E.E), Portuguese /e/ (P.e) and /ɛ/ (P.E) produced by the L2 and L3 groups. Graphed values are those measured at the midpoint of the vowel.

Table 3.20: Results from two linear mixed-effects models on logF1 and logF2 values of English /e/, English /ɛ/, Portuguese /e/ and /ɛ/ as produced by the L2 and L3 groups with L2 Portuguese /ɛ/ as the intercept.

	logF1		logF2	
	β	t	β	t
Intercept (= L2 Port. /ɛ/)	2.704		3.262	
L3 group	0.017	0.84	0.026	1.79
Engl. /e/	-0.095	-3.83*	0.015	0.71
Engl. /ɛ/	0.023	0.95	-0.099	-4.71*
Port. /e/	-0.017	-1.08	0.006	0.38
L3 group x Engl. /e/	-0.006	-0.19	-0.008	-0.41
L3 group x Engl. /ɛ/	0.011	0.37	-0.025	-1.22
L3 group x Port. /e/	-0.005	-0.44	-0.004	-0.51

is thus explained. Aside from this interaction, the results in Table 3.19, supported by Figure 3.18, also suggest that these learners' English /e/ does not differ from any of their own Portuguese mid-front vowels in backness.

The second set of tests exploring the differences between the English and Portuguese mid-front vowels produced by the two learner groups used the L2 groups' English /e/ as the intercept. These two 2x4 linear mixed-effects models rendered the results in Table 3.20. No 'group' by 'vowel' interactions were found significant so it is assumed here that both learners maintain the same pattern of differences and/or similarities in spectral quality between their Portuguese /ɛ/ and their other three mid-front vowels. This assumption is supported by the graphed data in Figures 3.17 and 3.18. Table 3.20 also shows that the Portuguese /ɛ/ produced by the two learner groups was neither found to be different in height from English /ɛ/ nor from Portuguese /e/. In backness, their Portuguese /ɛ/ was not found to be different from Portuguese /e/, but was found to significantly differ from their own English /ɛ/ (β= -0.099, t= -4.71).

As a summary of this subsection on the comparison of the English and Portuguese mid-front vowels as produced by the L2 and L3 groups, it was found that the L3 group produces their English /e/ and /ɛ/ further apart on the logF2 axis than the

L2 group does (see Figure 3.18), although the two groups do not differ in the way they distinguish their two English vowels in height (see Figure 3.17). It was also found that both these groups' English /e/ differed in height from both their own Portuguese /e/ and /ɛ/ (see Figure 3.17), but, most interestingly, they did not differ from their Portuguese mid-front vowels on the logF2 dimension (see Figure 3.18).

The two groups also produce their Portuguese /ɛ/ at a similar height as their own Portuguese /e/ and, quite relevantly, as their English /ɛ/, although their Portuguese /ɛ/ differs from their English /ɛ/ in backness, producing their Portuguese open mid-front vowels in a more fronted position. Their Portuguese /ɛ/ did not differ from their Portuguese /e/ in backness either (again, refer to Figures 3.17 and 3.18 for visual aids in the interpretation of these findings).

The results from the comparison of the spectral quality at their midpoint of the four mid-front vowels of English and Portuguese produced by the two learner groups suggest that both groups have created a new category for the Portuguese mid-front vowels that differs from those of English. It is interpreted here that these learners have created one new phonetic category for Portuguese /e/-/ɛ/ and not two, as their productions of Portuguese /e/ and /ɛ/ do not differ significantly in height nor backness from each other. Nevertheless, their new Portuguese mid-front vowel category is located at similar height as their own English /ɛ/, but at a similar location along the logF2 axis as their English /e/. The vowel space of the L2 and L3 groups was graphed in Figures 3.19 and 3.20. Ultimately, the new mid-front vowel category created by the L2 and L3 groups for Portuguese differs from both the native speakers' Portuguese /e/ and /ɛ/. Figure 3.1 above presents the Portuguese vowels of all native speakers, L2 and L3 groups.

In addition to differences and similarities in spectral quality, which 'equated' their new Portuguese vowel with English /e/ in backness and with English /ɛ/ in height, the mid-front vowels of Portuguese produced by the L2 and L3 groups were investigated for potential differences and similarities in degree and direction of diphthongization both as compared to the target native Portuguese categories, as

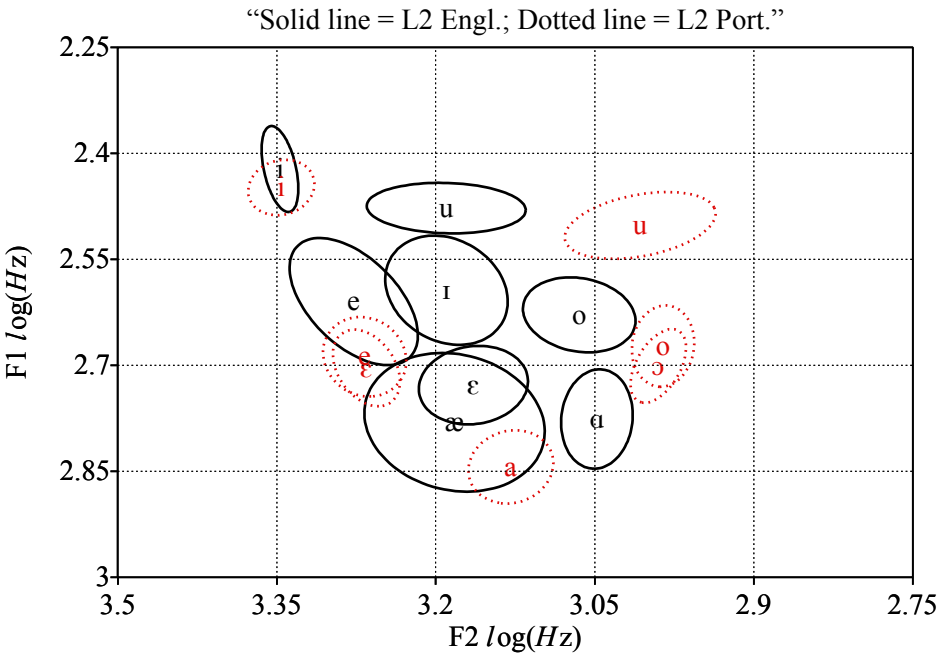

Figure 3.19: Vowel space of participants in L2 group with vowels of English and Portuguese in it.

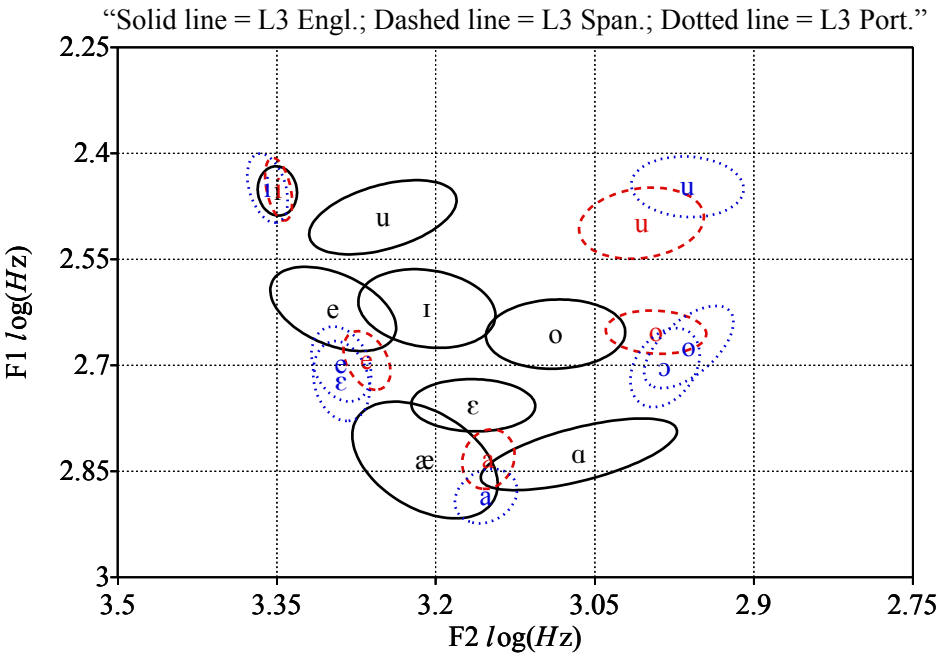

Figure 3.20: Vowel space of participants in L3 group with vowels of English, Spanish and Portuguese in it.

Table 3.21: Results from two linear mixed-effects models on Diff1 and Diff2 values of English /e/, English /ɛ/, Portuguese /e/ and /ɛ/ as produced by the L2 and L3 groups with L2 English /e/ as the intercept.

	Diff1		Diff2	
	β	t	β	t
Intercept (= L2 Engl. /e/)	-0.059		0.051	
L3 group	-0.010	-0.89	-0.010	-0.99
Engl. /ɛ/	0.054	4.41*	-0.063	-4.82*
Port. /e/	0.094	5.94*	-0.079	-5.22
Port /ɛ/	0.095	5.98*	-0.074	-4.86*
L3 group x Engl. /ɛ/	0.013	1.47	0.003	0.30
L3 group x Port. /e/	0.006	0.39	0.022	1.49
L3 group x Port. /ɛ/	0.007	0.43	0.016	1.12

well as to their own English mid-front vowels.

In order to investigate the differences and similarities in diphthongization pattern among their own four mid-front vowels, two sets of 2x4 linear mixed-effects models were run with 'group' (L2, L3) and 'vowel' (English /e/, English /ɛ/, Portuguese /e/, Portuguese /ɛ/) as the fixed factors, 'word' and 'speaker' as random intercepts, and Diff1 and Diff2 measures as the response.

The first of these models took the English /e/ produced by the L2 group as the intercept. The results from this first test are presented in Table 3.21. The results show no 'group' by 'vowel' interactions on any of the two diphthongization measures. Thus, it is assumed here that the relationship among the four vowels compared in this test is the same across groups. The Diff1 and Diff2 values of the vowels compared here are graphed in Figures 3.21 and 3.22. The results in Table 3.21, supported by Figures 3.21 and 3.22, suggest that the English /e/ of these learners differs in diphthongization degree, both in Diff1 and in Diff2, from all three other mid-front vowels they produced: their own English /ɛ/ and their own Portuguese /e/ and /ɛ/. Since English /e/ was found to be the most diphthongizing of the four mid-front vowels in the comparison among native speaker inventories in the previous chapter, this finding is not surprising.

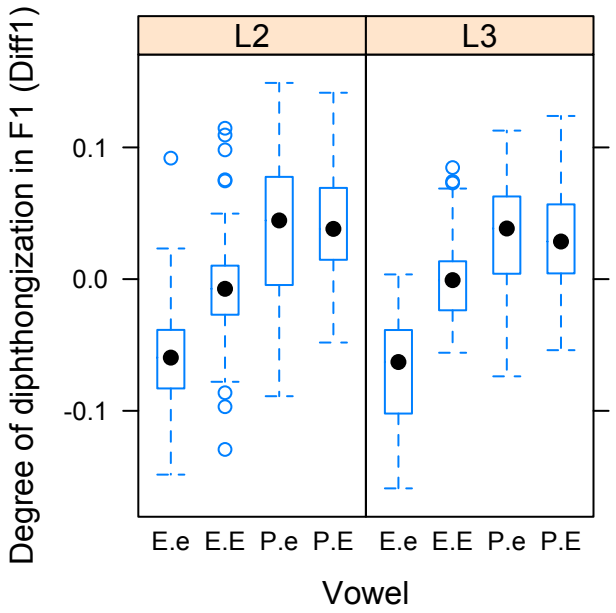

Figure 3.21: Degree of diphthongization in height of English /e/ (E.e), English /ɛ/ (E.E), Portuguese /e/ (P.e) and Portuguese /ɛ/ (P.E) as produced by the L2 and L3 groups. Values graphed correspond to mean Diff1 values.

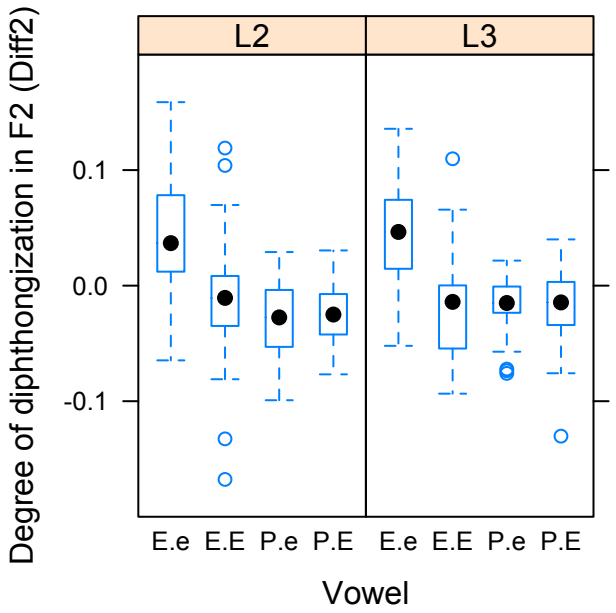

Figure 3.22: Degree of diphthongization in backness of English /e/ (E.e), English /ɛ/ (E.E), Portuguese /e/ (P.e) and Portuguese /ɛ/ (P.E) as produced by the L2 and L3 groups. Values graphed correspond to mean Diff2 values.

Table 3.22: Results from two linear mixed-effects models on Diff1 and Diff2 values of English /e/, English /ɛ/, Portuguese /e/ and /ɛ/ as produced by the L2 and L3 groups with L2 Portuguese /ɛ/ as the intercept.

	Diff1		Diff2	
	β	t	β	t
Intercept (= L2 Port. /ɛ/)	0.036		-0.023	
L3 group	-0.003	-0.27	0.006	0.61
Engl. /e/	-0.095	-5.98*	0.074	4.86*
Engl. /ɛ/	0.041	-2.58*	0.011	0.74
Port. /e/	-0.001	-0.10	-0.005	-0.37
L3 group x Engl. /e/	-0.007	-0.43	-0.016	-1.12
L3 group x Engl. /ɛ/	0.006	0.36	-0.013	-0.92
L3 group x Port. /e/	-0.001	-0.07	0.005	0.53

An additional set of 2x4 linear mixed-effects model of similar structure was run on the Diff1 and Diff2 data with Portuguese /ɛ/ as the intercept. The results from this test are presented in Table 3.22. The results in Table 3.22 support that, as the previous test suggested, the relationship among all vowels compared is the same across both groups of learners. The results show that the Portuguese /ɛ/ produced by the learners differs in Diff1 from their own two English vowels but not from their Portuguese /e/, and in Diff2 from their English /e/ but not from their own English /ɛ/ or Portuguese /e/. These results are also observable in Figures 3.21 and 3.22.

So far, the results from these two sets of statistical models suggest that these learners' English /e/ and Portuguese /e/ differ both in degree and direction of diphthongization and their English /ɛ/ and Portuguese /ɛ/ also differ in direction and degree of vowel movement in height, but not in backness. Their Portuguese /e/-/ɛ/, on the other hand, differ from each other in movement along the logF2 axis but not in height. These findings suggest that the learners have learned that the diphthongization patterns of Portuguese mid-front vowels are different from those they use for their native English ones. However, they seem to diphthongize equally in their two Portuguese vowels, which raises the question of whether they might have indeed learned this acoustic feature in a target-like fashion.

Table 3.23: Results from two linear mixed-effects models on Diff1 and Diff2 values of Portuguese /e/ (="15") and /ɛ/ (="16") as produced by the native (="N"), L2 (="B") and L3 ("T") groups with L2 Portuguese /e/ as the intercept.

	logF1		logF2	
	β	t	β	t
Intercept (= L2 Port /e/)	0.035		-0.027	
Native group	-0.002	-0.15	0.002	0.16
L3 group	-0.004	-0.26	0.011	1.03
L2 Port /ɛ/	0.000	0.04	0.006	0.84
Native group x Port. /ɛ/	-0.005	-0.59	-0.006	-0.93
L3 group x Port. /ɛ/	0.001	0.18	-0.006	-0.91

In order to investigate this possibility, one additional set of 2x3 linear mixed-effects models was run with 'group' (Native, L2, L3) and 'vowel' (Portuguese /e/, Portuguese /ɛ/) as the fixed factors, 'word' and 'speaker' as random intercepts, and Diff1 and Diff2 as the response. The Diff1 and Diff2 data analyzed in this set of tests was graphed resulting in Figures 3.23 and 3.24. The statistical analyses, whose results are shown in Table 3.23, took the Portuguese /e/ produced by the L2 group as the intercept.

The results in Table 3.23 found no interaction between the fixed factors in any of the two measures. The results also show that neither of the learner groups differs in any of the two measures of spectral movement from the Native group, implying, most relevantly, that the learners have indeed acquired the diphthongization patterns of the Portuguese /e/ and /ɛ/ in a native-like fashion.

3.3.2 Learners vs. Native speakers' Portuguese vowels: an alternative analysis

The previous section sought to search for potential differences in the production of the seven Portuguese stressed oral vowels between (a) either of the two learner groups in this dissertation (L2 and L3) and the native Portuguese-speaking group (NP), as well as (b) between the L2 and L3 group. While several interesting findings emerge from the multiple comparisons carried out by comparing the logF1 and logF2 values

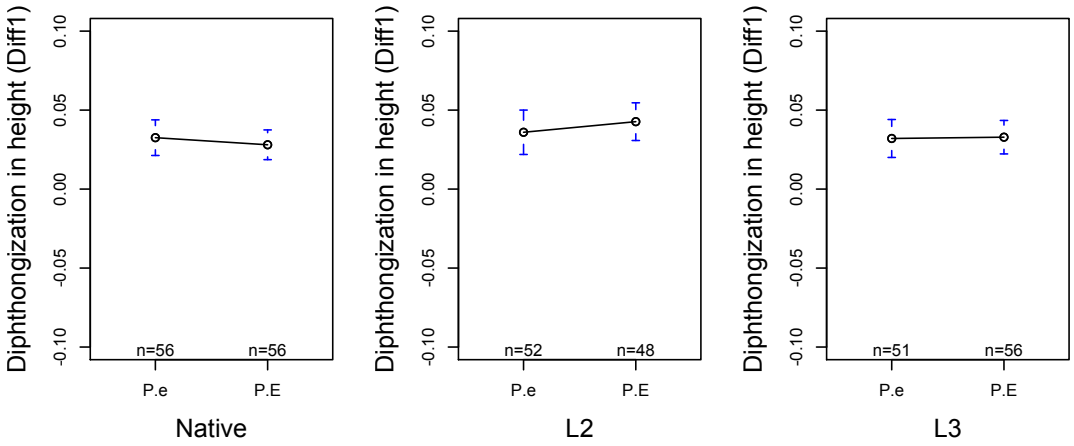

Figure 3.23: Degree of diphthongization in height of Portuguese /e/ (P.e) and Portuguese /ɛ/ (P.E) produced by the Native, L2 and L3 groups. Graphed values are mean Diff1 values.

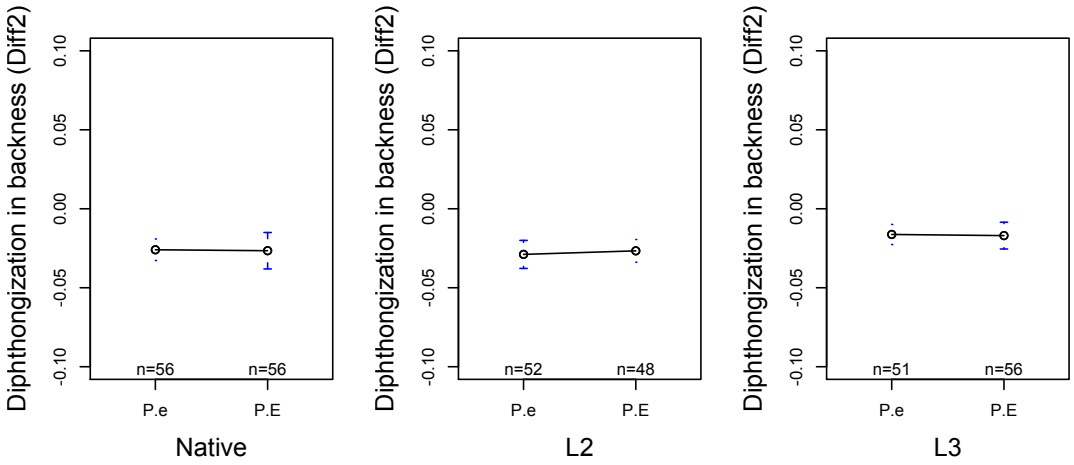

Figure 3.24: Degree of diphthongization in backness of Portuguese /e/ (P.e) and Portuguese /ɛ/ (P.E) produced by the Native, L2 and L3 groups. Graphed values are mean Diff2 values.

of the Portuguese vowels produced by each of the groups, most of the differences were found, as it was to be expected, between the learner groups' and the native speakers' productions of the Portuguese vowels.

Despite the more limited number of results suggesting a difference between the L2 and the L3 groups' productions of Portuguese vowels, it is important to observe the nature of the differences that were actually found between them. On the one hand, the L2 and L3 groups differed from each other in their production of Portuguese /u/, as the L3 group's was found to be located in a position as back in the vowel space as the NP's, while the L2 group's was located in a much more fronted position. Moreover, the L3 groups' Portuguese /u/ was found to be located even higher in the vowel space than the one produced by the NP group (which was, together with the position of the L3 group's English /i/, interpreted as being a sign of phonetic dissimilation). On the other hand, the L2 and L3 groups' Portuguese /a/ also differed from one another in that the L3 group's, as well as their English /æ/ and /ɑ/, was found to be located lower in the vowel space than the L2 group's.

These observations evoke a close resemblance to the claim, explored and partially supported by Bradlow (1995) among others, that acoustic vowel spaces of languages with relatively crowded vowel inventories may be expanded with respect to acoustic vowel spaces of languages with smaller numbers of vowels in their inventories. While studies on the effect of inventory size on acoustic vowel space usually focus on the study and comparison of vowels as produced by monolingual speakers of different languages, the idea of larger vowel inventories having a larger vowel space could be, in principle, applied to the context of second/third language acquisition, where speakers of three languages most likely have a more crowded vowel space than speakers of one or two of the same languages.

The two significant differences between the L2 and L3 groups in their production of Portuguese vowels discussed above, together with the abovementioned effect of inventory size on acoustic vowel space, seem to suggest that it might be the case that, due to the fact that the L3 group has five more vowel categories in their vowel inventory (namely those of Spanish, which the L2 group lacks), the L3 group's vowels

have suffered greater dispersion in the vowel space and are, thus, further out in the vowel space than those produced by the L2 group.

In order to put this hypothesis to the test, an alternative analysis of the vowels produced by the L2, L3 and NP groups (studied in the previous sections in this chapter) was carried out. This time, the distance from each vowel token to the centroid of each speaker's vowel space was calculated and the potential differences in vowel location among the three participant groups were investigated. Materials, Participants, and Procedure for this alternative analysis are, thus, exactly the same as for the main analysis in this chapter and described under sections 3.2.1, 3.2.2 and 3.2.3 above. The data analysis was, however, different.

Alternative data analysis

For this alternative investigation of the potential differences in Portuguese vowel production among the NP, L2 and L3 groups, first, the centroid of the Portuguese vowel space was calculated (in logF1 and logF2 values) for each participant separately. The centroid for each participant equalled the mean of all Portuguese vowel tokens produced by said participant. Next, also for each participant separately, the distance between their centroid and each of the Portuguese vowels they produced was calculated. This distance consisted of the Euclidean Distance (for two dimensions) between each token and the centroid, a value obtained by applying the basic formula in equation 3.1, where "ED (x, c)" refers to the Euclidean Distance (ED) between a specific token ("x") and the centroid ("c"), x_{logF1} and c_{logF1} refer to the logF1 values of "x" and the centroid, and x_{logF2} and c_{logF2} refer to their respective logF2 values:

$$ED(x, c) = \sqrt{(x_{logF1} - c_{logF1})^2 + (x_{logF2} - c_{logF2})^2} \tag{3.1}$$

Results of alternative data analysis

With the objective of investigating the potential differences among the three groups of participants in their production of each of the seven Portuguese vowels with regards to how far away in the vowel space they were located, a linear mixed-effects regression was run. This linear mixed-effects regression took the Portuguese /i/ produced by the L2 group as the base of the comparison (intercept). It had 'distance to centroid' as response, 'group' and 'vowel' as fixed factors, and 'speaker' and 'word' as random factors. The 'group' fixed factor had three levels (NP, L2 and L3), while the 'vowel' factor had seven levels, one for each Portuguese vowel category. The results from this linear mixed-effects regression model are presented in Table 3.24.

Table 3.24 shows different kinds of information. First of all, the top section of the table reports that there is a significant difference in distance to the centroid between the Portuguese /i/ produced by the L2 group (β= 0.297) and the one produced by the NP group (β= -0.042, t = -4.08). The L2 group's Portuguese /i/ is not, however, significantly different from the one produced by the L3 group as far as distance from the centroid is concerned. The second section of Table 3.24 reports a significant effect of vowel in the L2 group, as all possible vowel comparisons within the group were found statistically significant. The third section of the table reports a significant group by vowel interaction, as the model revealed that the effect of 'vowel' is not consistent across all levels of the 'group' factor (see, for example, that the L2 group's Portuguese /i/ is rendered significantly different from the NP group's Portuguese /e/ but not from the L3 group's, while it is found significantly different from both the NP and L3 groups' Portuguese /a/). The vowel by group interaction reported in Table 3.24 is also visible in Figure 3.25.

To further explore the group by vowel interaction revealed by the linear mixed-effects regression model, seven additional sets of follow-up linear mixed-effects regressions were carried out to investigate the main effect of 'group' for each of the Portuguese vowels. These tests had distance form the centroid (Euclidean Distance,

Table 3.24: Results from a two-factor linear mixed-effects regression model on the Portuguese vowels produced by the NP, L2 and L3 groups, with 'distance to centroid' (ED) as the response, 'group' (NP, L2, L3) and 'vowel' (Portuguese /i/, /e/, /ɛ/, /a/, /ɔ/, /o/, /u/) as fixed factors, and 'speaker' and 'word' as random factors. The L2 group's Portuguese /i/ was the intercept.

		β	t
ED	Intercept (=L2 Port /i/)	0.297	
	NP Port /i/	-0.042	-4.08*
	L3 Port /i/	0.010	1.01
	L2 Port /e/	-0.153	-20.27*
	L2 Port /ɛ/	-0.155	-20.76*
	L2 Port /a/	-0.106	-14.47*
	L2 Port /ɔ/	-0.137	-18.39*
	L2 Port /o/	-0.137	-18.28*
	L2 Port /u/	-0.084	-11.28*
	NP Port /e/	0.044	4.97*
	L3 Port /e/	0.008	0.87
	NP Port /ɛ/	0.041	4.71*
	L3 Port /ɛ/	0.016	1.81
	NP Port /a/	0.047	5.51*
	L3 Port /a/	0.031	3.53*
	NP Port /ɔ/	0.042	4.77*
	L3 Port /ɔ/	-0.003	-0.31
	NP Port /o/	0.028	3.18*
	L3 Port /o/	0.019	2.08*
	NP Port /u/	0.057	6.50*
	L3 Port /u/	0.061	6.78*

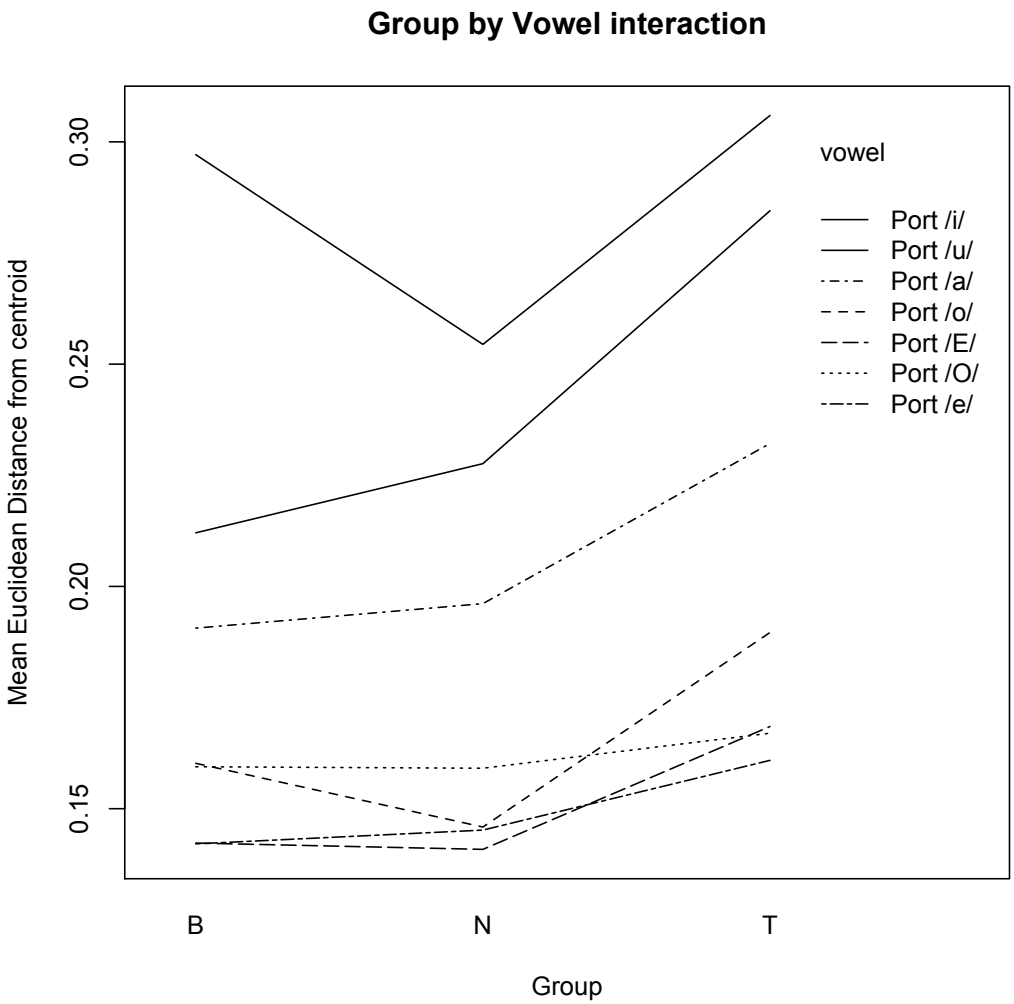

Figure 3.25: Means of Euclidean Distances from the centroid for each of the Portuguese vowels produced by the participants in the study and separated by group, where B refers to the L2 group, N refers to the NP group and T refers to the L3 group. The graph shows a significant group by vowel interaction.

Table 3.25: Results from a linear mixed-effects regression model testing for differences in distance of Portuguese /i/ from the centroid among the three groups of participants (NP, L2, L3).

		β	t
ED	Intercept (= L2 Port /i/)	0.297	
	NP Port /i/	-0.043	-3.23*
	L3 Port /i/	0.011	0.85
	Intercept (= NP Port /i/)	0.254	
	L3 Port /i/	0.054	4.07*

ED) as the response, 'group' as the fixed factor, and both 'speaker' and 'word' as random factors. The results from the first set of tests, namely those testing the Portuguese /i/ tokens produced by each of the three participant groups for differences in distance from their corresponding centroids, are presented in Table 3.25. Table 3.25 shows that the Portuguese /i/ produced by the participants in the L2 group does not differ from the ones produced by the participants in the L3 group, but that both are significantly further out in the vowel space than those produced by the NP group. These results are consistent with the interpretation of the first section of Table 3.24 above, as well as with the representation of the Portuguese /i/ vowel in Figure 3.25.

A second set of linear mixed-effects models with similar structure as the one for Portuguese /i/ above tested for differences in distance from the centroid among the Portuguese /e/ tokens produced by the three groups of participants. These results are presented in Table 3.26, where it is shown that no significant differences were found among the three groups with regards to how far away from the centroid they produced their Portuguese /e/ vowels. This is also visible in Figure 3.25 above.

A third set of linear mixed-effects models with similar characteristics tested the Portuguese /ɛ/ produced by the three groups for differences with regards to distance from their corresponding centroids. The results from this set of tests are shown in Table 3.27. The results show that the Portuguese /ɛ/ produced by the L2 group differs significantly from the ones produced by the L3 group but not from the

Table 3.26: Results from a linear mixed-effects regression model testing for differences in distance of Portuguese /e/ from the centroid among the three groups of participants (NP, L2, L3).

		β	t
ED	Intercept (= L2 Port /e/)	0.143	
	NP Port /e/	0.002	0.13
	L3 Port /e/	0.020	1.42
	Intercept (= NP Port /e/)	0.254	
	L3 Port /e/	0.018	1.29

Table 3.27: Results from a linear mixed-effects regression model testing for differences in distance of Portuguese /ɛ/ from the centroid among the three groups of participants (NP, L2, L3).

		β	t
ED	Intercept (= L2 Port /ɛ/)	0.142	
	NP Port /ɛ/	-0.001	-0.09
	L3 Port /ɛ/	0.026	2.27*
	Intercept (= NP Port /ɛ/)	0.141	
	L3 Port /ɛ/	0.028	2.36*

Portuguese /ɛ/ produced by the NP group. Portuguese /a/ tokens produced by the L2 and NP groups differ significantly from the L3 group's, as the latter are produced in a position further away from their centroid. This pattern is also observable in Figure 3.25.

The three groups' Portuguese /a/ productions were tested through a fourth set of linear mixed-effects regressions. The results from these tests are presented in Table 3.28 where it is shown that the Portuguese /a/ produced by the L2 group is, once again, different from the Portuguese /a/ produced by the L3 group, which is located further away from their centroid, while the Portuguese /a/ produced by the L2 group does not significantly differ from that produced by the NP group. The latter, however, does significantly differ in this measure from the Portuguese /a/ produced by the L3 group. The L3 group's Portuguese /a/ is, as it can be observed

Table 3.28: Results from a linear mixed-effects regression model testing for differences in distance of Portuguese /a/ from the centroid among the three groups of participants (NP, L2, L3).

		β	t
ED	Intercept (= L2 Port /a/)	0.190	
	NP Port /a/	0.006	0.33
	L3 Port /a/	0.042	2.45*
	Intercept (= NP Port /a/)	0.196	
	L3 Port /a/	0.036	2.12*

Table 3.29: Results from a linear mixed-effects regression model testing for differences in distance of Portuguese /ɔ/ from the centroid among the three groups of participants (NP, L2, L3).

		β	t
ED	Intercept (= L2 Port /ɔ/)	0.159	
	NP Port /ɔ/	-0.000	-0.04
	L3 Port /ɔ/	0.007	0.80
	Intercept (= NP Port /ɔ/)	0.159	
	L3 Port /ɔ/	0.008	0.85

in Figure 3.25, the Portuguese /a/ that has its location furthest from its centroid among the three vowel spaces under study.

A fifth set of linear mixed-effects regressions focused on the Portuguese /ɔ/ produced by these three groups. As Table 3.29 shows, these tests revealed no significant differences in the distance between the Portuguese /ɔ/ tokens produced by the three participant groups and their corresponding centroids. This lack of differences is observable in the close to flat line that describes these distances in Figure 3.25 above.

The distance from the centroid of the Portuguese /o/ tokens produced by the three groups were tested through a sixth set of linear mixed-effects models that revealed that, as shown in Table 3.30, the Portuguese /o/ tokens produced by the L3 group significantly differ from those produced by the L2 and NP groups,

Table 3.30: Results from a linear mixed-effects regression model testing for differences in distance of Portuguese /o/ from the centroid among the three groups of participants (NP, L2, L3).

		β	t
ED	Intercept (= L2 Port /o/)	0.160	
	NP Port /o/	-0.014	-1.21
	L3 Port /o/	0.029	2.47*
	Intercept (= NP Port /o/)	0.146	
	L3 Port /o/	0.044	3.68*

Table 3.31: Results from a linear mixed-effects regression model testing for differences in distance of Portuguese /u/ from the centroid among the three groups of participants (NP, L2, L3).

		β	t
ED	Intercept (= L2 Port /u/)	0.212	
	NP Port /u/	0.015	1.25
	L3 Port /u/	0.072	5.89*
	Intercept (= NP Port /u/)	0.228	
	L3 Port /u/	0.057	4.65*

although the latter do not significantly differ from each other. Once again, the L3 group produces Portuguese vowel tokens that are further from their centroid than the ones produced by either the L2 group or the NP group.

The last set of linear mixed-effects models test the Portuguese /u/ tokens of the three participant groups for differences in distance from the centroid. The results, shown in Table 3.31, suggest that, consistent with the analysis of the high back vowel data in the main acoustic analysis in this chapter, that the Portuguese /u/ produced by the L3 group is furthest of all from their corresponding centroid, differing significantly from both the L2 and the NP groups' Portuguese /u/. The L2 and the NP groups' Portuguese /u/ tokens, on the other hand, were not found to differ from each other in this measure.

To sum up the results from this alternative analysis of the data, the Portuguese

/i/ produced by both the L2 and L3 groups were found to be located much further from the centroid than the NP group's. In the case of Portuguese /ɛ/, /a/, /o/ and /u/, the vowels produced by the native speakers were found to be located closer to the centroid than the L3 group's but not differing from the L2 group's. No differences were found among the groups for Portuguese /e/ nor /ɔ/.

With regards to the specific comparisons between the L2 and L3 groups, the tests on Portuguese /i/, /ɛ/, /a/, /o/ and /u/ revealed a significant difference, suggesting that the L3 Portuguese vowels are located further out in the vowel space than the L2 group's. The Portuguese /e/ and /ɔ/ of these two groups were not found to differ from each other, nor from the NP group, with regards to the distance to their respective centroids. Most importantly, this alternative analysis of Euclidean Distances succeeds in revealing differences between the two learner groups in their production of Portuguese vowels, unlike the first and main analysis of the data, suggesting that the L3 group's Portuguese vowels are more dispersed than the L2 group's (as well as the native speakers').

3.4 Summary of results and discussion of the predictions for L2 learning

The multiple vowel comparisons performed in the main analysis of the acoustic data in this chapter sought to investigate whether the predictions made for the learning of the different stressed vowel categories of Portuguese by native speakers of English, which were derived from SLM and PAM, would be fulfilled. This section summarizes the findings from the multiple statistical comparisons by vowel group performed above and revisits the predictions made for the learning of each of the target vowels.

The predictions for the learning of the Portuguese high front vowel by the learners in this study stated that no spectral differences should be found between these learners' Portuguese /i/ and the native speakers' Portuguese /i/. This lack of spectral differences expected originated in the findings that English /i/ did not differ acoustically from Portuguese /i/ in the native speakers' speech. The results from the

comparisons performed on the learners' and Portuguese native speakers' data indicated that the learners' Portuguese /i/ did indeed differ from the native speakers' in fronting. Nevertheless, a follow-up comparison between the learners' Portuguese /i/ and their own English /i/ rendered no spectral difference between them, indicating that the learners' Portuguese /i/ was still acoustically similar to their English /i/. The difference initially found happened to result from an inescapable variability inherent to individual and unpredictable differences among speakers.[8] The predictions for the learning of the Portuguese high front vowel by both groups of learners was thus, fulfilled.

The predictions for the learning of Portuguese high back vowel by these learners stated that a new category was to be created by both groups, while the L3 group's Portuguese /u/ was expected to be closer to the target Portuguese /u/ than that produced by the L2 group. The results from the multiple comparisons in previous sections show that, as expected, both groups of learners had created a new category for Portuguese /u/ different from their English /u/. The results also showed that the Portuguese /u/ produced by the L3 group matched the target in backness but it was higher than the target and further apart from their own English /u/. This finding was interpreted as evidence supporting the occurrence of phonetic category dissimilation. The Portuguese /u/ produced by the L2 group, however, approximated the target Portuguese /u/ but differed from it in backness with a mean logF2 value located in between their English /u/ and the target in the horizontal dimension. This is interpreted as the L2 group having learned a new category but still not being able to produce it in a native-like fashion. The finding that their Portuguese /u/ was located in between the target and their English /u/ could also be interpreted as a trace of some previous phonetic category assimilation process. In conclusion, the predictions for the learning of this vowel category were met by the L2 group, while the L3 group unexpectedly provided acoustic evidence supporting what is here interpreted as a phonetic category dissimilation mechanism in second (in this case third) language learning.

[8] They were, after all, different people.

With regards to the low vowel of Portuguese, it was expected that both learner groups would show evidence of having created a new phonetic category that would fall in between their own two English low vowels in the logF2 dimension. Also, it was expected to find that the L2 group would show a wider distribution for their Portuguese /a/ than the L3 group, who in turn would show a wider distribution than the Natives. It was also predicted that the mean of the Portuguese /a/ category produced by the L3 group would be located in a more fronted position than the mean of the Portuguese /a/ produced by the L2 group.

The results from multiple statistical analyses of the data showed that, as predicted, both groups have created a new phonetic category, in between their own two English low vowels, for Portuguese /a/. The results also show that the Portuguese /a/ produced by both groups is closer to English /æ/ than to English /ɑ/. In addition, the results showed that the Portuguese /a/ produced by the L3 group was located lower in the vowel space than that produced by the L2 group. Despite this finding, the Portuguese /a/ of each of these groups was found to differ equally from their own English /æ/ and /ɑ/, which suggested that the L3 group's English low vowels were also located in a lower location in the vowel space than those of the L2 group. As for the width of the distribution of the Portuguese /a/ tokens produced by the two learner groups, the predictions were partially fulfilled. As predicted, the L2 group was found to have created a Portuguese /a/ with a wider distribution than that of the native speakers, which is here interpreted as evidence of a rather initial stage of target category development after new category establishment in the L2. The L3 group, however, exceeded this author's expectations by showing a new Portuguese /a/ category with a distribution width statistically undistinguishable from that of the native speakers, showing that they had learned to use one additional acoustic cue in the production of this vowel in a native-like fashion.

As for the learning of the Portuguese /o/ and /ɔ/ by the English speaking learners, it was predicted that they would have created one, and not two, single new category for Portuguese /o/-/ɔ/, with a possible spectral difference between the new category created by each of the learner groups. The findings with regards

to the Portuguese /o/ and /ɔ/ produced by the learners did not fulfill any of the predictions, and are, indeed, interesting and unexpected.

On the one hand, both groups of learners were found to have created two distinct new categories, one for Portuguese /o/ and one for /ɔ/, contrary to the one single category that was expected according to the predictions. The two groups of learners were not found to differ in the spectral quality of the new categories they produce. The two new categories differ in height, just as the native speakers' Portuguese /o/ and /ɔ/ do, but the learners maintain a much smaller difference between the two. While the Native speakers' /o/ and /ɔ/ differ both in height and backness but not in diphthongization pattern, the learners were found to differentiate their two newly acquired categories by using different degrees of diphthongization for each of them. This different degree of diphthongization was concluded not to result from transfer from their own English /o/, as the latter shows a diphthongization pattern that differs considerably from the pattern by which the learners distinguish their two newly created Portuguese mid-back vowel categories. Since the diphthongization pattern by which they distinguish the two new categories is not likely to have been acquired due to the native speakers' input nor due to transfer from their own English /o/, it is here interpreted that the two groups of learners must have perceived that there is an acoustic difference between the two target vowel categories but have not identified which one it is. This way, the learners use a phonetic cue that is incongruent with the phonetic cue used by the native speakers to mark the phonetic distinction between the two Portuguese mid-back vowels.

With regards to the learning of the Portuguese mid-front vowels, the predictions were that two new categories were to be created by the learners. These two categories would show traces of assimilation to the two mid-front vowel categories of English and would approximate target Portuguese /e/ and /ɛ/ categories. Similar spectral and diphthongization patterns from English /e/-/ɛ/ were expected to be found in the learners' Portuguese /e/ and /ɛ/, but no acoustic differences were predicted between the L2 and L3 groups. The results from the multiple comparisons performed in this chapter contradict, once again, the predictions made for the learning of these two

vowels by these learners.

The two groups of learners showed evidence of the acquisition of only one phonetic category for the Portuguese /e/-/ɛ/ contrast pair. The new category was found to be similar to the English /ɛ/ in height but similar to English /e/ in frontedness at the midpoint. Additional tests suggested that the Portuguese /e/ and /ɛ/ produced by the learners differed in degree of diphthongization in the logF2 axis. However, the direction of diphthongization of their Portuguese mid-front vowels seems to suggest that the learners might have acquired a native-like directionality in the pattern of diphthongization. Despite these similarities, overall, the new category was found to be a distinct acoustic entity from both English /e/ and /ɛ/ as well as from the target native Portuguese /e/ and /ɛ/.

These findings are surprising but not completely unpredicted. The possibility that the learners would create a single phonetic category encompassing both Portuguese /e/ and /ɛ/ tokens was mentioned as an alternative prediction in Section 1.5 above. This alternative prediction bears from the possibility that, in PAM terms, the two Portuguese mid-front vowels are perceptually "uncategorized" by the learners. In PAM terms, if the contrasting L2 phones are not perceived as belonging clearly to a single L1 phonological category, but rather as each having a mixture of more modest similarities to several L1 phonological categories, then one or two new L2 phonological categories may be relatively easy to learn perceptually. However, if the L2 phones are perceived as similar to the same set of L1 phonemes, that is, if they are close to each other in phonological space, then the listener will have a hard time discriminating them. In such cases, a single new phonetic/phonological category would be expected to be learned (Best & Tyler (2007)).

Finally, and most interestingly, the alternative analysis carried out on the Euclidean Distances from the centroid on the Portuguese vowels produced by the two learner groups (L2 and L3) and the native speakers (NP) revealed statistically significant signs of vowel dispersion in the case of the L3 speakers, whose vowel inventory is, in principle, at least five (Spanish) vowels bigger than that of the L2 group's. These findings support, more so than the results from the main analysis in this

chapter alone, Flege's claim that (a) phonetic category dissimilation is a mechanism that occurs with the establishment of new phonetic categories, and that (b) a bigger inventory size has an effect on vowel space, as vowel categories tend to disperse in it to preserve contrast.

In order to investigate (a) the relationship between perceptual and acoustic differences that has been assumed to exist in this dissertation up to this point, and (b) to further investigate the perceptual discriminability of mid-front vowel categories and its relation to the acoustic findings reported in this chapter, an auditory categorical discrimination experiment involving mid-front vowels of English, Spanish and Portuguese will be presented in the following chapter.

CHAPTER 4

PERCEPTION OF PORTUGUESE /e/-/ɛ/ BY L2 AND L3 LEARNERS

Previously discussed acoustic data in Chapter 3 showed that learners in the L2 and L3 groups fail to produce native-like Portuguese /e/ and /ɛ/ vowels. Instead, the results from the analysis of acoustic data in Chapter 3 showed that both groups had created one new single phonetic category for both Portuguese /e/ and /ɛ/ that was a distinct acoustic entity from both English /e/ and /ɛ/ as well as from the target native Portuguese /e/ and /ɛ/.

The findings from the analyses of acoustic data in Chapter 3 summarized above contradict the original predictions created, based on the PAM model, for the production of the Portuguese mid-front vowels by the two Portuguese learner groups. It was concluded that the two Portuguese mid-front vowels constituted a pair of "Uncategorized" L2 phones, which are described under PAM as speech sounds that fall somewhere in between native phonemes. In these kinds of cases, PAM predicts poor to good discrimination of the two target L2 phones. Nevertheless, as both SLM and PAM make their predictions on the assumption that accurate perception precedes accurate production of speech sounds, the question remains of whether the difficulty demonstrated by the learners' production data might have a perceptual basis.

Several research questions arise from this reflection. On the one hand, how similar do the learners perceive Portuguese /e/ and /ɛ/ and English /e/ and /ɛ/ to be? Do they perceptually assimilate each of the Portuguese vowels to a different English vowel (as the originally proposed Two Category pair classification would predict)? Do they assimilate both Portuguese mid-front vowels to a single English vowel (following the pattern proposed for a Single Category pair)? Do they show signs of a specific assimilation pattern at all? Ultimately, how do the Portuguese mid-front vowels relate to each of the English mid-front vowels perceptually for these

speakers? On the other hand, do the learners perceptually discriminate between Portuguese /e/ and /ɛ/? And if so, how effectively?

With the objective of looking further into these questions, the current chapter reports on two experiments involving the two groups of Portuguese learners mentioned above and their perception of the Portuguese mid-front vowels. Experiment 4.1 seeks to investigate the patterns of assimilation of the Portuguese mid-front vowels to English vowels by the two groups of Portuguese learners via a phonetic category assimilation test (also known in the literature as a test of "goodness of fit"). Experiment 4.2 seeks to investigate their perceptual cross-linguistic discrimination of the mid-front vowels of English, Spanish and Portuguese through a 4-item oddity task.

4.1 Experiment 4.1

The phonetic category assimilation test in Experiment 4.1 was designed with the objective of investigating the pattern of perceptual assimilation between Portuguese /e/ and /ɛ/ and the English mid-front vowels in the two learner groups. The test explored the degree of similarity between different tokens of Portuguese /e/ and /ɛ/ vowels and the learners' English mid-front vowels by making them listen to specific Portuguese mid-front vowel tokens and asking them whether the said token was more similar to the vowel in the English word "bait" or to the vowel sound in the word "bet." Thus, each learner would classify each Portuguese token as most similar to English /e/ or to English /ɛ/, revealing a pattern of assimilation that would help in the interpretation of the production data results in the previous chapter.

4.1.1 Method

Participants

Three groups of native English speakers participated in Experiment 4.1. One group was composed of 12 (8 male, 4 female) monolingual English speakers (NE hereafter), 7 male native English speaking learners of Portuguese as an L2 (L2 group), and 5

Table 4.1: Participants' biographic and linguistic background where M/F represents their gender; Age is the mean age at time of testing; SpanAOA represents age of acquisition of Spanish and AOAPort represents age of acquisition of Portuguese.

Group	M/F	Age	SpanAOA	PortAOA
NE	8M/4F	19 (18-21)	N.A.	N.A.
L2	5M	23 (21-25)	N.A.	19 (18-19)
L3	7M	22 (21-26)	13 (8-15)	20 (18-26)

male native English speaking learners of Portuguese as an L3 who learned Spanish as an L2 prior to the onset of Portuguese learning (L3 group). The participants in the NE group were monolingual English speaking undergraduate students at the University of Arizona. The participants in the L2 and L3 groups are the same as those composing the L2 and L3 groups in Chapter 3, except only 5 of the original 7 participants in the L2 group were able to return to participate in the perception experiments. The participants' basic biographic and linguistic background information is included in Table 4.1. All participants were living in the US at the time of testing.

Stimuli

A total of 24 Portuguese vowel stimuli were obtained in the following way. Of the Portuguese words used in Chapter 2, those with a stressed /e/ or /ɛ/ preceded and followed by a non-nasal coronal consonant were selected (13 words: 7 with /e/ and 6 with /ɛ/). Three of the seven native Portuguese speakers' productions of those words from Chapter 2 were used, from which the Portuguese /e/ and /ɛ/ were extracted (39 vowel productions in total). The beginning and end of each vowel were set between the second and third periodic pulses and between the third-to-last and the second-to-last periodic pulses on the original vowel waveform.

A female native speaker of Brazilian Portuguese selected the most representative Portuguese /e/ and /ɛ/ tokens from the 39 tokens presented to her auditorily through headphones in a quiet room. Two different productions of each vowel cat-

egory were selected from each of the three speakers. Thus, a total of six tokens for Portuguese /e/ and six tokens for Portuguese /ɛ/ were used as stimuli for this perceptual assimilation experiment. The 24 vowel stimuli were obtained by setting the stimuli to run in two independently randomized cycles (2 tokens x 2 vowels x 3 speakers x 2 cycles = 24 vowel stimuli).

Procedure

The participants in all three groups were tested individually in a sound-treated booth, where stimuli were presented via high-quality headphones at a self-selected volume and pace. The 24 vowel stimuli were presented one by one in a single block. The participants were not allowed to listen to each stimulus-string more than once. The participants would hear one Portuguese vowel token as stimuli and they would answer by choosing the leftmost button in a response box if they thought the vowel they heard was more similar to the vowel in the English word "bait," or by choosing the rightmost button in the response box if they thought the vowel they heard was more similar to the English vowel in the word "bet." The words "bait" and "bet" were shown orthographically on a computer screen as they listened to the stimuli, but no auditory presentation of the English words was offered. This way, long-term representations of the vowels in these words were assumed to be triggered in the participants[1].

4.1.2 Results

Figure 4.1 represents the proportion of English /ɛ/ response for each of the three groups when listening to both Portuguese /e/ and /ɛ/ vowels. The graph shows

[1]Note that, while all Portuguese vowel tokens in the stimuli were extracted from a coronal context, "bait" and "bet" are not representative of this environment. Although a matching consonantal environment would have been more appropriate when selecting the English words representing English /e/ and /ɛ/ for this experiment, I do not believe that this should have a significant effect on the participants' responses, as they are solely used here to trigger the participants' access to their long-term representations of the English vowels in question, which should be context independent.

that, on the one hand, monolingual English speakers chose the vowel in the English word 'bet' around 60-65% of the time regardless of which Portuguese vowel they heard. On the other hand, the L2 and L3 groups selected 'bet' more than 90% of the time when hearing Portuguese /ɛ/, and about 30-35% of the time when hearing Portuguese /e/. These results indicate that, although Portuguese /ɛ/ seems to be identified with English /ɛ/ almost categorically, the two learner groups still assimilated Portuguese /e/ tokens to English /ɛ/ around 30-35% of the time.

A 2x3 mixed-design ANOVA with 'vowel' (Portuguese /e/, Portuguese /ɛ/) as the within-subjects factor and 'group' (L2, L3, NE) as the between-subjects factor was performed on the proportion of 'bet' responses given by the participants. Response scores were averaged over items for this analysis. The analysis yielded a non-significant main effect of language group, $F(2, 21) = .118$, $p=.889$, a significant main effect of vowel, $F(1, 21) = 24.582$, $p= .000$, and a significant two-way interaction, $F(2, 21) = 6.114$, $p= .008$. The two-way interaction was explored by performing a series of follow-up within-subjects ANOVAs by group, testing the simple effect of Portuguese vowel for each language group separately. The results from these analyses showed that the effect of vowel was significant for the L2 group ($F(1, 4) = 24.224$, $p= .008$) as well as for the L3 group ($F(1, 6) = 37.688$, $p= .001$), but not for the NE group ($F(1, 11) = .129$, $p= .726$). Thus, both groups of learners identified Portuguese /ɛ/ as similar to English /ɛ/ and Portuguese /e/ as most similar to English /e/, while monolingual English speakers identified both Portuguese vowels as equally similar to English /e/ or /ɛ/.

The results from the statistical analyses and Figure 4.1 seem to agree in suggesting that the two groups of learners assimilate each of the two Portuguese categories to a different English category in a similar manner: Portuguese /ɛ/ is assimilated to English /ɛ/ and Portuguese /e/ is assimilated to English /e/. In addition, Figure 4.1 suggests that Portuguese /ɛ/ is a better fit to English /ɛ/ than Portuguese /e/ is to English /e/ for these learners. Both groups seem to assimilate each Portuguese vowel to a different native category, but not equally well.

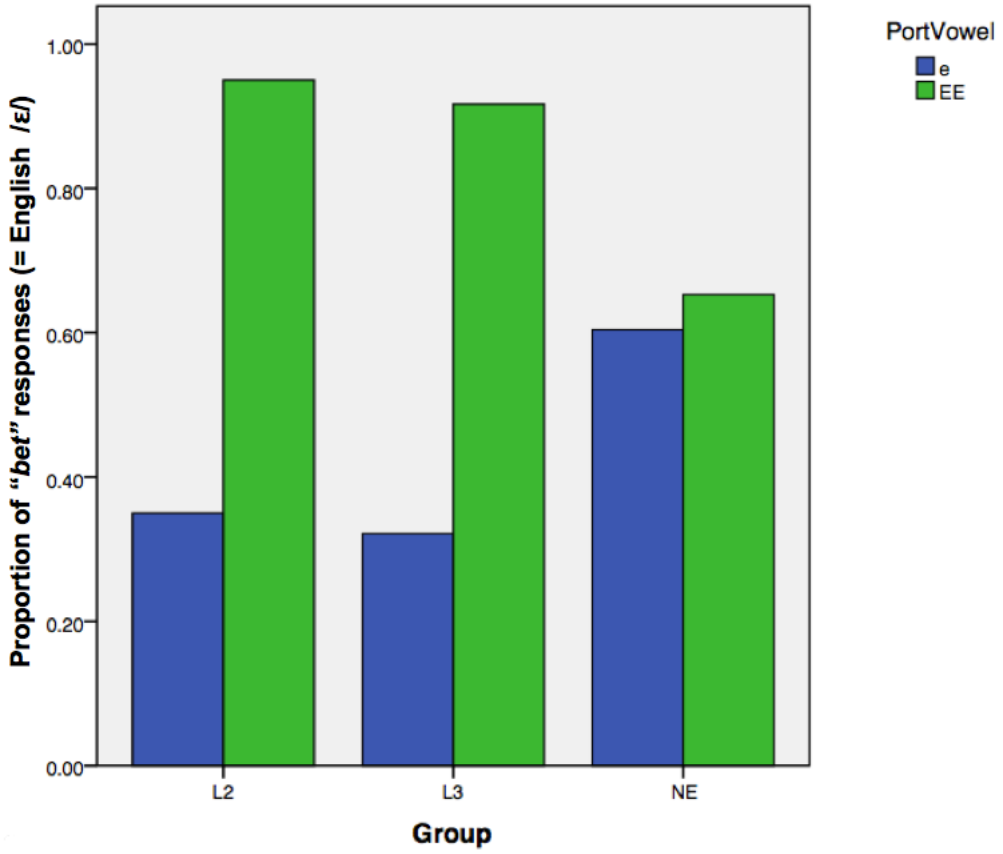

Figure 4.1: Proportion of Portuguese /e/ (labeled 'e' and represented by dark blue bars) and /ɛ/ tokens (labeled 'EE' and represented by green bars) identified by the native English (NE), L2 group and L3 group as the English /ɛ/ in the word 'bet.' A score of 1.0 indicates perfect fit between the Portuguese vowel and English /ɛ/ while a score of 0.5 indicates insensitivity to similarities between the Portuguese vowel heard in each case and English /ɛ/.

4.2 Experiment 4.2

The purpose of Experiment 4.2 was to investigate the potential differences in discriminability among within-language as well as cross-linguistic mid-front vowel category pairs of English, Spanish and Portuguese, by speakers of Portuguese as an L2 (L2 group) and speakers of Portuguese as an L3 (L3 group). As experimental controls, a group of monolingual speakers of English (NE) and a group of L1-dominant native speakers of Portuguese (NP) also participated in this multilingual categorical discrimination test.

Given that no significant differences were found between the two learner groups (L2 and L3 groups) in the production of Portuguese /e/ and /ɛ/ in Chapter 3, and that the two groups seem to be assimilating the Portuguese /e/ and /ɛ/ vowels to their English /e/ and /ɛ/ in a similar fashion, no differences are expected to be found between the two learner groups in this mid-front vowel categorical discrimination test.

4.2.1 Method

Participants

Four groups of speakers participated in Experiment 4.2. The participants in the NE, L2 and L3 groups who participated in Experiment 4.1 also participated in Experiment 4.2. An additional group, consisting of 8 (4 female, 4 male) native speakers of Brazilian Portuguese (NP group hereafter), participated together with the NE, L2 and L3 groups in Experiment 4.2. The NP group was composed Portuguese (L1) dominant speakers of English who resided in the USA at the time of testing. Note that the NE group was nearly monolingual in English while the NP group was bilingual in Portuguese and English. An effort was made to include only native Portuguese Brazilian Portuguese speakers that were highly dominant in their L1 in the NP group despite the fact that they were also speakers of English as an L2 given their current place of residence in the US. Since exposure to foreign language/L2 experience is assumed to have an effect on perception of foreign categories

Table 4.2: Participants' biographic and linguistic background where M/F represents their gender; Age is the mean age at time of testing; SpanAOA represents age of acquisition of Spanish and AOAPort represents age of acquisition of Portuguese.

Group	M/F	Age	SpanAOA	PortAOA
NE	8M/4F	19 (18-21)	N.A.	N.A.
L2	5M	23 (21-25)	N.A.	19 (18-19)
L3	7M	22 (21-26)	13 (8-15)	20 (18-26)
NP	4M/4F	35 (27-42)	N.A.	Native

(see SLM and PAM review in Chapter 1), this difference in bilingual vs. monolingual speaker groups might affect the results of this experiment. The biographic and linguistic information of the NP group is repeated here in Table 4.2 together with the information regarding the participants in the NE, L2 and L3 groups.

Stimuli

A total of 12 English mid-front vowel tokens were extracted and selected from the productions of the native English speaking group who participated in the production experiment described in Chapter 2 of this dissertation to create the stimulus-strings for this experiment. The 12 vowel tokens included two English /e/ and two English /ɛ/ vowel tokens produced by three different male native speakers in Chapter 2. Similarly to the stimuli in Experiment 4.1, the beginning and end of each vowel were set between the second and third periodic pulses and between the third-to-last and the second-to-last periodic pulses on the original vowel waveform. The specific 12 English vowel tokens to be used as stimuli were selected by a female native English speaker who identified them as the most representative of the two mid-front vowels of American English out of a total of 28 vowel sounds previously extracted from the native English speakers' acoustic data and played to her through headphones in a quiet room.

A total of six Spanish /e/ vowel tokens were selected to create the stimulus-strings in this experiment from the productions of the native Spanish speaking

group who participated in the experiments in Chapter 2. The six Spanish /e/ tokens included two Spanish /e/ tokens spoken by three different male native speakers. The beginning and end of the vowels were determined following the same guidelines as for the Portuguese vowels described in the previous paragraph. The six specific vowel tokens to be used as stimuli were selected by a female speaker of Spanish who identified them as the most representative of Spanish /e/ out of a total of 14 potential tokens of Spanish /e/ she was exposed to. In addition to the Spanish and English vowel tokens just described, the same 12 Portuguese vowel tokens used as stimuli in Experiment 4.1 were used to create the stimulus-strings in Experiment 4.2.

The stimulus-strings were generated by creating sequences of four vowel tokens that included either four tokens of the same vowel category ("no-change" trials hereafter) or four tokens in which either the second or third vowel sound in the sequence was a different vowel category from the rest ("change" trials hereafter). The decision to create a stimulus set consisting of both change and no-change trials (Flege et al. (1999a), Guion et al. (2000a), Rauber et al. (2005)) was motivated by the view (Liberman (1957), Kuhl (1980), Guenther et al. (1999)) that two things happen when a new phonetic category is established during language acquisition: on the one hand, there is an increase in sensitivity to differences between instances of the newly formed category and the already existing categories; and on the other, there is a decrease in sensitivity to token-to-token variation within the newly formed category. Thus, change trials were included to assess the participants' ability to detect the relevant differences that make two phones recognizable as instances of different phonetic categories, while no-changes were included to assess the participants' ability to ignore audible but phonetically irrelevant acoustic differences between tokens drawn from a single vowel category.

The four-item oddity task used here for Experiment 4.2 is similar to the widely used AXB task in which the participant is forced to hold two stimuli in auditory short-term memory while perceptually assessing a third stimulus. Triadic tests such as the AXB and ABX tasks, where participants are asked to decide whether

the token in the "X" position is most similar to the one in the "A" or in the "B" position, have been widely used in the past to assess perceived cross-linguistic similarity between the sounds of two different languages (Polka (1995), Gottfried (1984), Flege (2003b)). Oddity tasks are also common (Strain (1963), MacKain et al. (1981), Flege et al. (1994), Guion et al. (2000a)) and are similar to the AXB and ABX tasks in that the participant must identify the one stimulus that differs from the other two. In oddity tasks, however, the position of the odd item is not fixed, which intensifies task uncertainty and increases the overall difficulty of the task.

After considering the fact that the inclusion of sound categories from three different languages in such a discrimination test would most likely imply a much heavier processing load for the participants, an effort was made to limit task uncertainty by designing a four-item oddity task that restricts the position of the possible odd-item out to two positions alone (second or third in the sequence) while providing the listener with two physically different examples of the target of the comparison in each case (in first and fourth position in the sequence).

In all change and no-change trials three different male voices were used, with the first and fourth vowel token being physically different tokens of the same vowel produced by the same speaker. The use of three different speakers and two physically different tokens of each vowel by each of the three speakers seeks to prevent the participant from using memorization of speaker voice or physical characteristics of specific vowel tokens alone when responding to the task. The use of two different tokens of the same vowel in first and last position in each sequence also sought to provide the participant with more opportunities to identify relevant features, and ignore irrelevant ones, when responding to the task.

Ten contrast pairs were tested in this vowel category discrimination test. The 10 vowel category pairs tested are listed in Table 4.3.

Table 4.3: Contrast pairs tested in the category discrimination test in Experiment 4.2.

1. English /ɛ/ - Portuguese /e/
2. English /ɛ/ - Portuguese /ɛ/
3. English /e/ - English /ɛ/
4. English /e/ - Portuguese /e/
5. English /e/ - Portuguese /ɛ/
6. Portuguese /e/ - Portuguese /ɛ/
7. Spanish /e/ - English /ɛ/
8. Spanish /e/ - English /e/
9. Spanish /e/ - Portuguese /e/
10. Spanish /e/ - Portuguese /ɛ/

Architecture of vowel sequences (stimulus-strings)

The following method was used to create the sequences composing the stimulus-strings in Experiment 4.2. For each contrast pair in Table 4.3, 12 change trials and 6 no-change trials were built. Table 4.4 summarizes this basic building structure. An exemplified explanation of the method followed to build this structure is included below.

Consider a single mid-front vowel contrast such as the within-English contrast English /e/ - English /ɛ/. Remember that the sequences consist of four vowel tokens in which the first and last are always physically different tokens of the same vowel category (token "a" or token "b") produced by the same speaker (Speaker 1 for the purpose of this example). These two vowels constitute the basis to which Vowel #2 and Vowel #3 will be compared in order for the listener to decide whether all vowel tokens in the sequence are instances of a single vowel or whether there is one (either Vowel #2 or Vowel #3) that differs from the rest. Since each contrast contains two different sounds, there are two possible bases for building each stimulus-string. In the present example in particular, one set of stimulus-strings will have English /e/ tokens as Vowel #1 and Vowel #4, while another set of stimulus-strings will have English /ɛ/ tokens on these positions in the sequence. This first step into the

Table 4.4: Architecture for the stimulus-strings in Experiment 4.2. where capital letters refer to the first ("A") or second ("B") member of the vowel pair, numbers ("1", "2" or "3") refer to the Speaker producing the specific vowel token, and lowercase letters ("a" or "b") refer to the specific token produced by the Speaker in question. 12 change trials were built for each contrast pair, 6 with base A (1-6) and 6 with base B (7-12). 6 more no-change trials were built for each contrast pair, 3 for each member of the pair.

1.	A.1.a.	A.2.a.	B.3.a.	A.1.b.
2.	A.2.a.	A.3.a.	B.1.b.	A.2.b.
3.	A.3.a.	A.1.a.	B.2.a.	A.3.b.
4.	A.1.a.	B.3.b.	A.2.a.	A.1.b.
5.	A.2.a.	B.1.a.	A.3.a.	A.2.b.
6.	A.3.a.	B.2.b.	A.1.a.	A.3.b.
7.	B.1.a.	B.2.a.	A.3.a.	B.1.b.
8.	B.2.a.	B.3.a.	A.1.b.	B.2.b.
9.	B.3.a.	B.1.a.	A.2.a.	B.3.b.
10.	B.1.a.	A.3.b.	B.2.a.	B.1.b.
11.	B.2.a.	A.1.a.	B.3.a.	B.2.b.
12.	B.3.a.	A.2.b.	B.1.a.	B.3.b.
13.	A.1.a.	A.2.b	A.3.b.	A.1.b.
14.	A.2.a.	A.3.b.	A.1.b.	A.2.b.
15.	A.3.a.	A.1.b.	A.2.b.	A.3.b.
16.	B.1.a.	B.3.b.	B.2.b.	B.1.b.
17.	B.2.a.	B.1.b.	B.3.b.	B.2.b.
18.	B.3.a.	B.2.b.	B.1.b.	B.3.b.

Table 4.5: First step into building an architecture for the stimulus-strings representing the change trials for the English /e/ - English /ɛ/ vowel pair in Experiment 4.2.

Speaker 1 token "a" English /e/	*Vowel #2*	*Vowel #3*	Speaker 1 token "b" English /e/

Table 4.6: Second step into building an architecture for the stimulus-strings representing the change trials for the English /e/ - English /ɛ/ vowel pair in Experiment 4.2.

Speaker 1 token "a" English /e/	Speaker 2 token "a" English /e/	Speaker 3 token "a" English /ɛ/	Speaker 1 token "b" English /e/
Speaker 1 token "a" English /e/	Speaker 3 token "b" English /ɛ/	Speaker 2 token "a" English /e/	Speaker 1 token "b" English /e/

construction of the structure of the change trials is presented in Table 4.5.

In the basic structure of change trials presented in Table 4.5, Vowel #2 and Vowel #3 will be vowel tokens produced by two different speakers (Speaker 2 and Speaker 3), but only one them will be an instance of English /e/ while the other will be an instance of English /ɛ/, the odd-item out in the sequence. Thus, the odd-item out could be in two possible positions, resulting in two possible additional structures for these sequences, which are here represented in Table 4.6.

In addition, due to the fact that three speakers of each of the languages were used to construct these stimulus-strings, three different sequences were built for each of the two possible structures presented in Table 4.6, one with each of the speakers as the base. The resulting six sequences are presented in Table 4.7.

Furthermore, the six structures just described are examples of sequences in which the odd-item out was always an instance of English /ɛ/. Consequently, six additional change trials were built by switching the position of English /e/ and English /ɛ/

Table 4.7: Architecture for the stimulus-strings representing the change trials in the category discrimination test in Experiment 4.2 for the within English vowel pair, where English /ɛ/ was the odd-item out.

Speaker 1 token “a” English /e/	Speaker 2 token “a” English /e/	Speaker 3 token “a” English /ɛ/	Speaker 1 token “b” English /e/
Speaker 2 token “a” English /e/	Speaker 3 token “a” English /e/	Speaker 1 token “b” English /ɛ/	Speaker 2 token “b” English /e/
Speaker 3 token “a” English /e/	Speaker 1 token “a” English /e/	Speaker 2 token “a” English /ɛ/	Speaker 3 token “b” English /e/
Speaker 1 token “a” English /e/	Speaker 3 token “b” English /ɛ/	Speaker 2 token “a” English /e/	Speaker 1 token “b” English /e/
Speaker 2 token “a” English /e/	Speaker 1 token “a English /ɛ/	Speaker 3 token “a” English /e/	Speaker 2 token “b” English /e/
Speaker 3 token “a” English /e/	Speaker 2 token “b English /ɛ/	Speaker 1 token “a” English /e/	Speaker 3 token “b” English /e/

Table 4.8: Architecture for the stimulus-strings representing the change trials in the category discrimination test in Experiment 4.2 for the within English vowel pair, where English /e/ was the odd-item out.

Speaker 1 token "a" English /ɛ/	Speaker 2 token "a" English /ɛ/	Speaker 3 token "a" English /e/	Speaker 1 token "b" English /ɛ/
Speaker 2 token "a" English /ɛ/	Speaker 3 token "a" English /ɛ/	Speaker 1 token "b" English /e/	Speaker 2 token "b" English /ɛ/
Speaker 3 token "a" English /ɛ/	Speaker 1 token "a" English /ɛ/	Speaker 2 token "a" English /e/	Speaker 3 token "b" English /ɛ/
Speaker 1 token "a" English /ɛ/	Speaker 3 token "b" English /e/	Speaker 2 token "a" English /ɛ/	Speaker 1 token "b" English /ɛ/
Speaker 2 token "a" English /ɛ/	Speaker 1 token "a" English /e/	Speaker 3 token "a" English /ɛ/	Speaker 2 token "b" English /ɛ/
Speaker 3 token "a" English /ɛ/	Speaker 2 token "b" English /e/	Speaker 1 token "a" English /ɛ/	Speaker 3 token "b" English /ɛ/

so that an equal number of stimulus-strings were obtained with each member of the contrast pair as the base (see Table 4.8). This way, the following six sequence structures were added to the six described above, constituting, thus, 12 change trials for testing, in this specific example, the discrimination of English /e/ - English /ɛ/ pair.

A similar method was used to construct the no-change trials corresponding to each vowel pair. Three different sequences were built for testing each member of the vowel pair, one with each different speaker as the base. The six no-change trials for the within-English vowel pair are, thus, represented in Table 4.9[2]. The

[2]Note that, because each vowel appears in more than one contrast-pair (i.e. English /e/ appears in pairs 3, 4, 5 and 8 in Table 4.3) a portion of the sequences built from the specific structure described above will appear more than once in the experiment. To ensure that each stimulus-

Table 4.9: Architecture for the stimulus-strings representing the no-change trials in the category discrimination test in Experiment 4.2.

Speaker 1 token “a” English /e/	Speaker 2 token “b” English /e/	Speaker 3 token “b” English /e/	Speaker 1 token “b” English /e/
Speaker 2 token “a” English /e/	Speaker 3 token “b” English /e/	Speaker 1 token “b” English /e/	Speaker 2 token “b” English /e/
Speaker 3 token “a” English /e/	Speaker 1 token “b” English /e/	Speaker 2 token “b” English /e/	Speaker 3 token “b” English /e/
Speaker 1 token “a” English /ɛ/	Speaker 3 token “b” English /ɛ/	Speaker 2 token “b” English /ɛ/	Speaker 1 token “b” English /ɛ/
Speaker 2 token “a” English /ɛ/	Speaker 1 token “b” English /ɛ/	Speaker 3 token “b” English /ɛ/	Speaker 2 token “b” English /ɛ/
Speaker 3 token “a” English /ɛ/	Speaker 2 token “b” English /ɛ/	Speaker 1 token “b” English /ɛ/	Speaker 3 token “b” English /ɛ/

12 change trials represented in Tables 4.7 and 4.8, together with the 6 no-change trials represented in Table 4.9, constitute an example of the 18 stimulus-strings that were created to test the discriminability of each vowel pair and whose basic structure was presented in Table 4.4 above.

Procedure

Each participant responded to a total of 180 different trials. 120 of them (66%, 12 trials per contrast pair) were change trials while 60 of them (33%, 6 trials per contrast pair) were no-change trials. Within the change trials, the odd item was the second vowel sound in the sequence in half of them (33%). The third vowel was the

string included in the experiment was unique, the duplicated sequences were altered by modifying the specific tokens (token “a for token “b and vice versa) that were used in each sequence, thus producing 6 truly unique no-change sequences for each vowel pair under study.

odd item in the sequence in the other half (33%). The interval between the tokens inside each trial was 500ms. The inter-stimulus interval was 1s as triggered by the participants' response to the previous trial.

The participants in all four groups were tested individually in a sound-treated booth, where stimulus-strings were presented via high-quality headphones at a self-selected volume, regulated by a desktop computer. The participants were told to press the center button in a response box if they thought all of the tokens in the sequence were examples of a single vowel category, the leftmost button in the box if they thought vowel #2 in the sequence was different from the rest, and the rightmost button if they thought vowel #3 was different from the rest. The participants were also informed that the response times would be collected and they should respond as fast as they could. The participants were familiarized with the task during a short practice in which non-mid vowels of English were used to form the sequences (e.g. /a/ - /a/ - /u/ - /a/). The main section of the experiment was administered in two independently randomized blocks of 90 trials each, between which a break was inserted. The participants were asked to remain in the sound booth while they rested and to press any button to continue with the second portion of the experiment. Both response times and response given for each trial were collected[3].

4.2.2 Results

The change and no-change trials were analyzed separately. Once again, change trials were those stimulus-strings or sequences that contained an odd-item out, while no-change trials were those in which all four vowel tokens in the sequence were physically different examples of a single vowel category. The analysis of the responses to change trials seeks to investigate the participants' ability to identify the phonetically relevant differences that exist between distinct phonetic categories, while the analysis of responses to no-change trials serves to examine the participants' ability to ignore audible but phonetically irrelevant differences between different instances of a single phonetic category.

[3]Response times will not be discussed in this dissertation.

Results of analyses of responses to change trials

Results of responses to the change trials in Experiment 4.2 are presented in Figure 4.2 below. In Figure 4.2, contrast-pairs are labeled according to the vowel categories and languages they consist of and follow, left to right, the same order as the enumeration of contrast pairs listed in Table 4.3 above. Thus, "Engl_EE_Port_e" represents the English /ɛ/ - Portuguese /e/ pair, "Eng_EE_Port_EE" represents the English /ɛ/ - Portuguese /ɛ/ pair, "Engl_ei_EE" represents the English /e/ - English /ɛ/ pair, etc. Different language groups in this figure are represented by the different colored bars. Following their order from left to right, blue (medium dark shade) is used for the L2 group, green (medium light shade) for the L3 group, yellow (lightest shade) for the Native English (NE) group and purple (darkest shade) for the Native Portuguese (NP) group.

A 4x10 mixed-design ANOVA was performed on the correct response data from the change trials. The within-subjects factor was 'pair,' with 10 levels (listed above), and the between-subjects factor was 'group' with 4 levels: NP group, NE group, L2 group, and L3 group. The repeated-measures ANOVA[4] yielded a significant effect of 'pair' ($F(9, 252) = 33.962$, $p = .000$), a non-significant effect of 'group' ($F(3, 28) = 1.416$, $p = .259$) and a significant two-way interaction ($F(27, 252) = 2.407$, $p = .003$).

The group by pair interaction was further explored by carrying out a series of repeated-measures ANOVAs in order to investigate the effect of language group (L2, L3, NE, NP) for each vowel pair separately. 10 ANOVAs were conducted, one for each vowel pair under study (see list in Procedure subsection above). A main effect of language group was found for the English /e/ - Portuguese /e/ vowel pair ($F(3, 28) = 6.910$, $p = .001$) and for the Spanish /e/ - English /e/ pair ($F(3, 28) = 3.246$, $p = .037$).

[4]In this chapter, whenever a violation of the assumption of sphericity is detected by Mauchly's Test of Sphericity, the Greenhouse-Geisser correction was applied and the p values reported are those after the application of such correction. The degrees of freedom will, however, be those consistent with the uncorrected analysis of variance, for reference.

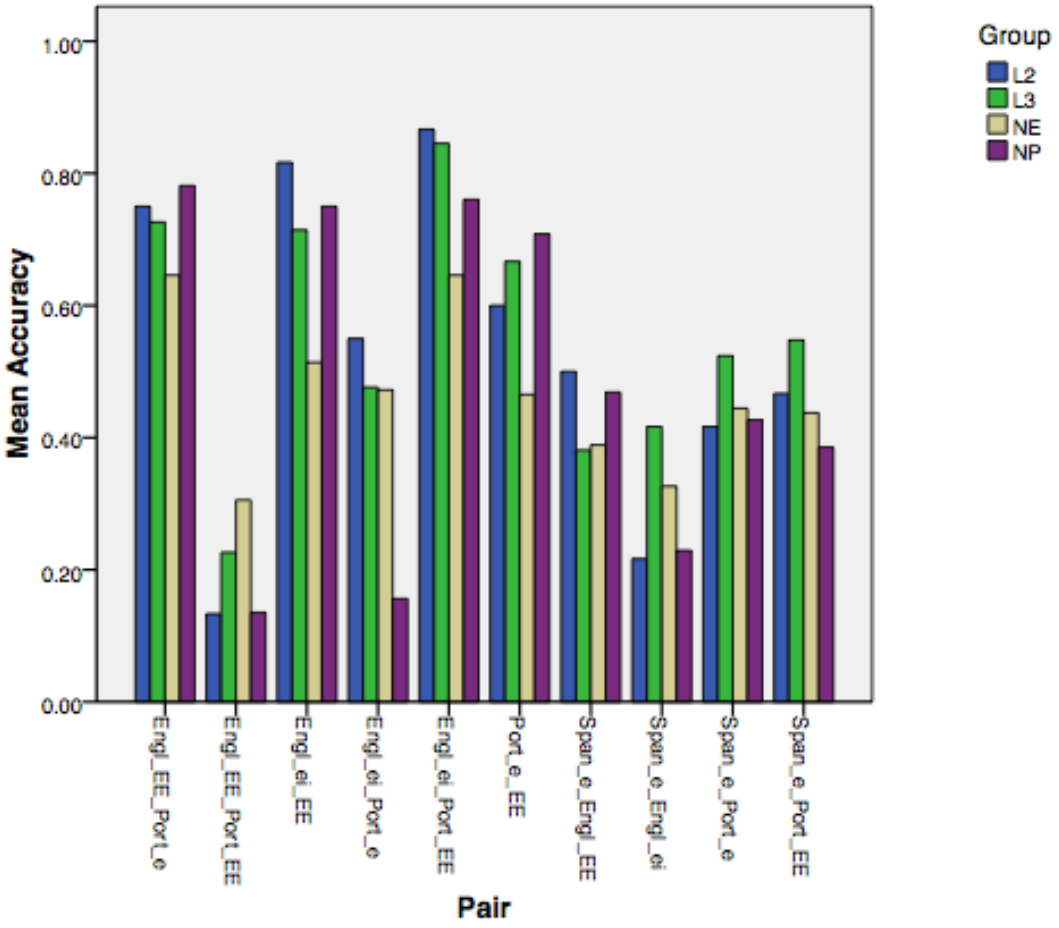

Figure 4.2: Proportion of mean accurate responses in change trials for each vowel contrast-pair (see list in Table 4.3) separated by language group (L2, L3, NE and NP). Contrast-pairs, which follow, left to right, the same order as the vowel pairs in Table 4.3 above, are labeled according to the vowel categories and languages they consist of. A score of 1.0 indicates perfect discrimination, and a score of 0.33 indicates insensibility to a contrast.

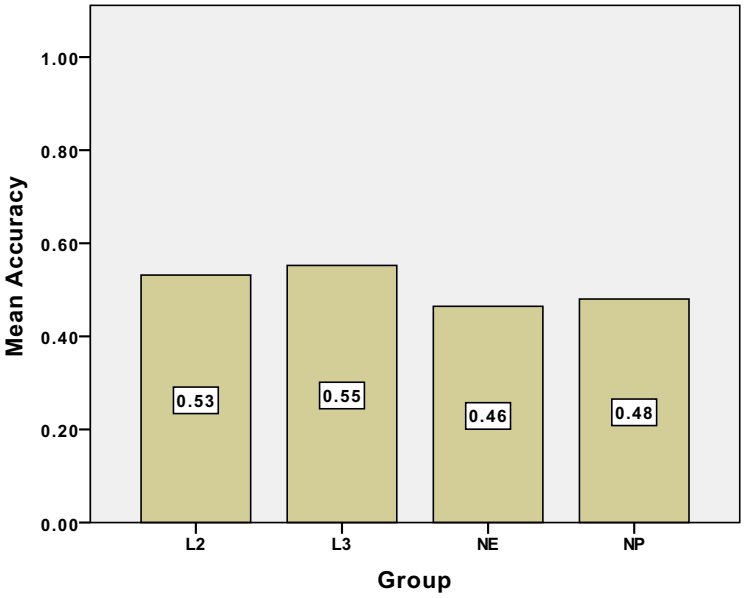

Figure 4.3: Proportion of mean accurate responses in change trials for each language group (L2, L3, NE and NP). Numbers inside bars represent Mean accurate response scores for each group. A score of 1.0 indicates perfect discrimination while a score of 0.33 indicates insensitivity to a contrast. No statistically significant differences were found among groups.

A one-way analysis of variance tested the simple effect of language group across all change trials, but it revealed no significant differences among the groups. Furthermore, a violation of the assumption of equality of variances was revealed after conducting Levene's Test of Equality of Error Variances ($F(3, 316) = 2.062$, $p = .105$) to investigate the potential effect of having such diverse number of participants in the language groups. The proportion of mean accurate responses to change trials by each language group are shown in Figure 4.3.

The group by pair interaction revealed by the 4x10 mixed-design ANOVA above is also observable in Figure 4.2. Chance level is 33%, as there was only one correct answer in each trial out of three possible ones. Figure 4.2 shows that the contrast-pairs that received the fewest accurate responses are English /ɛ/ - Portuguese /ɛ/ and English /e/ - Portuguese /e/, although the latter received more accurate responses than the former. This observation is consistent with the results

from Experiment 4.1, in which it was shown that Portuguese /ɛ/ is perceptually assimilated to English /ɛ/, for which it was considered to be a good fit. Accurate responses for English /e/ - Portuguese /e/ pairs were slightly more frequent. This is consistent with the more limited goodness of fit between Portuguese /e/ and English /e/ showed in Experiment 4.1.

It is initially surprising to observe that the NE group obtained the lowest accuracy score in the within-English contrast-pair, namely English /e/-English/ɛ/. One could predict that the NE group should perform best in discriminating the members of this pair. However, the results from this experiment seem to suggest that exposure to other languages improves the participants' discriminability of the different vowel categories. The NE group being the most monolingual of all four groups, its participants showed greater difficulty to discriminate between vowel tokens drawn from three different languages. In fact, the load imposed on memory by the 4-item oddity task format may have been sufficient to cause errors in the perception of even L1 vowels for all groups. The lack of near-perfect accuracy scores in neither of the vowel pairs by neither of the language groups supports this interpretation. It is nevertheless interesting to point out that those groups with experience in more than one language seem to perform better, in general, in the discrimination of the members of the different vowel pairs presented in this experiment.

The results from the pairs that included an English-Portuguese cross-linguistic contrast in which the members of the contrast were not those identified as the best fitting of all, namely English /ɛ/-Portuguese /e/ and English /e/-Portuguese /ɛ/, also seem to be revealing. Figure 4.2 suggests that participants in the L2 and L3 groups performed better at discriminating between the vowels of the English /e/-Portuguese /ɛ/ pair (fifth set of bars from left in Figure 4.2) than at discriminating between the members of the English /ɛ/-Portuguese /e/ pair (leftmost set of bars). This difference in discriminative abilities is consistent with the results from Experiment 4.1, as the better fit between Portuguese /ɛ/ and English /ɛ/ seems to be intervening to facilitate the discrimination between the English /e/-Portuguese /ɛ/ more than in the case of the English /ɛ/-Portuguese /e/ pair.

In order to explore potential differences in discrimination of specific contrast pairs between the L2 and L3 groups alone, a series of planned comparisons were performed. A separate one-factor ANOVA was run for each vowel pair with L2 and L3 as the two levels of the between-subjects language group factor. From the 10 statistical tests, only one reported a significant difference between the L2 and L3 groups, namely that of Spanish /e/ - English /e/ ($F(1, 12) = 6.176$, $p = .032$), with the L3 group obtaining higher accuracy scores than the L2 group. This significant result also supports the idea that the more varied the linguistic input the speakers have received, the better discriminative abilities they develop. On the other hand, while the lack of significant results in most of these tests might reflect a non-existent difference between the L2 and L3 groups' performance, it is also possible that the small number of subjects in these two groups (5 in the L2 group and 7 in the L3 group) might have occasioned not enough power for the statistical analyses to find such differences.

Returning to the interpretation of the connection between the results from Experiment 4.1 and the results from responses to change trials in the current experiment, the difference in goodness of fit of the Portuguese /e/ and /ɛ/ to their best-fitting English mid-front vowel found in Experiment 4.1 also seems to be yielding a different degree of discriminability between the cross-linguistic vowel pairs not involving Spanish for the L2 and L3 groups. Figure 4.2 shows that the contrast pairs involving Spanish /e/ receive low accuracy scores in comparison with the pairs not involving the Spanish vowel. Note that only the L3 group had experience with Spanish speech patterns. Since contrast pairs involving Spanish /e/ were observed to be more poorly discriminated than the pairs not involving the Spanish vowel, an exploratory 2x2 repeated measures ANOVA was performed with 'language group' (L2, L3) as the between-subjects factor, and 'contrast-type,' with two levels (that of pairs with Spanish involved and that of pairs with no Spanish involved), as the within-subjects factor.

The analysis returned a significant main effect of contrast-type ($F(1, 10) = 36.673$, $p = .000$), a non-significant main effect of language group ($F(1,10) = .509$,

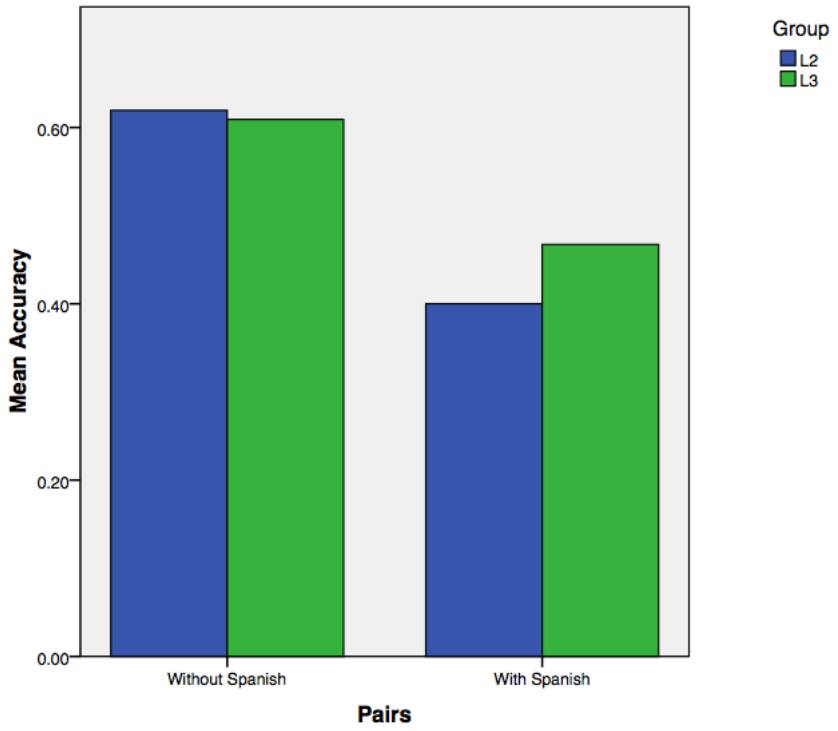

Figure 4.4: Proportion of accurate responses in change trials by the L2 (in blue) and L3 group (in green), separated by trials with (right side) and without (left side) the Spanish mid-front vowel involved. Graph shows that both groups obtained significantly higher accuracy scores when discriminating between vowel pairs which did not contain the Spanish vowel than when discriminating between pairs that involved Spanish.

p= .492), and a non-significant two-way interaction (F (1,10) = 1.691, p= .223), indicating that both groups discriminate better between the members of pairs that do not contain a Spanish vowel than between the members of pairs that contain the Spanish vowel. This difference is also visible in Figure 4.4, where the left-side columns, that is, those representing discrimination accuracy between the members of a vowel contrast-pair not involving Spanish, are significantly higher than the right-side columns, which represent the accurate discrimination score obtained by the two groups when responding to stimulus-strings that contained the Spanish mid-front vowel.

On the other hand, the 2x2 repeated measures ANOVA reported a non-significant effect of language group, which indicates that no statistically significant difference was found between the L2 and L3 groups in the proportion of accurate discrimination scores they obtained. While the validity of these results should not be underestimated, it is important to reconsider the possibility that the low number of participants in these two groups be causing the statistical model to have insufficient power to find any relevant differences between them. In accordance with this alternative explanation of the results, Figure 4.4 also suggests that the L3 group has a tendency to discriminate better between the members of the pairs that contain the Spanish vowel than the L2 group does, which would support the idea that a more varied input has positive effects on the speakers' cross-category discriminative abilities.

To summarize, the results of the analyses performed on the responses to change trials were consistent with the results from the goodness of fit test in Experiment 4.1 in that the more limited goodness of fit between Portuguese /e/ and English /e/ favored the discriminability of these two vowel sounds while the close to perfect fit between the Portuguese /ɛ/ and the English /ɛ/ hindered their discriminability for the L2 and L3 groups. Consistent with these results, the better fit between Portuguese /ɛ/ and English /ɛ/ seems to be intervening to facilitate the discrimination between the English /e/ - Portuguese /ɛ/ more than in the case of the English /ɛ/ - Portuguese /e/ pair.

Additionally, a difference was found in the way the L2 and L3 groups responded to the contrast pairs containing or not containing the Spanish vowel. Both groups showed a higher proportion of discrimination scores when exposed to stimulus-strings where no Spanish vowel was involved than when responding to sequences that did contain the Spanish vowel.

Lastly, the results from the analysis of the responses to the change trials in Experiment 4.2 also suggest that those groups with experience in more than one language (the L2 and L3 groups, and to a lesser degree, the NP group) seem to perform better in discriminating between the members of the different contrast

pairs presented in this experiment in general, in contrast with the truly monolingual native English speaker group (NE), whose discrimination scores were consistently lower than the other three groups'.

Results of analyses of responses to no-change trials

The analysis of the responses to no-change trials in Experiment 4.2 was carried out here separately from that of the change trials above. The reason behind this decision responds to the idea that different skills are at work when responding to these two different kinds of stimulus-strings. While both change and no-change trials were randomly presented to the participants during the experiment, in the case of change trials, the participants' task consisted of identifying the phonetically relevant differences that exist among the five mid-front vowel categories under study. In the case of the no-change trials, however, the participants' task consisted of ignoring the audible but phonetically irrelevant differences that exists among different realizations of a single phonetic category. The participants' responses to the no-change trials are presented in Figure 4.5.

The potential differences between groups in their accurate responses to no-change trials were tested in a 4x5 mixed-design ANOVA where 'group' was the between-subjects factor with four levels (NP, NE, L2, L3) and 'vowel' as the within-subjects factor with five levels (English /ɛ/, English /e/, Portuguese /e/, Portuguese /ɛ/, Spanish /e/). This analysis yielded a significant effect of 'vowel' ($F(4, 112) = 38.265$, $p = .000$), a significant effect of 'group' ($F(3, 28) = 5.703$, $p = .004$), and a significant vowel by group interaction ($F(12, 112) = 2.617$, $p = .004$).

The interaction between 'group' and 'vowel' can be observed in Figure 4.5. Figure 4.5 suggests that all four groups do worse in identifying the trials as no-change trials when they are exposed to vowel sequences containing tokens of Spanish /e/. It also seems to suggest that the NE group is the least accurate in responding to no-change trials of all vowel categories except in the case of Spanish /e/, where the other language groups find it approximately as difficult.

A series of follow-up univariate analyses of variance were performed in order

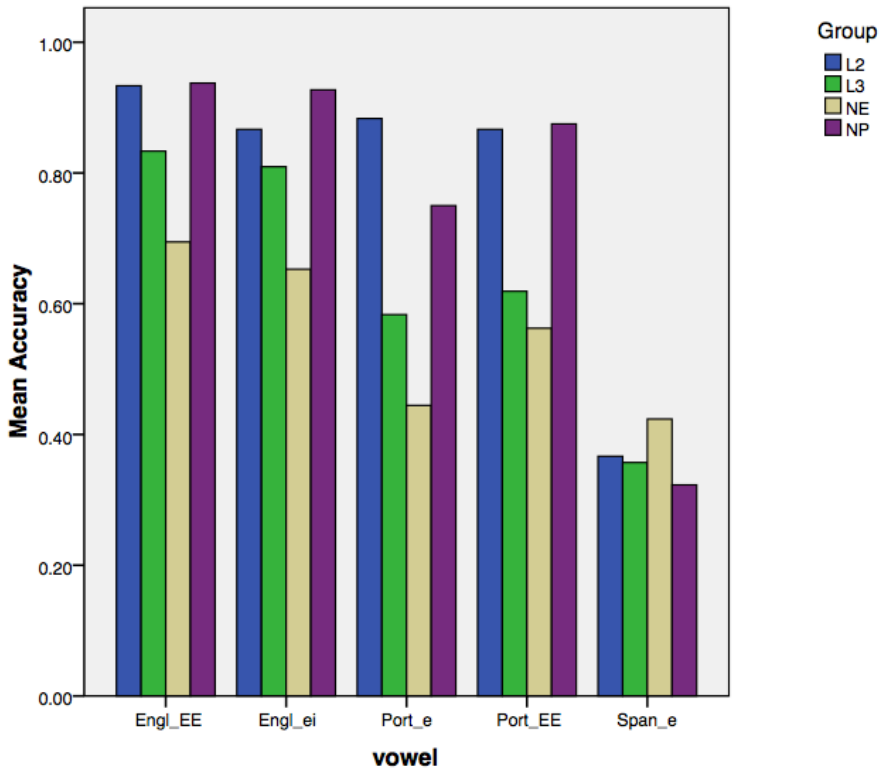

Figure 4.5: Proportion of accurate responses in no-change trials consisting of each of the five mid-front vowels of English, Portuguese and Spanish as provided by the NP, NE, L2 and L3 groups. A score of 1.0 indicates perfect identification of the stimulus-strings as representing a single vowel category while 0.33 represents chance level.

to explore the effect of language group across each vowel. These analyses yielded a significant effect of language group for English /ɛ/ no-change trials ($F(3, 32) = 4.856$, $p = .008$), as well as for English /e/ ($F(3, 32) = 4.849$, $p = .008$), Portuguese /e/ ($F(3, 32) = 5.622$, $p = .004$), and Portuguese /ɛ/ ($F(3, 32) = 4.387$, $p = .012$). A non-significant effect was found for Spanish /e/, which is consistent with the interpretation of the generalized low accuracy scores for Spanish /e/ for all language groups in Figure 4.5 above.

The significant group effects in the previous series of univariate analysis of variance were further explored to investigate the potential difference between the L2 and L3 groups' performance specifically. Planned comparisons of the L2 and L3 groups' accuracy scores were conducted for each vowel except for Spanish /e/, as no effects of language group were found in the univariate ANOVAs reported above. The four separate univariate ANOVAs conducted on the accuracy scores of the two learner groups for the vowels of English and Portuguese rendered one statistically significant difference, namely in the case of no-change trials consisting of Portuguese /e/ vowel tokens ($F(1, 12) = 6.608$, $p = .028$), where, as it is observable in Figure 4.5, the L2 group performed significantly better than the L3 group. The analysis also shows a non-significant trend in the same direction (L2 group performing better than L3 group) for Portuguese /ɛ/, and to a smaller degree for both English vowels. This trend is also visible in Figure 4.5.

4.3 Summary of overall results and Discussion

The results from Experiments 4.1 and 4.2 in this chapter suggest that the learners of Portuguese assimilate the two mid-front vowels of Portuguese to different English mid-front vowels, Portuguese /ɛ/ to English /ɛ/ and Portuguese /e/ to English /e/, but the assimilation of the Portuguese vowels to their best fitting English vowel is not equally good. More specifically, the fit of Portuguese /ɛ/ to English /ɛ/ is almost categorical (90-95% of the time), while they identify Portuguese /e/ with English /e/ only 60-65% of the time. These results suggest that both groups have

perceptually learned to differentiate between the two Portuguese mid-front vowels in comparison to the NE group, who were not found to be sensitive to that difference, as expected from their lack of previous exposure to Portuguese speech patterns.

This above mentioned difference in goodness of fit between the two Portuguese mid-front vowels to the English ones is reflected in the pattern of discrimination between cross-linguistic vowel pairs involving these two languages in Experiment 4.2. More specifically, the contrast pairs composed of the two assimilated vowels were found to be especially hard to discriminate with the best fitting pair (English /ɛ/-Portuguese /ɛ/) being the hardest. The more limited goodness of fit between Portuguese /e/ and English /e/ favored the discriminability of these two sounds while the close to perfect fit between Portuguese /ɛ/ and English /ɛ/ hindered their discriminability for the L2 and L3 groups. This difference in goodness of fit also seemed to be intervening to facilitate the discrimination between English /e/ and Portuguese /ɛ/, more so than in the case of the English /ɛ/ - Portuguese /e/ pair.

The results from the analysis of responses to change trials in Experiment 4.2 also revealed, on the one hand, that both the L2 and L3 groups discriminated more accurately between sounds that did not involve Spanish /e/ than they did when discriminating between vowel pairs that did not contain Spanish. On the other, the results from these analyses also suggested that those groups with experience in more than one language seem to perform better in discriminating between the members of different contrast pairs. This was especially true for the L2 and L3 groups, and, to a lesser degree, for the NP group, who was bilingual although dominant in their L1, Portuguese. The NE group, however, representing a truly monolingual case, consistently obtained lower scores than the L2, L3 and NP groups. These results support the view that, when a new phonetic category is established during language acquisition, there's an increase in sensitivity to differences between instances of the newly formed category and other categories.

Lastly, the analysis of the participants' responses to the no-change trials in Experiment 4.2 also yielded interesting results. The analysis of this data showed that all four groups did worse in identifying the sequences as no-change trials when they

were exposed to vowel sequences containing tokens of Spanish /e/. This finding is consistent with the conclusions drawn in Chapter 2 that Spanish /e/ is the vowel that (a) is the most stable of all five mid-front vowels tested and (b) falls in the middle of all other mid-front categories in the vowel space. Additionally, the NE group proved to be the least accurate in responding to no-change trials of all vowel categories except in the case of Spanish /e/, where all groups found it similarly difficult. Most relevantly, a follow-up analysis of the L2 and L3 groups' responses to no-change trials alone revealed that the L2 group did a significantly better job at identifying Portuguese /e/ sequences as no-change trials than the L3 group did, with a non-significant trend in the same direction for Portuguese /ɛ/, and to a lesser degree, for English /e/ and /ɛ/.

Note that the same two Portuguese learner groups who participated in the perception experiments reported in this chapter showed that they had created a single new acoustic category in the production study reported in Chapter 3. This category was found to be different from both their English /e/ and /ɛ/ categories and the target Portuguese /e/ and /ɛ/ categories. Since both SLM and PAM make their predictions on the assumption that accurate perception precedes accurate production of foreign speech sounds, it is concluded here that a considerable degree of learning has occurred for these learners with regards to perceiving the mid-front vowels of Portuguese but that they are still in the process of learning to articulate them in a native-like fashion. According to SLM and PAM-L2, with increasing exposure to adequate meaningful input, the learners will be more likely to break their new single Portuguese mid-front category into two phonetic categories that approximate Portuguese /e/ and /ɛ/ targets.

CHAPTER 5

DISCUSSION AND CONCLUSIONS

5.1 Introduction

Through the course of this dissertation, the acquisition of the seven stressed oral vowels of Brazilian Portuguese by English native speakers has been investigated. The first chapter presented a comparative analysis of the native stressed vowels of all three languages: English, Spanish and Portuguese. The acoustic similarities and differences identified from these comparisons served to create predictions for the acquisition of Portuguese by native speakers of English who had, or had not, previous and more extensive experience with Spanish speech patterns. The acquisition of the Portuguese vowels by these speakers was researched by analyzing the Portuguese vowel productions of the two groups, as well as their perception of the Portuguese mid-front vowel categories.

5.2 Summary of Results

5.2.1 Comparison between native vowel inventories

Chapter 2 in this dissertation included a comparison of the vowel inventories of American English, Mexican Spanish and Brazilian Portuguese based on the productions by three groups of native speakers of these languages. As expected, each of the traditionally defined vowels tested within each language were found to be distinct phonetic categories, occupying different areas in the vowel space and never overlapping each other. In addition, the results from the multiple cross-linguistic vowel comparisons showed that the relationships among the stressed vowels of English, Spanish and Portuguese are varied. On the one hand, some vowel categories were not found to differ from one another with respect to the parameters studied in

this dissertation, while others were found to differ in varying measures and degrees. The cases of the English, Spanish and Portuguese /i/, the Spanish and Portuguese /u/, or Spanish and Portuguese /a/ are three of the cases where these vowels did not prove to be acoustically distinct. The comparisons among the mid-back vowels in these three languages on the one hand, and the mid-front vowels on the other, revealed a diversity of ways in which these vowels differ. More specifically, amount and directionality of formant movement, as well as differences and similarities in formant structure at different points through the vowel, were found to be relevant factors interacting to create significant phonetic category differentiation. One additional interesting finding was that American English /u/ is located in a position much more fronted than expected, almost as fronted as English /e/, in agreement of previous accounts of this relatively recent phenomenon.

5.2.2 L2 and L3 speakers' production of Portuguese vowels

The results from the acoustic comparisons between vowel inventories in Chapter 2 were used to create predictions for the acquisition of Portuguese vowels by two groups of native speakers of English, using the principles of PAM and SLM as the basis. One group of participants represented English speakers who learned Portuguese as their L2 (L2 group), while the second group had learned Portuguese as their L3, having learned Spanish earlier as their L2 (L3 group).

The predictions made with basis on PAM and SLM were only partially fulfilled in the case of most of the vowel groups. In the case of the acquisition of Portuguese /u/, the L2 groups showed signs of phonetic category assimilation, as predicted, while the L3 group presented a case of category dissimilation. For Portuguese /a/, it was predicted that both groups would produce a single Portuguese /a/ located in between the two English low vowels, which both groups did. However, the both groups were expected to produce a Portuguese low vowel with a wider distribution than the Natives', and the L3 group exceeded once again these expectations by producing a low Portuguese vowel that did not differ from the Natives' in width of distribution.

The predictions for the mid-front and mid-back vowels of Portuguese, on the other hand, were not met, and no significant acoustic differences were found between the two learner groups' productions. For Portuguese /o/-/ɔ/, both groups produced two distinct mid-back vowels differing in height instead of the single vowel that was predicted, even though the height difference between the two new categories did not reach target-like extent. The two new categories, however, did differ in degree of diphthongization, a measure not used by Native speakers of Portuguese to distinguish them. In the case of Portuguese mid-front vowels, the two learner groups were expected to produce Portuguese /e/ and /ɛ/ vowels that showed similar acoustic features to their English mid-front vowels. The results, however, contradict this prediction. Both learner groups prove to have created a single new phonetic category, with similar diphthongization pattern as the target Portuguese mid-front vowels, but acoustically different from both target Portuguese vowels on the one hand, and from both English mid-front vowels on the other. To investigate the potential perceptual basis for these results, the experiments in Chapter 4 were performed.

5.2.3 L2 and L3 speakers' perception of Portuguese /e/-/ɛ/

In Chapter 4, a 'goodness of fit' task tested the perceptual assimilation of the Portuguese mid-front vowel categories to English ones, showing that both groups of learners assimilate the two mid-front vowels of Portuguese to different English mid-front vowels, Portuguese /ɛ/ to English /ɛ/ and Portuguese /e/ to English /e/, but not equally well, as Portuguese /ɛ/ was found to be a better fit to English /ɛ/ than Portuguese /e/ was to English /e/. The results from the phonetic assimilation experiment also indicate that the two learner groups were able to distinguish between the two Portuguese vowels in contrast with the English monolingual group, who were not found to be sensitive to this difference. The difference between the learner groups and the English monolingual group was interpreted to respond to a difference in amount and quality of relevant input, as the English monolingual group lacked previous exposure to Portuguese speech patterns.

A '4-item oddity' task investigated the perceptual cross-linguistic discrimination

of the mid-front vowels of English, Spanish and Portuguese by the same four groups that participated in the 'goodness of fit' task above: namely the Native English, Native Portuguese, L2 and L3 groups. The results from this experiment further support the findings from the 'goodness of fit' test as they show that the stimulus-strings consisting of the best fitting mid-front vowel pairs were the hardest to discriminate, and the discriminability decreased as the perceptual similarity increased. In addition, participants did worse in discriminating between vowel pairs that involved Spanish /e/, with which only the L3 group had experience.

Most interestingly, the results from the analysis of change trials in this experiment suggest that those participants with experience in more than one language did better than those with less varied linguistic background. This was especially true for the L2 and L3 groups, and to a lesser degree for the Native Portuguese group, who were L1-dominant speakers of English as an L2. The truly monolingual Native English group showed the least sensitivity to distinctions between the mid-front vowels in this experiment. All in all, these findings support the idea that the more varied one's linguistic experience is, the more sensitive one becomes to variations in the speech signal.

5.3 The importance of input in L2 and L3 speech learning

The results from the learners' production experiments reported in Chapter 3 showed that, whenever the Spanish and Portuguese native categories were acoustically indistinguishable (as suggested by the native speakers' production data in Chapter 2), the L3 group, that is, the group that had had exposure to Spanish speech patterns from a younger age and for longer, tended to differ from the L2 group in their production of such Portuguese vowel categories. One example of this difference in production patterns is the case of the acquisition of Portuguese /u/, which was found to be acoustically indistinguishable from Spanish /u/. In this case, the L3 group produced a Portuguese /u/ that was higher than the target and further away from their own English /u/ than the Natives'. The L2 group's Portuguese /u/, on

the other hand, was at the same height as the target but did not reach target-like position in backness. It was concluded that the L3 group had undergone a process of phonetic category dissimilation while the L2 group showed traces of phonetic category assimilation.

In the case of the Portuguese /a/ produced by the two learner groups, which was also found not to differ from Spanish /a/ in Chapter 2, despite the fact that both groups had created a new category for Portuguese /a/ that was similarly related to their own English /æ/ and /ɑ/, the L3 group produced their Portuguese /a/ vowels with a target-like within-category distribution, while the L2 group showed a wider distribution overall. It is thus concluded here that it is the more extensive exposure of the participants in the L3 group to the specific Spanish/Portuguese acoustic category /a/ and/or /u/ that causes a difference in production between the two learner groups.

In those cases in which the Spanish vowel was found to be a different acoustic category from the target Portuguese one(s), the two learner groups did not differ in their production of the target category/categories. Such is the case of Portuguese /o/ and /ɔ/. In this case, Spanish /o/ was found to be located right between the two Portuguese mid-back vowels in the vowel space, so exposure to this vowel for a longer period was predicted not to aid in reaching target-like production of the Portuguese mid-back vowels. In fact, the L2 and L3 groups were found not to differ in their production of the Portuguese /o/-/ɔ/ pair as predicted. In the case of the Portuguese /e/-/ɛ/ pair, no statistically significant acoustic differences were found among the productions of the two learner groups. The case of the mid-front vowels of Portuguese is similar to that of the mid-back vowels. Spanish /e/ is a different phonetic category from Portuguese /e/ and /ɛ/, and the learners' production of the mid-front vowels of Portuguese did not differ.

One of the main conclusions in this dissertation is thus that, at least as it regards to the acquisition of the sound system of a new language, it is not the case that the more languages one learns, the easier it will be to produce a native-like sound in a new language, as it is commonly believed. It is claimed, however, that the longer

the learner has been exposed to a specific sound, whether that sound was heard as part of one's first, second or third language, the more likely it is that the learners will be able to reach native-like realization of that specific sound.

According to SLM, "the quantity and quality of L2 input received by an L2 learner is an important determinant of ultimate degree of attainment in an L2" (Flege (2007)). Flege and his colleagues explore the relationship between input and age effects in various large-scale research projects (Flege et al. (1999b, 1995b)). The results from these projects were often explained in a way that age of acquisition of the foreign language, and/or age of arrival in the foreign country in the case of immigrants, were the most significant factors in determining the L2 speakers' success in the accurate production and perception of the L2.

Nonetheless, when the variable of percentage of use of the L1 is introduced in these research studies, the authors observe that the amount of L1 used by the L2 speakers in their daily lives was inversely correlated with the amount of L2 they used, and that the greater the percentage of L1 used, the stronger foreign accent the learners showed in the L2 and viceversa. These factors were reported to be closely related to age of arrival in the foreign country, with L2 speakers who arrived later in life using more L1 than those who arrived earlier in life (Flege et al. (1999b, 1995b)). Interestingly, Flege also suggests that, in these research studies, the participants who reported using greater amounts of L1 in their daily lives were also more prone to receiving smaller quantities of native input in the L2 than the ones who reported using less L1, as the interactions in their daily lives were likely to happen less frequently with native speakers of the L2, and/or they were more likely to interact more frequently with non-native speakers of the L2. Under these circumstances, the participants who used their L1 more frequently were likely to have received less meaningful input in their L2 than those who reported not to use their L1 often.

This interpretation of Flege's data is consistent with the conclusion from this dissertation that the longer the learner has been exposed to a specific sound, the more likely it is that they will be able to reach native-like pronunciation of that

sound, regardless of whether the specific sound has been heard as belonging to the L1, L2 or L3. In the case of the L3 group studied in this dissertation, they had been receiving Spanish/Portuguese /u/ and /a/ input for an average of 5 years longer than the L2 group[1], and this had an effect on their production of these sounds in Portuguese.

In the case of the Portuguese /u/, the L3 group showed evidence of phonetic category dissimilation, with their Portuguese /u/ being located further up in the vowel space than the target Portuguese /u/ and further from their own English /u/ than in the case of the L2 group. According to SLM, phonetic category dissimilation is the mechanism by which speakers assure distinguishability among native and non-native sound categories in more advanced stages of L2 acquisition. On the other hand, the L2 group showed traces of phonetic category assimilation, with their Portuguese /u/ located between their English /u/ and the target native Portuguese /u/ in the vowel space. The process of phonetic category assimilation is described under SLM as occurring in initial stages of L2 speech acquisition, when the learners assimilate the new L2 sounds to the closest L1 sounds depending on perceived phonetic similarities among them. Under this perspective, the L3 group can be assumed to have received more Portuguese /u/ and /a/ input due to their more extensive exposure to these sounds through Spanish, while both the L2 and L3 groups can be assumed to have received similar amounts of input with regards to the mid vowels of Portuguese, as these vowels are acoustically different from any other vowel category any of the two groups were exposed to before the onset of Portuguese language acquisition.

[1]note that the L3 group's onset of Spanish language acquisition was around the age of 13, while both learner groups had started learning Portuguese around the age of 18.

5.4 Speech perception, production and signs of learning in the acquisition of a new sound inventory

The relationship between speech perception and speech production in second language acquisition has been widely debated over the last few decades: Can learners accurately produce sounds that are not accurately perceived, or is accurate perception a prerequisite to accurate pronunciation? Although this dissertation follows the principles outlined by the SLM (and PAM), which makes predictions based on the hypothesis that the accuracy with which L2 sounds are produced is limited by how accurately they are perceived, this topic was researched here by including the perception experiments in Chapter 4. These experiments were conducted following the surprising results of the production experiments in Chapter 3, which showed that the L2 and L3 groups inaccurately produced a single mid-front Portuguese vowel, different from the two English mid-front vowels as well as the two target Portuguese vowels, but a single phonetic category, and not two, nonetheless. The question was raised, then, of whether there could be a perceptual basis for this phenomenon in production.

While no conclusive answer to the issue at hand can be drawn by starting from the inaccurate production of specific sounds (this starting point does not allow for the consideration of the opposite alternative, namely that of accurate production being possible regardless of inaccurate perception), the results from the current study support the idea that perception precedes production. More specifically, the results from the 'goodness of fit' experiment suggest that the two learner groups have undergone perceptual learning of the Portuguese mid-front vowels, as their performance in this perceptual assimilation test is much better than the Native English group's. Despite the inaccurate production of these two Portuguese vowels by the two learner groups, the perception data shows that both groups have learned to distinguish the two vowels of Portuguese and they assimilate each of them to a different English vowel, though not perfectly nor equally well. By contrast, the Native English group, with no experience in Portuguese at all, performed very poorly in this task. The fact

that the L2 and L3 groups do such a good job in this assimilation test in comparison with the Native English group indicates that perceptual learning must have taken place, but that they might not have reached the point in which they are able to realize this learning in production.

Other signs of learning were found in the results of this study, despite the fact that native-like vowel production was not attained in any case. Some of these signs relate to the predictions by the SLM that phonetic category assimilation is more likely to occur in initial stages of acquisition while phonetic category dissimilation is more likely in more advanced stages. Such is the case of the L2 group's (a case of assimilation) and the L3 group's (a case of dissimilation) Portuguese /u/. Since Portuguese /u/ was found to be acoustically very similar to Spanish /u/, the L3 group represents a case of a more advanced stage of acquisition for having been exposed to this phonetic category for a longer period, that is, since the onset of learning Spanish. The L2 group, on the other hand, represents an earlier stage of acquisition for having been exposed to the Portuguese/Spanish /u/ for a shorter period, and thus, having received more limited meaningful input.

The results from the production of Portuguese /o/ and /ɔ/ by these two groups is of a different nature. While the L3 group produces Portuguese /o/ and /ɔ/ instances that differ in both height and backness from each other, like the Native Portuguese group does, they do so in a smaller degree. The L2 group, on the other hand, produces a difference in height but not in backness between the two Portuguese mid-back vowels. These results suggest that both groups have noticed that these are two distinct phonetic categories, but there is evidence only from the L3 group of them picking up on the fact that the distinction between the two acoustic entities is characterized by both height and backness differences. In fact, both groups of learners produce mid-back vowels that differ from each other in degree of diphthongization. This is not an acoustic feature that the Native speakers of Portuguese use to distinguish between the two sounds and it differs from the direction and degree of diphthongization of their Native English /o/, so it is interpreted here that the learners are trying to realize their knowledge of the difference between the

two vowels but that they do so by using the wrong phonetic cues. In this case, the inaccurate implementation of an unidentified but nevertheless perceived phonetic distinction is interpreted as an indicator of learning. In connection with the discussion in the previous section, it is likely that they will need to be exposed to more adequate input in order to be able to identify the appropriate source of phonetic difference between the mid-back vowels of Portuguese before they are able to realize it accurately in production.

Additionally, the finding that the L3 group produces a Portuguese /a/ category that does not differ significantly from the Native Portuguese group's in within category variance (as indicated by a similar width of distribution), while the L2 group differs from both of them, is also interpreted here as a sign that learning has occurred. Again, the fact that Portuguese /a/ and Spanish /a/ were found to be acoustically the same is here interpreted as having facilitated the more target-like production of this vowel by the L3 group in contrast with the L2 group, who had been exposed to this category for a shorter period of time. However, the L2 group also produced a Portuguese /a/ which, despite not reaching target-like width of distribution, proved to be different from both their English low vowels and target-like in location in the vowel space. Learning, in the case of the L2 group, has thus undoubtedly occurred.

In conclusion, whether the observed phenomenon included a case of phonetic category interaction (assimilation or dissimilation), consisted of misinterpreting and misplacing the source of a phonetic contrast, or involved the accurate identification and partial or complete realization of relevant phonetic detail, it is concluded here that all of these are signs that the process of learning is in process. This is consistent with the multiple previous accounts of phonetic category interactions in L2 learning.

5.5 Searching for a model of L2 and L3 phonetic acquisition

The two models of cross-linguistic speech perception or speech learning discussed in this dissertation, namely SLM and PAM, were found to provide certain amount

of predictability in the case of the acquisition of the Portuguese stressed vowels discussed in this study. However, SLM was designed to address quite proficient L2 speakers and most frequently serves to study acoustic data, while PAM was designed to address non-native perception and discriminability of foreign sound contrasts by naïve listeners. While SLM makes predictions based on acoustic data that could best be tested on acoustic data, PAM bases its predictions on perception data and its predictions strictly address perceptual learning.

In this dissertation, it has been assumed that relevant acoustic differences in the production data were to some extent related to differences relevant for perception and contrast discriminability. However, this relationship has not explicitly been tested, and it might be the case that acoustic differences that are relevant statistically might not be relevant for perceptual discrimination of L2 phones. Given the common assumption to these models that accurate perception precedes accurate production of L2 sounds, a more adequate model of L2 and post-L2 speech learning should address both production and perception data and make predictions for both.

A new post-L2 speech learning model should also address the importance of input in post-L2 language learning, embracing the conclusion from this study that accurate L3 phonetic input might be received not only from exposure to L3 but also from exposure to other languages that contain phonetic segments with similar acoustic characteristics as the target L3 phones.

5.6 Vowel dispersion and L3 speech learning

As it was mentioned in Chapter 3, production studies have shown that the crowdedness of vowel inventories can affect the acoustic distance between their phonetic categories, with categories in bigger inventories being more dispersed in the vowel space than those in smaller inventories (see Bradlow (1995) for an analysis of the English and Spanish vowel inventories and Jongman et al. (1989) for Greek and German). The possibility was explored in this dissertation that the same kind of effect might be found in the case of the vowel inventories of speakers of multiple

languages, as their vowel inventories would potentially be more crowded than the vowel space of monolingual speakers. If this were the case, we would expect the Portuguese vowels of the participants of the L3 group in this dissertation to be located further out in the vowel space than those of the L2 group, which could be more dispersed than those of the Native Portuguese speakers. The results from the different comparisons made in this study suggest that this is indeed the case.

First of all, the Portuguese /u/ produced by the L3 group is as far back but interestingly higher in the vowel space than the one produced by the Native speakers of this language. Second, the Portuguese /i/ produced by both the L2 and L3 groups, which happened to be acoustically similar to their own English /i/, was also found to be located in a more fronted position than that of the Native Portuguese speakers. And third, the Portuguese /a/ produced by the L3 group, as well as their own English /a/ and /ɑ/, were found to be located lower than the L2 group's in the vowel space. The more spread out location of the cardinal vowels of Portuguese as produced by the L3 group in comparison with the L2 group's, support the claim that more crowded vowel inventories may be expanded with respect to acoustic spaces containing a smaller number of vowel categories. In the same line of thought, these results support the idea that phonetic category interaction in the process of L2 and/or L3 speech learning may affect the size of the vowel space of the learners, as more cases of phonetic category dissimilation occur and the categories get more dispersed to maintain contrast.

5.7 Acquiring multiple languages in different orders

Order of acquisition of the different languages has been proposed to be an important factor in the acquisition of L2 and L3 speech patterns. According to Smith & Baker (2007) in his study of English(L1)-German(L2) and English(L1)-French(L2) bilinguals exposed to French and German vowels as an L3 respectively, the more diverse the existing phonetic category inventory is, the better new sounds will be perceived. The same conclusion was drawn by Iverson & Evans (2007) in their study

of the perception of English vowels by individuals with native vowel inventories differing in size.

This finding is, a priori, counterintuitive under the SLM. According to the predictions made by the SLM, it seems to be the case that the more crowded the perceptual space of the speaker is, the least likely it is that they will be able to perceive differences between new phonetic categories and the closest already existing ones. The conclusion from the above mentioned studies, on the other hand, brings into play the fact that, the more experience one has with different kinds of phonetic cue weighting and with evaluating the phonological significance of audible acoustic distinctions, the more likely it is that new contrastive acoustic differences will be noticed when encountering a new set of sounds. Once again, as in the discussion of the importance of input in L2 and L3 speech learning above: the more, the better. While the SLM seems to initially contradict this view, if the 'perceived phonetic distance' that forms the basis of the SLM's predictions for establishing new phonetic categories is understood as taking into account all acoustic information that the speaker is able to evaluate, then the two views presented in this section cease to differ from each other.

While the present study presents a single order of acquisition and thus does not lend itself to drawing conclusions in this respect, the results of this project may serve as comparison for a future study in which the order of acquisition of English, Spanish and Portuguese differ. It would be interesting to compare the results of the current project with data from native speakers of Spanish who are learners of Portuguese with or without English as the L2. In such a case, the L2 group's vowel inventory at the onset of learning Portuguese would consist of just the five vowels of Spanish, while the L3 group's vowel inventory at the same stage would include the five vowels of Spanish plus the vowels of English that were acquired as an L2 prior to being exposed to Portuguese speech patterns. If that were the case, the much more limited experience of the L2 group with vowel sounds should, following the conclusions by Iverson & Evans (2007) and Smith & Baker (2007) above, constitute a big disadvantage at the time of identifying relevant acoustic

differences to accurately perceive the Portuguese speech sounds. The L3 group, on the other hand, would be able to use their knowledge of the differences relevant to distinguish between the vowels of Spanish as well as English when perceiving Portuguese vowel sounds, which should thus allow them to perceive them better than the L2 group. The study of such speaker groups would, at the same time, allow for the investigation of the role of typology in L3 speech learning, as Spanish and Portuguese are perceived as typologically closely related languages in contrast with English, which is typologically more distant.

An alternative way of exploring the effect of order of acquisition with speakers of these same languages would be to study the Spanish vowels of native speakers of English who may or may not have studied Portuguese as the L2 prior to the learning of Spanish. In this case, the acquisition of each of the languages by the native English speakers would involve the acquisition of a smaller vowel inventory, which might involve different processes than those observed with the participant groups in the current study. The last possible comparative study would involve the English speech of native speakers of Spanish who may or may not have Portuguese as their L2, in which case the acquisition of each new language would involve an attempt to learn a new vowel inventory that is bigger than the existing one.

In a perfect world, it would be ideal to study all the possible orders of acquisition of a specific set of languages (also following the same methods) in order to be able to draw more reliable conclusions about the effect of order of acquisition in post-L2 speech learning. The probability that we will find speaker populations representing all possible orders of acquisition with similar linguistic experiences is, unfortunately, practically nonexistent. Nevertheless, an effort can be made to shed some light on this issue by studying bilingual populations learning an L3, which may differ in order of acquisition of their first two languages, such would be the case of Spanish-Basque bilinguals learning English as the L3 in the Basque country, Spanish-Catalan bilinguals learning English as their L3 in Catalunya, or French-English bilinguals learning an L3 in Québec, Canada.

5.8 Pedagogical applications of current research

The results from this dissertation have applications for the pedagogy of foreign languages. On the one hand, the surprising findings in the current study, including the fact that the L3 group did not accurately produce the Portuguese mid-front vowels despite the fact that they were predicted to be an easy vowel group to acquire by native speakers of English, suggest that there might be specific phonetic issues that are being overlooked in the Portuguese (and in general foreign) language classroom that are indeed problematic in the learning process. In accordance with the current findings, it is suggested here that more attention be paid to the speech patterns being produced by the students in order to identify previously unnoticed areas of difficulty and address them accordingly. On the other hand, the importance of providing the learners with as many opportunities to be exposed to authentic and varied native speech input, both inside and outside the classroom, cannot be emphasized enough. With the availability of all kinds of multimedia materials via the internet nowadays, this should not be a very difficult task. As discussed above, with exposure to more meaningful input, a faster and more efficient learning process is predicted.

As for the specific case of learners of Portuguese who already speak Spanish, as is the focus of several rapidly growing Portuguese programs in universities across the US, I agree with Carvalho et al. (2010) in suggesting that "pedagogical materials need to address both the positive and negative crosslinguistic influences between Portuguese and Spanish; and [...] activities need to [...] promote metalinguistic awareness with a goal of helping learners attend to both congruent and divergent structural characteristics between the languages" (p.71). While these recommendations are generally valid for any foreign language program, the case of Portuguese for Spanish speakers involves two languages that are both typologically and perceptually considered as closely related, which can facilitate the rapid progress in some aspects of the language acquisition process, such as listening and reading comprehension, but which can also hinder the native-like acquisition of the formal features of the target language, such as accurate grammatical or phonetic patterns, risking

a "rapid stabilization of structural errors due to speakers' ability to communicate meaning at early stages of learning" (p.71). For this reason I follow Carvalho & Silva (2006) in suggesting that metalinguistic awareness be made an intrinsic part of courses of Portuguese for Spanish speakers, as a mainly communicative approach in the classroom will most likely be insufficient to draw the learners' attention to the subtle differences between the two languages.

5.9 Future research

The effects of input and age are difficult to tell apart in the current study. Whether the better performance of the L3 group in the production of the above specified Portuguese vowels is due solely to the fact that they have received more meaningful input than the L2 group or to the fact that they were exposed to these sounds at an earlier age (13 versus 18 years old), it is impossible to conclude from this study. After a thorough discussion of the four theoretical hypotheses that attempt to explain age effects in second language acquisition (see Flege & MacKay (2010) for the complete discussion), Flege and MacKay recently conclude that none offer a convincing explanation of age effects and none was able to account for all the evidence considered. They infer that all four hypotheses might help explain different aspects of age-related effects reported in the literature at varying degrees. However, these authors suggest that, in order to ultimately learn about the true bases of age-related differences in L2 proficiency, a stronger effort must be made to investigate potentially "causative" variables such as age-related changes in the perceived relation between sounds in the L1 and L2, variations in the kind of input received, and measured (not estimated) amount of L2 input received.

The conclusions drawn from the current study support a more thorough investigation of amount and kind of input received in L2 acquisition. However, this study relied on age and order of acquisition of each of the foreign languages (Spanish and Portuguese) as the most decisive characteristic that defined the two learner groups. Without controlling for age and order of acquisition, the inherent variation in lin-

guistic background and experience among the potential participants made it very difficult to build any kind of comparison group, as differences in experience abroad, amount of L1 and/or L2 (and/or L3) used in daily life, and possibly amount and kind of education received in each language (although this information was not explicitly gathered) varied considerably from individual to individual.

I agree with Flege and his colleagues' recommendation that a more exhaustive examination of the effect of amount and quality of input received by L2 learners is needed in order to expand our knowledge about the true bases for age-related differences in ultimate attainment of L2 speech patterns. However, the field of post-L2 speech learning is still at its beginnings and more research on the relationship between L2 and L3 acquisition processes is still required.

First, the reliability of the conclusions drawn from the current study should be tested by replicating this study in different learner populations. This study focused on native English native speakers learning Portuguese who may or may not have had experience with Spanish before. Portuguese and Spanish, however, are two closely related languages, they are known to share around 70% of their lexicon and they have a common Latin origin, which is distinct from the Germanic origin of the native language of the participants. It would be revealing to replicate this study with participants whose L1, L2 and L3 are all typologically distant, or all typologically close, from one another. In addition, there is still a lot of work to be done in order to shed some light on the relative effect of order of acquisition and typology in L3 speech learning.

Second, this study could be replicated with bigger comparison groups. While the seven participants composing each of the five groups studied in this dissertation (the native speakers of English, of Spanish and of Portuguese, the L2 and the L3 group) seemed to provide enough data for the production experiments to render enough statistically significant differences among the acoustic categories produced by the participants, more participants would give the perceptual data greater statistical power. In addition, two of the seven participants in the initial L2 group were not available for taking part in the perception experiments (a phenomenon commonly

known as *subject mortality* from its use in medical research), which decreased the power of the statistical analyses on the data. It would be advisable to replicate the study with bigger participant groups in order to increase the power of the analyses.

Ultimately, longitudinal studies of the processes of L2 and L3 acquisition should be carried out. Contingent on availability of funds and the required infrastructure, a long-term study of the process of acquisition of the L2 and L3 of a specific group of individuals undergoing similar L2 and L3 acquisition processes (including kind and amount of input, linguistic and cultural backgrounds, etc) would be most adequate to investigate changes in the perceived relation between sounds in the L1, L2 and L3 through regularly scheduled perception studies, while controlling for variations in the quality and quantity of input received. Accompanying production studies should also be carried out with the same regularity. Such an enterprise would potentially require long-term collaboration with educational or cultural institutions in large immigrant-populated urban centers with multilingual populations and well-established ties to local immigrant populations so that *subject mortality* is reduced as much as possible through the course of the project. Such a project would logically require considerable financial and human resources.

5.10 Final words

All in all, the main conclusion of this study is that the longer one has being exposed to a specific sound, regardless of whether this sound belongs to one's L1, L2, L3 or additional language, the more likely it is that one will be able to reach native-like pronunciation of that sound. In the case of the Portuguese learners studied in this dissertation, they had been exposed to Spanish/Portuguese /u/ and /a/ (which were found to share the same acoustic properties) for a period 5 years longer than the L2 group, who had not learned Spanish before. This extra input materialized for the L3 group in their more native-like production of Portuguese /u/ and /a/.

In addition, this study also found that L2/L3 language learners' speech displays a very wide variety of signs that learning is indeed taking place. These signs of learning

might be observable in the accurate use of formant structure in production, in the accurate implementation of diphthongization direction and/or degree, as well as in the sources of other kinds of inaccuracies, such as the implementation of non-target-like diphthongization patterns to realize a still unidentified but perceived acoustic contrast, among other examples. It was also suggested that learning of speech patterns often involves an interaction of perception and production abilities and that production inaccuracies do not necessarily denote an absence of learning, providing support for the view that accurate perception precedes accurate production in post-L1 speech learning.

APPENDIX A

LANGUAGE BACKGROUND QUESTIONNAIRE

LANGUAGE BACKGROUND QUESTIONNAIRE

The purpose of this questionnaire is to learn about your language history. We would like to find out what languages you know, when you first learned them, and how much you use them.

1. Gender: M / F **2.** Today's date: _____

3. City and State of birth: _____

4. Age: _____

5. Do you speak any of the following languages? Please circle all that apply:

English	Spanish	Portuguese	Italian	Russian
Chinese	Japanese	Korean	French	German

Other: _____

6. What is your 1st language? _____
What is your 2nd language? _____
What is your 3rd language? _____
4th? 5th? (if applicable) _____

7. In what order did you learn/study them?
(a) _____ is my native language.
(b) _____ started learning it at age _____
(c) _____ started learning it at age _____
(d) _____ started learning it at age _____
Other (if applicable)? _____

8. Have you ever lived in a country where any of your non-native languages is spoken? For how long?
In _____ for _____
In _____ for _____
In _____ for _____
In _____ for _____

9. Please estimate, using a percentage as an index (%), what percentage of your time you have spoken each of the languages you mentioned in question 7:

...(7.a): in the past 5 years _____ ; in the past 5 months _____ ; in the past 5 weeks _____

...(7.b): in the past 5 years _____ ; in the past 5 months _____ ; in the past 5 weeks _____

...(7.c): in the past 5 years _____ ; in the past 5 months _____ ; in the past 5 weeks _____

...(7.d): in the past 5 years _____ ; in the past 5 months _____ ; in the past 5 weeks _____

10. For each of the languages in question 7, please estimate to the nearest 10% how much you use each of these languages at present in the following places or situations:

For language in (7.a.): _____

	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
While at home											
Visiting family members											
At work (including volunteer work)											
At church or church functions											
Visiting friends											
On the telephone											
While on vacation											
While shopping											
At parties and social gatherings											

For language in (7.b.): _____

	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
While at home											
Visiting family members											
At work (including volunteer work)											
At church or church functions											
Visiting friends											
On the telephone											
While on vacation											
While shopping											
At parties and social gatherings											

For language in (7.c.): _____

	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
While at home											
Visiting family members											
At work (including volunteer work)											
At church or church functions											
Visiting friends											
On the telephone											
While on vacation											
While shopping											
At parties and social gatherings											

For language in (7.d.): _____

	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
While at home											
Visiting family members											
At work (including volunteer work)											
At church or church functions											
Visiting friends											
On the telephone											
While on vacation											
While shopping											
At parties and social gatherings											

11. Please estimate your ability to speak, understand, read, and write each of the languages you mentioned in (7). Use the number "1" if your ability is poor, "7" if your ability is good, and numbers in between for ability levels that are in between.

	For language in (7.a.): _____								For language in (7.b.): _____						
	1 Poor	2	3	4	5	6	7 Good		1 Poor	2	3	4	5	6	7 Good
Speaking															
Understanding															
Reading															
Writing															

	For language in (7.c.): _____								For language in (7.d.): _____						
	1 Poor	2	3	4	5	6	7 Good		1 Poor	2	3	4	5	6	7 Good
Speaking															
Understanding															
Reading															
Writing															

APPENDIX B

RESULTS FROM CROSS-LINGUISTIC COMPARISONS OF MID VOWELS OF ENGLISH SPANISH AND PORTUGUESE AT DIFFERENT POINTS OF MEASUREMENT

Table B.1: Results from a set of linear mixed-effects models on each of the logF1 and logF2 measures of the mid back vowels of English, Spanish and Portuguese, at the three points of measurement, with Portuguese /ɔ/ as the intercept.

			25%		50%		75%
		β	t	β	t	β	t
logF1	Intercept (= Port. /ɔ/)	-2.721		2.741		2.738	
	Portuguese /o/	-0.134	-14.48*	-0.144	-17.68*	-0.124	-10.06*
	Spanish /o/	-0.051	-2.49*	-0.068	-3.43*	-0.067	-3.03*
	English /o/	-0.052	-2.56*	-0.086	-4.33*	-0.111	-5.04*
logF2	Intercept (= Port. /ɔ/)	3.010		3.015		3.045	
	Portuguese /o/	-0.003	-0.22	-0.021	-1.37	-0.006	-0.31
	Spanish /o/	0.009	0.46	-0.010	-0.40	-0.034	-1.19
	English /o/	0.115	5.47*	0.069	2.63*	0.030	1.03

Table B.2: Results from a set of linear mixed-effects models on each of the logF1 and logF2 measures of the mid back vowels of English, Spanish and Portuguese, at the three points of measurement, with Spanish /o/ as the intercept.

			25%		50%		75%
		β	t	β	t	β	t
logF1	Intercept (= Span. /o/)	2.671		2.673		2.671	
	English /o/	-0.001	-0.07	-0.018	-0.90	-0.044	-2.02*
	Portuguese /ɔ/	0.051	2.49*	0.068	3.43*	0.067	3.03*
	Portuguese /o/	-0.083	-4.07*	-0.076	-3.82*	-0.057	-2.58*
logF2	Intercept (= Span. /o/)	3.019		3.004		3.011	
	English /o/	0.015	5.01*	0.079	3.02*	0.063	2.22*
	Portuguese /ɔ/	-0.009	-0.46	0.010	0.40	0.034	1.19
	Portuguese /o/	-0.012	-0.59	-0.010	-0.39	0.028	0.98

Table B.3: Results from a set of linear mixed-effects models on each of the logF1 and logF2 measures of the mid back vowels of English, Spanish and Portuguese, at the three points of measurement, with Portuguese /o/ as the intercept.

			25%		50%		75%
		β	t	β	t	β	t
logF1	Intercept (= Port. /o/)	2.588		2.598		2.614	
	Spanish /o/	0.083	4.07*	0.076	3.82*	0.057	2.58*
	English /o/	0.081	4.00*	0.058	2.92*	0.012	0.57
	Portuguese /ɔ/	0.134	14.48*	0.144	17.68*	0.124	10.06*
logF2	Intercept (= Port. /o/)	3.007		2.994		3.039	
	Spanish /o/	0.012	0.59	0.010	0.39	-0.028	-0.98
	English /o/	0.117	5.60*	0.089	3.41*	0.035	1.24
	Portuguese /ɔ/	0.003	0.22	0.021	1.37	0.006	0.31

Table B.4: Results from a set of linear mixed-effects models on each of the logF1 and logF2 measures of the mid front vowels of English, Spanish and Portuguese, at the three points of measurement, with Portuguese /ɛ/ as the intercept.

			25%		50%		75%
		β	t	β	t	β	t
logF1	Intercept (= Port. /ɛ/)	2.713		2.734		2.741	
	Portuguese /e/	-0.140	-14.42*	-0.141	-13.59*	-0.135	-10.11*
	Spanish /e/	-0.057	-3.16*	-0.079	-4.28*	-0.089	-4.41*
	English /ɛ/	0.040	2.21*	0.017	0.93	-0.002	-0.09
	English /e/	-0.072	-4.00*	-0.118	-6.34*	-0.152	-7.55*
logF2	Intercept (= Port. /ɛ/)	3.235		3.229		3.209	
	Portuguese /e/	0.035	2.25*	0.038	3.13*	0.037	2.86*
	Spanish /e/	0.025	1.19	0.038	1.98	0.053	2.77*
	English /ɛ/	-0.062	-2.86*	-0.061	-3.16*	-0.049	-2.58*
	English /e/	0.009	0.44	0.060	3.10*	0.095	5.01*

Table B.5: Results from a set of linear mixed-effects models on each of the logF1 and logF2 measures of the mid front vowels of English, Spanish and Portuguese, at the three points of measurement, with Spanish /e/ as the intercept.

			25%		50%		75%
		β	t	β	t	β	t
logF1	Intercept (= Span. /e/)	2.656		2.654		2.652	
	English /ɛ/	0.097	5.37*	0.097	5.21*	0.087	4.31*
	English /e/	-0.015	-0.84	-0.038	-2.06*	-0.063	-3.14*
	Portuguese /ɛ/	0.057	3.16*	0.080	4.28*	0.089	4.41*
	Portuguese /e/	-0.083	-4.61*	-0.061	-3.31*	-0.046	-2.31*
logF2	Intercept (= Span. /e/)	3.261		3.267		3.262	
	English /ɛ/	-0.088	-4.05*	-0.099	-13.59*	-0.135	-10.11*
	English /e/	-0.016	-0.75	0.021	1.11	0.042	2.23*
	Portuguese /ɛ/	-0.026	-1.19	-0.038	-1.98	-0.053	-2.77*
	Portuguese /e/	0.009	0.44	-0.000	-0.03	-0.016	-0.85

Table B.6: Results from a set of linear mixed-effects models on each of the logF1 and logF2 measures of the mid front vowels of English, Spanish and Portuguese, at the three points of measurement, with English /e/ as the intercept.

			25%		50%		75%
		β	t	β	t	β	t
logF1	Intercept (= Engl. /e/)	2.641		2.615		2.589	
	Spanish /e/	0.015	0.84	0.038	2.06*	0.063	3.14*
	English /ɛ/	0.112	11.38*	0.135	12.83*	0.151	11.01*
	Portuguese /ɛ/	0.072	4.00*	0.118	6.34*	0.152	7.55*
	Portuguese /e/	-0.068	-3.77*	-0.023	-1.24	0.017	0.86
logF2	Intercept (= Engl. /e/)	3.245		3.289		3.304	
	Spanish /e/	0.016	0.75	-0.021	-1.11	-0.042	-2.23*
	English /ɛ/	-0.072	-4.46*	-0.121	-9.78*	-0.144	-10.98*
	Portuguese /ɛ/	-0.009	-0.44	-0.060	-3.10*	-0.095	-5.01*
	Portuguese /e/	0.026	1.20	-0.022	-1.15	-0.058	-3.10*

Table B.7: Results from a set of linear mixed-effects models on each of the logF1 and logF2 measures of the mid front vowels of English, Spanish and Portuguese, at the three points of measurement, with Portuguese /e/ as the intercept.

			25%		50%		75%
		β	t	β	t	β	t
logF1	Intercept (= Port. /e/)	2.573		2.593		2.606	
	Spanish /e/	0.083	4.61*	0.061	3.31*	0.046	2.31*
	English /ɛ/	0.180	9.99*	0.158	8.53*	0.133	6.65*
	English /e/	0.068	3.77*	0.023	1.24	-0.017	-0.86
	Portuguese /ɛ/	0.140	14.42*	0.141	13.59*	0.135	10.11*
logF2	Intercept (= Port. /e/)	3.271		3.267		3.246	
	Spanish /e/	-0.009	-0.44	0.000	0.03	0.016	0.85
	English /ɛ/	-0.098	-4.55*	-0.099	-5.16*	-0.086	-4.56*
	English /e/	-0.026	-1.20	0.022	1.15	0.059	3.10*
	Portuguese /ɛ/	-0.035	-2.25*	-0.038	-3.13*	-0.037	-2.86*

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